Euler's Constant:
Complete Integrated Research Manuscript

Exponential-Integral Foundations, Positive Moments, Infinite Families, Local Arithmetic, and Transfer Obstructions

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Cumulative edition — 10 September 2026

Abstract

This cumulative research manuscript brings together the gamma-related foundations of companion manuscripts [001] and [002], the earlier 35-page positive-moment paper, the subsequent analytic and infinite-family investigations, the algebraic/transcendental alternative, and the later 18-page arithmetic-transfer paper. Its purpose is to preserve the actual mathematical content across these stages, with consistent hypotheses and an explicit distinction between auxiliary theorems, published inputs, finite certificates, conditional implications, and unfinished targets. The constructions involve logarithmic elimination, positive moments, primitive lattice norms, orthogonal polynomials, denominator arithmetic, differential Galois theory, local root counts, probability, and arithmetic Gevrey series. They yield concrete identities and obstructions for the stated classes of constructions. They do not prove the irrationality or transcendence of Euler's constant. A source-coverage ledger and the original research packages accompany this edition so that corrections and retained historical claims can be traced.

Editorial status. Research manuscript; no assertion of external peer review or of novelty for every auxiliary result. The word *complete* refers to the identified source corpus and its gamma-related mathematics, not to a completed transcendence proof or to all mathematics in the two large companion manuscripts.

Contents

1 Scope, status, and how the parts fit together 5

I Foundations from the companion manuscripts 7

2 Analytic and arithmetic foundations from the companion papers 7

3 Moving levels, zero traces, and exact arithmetic detectors 9

4 Reciprocal witnesses and the distinction between a level and its carrier 11

5 Stokes data, Frobenius splitting, and the comparison gap 11

6 The earlier interior-zero and Stieltjes attacks 13

7 Positive rational kernels: the corrected infinite-family target 15

II Positive moments and primitive logarithmic elimination 19

8 Positive moments: framework and scope 19

9 The integral, its integer coefficients, and nonvanishing 20

10 Prime support and canonical logarithm elimination 22

11 A fixed-order recurrence obstruction 24

12 Normalization, short relations, and survival of target directions 26

13 Primitive targets and an exact quotient norm 29

14 Nonvanishing from polynomial positivity 30

15 Two-sided bounds and an affine feasible interval 33

16 Exact finite certificates at two levels 35

17 Positive differences of moving order 36

18 Rational moment approximants and graded denominator bounds 39

19 Companion constructions and the specialization problem 44

20 What an infinite construction must establish 48

21 Reproducibility and the theorem boundary 50

III Additional early constructions and arithmetic audits 51

22 Early constructions not contained in the first gamma paper 51

23 A full-rank logarithmic Laguerre compression 51

24 Exact logarithmic denominators and the limits of a fixed transform 52

25 The Vacca generator and a rational Markov realization 53

26 The full logarithm-free floor-integral space 54

27 Gamma filters: positivity, integer sampling and algebraic elimination 56

28 Primitive exterior elimination in the Euler–Gompertz construction 60

29 Gregory arithmetic: exact cancellation before integration 61

30 Moving factorial parameters and the exact target they approach 64

31 Sign variation, normal indices and growing prime windows 65

32 A raw quotient index and why primitive cancellation remains separate 66

33 Retained auxiliary routes and their exact status 67

IV Analytic variations and imported methods 69

34 Automatic series, periodic identities, and normalization 69

35 Divisor sums, exact dyadic denominators, and logarithmic jets 72

36 Exact filter optimization and continuous divisor correlations 75

37 Reciprocity, special-function conversion, and arithmetic transfer tests 78

38 Residue-first positivity, convex tensor tests, and two-pole escape 81

39 Ten further routes: completed algebra and the remaining value problem 85

V Infinite polynomial families and exact targets 92

40 Infinite polynomial families and exact arithmetic targets 92

41 Positive realization and the complete two-pole integer lattice 99

42 An all-order Bernoulli–Stieltjes family and its rate obstruction 102

43 Bounded rational orbits and a vanishing approximation margin 104

44 Bernoulli convolutions: an explicitly conditional arithmetic branch 107

VI The two alternatives and local arithmetic 111

45 The two arithmetic cases and the obligation to preserve relations 111

46 One-prime character counts and optimal finite camouflage 114

47 Real limits, arithmetic places, and invariant analytic data 117

48 A feasibility obstruction for replication near zero 122

49 Effective nonvanishing tests and the limits of generic world search 123

VII Arithmetic transfer and its obstructions 126

50 Arithmetic transfer: notation and scope 126

51 Explicit real and 2-adic approximation 126

52 What is sufficient to transfer a relation? 128

53 Globally nilpotent objects in the tensor category of the Ein connection 129

54 Independent samples, exact cancellation, and positivity 132

55 Exact denominator growth after division of Euler's series 135

56 Remaining arithmetic assertions 140

57 Transfer certificates and reproducibility 141

58 Transfer revision record 141

VIII Unified targets, corrections, and the source record 143

59 What would complete an arithmetic argument? 143

60 Corrections carried into the cumulative edition 144

61 Coverage and reproducibility 145

References 146

1 Scope, status, and how the parts fit together

The common target is the following statement:

$$P(\gamma) \neq 0 \qquad \text{for every } 0 \neq P \in \mathbb{Z}[X], \qquad \gamma = \lim_{n\to\infty}(H_n - \log n). \tag{1.1}$$

An exact identity, a positive integral, or a finite rational certificate does not by itself establish this universally quantified assertion. The purpose of integrating the research is to display which implications have been proved and which additional implication would still be needed.

The source corpus. This edition uses 59 identified research files and packages, including the complete source of the 35-page paper of 8 September and the 18-page arithmetic-transfer revision of 10 September. The companion sources [001] and [002] are the versions preserved in the earlier complete paper archive. Their gamma-relevant foundations are integrated here; their complete original texts are preserved in that archive. The earlier document recalled as 27 pages has not been uniquely identified in the available corpus, so no claim is made to have compared that particular pagination. The source manifest records exactly what was available for this edition.

The integration rule. The earlier 35-page and 18-page mathematical bodies are retained, with their labels and citations renamed to avoid collisions. Intervening notes are synthesized by subject, with missing early results added separately. Duplicate arguments are cross-referenced or shortened; statements corrected in a later source are presented in their corrected form. The preserved source packages contain the original wording and finite data. When a broad companion assertion is not re-proved here, it is identified as a source input and is not silently used to close the arithmetic problem.

Part	Mathematical role
I	Gamma-related foundations from [001], [002], and the earliest certificates.
II	The complete positive-moment, Sondow, primitive-elimination, and finite-certificate development.
III	Additional early results absent from the 35-page synthesis.
IV	Later analytic variations, filters, residues, and comparisons with other methods.
V	Polynomial infinite families, exact arithmetic targets, and conditional transcendence criteria.
VI	Algebraic/transcendental alternatives, local counts, finite camouflage, and logical limits.
VII	Simultaneous approximation, tensor reduction, statistical cancellation, and Euler-quotient denominators.
VIII	Unified open implications, corrections, source coverage, and reproducibility.

Five different statuses. An *auxiliary theorem* has the displayed hypotheses and a mathematical proof. A *published input* is attributed to its external source. A *finite certificate* concerns explicitly bounded data and can be verified by exact arithmetic. A *conditional implication* proves "hypothesis implies conclusion" without asserting its hypothesis for γ. An *open target* is a statement not proved by the construction. Historical observations or approximate experimental exponents are not silently promoted from the last two statuses to the first.

Notation is local when necessary. Throughout, $H_n = \sum_{j=1}^{n} 1/j$, $H_0 = 0$, and $d_n = \operatorname{lcm}(1,\dots,n)$. When a symbol such as D_N, P_N, or E is re-used in a different construction, its definition is local to that section. The entire function

$$\operatorname{Ein}(z) = \sum_{n\geq 1} \frac{(-1)^{n+1} z^n}{n\, n!}$$

is kept distinct from a plus-sign polyexponential series. On the positive real branch $\mathrm{Ein}(x) = \gamma + \log x + E_1(x)$, where $E_1(x) = \int_x^\infty e^{-t}\,dt/t$. Thus $\mathrm{Ein}(1) = \gamma + \delta/e$ for $\delta = eE_1(1)$. No independence theorem for $\mathrm{Ein}(1)$ is interpreted as an independence theorem for γ without an additional argument.

What counts as progress toward the target. Three gaps recur: nonzero coefficient vectors need not have nonzero values; fast analytic decay can disappear after taking primitive arithmetic heights; and an identity of functions need not transfer its arithmetic information to a chosen constant. The parts below address specific forms of these gaps. None is a universal impossibility theorem for other methods.

Part I
Foundations from the companion manuscripts

2 Analytic and arithmetic foundations from the companion papers

This section integrates the gamma-related inputs of the companion manuscripts [15, 16]. Their broader results on independent spectra, inverse monodromy, and algebraic models are retained in the source archive; they are not additional proofs of the arithmetic nature of Euler's constant. The order-one independence statement needed here is supplied with its argument. Statements involving all higher polyexponential orders are identified separately as companion inputs.

We use

$$\gamma = \lim_{n\to\infty}(H_n - \log n), \qquad \mathrm{Ein}(z) = \sum_{n\ge 1} \frac{(-1)^{n-1} z^n}{n\, n!}, \qquad \delta = \int_0^\infty \frac{e^{-t}}{1+t}\, dt, \tag{2.1}$$

where $H_n = \sum_{j=1}^{n} 1/j$ and $H_0 = 0$. On the principal slit plane,

$$E_1(z) = \mathrm{Ein}(z) - \gamma - \log z, \qquad \mathrm{Ein}'(z) = \frac{1 - e^{-z}}{z}, \qquad \eta := \mathrm{Ein}(1) = \gamma + \delta/e. \tag{2.2}$$

The normalization agrees with [1, §6.2]. The entire function Ein is an E-function: its normalized Taylor coefficients are $(-1)^{n-1}/n$, with exponentially bounded common denominators. The function E_1 has a logarithmic singularity at zero and is not an E-function. In particular an E-function value theorem must not be applied to $E_1(1)$ merely because (2.2) expresses it through an E-value and γ.

2.1 The finite-point independence input and its exact scope

Put

$$K_{\exp} = \overline{\mathbb{Q}}(e^{\alpha} : \alpha \in \overline{\mathbb{Q}}).$$

Every coefficient of a polynomial over this field involves only finitely many exponential values.

Lemma 2.1 (Laurent derivative obstruction)**.** *Let $\nu_1, \dots, \nu_d \in \overline{\mathbb{Q}}$ be linearly independent over $\mathbb{Q}$, and endow*

$$R = \overline{\mathbb{Q}}(z)[X_1^{\pm 1}, \dots, X_d^{\pm 1}], \qquad K = \mathrm{Frac}(R)$$

with $DX_j = -\nu_j X_j$. If finitely supported constants $c_u \in \overline{\mathbb{Q}}$ satisfy

$$Dg = z^{-1} \sum_{u\in\mathbb{Z}^d} c_u X^u, \qquad g \in K,$$

then every c_u is zero.

Proof. First suppose an irreducible nonunit $P \in R$ divides DP. Writing $DP = AP$, comparison of Newton polytopes shows that $A \in \overline{\mathbb{Q}}(z)$; if $DP = 0$, the following coefficient argument applies directly. For two nonzero coefficients a_u, a_v of P,

$$(a_u/a_v)'/(a_u/a_v) = (u - v)\cdot\nu.$$

A rational function has no nonzero constant logarithmic derivative. Thus $u = v$, so P is a Laurent monomial times a unit, a contradiction. If a reduced denominator of g had a nonunit

factor P, differentiating would increase its pole order unless $P \mid DP$. Hence $g \in R$. Writing $g = \sum a_u(z)X^u$ gives

$$a'_u - (u \cdot \nu)a_u = c_u/z.$$

For $u = 0$, rational derivatives have zero residue. For $u \neq 0$, a finite pole of a_u creates a higher-order pole on the left; if it has no pole at zero, that side is regular there. A nonzero c_u/z is therefore impossible. The same argument with zero right side identifies the constant field of K as $\overline{\mathbb{Q}}$. □

Theorem 2.2 (Multipoint Ein input)**.** *For distinct $\lambda_1, \dots, \lambda_s \in \overline{\mathbb{Q}}^\times$, let*

$$d = \dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}}\{\lambda_1, \dots, \lambda_s\}.$$

Then

$$\operatorname{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(e^{\lambda_i}, \operatorname{Ein}(\lambda_i) : 1 \le i \le s) = d + s.$$

The only algebraic relations among the exponential and Ein *coordinates are the toric relations among the exponentials. In particular the family $\{\operatorname{Ein}(\lambda) : \lambda \in \overline{\mathbb{Q}}^\times\}$ is algebraically independent over $K_{\exp}$, in the finite-subset sense.*

Proof. Choose a $\mathbb{Z}$-basis $\nu_1, \dots, \nu_d$ of the group generated by the λ_i, and write $\lambda_i = m_i \cdot \nu$. Distinct exponential characters $e^{-u\cdot\nu z}$ are linearly independent over $\overline{\mathbb{Q}}(z)$, so $X_j = e^{-\nu_j z}$ realizes the field of Lemma 2.1. Put $Y_i(z) = \operatorname{Ein}(\lambda_i z)$. Then

$$DY_i = (1 - X^{m_i})/z.$$

The field generated by these functions embeds in $\overline{\mathbb{Q}}((z))$, so it has no new differential constants. The primitive Kolchin–Ostrowski theorem [2] would turn any algebraic dependence of the Y_i over K into an identity

$$Dg = \frac{\sum_i c_i - \sum_i c_i X^{m_i}}{z}$$

with some $c_i \in \overline{\mathbb{Q}}$ not all zero. Lemma 2.1 excludes this, since the m_i are distinct and nonzero.

The functional transcendence degree is unchanged on extending constants from $\overline{\mathbb{Q}}$ to $\mathbb{C}$: any relation involves finitely many monomials whose Taylor coefficients are algebraic, and scalar extension preserves the rank of their coefficient matrix. For specialization include 1, the required exponential characters, and all Y_i in a finite first-order system. Its coefficients lie in $\overline{\mathbb{Q}}(z)$ and have no finite pole except possibly zero. All entries are E-functions. The Siegel–Shidlovskii theorem, in the refined form of Beukers [10], therefore transfers the functional transcendence degree at $z = 1$. Algebraic independence of the Y_i over the exponential torus makes its binomial character relations the full relation ideal. Finally, to enlarge the exponential field to $K_{\exp}$, add to the finite set of arguments all algebraic exponents occurring in a hypothetical relation's coefficients. The finite result again excludes that relation. □

Corollary 2.3 (Full-rank linear images)**.** *If $Y = (\operatorname{Ein}(\lambda_1), \dots, \operatorname{Ein}(\lambda_s))^t$ and $B \in M_{r,s}(\overline{\mathbb{Q}})$ has row rank r, the entries of BY are algebraically independent over $K_{\exp}$. For arbitrary B, their relation ideal is generated by the linear relations in the left kernel of B.*

Proof. Theorem 2.2 identifies Y with independent indeterminates for the purpose of polynomial evaluation. Invertible linear changes of coordinates reduce B to a coordinate projection. □

Remark 2.4 (Higher orders retained as companion input). For $r \ge 1$ put

$$\operatorname{Ein}_r(z) = \sum_{n\ge1} \frac{(-1)^{n-1} z^n}{n^r n!}, \qquad z \operatorname{Ein}'_r(z) = \operatorname{Ein}_{r-1}(z), \quad \operatorname{Ein}_0(z) = 1 - e^{-z}.$$

Each fixed-order function is an E-function. The all-order statement in [15, "Multipoint one-level polyexponentials of arbitrary weight"] is

$$\operatorname{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(e^{\lambda_i}, \operatorname{Ein}_r(\lambda_i) : 1 \le i \le s, 1 \le r \le R) = d + Rs.$$

Its iterated-primitive proof is retained in the companion source. It is not a new theorem of this revision, and none of the gamma arithmetic conclusions below requires an unproved extension from order one to unbounded order. In particular, finite-point independence does not give a uniform estimate as R, the arguments, or their heights increase.

3 Moving levels, zero traces, and exact arithmetic detectors

Proposition 3.1 (Balanced dyadic levels)**.** *For $N \ge 1$ define*

$$Y_N = (N+1)\operatorname{Ein}(2^N) - N\operatorname{Ein}(2^{N+1}).$$

Then the Y_N are algebraically independent over $K_{\exp}$ in the finite-subset sense, and

$$Y_N = \gamma + \varepsilon_N, \qquad \varepsilon_N = (N+1)E_1(2^N) - NE_1(2^{N+1}) > 0, \qquad \varepsilon_N \sim \frac{N+1}{2^N} e^{-2^N}. \tag{3.1}$$

Moreover Y_N strictly decreases to γ.

Proof. The rows $(N+1)e_N - Ne_{N+1}$ have full rank for every finite selection: inspect the smallest occurring coordinate and proceed inductively. Corollary 2.3 gives independence. The connection formula cancels the logarithms exactly. Positivity follows from the strict decrease of E_1 on the positive axis. Put $g(t) = E_1(2^t)$; then $g''(t) = (\log 2)^2 2^t e^{-2^t} > 0$, and

$$\varepsilon_N - \varepsilon_{N+1} = (N+1)(g(N) - 2g(N+1) + g(N+2)) > 0.$$

Integration by parts gives $E_1(x) = e^{-x}x^{-1}(1+O(x^{-1}))$, proving the asymptotic and the limit. □

Proposition 3.2 (Critical slope is root multiplicity)**.** *For every nonzero $P \in \mathbb{C}[X]$,*

$$\lim_{N\to\infty} \frac{-\log|P(Y_N)|}{2^N} = \operatorname{ord}_{X=\gamma} P. \tag{3.2}$$

Consequently γ is algebraic if and only if there exists $0 \ne P \in \mathbb{Q}[X]$ for which the limit equals 1.

Proof. Write $P(X) = (X-\gamma)^m Q(X)$ with $Q(\gamma) \ne 0$. Equation (3.1) gives $-\log \varepsilon_N = 2^N + O(N + \log(N+1))$; the contribution of $\log|Q(Y_N)|$ stays bounded. This proves the limit. An irreducible polynomial over $\mathbb{Q}$ is separable, so the minimal polynomial of an algebraic γ has multiplicity one. □

This is an exact detector, not an estimate showing that its positive alternative cannot occur. It also explains why a lower bound uniform in moving arguments cannot be inferred from a fixed-function, fixed-point E-value theorem.

For $c \ne 0$ let Z_c be the zero multiset of $\operatorname{Ein}(z) - c$. Define

$$P_m(X) = -m[z^m]\log(1 - X\operatorname{Ein}(z)), \qquad m \ge 2.$$

Proposition 3.3 (Trace recovery and finite-stage closure)**.** *The inverse power sums converge absolutely and satisfy*

$$T_m(c) := \sum_{\rho \in Z_c} \rho^{-m} = P_m(c^{-1}).$$

The polynomials $P_m \in \mathbb{Q}[X]$ are monic of degree m. In particular

$$P_2 = X^2 - X/2, \qquad P_4 = X^4 - X^3 + 25X^2/72 - X/24,$$

and

$$c^{-1} = 144T_4(c) - 144T_2(c)^2 - 14T_2(c), \qquad K[T_m(c) : m \geq 2] = K[c^{-1}] \tag{3.3}$$

for every characteristic-zero subfield $K \subset \mathbb{C}$.

Proof. The growth estimate $|\operatorname{Ein}(z)| \leq |z|e^{|z|}$ gives order at most one; its negative-axis growth gives order one. Genus-one Hadamard factorization of $\operatorname{Ein}(z) - c$ gives

$$[z^m] \log(1 - \operatorname{Ein}(z)/c) = -\frac{1}{m} \sum_{\rho \in Z_c} \rho^{-m} \quad (m \geq 2).$$

This proves the trace formula, including multiplicities. Expanding the formal logarithm shows monicity and the displayed polynomials. The identity $144P_4 - 144P_2^2 - 14P_2 = X$ proves the ring equality. □

One trace chosen at each independent Y_N remains algebraically independent, because a nonconstant polynomial image is an algebraic extension in the reverse direction. In contrast, all orders at one level generate only $K(Y_N)$: increasing trace order at a fixed level adds no new transcendence degree. Their limits are $S_m = P_m(\gamma^{-1})$. For $S_2 = \gamma^{-2} - (2\gamma)^{-1}$, set

$$L_N(s) = 2sY_N^2 + Y_N - 2.$$

The identity

$$L_N(S_2) = (Y_N - \gamma)\{2S_2(Y_N + \gamma) + 1\}$$

shows slope one, whereas for fixed $s \neq S_2$, $L_N(s)$ tends to a nonzero constant and has slope zero. Proving that no algebraic s can have the first behavior would prove transcendence, but is not provided by the construction.

3.1 A common-shift alternative with an independent anchor

Proposition 3.4 (Common-shift defect)**.** *Let $z_0, z_1, \ldots, z_r$ be algebraically independent over a field $K \subset \mathbb{C}$, let $c, e_0 \in K^\times$, and let $g \in \mathbb{C}$. Put $T_j = c(z_j - g)$ and $\Delta = e_0(z_0 - g)$. Then*

$$\operatorname{trdeg}_K K(T_1, \ldots, T_r) \geq r - 1.$$

If the T_j are algebraically dependent, g and Δ are algebraically independent over K.

Proof. The $r - 1$ differences $T_j - T_1 = c(z_j - z_1)$ are independent. Under dependence put $L = K(T_1, \ldots, T_r)$, of transcendence degree $r - 1$. The equality $L(g) = K(z_1, \ldots, z_r, g)$ forces its transcendence degree to equal r. Thus g is transcendental over L and algebraic over $K(z_1, \ldots, z_r)$. Independence of the anchor z_0 makes it transcendental over the algebraic closure of this last field, hence over $K(g)$. Therefore g and $e_0(z_0 - g)$ are independent. □

Apply this with $K = K_{\exp}$, $g = \gamma$, $z_0 = \eta$, and $c = e_0 = e$. The remaining coordinates may be any finite selection of Y_N and $\operatorname{Ein}(-n)$, whose joint independence follows from disjoint supports and Corollary 2.3. Thus either all the corresponding residuals are independent, or a finite relation proves algebraic independence of γ and δ. This alternative does not construct that relation; expanding the family alone cannot rule out its independent branch.

4 Reciprocal witnesses and the distinction between a level and its carrier

Let $J(z) = z + z^{-1}$. With $C_0(w) = 2$, $C_1(w) = w$ and $C_{n+1} = wC_n - C_{n-1}$, define

$$H(w) = \sum_{n\geq 1} \frac{(-1)^{n-1} C_n(w)}{n\, n!}.$$

Both roots of $z^2 - wz + 1$ remain bounded on compact w-sets, so this series is locally uniformly convergent. It gives an entire function with

$$H(J(z)) = \mathrm{Ein}(z) + \mathrm{Ein}(z^{-1}), \qquad H'(w) = 2e^{-w/2} \frac{\sinh(\sqrt{w^2-4}/2)}{\sqrt{w^2-4}}. \tag{4.1}$$

The apparent singularities at $w = \pm 2$ are removable.

Proposition 4.1 (Arithmetic reciprocal witness)**.** *Distinct algebraic $w_1, \ldots, w_s$ give algebraically independent values $H(w_i)$ over $K_{\exp}$. Hence an algebraic solution of $H(w) = 2\gamma$ would make γ transcendental. The real equation has a unique solution $w_* \in (-2, 2)$. If w_* is algebraic, then γ and δ are algebraically independent over $K_{\exp}$.*

Proof. The reciprocal pairs above distinct algebraic w_i are disjoint sets of nonzero algebraic numbers. Their sums of Ein values are nonzero linear forms on disjoint independent supports; repeated roots at ± 2 have coefficient two. Corollary 2.3 applies. The derivative in (4.1) is positive on $\mathbb{R}$: for $|w| < 2$ its trigonometric numerator has argument in $[0, 1]$, and for $|w| \geq 2$ the hyperbolic expression is positive. Its endpoint limits show $H(\mathbb{R}) = \mathbb{R}$. Since $\mathrm{Ein}(-1) < 0 < \gamma < \mathrm{Ein}(1)$, the Euler root lies in $(-2, 2)$. If it is algebraic, its roots a, a^{-1} are algebraic, distinct, and different from 1. The three numbers $\mathrm{Ein}(a), \mathrm{Ein}(a^{-1}), \eta$ are independent over $K_{\exp}$. The invertible transformation

$$\gamma = \tfrac{1}{2}(\mathrm{Ein}(a) + \mathrm{Ein}(a^{-1})), \quad \delta = e(\eta - \gamma), \quad \mathrm{Ein}(a) - \mathrm{Ein}(a^{-1})$$

proves the last assertion. □

The identity $H(w_*) = 2\gamma$ does not show that w_* is algebraic. Conversely, transcendence of w_* alone does not decide the nature of γ. The rational differential equation for H is unchanged by adding a constant; a differential carrier which forgets its level does not supply the missing arithmetic witness.

5 Stokes data, Frobenius splitting, and the comparison gap

Set $F(x) = -\mathrm{Ein}(x)$ and

$$\mathcal{L} = (\partial + 1) \circ \partial \circ x \circ \partial = x\partial^3 + (x+2)\partial^2 + \partial.$$

The bases $(1, \log x, E_1(x))$ and $(1, F(x), \log x)$ are related by

$$C_\gamma = \begin{pmatrix} 1 & 0 & 0 \\ -\gamma & -1 & -1 \\ 0 & 1 & 0 \end{pmatrix}. \tag{5.1}$$

Their Wronskian is nonzero, since $W(1, \log x, E_1(x)) = e^{-x}/x^2$.

Proposition 5.1 (The lateral transition is gamma-free)**.** *Across the negative real axis the principal lateral values satisfy*

$$\begin{pmatrix} 1 \\ \log_+ x \\ E_{1,+}(x) \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 2\pi i & 1 & 0 \\ -2\pi i & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ \log_- x \\ E_{1,-}(x) \end{pmatrix}.$$

The row $(-g, -1, -1)$ is fixed by this transition for every $g \in \mathbb{C}$. Adding an algebraic constant t to F preserves its E-function status and changes the entry $-\gamma$ in (5.1) to $t - \gamma$. The unframed lateral transition, its trace, determinant, and exterior characteristic data do not detect this entry.

Proof. The logarithm jumps by $2\pi i$, whereas (2.2) makes E_1 jump by $-2\pi i$. The two contributions cancel in the displayed row. The change of F adds t times the constant basis vector, which is precisely the stated shear of the connection matrix. □

Here a distinguished asymptotic coefficient such as $-\gamma$ may be called a framed Stokes constant, as in the arithmetic theory of E-operators [13]. It must be distinguished from a strict Stokes factor or an unframed conjugacy invariant. An arithmetic containment theorem for possible coefficients is not a separation theorem asserting that a particular coefficient is transcendental.

Proposition 5.2 (Bare Frobenius splits, filtration can retain an extension)**.** *Let K be complete non-archimedean, let σ be an isometric automorphism, and let $q \in K^\sigma$ satisfy $|q| \neq 1$. For every $c \in K$ there is a unique t solving $qt - \sigma(t) = c$, and*

$$\begin{pmatrix} 1 & -t \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & c \\ 0 & q \end{pmatrix} \begin{pmatrix} 1 & \sigma(t) \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & q \end{pmatrix}.$$

If a filtration line is $K(e_1 + he_0)$, its displacement from the Frobenius splitting is $\kappa = h - t$, and it is invariant under changing e_1 to $e_1 + ue_0$. A splitting preserving this line exists exactly when $\kappa = 0$.

Proof. For $|q| < 1$ use $t = -\sum_{n\geq 0} q^n \sigma^{-n-1}(c)$; for $|q| > 1$ use $t = \sum_{n\geq 0} q^{-n-1}\sigma^n(c)$. Both series converge and satisfy the equation. A difference of two solutions obeys $qu = \sigma(u)$, hence is zero by the norm. Matrix multiplication gives the splitting. Under the basis change, c becomes $c + \sigma(u) - qu$, while h and t both become their old values minus u. Thus their difference is unchanged. □

For example, over $\mathbb{Q}_p$, $\varphi = \mathrm{diag}(1, p)$ and $\mathrm{Fil}^1 = \mathbb{Q}_p(e_1 + ae_0)$, with the usual two Hodge weights $0, 1$, are weakly admissible for every $a \in \mathbb{Q}_p$. The total Hodge and Newton numbers are one; on the two Frobenius eigenlines the Hodge number is at most the Newton number. The filtered extension splits exactly at $a = 0$, but its Frobenius traces, including tensor and symmetric-power traces, do not depend on a. The example has no assumed identification of a with any p-adic Euler constant.

The companion construction of the Euler exponential motive uses the external input of Fresán–Jossen [3, §12.8]: a non-split extension of $\mathbb{Q}(-1)$ by $\mathbb{Q}(0)$ has adapted comparison matrix

$$P_\gamma = \begin{pmatrix} 1 & \gamma \\ 0 & 2\pi i \end{pmatrix}.$$

This motivic statement is cited, not reproved here. With columns in the de Rham basis and rows in the Betti basis,

$$\mathrm{comp}(xd_0 + d_1) = (x + \gamma)b_0 + 2\pi i b_1.$$

Consequently transverse lines defined over $\overline{\mathbb{Q}}$ are matched exactly when $x + \gamma = 2\pi i u$ for some $x, u \in \overline{\mathbb{Q}}$.

Proposition 5.3 (Conditional matched-line implication)**.** *If every such matched pair is induced by a subobject splitting the Euler extension, then* $1, \pi, \gamma$ *are linearly independent over* $\overline{\mathbb{Q}}$*. The version restricted to the line* $\overline{\mathbb{Q}}b_1$ *already implies transcendence of* γ*.*

Proof. A relation $\gamma \in \overline{\mathbb{Q}} + \pi\overline{\mathbb{Q}}$ supplies the matched pair above. Its hypothetical lift splits the non-split extension, a contradiction. For the restricted version an algebraic γ supplies $x = -\gamma$ and $u = 0$. □

The lifting premise remains unproved. Existence of realizations, non-splitting of the motive, uniqueness of a bare Frobenius splitting, and a numerical relation in the comparison matrix are different statements. Neither the matrix calculation nor known non-splitting permits one to delete the algebraic-γ alternative.

6 The earlier interior-zero and Stieltjes attacks

The first exploration [5] introduced signed-coefficient kernels while retaining positivity of their values. This escapes a restriction on coefficientwise-positive perturbations, but not the height cost of removing their logarithmic coordinates. Put

$$W(u) = \frac{1}{\log u} + \frac{1}{1-u}, \quad \omega(t) = \frac{1}{t\{\log^2(t^{-1}-1) + \pi^2\}}, \quad D(u,t) = 1 - (1-u)t.$$

The Markov representation used in the Prévost family is $W(u) = \int_0^1 \omega(t) D(u,t)^{-1} dt$; see [4] and [16, "Fixed-prime Rodrigues twists"].

Proposition 6.1 (Positive Rodrigues transducer)**.** *Let* $G \in \mathbb{Z}[u]$ *be nonnegative on* $[0,1]$ *with* $G(1) > 0$*, let* $n = \ell + \deg G$ *and* $0 \le m \le \ell$*. Put*

$$P(u) = \frac{(-1)^\ell}{\ell!} \frac{d^\ell}{du^\ell}\{u^\ell (1-u)^\ell G(u)\} = \sum_{j=0}^{n} c_j u^j, \qquad J = \int_0^1 u^{\ell-m} P(u) W(u)\, du.$$

Then $P \in \mathbb{Z}[u]$*,* $P(1) = G(1)$*, and*

$$J = G(1)\gamma - \sum_j c_j H_j + \sum_j c_j \log(\ell - m + j + 1), \tag{6.1}$$

$$(-1)^m J = \int_0^1 \int_0^1 \frac{u^\ell (1-u)^\ell G(u) (1-t)^{\ell-m} t^m}{D(u,t)^{\ell+1}} \omega(t)\, dt\, du > 0. \tag{6.2}$$

Proof. Dividing a derivative of a monomial by $\ell!$ gives an integer binomial coefficient. Integration by parts gives $\int_0^1 u^j P(u) du = 0$ for $0 \le j < \ell$. The identity

$$\int_0^1 u^r W(u) du = \gamma + \log(r+1) - H_r$$

then yields (6.1), since $\sum_j c_j (H_{j+\ell-m} - H_j) = 0$ by these moment equalities. For the positive form differentiate $u^{\ell-m}/D(u,t)$ exactly ℓ times:

$$\frac{d^\ell}{du^\ell} \frac{u^{\ell-m}}{D(u,t)} = (-1)^m \ell! \frac{(1-t)^{\ell-m} t^m}{D(u,t)^{\ell+1}}.$$

A further integration by parts proves (6.2); the integrand is nonzero on an open rectangle. □

For $n = 400k$, $\ell = 392k$, $m = 114k$, and $G(u) = (5u-2)^{4k}(2u-1)^{4k}$, the original note gives

$$J_k = 3^{4k}\gamma - A_k + L_k > 0, \qquad \lim_{k\to\infty} J_k^{1/(400k)} = \rho.$$

The constant is specified algebraically by

$$\rho^{200} = \max_{0\le u\le 1} \left(\frac{57}{196}\right)^{57} \left(\frac{139}{196}\right)^{139} u^{139}(1-u)^{196}(5u-2)^2(2u-1)^2.$$

Indeed maximizing in t first gives, for $\alpha = m/\ell$,

$$\max_t \frac{u(1-u)(1-t)^{1-\alpha}t^{\alpha}}{D(u,t)} = \alpha^{\alpha}(1-\alpha)^{1-\alpha}u^{1-\alpha}(1-u).$$

Laplace upper and lower bounds on compact neighborhoods of maxima give the root limit. The three interior maxima are roots of $3390u^3 - 4463u^2 + 1939u - 278$ on the intervals separated by $2/5$ and $1/2$. The source's rational interval certificate proves

$$-\log\rho > \log(3+2\sqrt{2}), \qquad -\log\rho = 1.7666899507\ldots.$$

The exact interval certificate is archived; this revision does not label its decimal evaluation as a proof. The surviving L_k in (6.1) is a sum of prime logarithms. The improvement is therefore an analytic decay improvement before logarithm elimination. It does not by itself furnish small integer forms in $1, \gamma$.

The companion Prévost ledger has optimal raw rate $\log(3+2\sqrt{2}) = 1.7627471740\ldots$. Since $\log \operatorname{lcm}(1,\ldots,n) = n + o(n)$ and every fixed finite set of primes contributes only $O(\log n)$ to this least common multiple, its favorable height-one Ridout comparison misses the threshold by $2 - \log(3+2\sqrt{2}) = 0.2372528259\ldots$. This is a ledger for the specified template with logarithms still present, not a universal impossibility theorem for their linear combinations. The first exact seven-column logarithm cancellation in [5] produced

$$-858624606598580579198302 8 + 152727767653564348273828 80\,\gamma,$$

whose coefficients and value are large. Its exact cancellation is preserved as a finite certificate, with no optimality or limiting claim.

6.1 A positive Padé error and its moving-scale denominator cost

The Binet measure and its moments are

$$d\mu(u) = \frac{du}{e^{2\pi\sqrt{u}} - 1}, \quad \mu_j = \frac{|B_{2j+2}|}{2j+2}, \quad S(s) = \int_0^{\infty} \frac{d\mu(u)}{s+u}. \tag{6.3}$$

Binet's formula [1, §5.9] gives $\gamma = H_n - \log n - (2n)^{-1} + S(n^2)$. For the monic orthogonal polynomial p_m and the Gaussian rational approximant R_m, polynomial division and orthogonality give

$$E_m(s) := S(s) - R_m(s) = \frac{1}{p_m(-s)^2} \int_0^{\infty} \frac{p_m(u)^2}{s+u}\, d\mu(u) > 0. \tag{6.4}$$

All zeros of p_m are positive, so E_m decreases on $s > 0$. For $n = 2^N$ put

$$r_{m,N} = (N+1)\{H_n - (2n)^{-1} + R_m(n^2)\} - N\{H_{2n} - (4n)^{-1} + R_m(4n^2)\}.$$

Then $r_{m,N} \in \mathbb{Q}$ and

$$\gamma - r_{m,N} = (N+1)E_m(n^2) - NE_m(4n^2) \ge E_m(n^2) > 0.$$

Proposition 6.2 (Slow-degree Stieltjes barrier)**.** *If $Q_{m,N}$ is the reduced denominator of $r_{m,N}$, there is an absolute constant $C > 0$ such that, for $m, N \geq 1$ and $n = 2^N$,*

$$\log\{Q_{m,N}(\gamma - r_{m,N})\} \geq \vartheta(2n) - \vartheta(n) - C\{m^2 \log(m+1) + (m+1)\log n\},$$

where $\vartheta(x) = \sum_{p \leq x} \log p$. Hence $Q_{m,N}(\gamma - r_{m,N}) \to \infty$ whenever the expression in braces is $o(n)$, in particular if $m = o(\sqrt{n/\log n})$.

Proof. Von Staudt–Clausen gives a common denominator D_m of the first $2m+1$ Binet moments with $\log D_m = O(m)$. Their integerized logarithmic heights are $O(m \log(m+1))$. Cramer's rule on the m by m moment matrix gives integer polynomials U_m, V_m such that $R_m(s) = U_m(s)/(D_m V_m(s))$ and coefficient logarithmic heights $O(m^2 \log(m+1))$. The same bound controls p_m. For $n < p \leq 2n$, unless p divides $D_m V_m(n^2) V_m(4n^2)$, the harmonic term $-NH_{2n}$ contributes the unique nonintegral p-part, $-N/p$; $p > N$ excludes cancellation. The total logarithm of these exceptional divisors is $O(m^2 \log(m+1) + m \log n)$, proving the required lower bound on $\log Q$. For the error, the density of μ is bounded below on $[1, 2]$ and

$$\int_1^2 p_m(u)^2 du \geq \frac{1}{(2m+1)\binom{2m}{m}^2} \geq \frac{1}{(2m+1)16^m}.$$

Insert this in (6.4) and use the coefficient bound on $p_m(-n^2)$. Combining the two inequalities proves the result. The asymptotic consequence follows from the prime number theorem. □

7 Positive rational kernels: the corrected infinite-family target

Let $h_m = H_{m-1} + (2m)^{-1}$ and $L_M = \operatorname{lcm}(1, \ldots, M)$. For real coefficients c_m, $2 \leq m \leq M$, set

$$F_c(u) = -\sum_{m=2}^{M} \frac{c_m}{u + m^2}.$$

The exact constraints

$$\sum c_m = -1, \qquad \sum c_m v_p(m) = 0 \quad \text{for every prime } p \tag{7.1}$$

yield

$$\int_0^\infty F_c(u) d\mu(u) = \gamma - r_c, \qquad r_c = -\sum c_m h_m. \tag{7.2}$$

If $F_c \geq 0$ it is not identically zero, since $uF_c(u) \to 1$; its integral is therefore strictly positive.

Proposition 7.1 (Target denominator, not residue denominator)**.** *Suppose (7.1) and $F_c \geq 0$ hold, and impose $r_c = P/(2L_M)$ with $P \in \mathbb{Z}$. Then*

$$2L_M \int F_c, d\mu = 2L_M \gamma - P > 0$$

regardless of the denominators of the individual c_m. If the last quantity is less than 1 for unbounded M, then γ is irrational.

Proof. The displayed equality is exact. If $\gamma = a/b$, then $b \mid L_M$ for every $M \geq b$, making its right side an integer strictly between zero and one. This is impossible. □

This is the correction introduced by [7]: integer residues are a sufficient design constraint, not a necessary one. The fixed-order Smith lattice of [6] is therefore not an obligatory gate. Its homogeneous spacing calculations neither solve the affine target condition nor supply an inner radius for the positive cone. An upper bound on available coefficient width must not be used as a lower bound ensuring successful rounding.

Proposition 7.2 (Compact feasible cone and exact intersection condition)**.** *For $M \geq 4$, the set $\mathcal{C}_M$ of real c satisfying (7.1) and $F_c \geq 0$ is nonempty and compact. Its minimum energy*

$$E_M^* = \min_{c \in \mathcal{C}_M} \int F_c \, d\mu$$

is attained and is positive. Put $P_M = \lfloor 2L_M\gamma \rfloor$. For every sufficiently large M, a feasible real kernel with $r_c = P_M/(2L_M)$ exists if and only if

$$E_M^* \leq \frac{\{2L_M\gamma\}}{2L_M}. \tag{7.3}$$

Proof. The kernel $2/(u+4) - 1/(u+16)$ is feasible and has $r_c = 13/24$. The set is closed. If it were unbounded, divide an unbounded sequence by its norm and pass to a unit limit d. Its kernel F_d is nonnegative, has $\sum d_m = 0$, and is log-balanced. Because its leading $1/u$ term vanishes, it is integrable against Lebesgue measure, and

$$\int_0^\infty F_d(u)du = 2\sum_m d_m \log m = 0.$$

Nonnegativity forces $F_d = 0$, hence $d = 0$ by uniqueness of partial fractions, a contradiction. The objective is continuous and positive at every feasible point, so its attained minimum is positive. Convexity makes the range of r_c an interval whose upper endpoint is $\gamma - E_M^*$. Since $P_M/(2L_M) \to \gamma > 13/24$, the target lies in that interval exactly when (7.3) holds. □

The proof does not assert that a boundary real solution has rational coefficients. This distinction does not affect Proposition 7.1, whose arithmetic target is the fixed rational value r_c. It does affect claims about rational certificates.

The exact finite construction of [7] uses $\mathcal{S} = \{2, 3, 4, 6, 8, 9, 12, 16, 18, 24\}$ and

$$F(u) = \frac{u^6(au^3 + bu^2 + cu + d)}{a\prod_{m\in\mathcal{S}}(u + m^2)},$$

where

$$a = 5967500403865, \qquad b = 40760598624375361,$$
$$c = 13281147167266501260, \qquad d = 161940004068861643104.$$

All four integers are positive, and exact partial fractions give

$$\int F \, d\mu = \gamma - \frac{3090544783}{5354228880}, \qquad 0 < 10708457760\gamma - 6181089566 < 1.$$

The final inequality is certified by independent rational Binet and logarithm bounds in the archived verifier. Later selected stages $M = 24, 32, 40, 48, 64, 80, 128$ have exact primal–dual certificates for optimization in the smaller cone of numerators with nonnegative monomial coefficients [8]. These are finite optimizations, not proofs for every intervening M or for an unbounded family.

7.1 Two explicit infinite families and their denominator limitations

For integers $a \geq 2$, $k \geq 1$,

$$\Phi_{a,k}(u) = \frac{k+1}{u + a^{2k}} - \frac{k}{u + a^{2k+2}} > 0.$$

Its residues satisfy the required logarithmic balance, and

$$\int \Phi_{a,k}\, d\mu = \frac{k+1-k/a^2}{12a^{2k}} + O_a(ka^{-4k}).$$

At the natural top node $M = a^{k+1}$, this is of order k/M^2, so multiplication by $2L_M$ makes it diverge. The multiscale butterfly identities in [6] construct this same two-pole family; they do not change its error scale.

A genuinely many-pole infinite family from [8] is as follows. Fix $q \geq 2$, put $a = q^2$, and for $s \geq 2$ define

$$C_s = \sum_{k=1}^{s-1} \frac{1}{a^k - 1}, \quad t_s = (a^s - a)(s + C_s), \quad F_s(u) = \frac{u^{s-2}(u + t_s)}{\prod_{j=1}^{s}(u + a^j)}.$$

Proposition 7.3 (Geometric positive family and reduced-denominator barrier)**.** *Writing* $F_s(u) = \sum_{j=1}^{s} R_j/(u + a^j)$ *gives*

$$\sum R_j = 1, \qquad \sum jR_j = 0, \qquad \gamma - r_s = \int F_s\, d\mu > 0, \quad r_s = \sum_{j=1}^{s} R_j h_{q^j} \in \mathbb{Q}.$$

If $r_s = P_s/Q_s$ *is reduced and* $M = q^s$*, then*

$$\log(\gamma - r_s) = -(\log q)s^2 + O_q(s \log(s+1)), \qquad \log Q_s \geq (1 - 1/q)M + o_q(M).$$

In particular $Q_s(\gamma - r_s) \to \infty$*.*

Proof. If A_j are the residues of $u^{s-1}/\prod(u + a^j)$, their polynomial is

$$W_s(z) = \sum A_j z^j = z \prod_{k=1}^{s-1} \frac{a^k z - 1}{a^k - 1}.$$

It is characterized by $W_s(1) = 1$ and the zeros a^{-k}. Thus $W_s'(1) = s + C_s$ and $a^{-1}W_s'(a^{-1}) = 1/(a^s - a)$. Substitution of $R_j = A_j(1 - t_s/a^j)$ proves both sums. Positivity is immediate.

A lower bound for the integral on $[1, 2]$ is $\exp\{-(\log q)s^2 + O_q(s)\}$. For an upper bound use $\prod(u+a^j) \geq a^{s(s+1)/2}$ and the moments (6.3), whose logarithms are $O(s \log(s+1))$. These bounds prove the stated leading error scale. The residue formula shows that a common denominator of the R_j, and the numerator of the nonzero R_s, have logarithmic size $O_q(s^2)$. For clarity a power of a times $\prod_{k=1}^{s-1}(a^k - 1)^3$ with exponent $O(s^2)$ on a clears all these residues. Outside divisors of these integers, a prime $M/q < p \leq M$ does not occur in the earlier h_{q^j}, while

$$ph_M \equiv H_{\lfloor M/p \rfloor} \pmod{p}.$$

Here the index is in $\{1, \ldots, q-1\}$; the finitely many possible nonzero rational harmonic numbers are units at all sufficiently large p. Thus each such unexceptional prime divides the reduced Q_s. The excluded primes have total logarithm $O_q(s^2)$, and the prime number theorem gives the denominator lower bound. □

The geometric denominator barrier applies to this free rational target r_s, not to all prescribed targets $P/(2L_M)$. Given any feasible rational kernel with $r_1 > P/(2L_M) > 13/24$, convex interpolation with the basic two-pole kernel realizes the prescribed rational target with positive rational coefficients. What remains is precisely the unbounded intersection in (7.3); small unconstrained energy does not bound the fractional part on its right away from zero.

Remark 7.4 (Historical corrections retained). The earlier multiscale note also records local fixed-order Smith laws, a two-by-two failure of total positivity for its atom kernel, and a cyclotomic denominator obstruction for a fixed-base flattening ansatz. Their detailed certificates and proof architecture remain in the archive; the stronger global rounding conclusion suggested there is not assumed. The later $v2$ relaxation supersedes the claim that integral residues are necessary. The $v3$ computations also refute the rule that maximal moment cancellation is always optimal: at $M = 80$ it is not optimal in the specified coefficientwise cone, and at $M = 100$ that particular candidate leaves the cone. Neither observation proves nonexistence of other positive kernels at those levels.

Part II
Positive moments and primitive logarithmic elimination

8 Positive moments: framework and scope

Euler's constant

$$\gamma = \lim_{n\to\infty}\left(\sum_{k=1}^{n}\frac{1}{k} - \log n\right)$$

appears both in convergent special-function identities and as a connection constant between analytic solutions and asymptotic expansions. The companion manuscripts [15, 16] develop these two viewpoints through exponential-integral periods, positive moment systems, and moving polyexponential levels. This part concentrates on the arithmetic step which those viewpoints leave unresolved: producing useful integer forms in 1 and γ after all auxiliary logarithms have been removed.

Our starting point is Sondow's positive double integral [19]. We combine its exact logarithmic expansion with rational linear programming, polynomial positivity on a compact interval, and the geometry of the primitive target lattice. These tools have distinct roles. Prime valuations describe which logarithms can cancel; positivity certifies that a resulting form does not vanish; rational optimization accounts for the common factor removed when the target is made primitive. Correlated moment inequalities improve the finite optimization without introducing new transcendental variables.

The principal conclusions are as follows. All unrestricted assertions below are proved in the indicated sections; numerical exponents refer to finite certificates, not asymptotic extrapolations.

1. The prime support of Sondow's logarithmic integer is exactly the primes at most $2n$. This gives the exact rank of the prefix logarithm matrix, explicit prime-adaptive multiplicative relations, and an obstruction to every fixed-order recurrence with algebraic rational-function coefficients.

2. For a primitive target (q, p), a rational quotient-norm problem is exactly equivalent to integer logarithm elimination followed by removal of the target common factor. Positive rational changes of column normalization do not change this primitive optimization problem.

3. Bernstein coefficients give sufficient certificates of nonvanishing which are strictly less restrictive than first-coordinate dominance. Two-sided rational bounds for the integrals lead to an exact interval model for γ, with rational primal and dual certificates.

4. Positive differences of moving order have a fully explicit integral representation and an exact exponential decay rate. Their best rate after the displayed integer normalization is

   $$\log 16 - 2 - \log(1 - e^{-2}) = 0.9180021801\ldots.$$

   These forms still contain prime logarithms. Their bounds can nevertheless be used as additional constraints in the logarithm-elimination problem.

5. The finite certificates include one primitive pair satisfying $0 < p - q\gamma < q^{-103/100}$. Adding fifteen correlated moment constraints contracts the computed intervals by factors of approximately 11.32 and 9.33 at levels 32 and 40, respectively.

6. A separate rational moment construction admits a graded Todd normalization, with an explicit improvement in its denominator upper bound. The mixed exponential-integral viewpoint supplies exact reduction statements, but its arithmetic specialization step remains open.

Boundary of the positive-moment results. It proves neither the irrationality nor the transcendence of γ. In particular, no infinite family with an unbounded primitive denominator and a fixed positive Roth margin is established. This distinction is part of the mathematical statement, not a qualification to be removed by further finite precision. Section 20 formulates the exact missing implications. The work also makes no claim that the constructions exhaust the approaches in the literature, or that the sufficient criteria used here are necessary for every possible proof of transcendence.

Relation to earlier work. The integral and its logarithmic decomposition are due to Sondow; the rational-function interpretation is developed by Sondow and Zlobin [19, 20]. Rational approximants associated with multiple orthogonal polynomials provide a different established approach; see Wolfs and Van Assche [21]. The present draft organizes the subsequent arguments and certificates developed after the companion manuscripts. Those manuscripts are unpublished sources, not external validation. Results requiring only the classical integral and elementary finite-dimensional arguments are proved here independently of their broader algebraic-independence claims. Historical priority for the particular combinations of methods has not been established.

Notation and status. All logarithms are natural. Put $H_0 = 0$ and $H_n = \sum_{k=1}^{n} 1/k$, and write $d_j = \operatorname{lcm}(1, \dots, j)$ for $j \geq 1$. The word *primitive* means $\gcd(q, p) = 1$, with $q > 0$ when p/q is viewed as an approximant. A *finite certificate* consists of exact integer or rational data and identities or inequalities which can be checked without evaluating γ. Rounded decimal summaries are not used in these checks.

9 The integral, its integer coefficients, and nonvanishing

For $n \geq 1$ define

$$I_n = \int_0^1 \int_0^1 \frac{[x(1-x)y(1-y)]^n}{(1-xy)(-\log(xy))}\, dx\, dy. \tag{9.1}$$

Let $t = x(1-x)y(1-y)$ and $h(x, y) = ((1-xy)(-\log(xy)))^{-1}$. Then $0 < t \leq 1/16$ in the open square and $h > 0$. The singular weight $h\, dx\, dy$ alone has infinite mass, so finite-measure arguments below always include a factor t before taking a pushforward.

The classical evaluation [19, 20] is

$$I_n = C_n\gamma - A_n + L_n, \qquad C_n = \binom{2n}{n}, \qquad A_n = \sum_{r=0}^{n} \binom{n}{r}^2 H_{n+r}, \tag{9.2}$$

where

$$L_n = 2\sum_{k=1}^{n} \left(\sum_{r=0}^{\min(k-1,n-k)} \binom{n}{r}^2 (H_{n-r} - H_r) \right) \log(n+k). \tag{9.3}$$

Equivalently,

$$L_n = 2\sum_{r=0}^{n} \binom{n}{r}^2 (H_r - H_{n-r}) \log((n+r)!). \tag{9.4}$$

For completeness, the equivalence of the two logarithmic expressions is purely finite. Set $c_r = 2\binom{n}{r}^2 (H_r - H_{n-r})$. Then $c_{n-r} = -c_r$ and $\sum c_r = 0$. On expanding each factorial logarithm, the terms $\log j$ for $j \leq n$ disappear, while the coefficient of $\log(n+k)$ is $\sum_{r=k}^{n} c_r$. Pairing indices r and $n-r$ reduces this last sum to the coefficient in (9.3). The integral evaluation itself is a cited classical input, not a new identity claimed here.

Every denominator occurring in A_n and in the coefficients of L_n divides d_{2n}. Consequently

$$e_n := d_{2n} I_n = a_n \gamma - b_n + \sum_{p \le 2n} \beta_{p,n} \log p, \quad a_n = d_{2n} C_n, \quad b_n = d_{2n} A_n, \tag{9.5}$$

with integer $a_n, b_n, \beta_{p,n}$; throughout a prime index is denoted by p only when no target numerator is in use. The integer

$$S_n = \exp(d_{2n} L_n) = \prod_{p \le 2n} p^{\beta_{p,n}} \tag{9.6}$$

is well-defined and positive.

For a finite set of indices $\mathcal{J}$, let $B = (\beta_{p,n})_{p \le 2 \max \mathcal{J},\ n \in \mathcal{J}}$, where the row index runs over all primes at most $2 \max \mathcal{J}$. Write $a = (a_n)_{n \in \mathcal{J}}$, $b = (b_n)_{n \in \mathcal{J}}$ and

$$\Phi(w) = (a \cdot w, b \cdot w).$$

An integer vector u in $\ker B$ gives the exact identity

$$\sum_{n \in \mathcal{J}} u_n e_n = Q\gamma - P, \qquad (Q, P) = \Phi(u), \qquad \prod_{n \in \mathcal{J}} S_n^{u_n} = 1. \tag{9.7}$$

Unique prime factorization proves the equivalence between the multiplicative identity and $Bu = 0$.

Lemma 9.1 (Two-sided rational integral bounds)**.** *For every $n \ge 1$,*

$$\ell_n := \frac{d_{2n}}{4(2n+1)^2 C_n^2} < e_n < h_n := \frac{d_{2n}}{n^2 C_n^2}. \tag{9.8}$$

In particular, all integrals in (9.1) are finite.

Proof. For $0 < u < 1$, one has $-\log u < 1/u$ and $-\log u \ge 1 - u$. It follows that $h(x, y) > xy$ and

$$h(x, y) \le \frac{1}{(1 - xy)^2} \le \frac{1}{(1 - x)(1 - y)}.$$

The comparisons are strict on a subset of positive measure. Thus

$$B(n+2, n+1)^2 < I_n < B(n+1, n)^2,$$

where $B(\ ,\ )$ denotes Euler's beta function only in this display. The two beta values equal $[2(2n+1)C_n]^{-1}$ and $(nC_n)^{-1}$, respectively. Multiplication by d_{2n} proves the assertion. □

Example 9.2 (The first two indices)**.** There is no singular starting case for the Sondow family:

$$I_1 = 2\gamma - \frac{5}{2} + 2 \log 2, \qquad I_2 = 6\gamma - \frac{131}{12} + 3 \log 12.$$

In particular $S_1 = 16$. The bounds and integer normalization above already hold at $n = 1$. The order N of the separate moment matrix in Section 18 is a different parameter.

Three logically distinct conditions must be maintained:

$$u \ne 0, \qquad \Phi(u) \ne (0, 0), \qquad Q\gamma - P \ne 0.$$

The first does not imply the second. The second does not imply the third without an additional argument. Positive integral representations and strict rational interval certificates will supply that argument in the finite examples below.

10 Prime support and canonical logarithm elimination

The logarithms in Sondow's evaluation have a particularly rigid support. This section determines that support and the rank of every prefix of the prime-exponent matrix. Prefixes and moving windows have different rank questions; we distinguish them throughout.

For $n \geq 1$, define

$$c_{n,k} = 2 \sum_{i=0}^{\min(k-1,n-k)} \binom{n}{i}^2 (H_{n-i} - H_i) \qquad (1 \leq k \leq n). \tag{10.1}$$

The evaluation of the integral in [19, 20] gives $L_n = \sum_{k=1}^{n} c_{n,k} \log(n+k)$. Put

$$S_n = \prod_{k=1}^{n} (n+k)^{d_{2n} c_{n,k}}, \qquad \beta_{p,n} = v_p(S_n). \tag{10.2}$$

In particular, $d_{2n} L_n = \log S_n = \sum_p \beta_{p,n} \log p$.

Lemma 10.1. *For $n \geq 1$, each $c_{n,k}$ is strictly positive and $d_{2n} c_{n,k}$ is a positive integer. Consequently S_n is a positive integer greater than one.*

Proof. In (10.1), the summation limit is at most $\lfloor (n-1)/2 \rfloor$. Thus every harmonic difference in that sum is positive. The denominators of all those harmonic numbers divide d_{2n}, proving the integrality assertion. □

Theorem 10.2 (Exact support and prefix rank)**.** *For every prime p and integer $n \geq 1$,*

$$\beta_{p,n} > 0 \quad \Longleftrightarrow \quad p \leq 2n. \tag{10.3}$$

Let $\pi(x)$ denote the number of primes at most x, and define

$$n(2) = 1, \qquad n(p) = \frac{p+1}{2} \quad (p \text{ odd prime}).$$

For every $M \geq 1$, the prefix matrix

$$B_{[1,M]} = (\beta_{p,n})_{p \leq 2M,\ 1 \leq n \leq M}$$

has rank $\pi(2M)$. Its columns with indices $n(p)$ form a basis of its column space over $\mathbb{Q}$, and

$$\dim_{\mathbb{Q}} \ker B_{[1,M]} = M - \pi(2M). \tag{10.4}$$

Proof. All bases in the product (10.2) lie in $(n, 2n]$, so no prime greater than $2n$ occurs. If $n < p \leq 2n$, the base p itself occurs. If $p \leq n$, then the interval of n consecutive integers $n+1, \ldots, 2n$ contains a multiple of p. Strict positivity of every exponent proves (10.3) in both cases.

Order the primes at most $2M$ increasingly. Their indices $n(p)$ are distinct and at most M. In the square submatrix formed by these columns, the entry in a row below the diagonal is zero: a larger prime has not yet appeared at that column. Every diagonal entry is positive by (10.3). The determinant is the product of the diagonal entries and is nonzero. The number of rows gives the reverse rank inequality, and rank–nullity proves (10.4). □

Corollary 10.3 (Canonical prime-adaptive relations)**.** *If n is not one of the indices $n(p)$, there is a unique rational expansion*

$$(\beta_{p,n})_{p \leq 2n} = \sum_{\substack{q \leq 2n \\ q \text{ prime}}} \rho_{q,n} (\beta_{p,n(q)})_{p \leq 2n}. \tag{10.5}$$

If D is any common positive denominator of the $\rho_{q,n}$, then

$$S_n^D \prod_{\substack{q \le 2n \\ q \text{ prime}}} S_{n(q)}^{-D\rho_{q,n}} = 1. \tag{10.6}$$

There are infinitely many such relations, constructed without the value of γ.

Proof. The square triangular submatrix in Theorem 10.2 is invertible, so back substitution gives the unique coefficients in (10.5). Clearing denominators makes the valuation at every prime of the left side of (10.6) zero. Unique factorization proves that identity. For example, every $n > 2$ with $n \equiv 2 \pmod 3$ is not an entry index, since $2n - 1 > 3$ is divisible by 3. There are infinitely many such n. □

The support of these relations can extend back to level 1. The construction supplies neither a short coefficient vector in a moving window nor a small primitive linear form in $1, \gamma$. Thus the existence of exact multiplicative identities and their usefulness for Diophantine approximation are separate questions.

For completeness, the two target coordinates already survive independently in a small prefix. This finite fact has an all-prefix consequence, but no corresponding all-window consequence.

Proposition 10.4 (Rank of the prefix target)**.** *For every $M \ge 8$, the map*

$$\Phi_M : \ker_{\mathbb{Q}} B_{[1,M]} \longrightarrow \mathbb{Q}^2, \qquad u \longmapsto \left(\sum_{n=1}^{M} a_n u_n, \sum_{n=1}^{M} b_n u_n \right)$$

has rank two. Its kernel has dimension $M - \pi(2M) - 2$.

Proof. The following two integer vectors, displayed by coordinates, give an explicit certificate at $M = 8$:

n	u_n	v_n
1	−3923225775	353653621155957622500
2	−158150125	−24892511625147813075
3	392805	−633334105016310555
4	−200868	−518196421884141
5	35150	0
6	0	57656027697450
7	0	−309266976225
8	0	5037311350

Substitution in (10.1)–(10.2), with prime rows $2, 3, 5, 7, 11, 13$, gives $B_{[1,8]}u = B_{[1,8]}v = 0$ and

$$\Phi_8(u) = (-16097728500, -8969752951),$$
$$\Phi_8(v) = (-27120467275073944200, -44055302128555509412).$$

The determinant of these two target vectors is the nonzero integer

$$4659264012778652479624113078 00.$$

These equalities involve only integer and rational arithmetic and are also checked by the accompanying verifier. Appending zero coordinates preserves the two vectors and their target images in every larger prefix. Rank–nullity and (10.4) finish the proof. □

Remark 10.5. Theorems about $B_{[1,M]}$ cannot be applied to $B_{[N,M]}$ merely by replacing 1 by N. Removing the early entry columns removes the triangular certificate. In particular, no full prime-row rank or rank-two target assertion for every window $[N, 2N]$ is claimed here.

11 A fixed-order recurrence obstruction

We use P-recursiveness in the following sense: a complex sequence f_n is P-recursive over a field $K \subseteq \mathbb{C}$ if it satisfies a nonzero recurrence $\sum_{j=0}^{r} p_j(n) f_{n+j} = 0$, with $p_j \in K[n]$, for all sufficiently large integers n. Its initial values need not belong to K.

Theorem 11.1. *Neither $(L_n)_{n\geq 1}$ nor $(\log S_n)_{n\geq 1}$ is P-recursive over $\overline{\mathbb{Q}}$. Equivalently, for either sequence f_n, an identity*

$$\sum_{j=0}^{r} c_j(n) f_{n+j} = 0, \qquad c_j \in \overline{\mathbb{Q}}(n), \tag{11.1}$$

valid for every sufficiently large integer n, forces all c_j to be identically zero. Their ordinary generating series are not D-finite over $\overline{\mathbb{Q}}(z)$.

Proof. First take coefficients in $\mathbb{Q}(n)$. Distinct prime logarithms are linearly independent over $\mathbb{Q}$: clear denominators in a putative relation, exponentiate, and apply unique factorization. Suppose (11.1) holds, and discard its trailing zero coefficients so that $c_r \neq 0$. For each sufficiently large odd prime p, put $n = (p+1)/2 - r$. The final term has index $n + r = (p+1)/2$, so its logarithmic expansion contains $\log p$ with a positive coefficient. Every preceding term has index at most $(p-1)/2$ and therefore contains no $\log p$. Linear independence forces $c_r(n) = 0$. This happens at infinitely many distinct integers, contradicting that c_r is a nonzero rational function. The argument applies to both sequences: division by d_{2n} changes coefficients but not support.

For algebraic coefficients, use Baker's linear-independence theorem [9]: logarithms of algebraic numbers that are linearly independent over $\mathbb{Q}$ remain linearly independent over $\overline{\mathbb{Q}}$. At every integer outside the finitely many poles of the c_j, their values are algebraic. The same isolated coefficient of $\log p$ again forces $c_r(n) = 0$ infinitely often.

Finally, comparison of coefficients in a nonzero linear differential equation with polynomial coefficients for the ordinary generating series produces a nonzero polynomial-coefficient recurrence on a tail. Such a recurrence has just been excluded. □

The isolation argument also constrains any individual relation.

Corollary 11.2. *Suppose $m_1 < \cdots < m_s$, $u_j \in \mathbb{Q} \setminus \{0\}$, and $\prod_{j=1}^{s} S_{m_j}^{u_j} = 1$. Then $s \geq 2$, and the interval $(2m_{s-1}, 2m_s]$ contains no prime.*

Proof. One factor cannot give a relation because $S_m > 1$. A prime in the stated interval would have positive valuation in the last factor and zero valuation in all other factors. After clearing denominators of the u_j, its valuation in the product would be nonzero. □

The non-P-recursiveness is also present before the lcm normalization and before the logarithmic part is separated from the integral.

Lemma 11.3. *The rational sequence $A_n = \sum_{k=0}^{n} \binom{n}{k}^2 H_{n+k}$ is P-recursive over $\mathbb{Q}$.*

Proof. Introduce the parameter polynomial

$$F_n(t) = \sum_{k=0}^{n} \binom{n}{k}^2 \frac{(1+t)_{n+k}}{(n+k)!}, \qquad (a)_m = \prod_{j=0}^{m-1} (a+j). \tag{11.2}$$

Vandermonde's identity and differentiation of the finite product give

$$F_n(0) = C_n, \qquad F_n'(0) = A_n. \tag{11.3}$$

For generic t, the summand is the proper hypergeometric term

$$T(n,k;t)=\frac{\Gamma(n+1)^2\Gamma(n+k+1+t)}{\Gamma(k+1)^2\Gamma(n-k+1)^2\Gamma(n+k+1)\Gamma(1+t)}.$$

Its two discrete shift ratios are

$$\frac{T(n+1,k;t)}{T(n,k;t)}=\frac{(n+1)^2(n+k+1+t)}{(n+1-k)^2(n+k+1)},$$
$$\frac{T(n,k+1;t)}{T(n,k;t)}=\frac{(n-k)^2(n+k+1+t)}{(k+1)^2(n+k+1)}.$$

The reciprocal Gamma factors give its terminating support $0\le k\le n$; outside this interval the term vanishes for generic t. The proper-hypergeometric summation theorem, over the parameter field $\mathbb{Q}(t)$, therefore supplies a nonzero recurrence for the finite sum [22, Sections 2.1 and 3.3]. Clearing coefficient denominators, write that recurrence as

$$P(t)F(t)=0,\qquad P(t)=\sum_{j=0}^{r}p_j(n,t)E^j,\qquad p_j\in\mathbb{Q}[n,t],\tag{11.4}$$

where E shifts n to $n+1$.

Remove the largest common power of t from the coefficients of P. Then $P(0)$ is a nonzero recurrence operator. Since every $F_n(t)$ is a polynomial, (11.4) extends to $t=0$ and may be differentiated there. By (11.3),

$$P(0)A=-P'(0)C.\tag{11.5}$$

Each ratio C_{n+j}/C_n is rational in n, so the right side is $C_nR(n)$ for some $R\in\mathbb{Q}(n)$. If $R=0$, $P(0)$ already annihilates A. Otherwise $C_nR(n)$ is annihilated, on a nonsingular tail, by

$$Q=E-\frac{2(2n+1)R(n+1)}{(n+1)R(n)}.$$

The product $QP(0)$ is nonzero: in the ring of recurrence operators over $\mathbb{Q}(n)$, the leading coefficient of a product is the product of a nonzero coefficient and a shift of a nonzero coefficient. This product annihilates A. Clearing its rational denominators completes the proof. □

Theorem 11.4. *The unscaled Sondow sequence $(I_n)_{n\ge1}$ is not P-recursive over $\overline{\mathbb{Q}}$, and its ordinary generating series is not D-finite over $\overline{\mathbb{Q}}(z)$.*

Proof. The sequence γC_n obeys

$$(n+1)\gamma C_{n+1}-2(2n+1)\gamma C_n=0$$

regardless of the arithmetic nature of γ. If I_n were P-recursive over $\overline{\mathbb{Q}}$, closure under addition and Lemma 11.3 would make $L_n=I_n-\gamma C_n+A_n$ P-recursive over that field. This contradicts Theorem 11.1. The closure property does not require algebraic initial values: the shifts of each sequence span a finite-dimensional space over $\overline{\mathbb{Q}}(n)$, and shifts of their sum lie in the sum of those spaces. The generating-series assertion follows by coefficient comparison. □

Remark 11.5. Theorems 11.1 and 11.4 exclude a homogeneous recurrence of fixed order, with coefficients in $\overline{\mathbb{Q}}(n)$, holding at every sufficiently large integer. They do not exclude growing-order transformations, prime-adaptive coefficients, exceptional subsequences, inhomogeneous relations, or other integral families. In particular, they are consistent with Corollary 10.3 and do not settle the arithmetic nature of γ.

12 Normalization, short relations, and survival of target directions

For a window $N \le n \le M$, write $B = B_{[N,M]}$ and

$$\Phi(u) = (Q, P) = (a \cdot u, b \cdot u).$$

When $Bu = 0$, the logarithms cancel and

$$\sum_{n=N}^{M} u_n e_n = Q\gamma - P. \tag{12.1}$$

The assertions $u \neq 0$, $\Phi(u) \neq 0$, and $Q\gamma - P \neq 0$ are different. The first two are statements about rational linear maps; the last is a statement about a value at γ.

Proposition 12.1 (Dependence of crude width bounds on normalization)**.** *Fix $w > 0$, put $M = \lfloor (1+w)N \rfloor$, and let $c = \log 16 - 2 > 0$. As $N \to \infty$,*

$$\max_{N \le n \le M} d_{2n} I_n \le \exp(-cN + o(N)), \tag{12.2}$$

$$\max_{N \le n \le M} d_{2M} I_n \le \exp(-(c-2w)N + o(N)). \tag{12.3}$$

Thus the restriction $w < c/2$ is sufficient for exponential decay in the second crude bound, but is not a restriction in the first.

Proof. Write $T = x(1-x)y(1-y)$ and $h = ((1-xy)(-\log(xy)))^{-1}$ for the integral kernel. For $0 < x, y < 1$, the inequality $-\log(xy) > 1 - xy$ gives

$$Th < \frac{xy(1-x)(1-y)}{(1-xy)^2} \le xy.$$

The last inequality follows from $1 - xy \ge 1 - x$ and $1 - xy \ge 1 - y$. Therefore $I_1 < 1/4$. Since $T \le 1/16$,

$$0 < I_n \le 16^{-(n-1)} I_1 < 4\,16^{-n} \qquad (n \ge 1). \tag{12.4}$$

The prime number theorem implies $\log d_m = m + o(m)$. On the fixed-width window this yields $\log d_{2n} = 2n + o(N)$ uniformly. Combining with (12.4) proves (12.2). Using d_{2M} for every column instead gives $2M - N \log 16 = -(c-2w)N + O(1)$, proving (12.3). □

The two presentations are related by positive rational column rescaling. To see exactly what is preserved, let $C = [B; a; b]$, let R be a positive rational diagonal matrix, and let $\epsilon_i > 0$ be error bounds for the columns of C. For a fixed target v, the substitution $x = Rw$ gives a bijection

$$\{w : (CR)w = (0, v)\} \longleftrightarrow \{x : Cx = (0, v)\}, \qquad \sum_i (\epsilon_i R_{ii})|w_i| = \sum_i \epsilon_i |x_i|.$$

Consequently the rational weighted optimization for that target is unchanged. An unweighted bound on integer coefficients, or on their relative sizes, is not preserved by this substitution. The width penalty in (12.3) must therefore not be combined with a coefficient bound from a different presentation.

Lemma 12.2 (An elementary kernel bound)**.** *Let an integer matrix have R rows and s columns, with entries of absolute value at most $H \ge 1$, and suppose $s > R$. It has a nonzero integer kernel vector u with*

$$\|u\|_\infty \le K := \left\lfloor (2sH+1)^{R/(s-R)} \right\rfloor. \tag{12.5}$$

No row-rank hypothesis is needed.

Proof. The $(K+1)^s$ points of $\{0,\dots,K\}^s$ have at most $(2sHK+1)^R$ images. Since

$$(K+1)^{s-R} > (2sH+1)^R, \qquad 2sHK+1 \le (2sH+1)(K+1),$$

two distinct grid points have the same image. Their difference is the required vector. □

Theorem 12.3 (Small relations can have zero target)**.** *Fix $w>0$ and $M=\lfloor (1+w)N\rfloor$. For every sufficiently large N there is a nonzero integer vector u, supported in $[N,M]$, such that*

$$Bu = 0, \qquad \Phi(u) = 0, \tag{12.6}$$

and

$$\log \|u\|_\infty \le \left(\frac{2(2+\log 4)(1+w)^2}{w} + o(1) \right) \frac{N}{\log N}, \tag{12.7}$$

$$\sum_{n=N}^{M} |u_n| e_n \le \exp(-cN + o(N)). \tag{12.8}$$

The signed sum in (12.1) is exactly zero for these vectors. The same size estimates, without the requirement $\Phi(u)=0$, also follow directly from the kernel bound.

Proof. The binomial identity $\sum_i \binom{n}{i}^2 = C_n \le 4^n$ gives $c_{n,k} \le 2H_n 4^n$. Therefore

$$0 \le \beta_{p,n} \le 2n\, d_{2n} H_n 4^n \log_2(2n) \le 4n^3 d_{2n} 4^n.$$

Here $H_n \le n$ and $\log_2(2n) \le 2n$ for $n \ge 1$. The last bound also dominates a_n and b_n, since $A_n \le H_{2n} C_n \le 2n4^n$. Thus the integer matrix $[B; a; b]$ has at most $R = \pi(2M)+2$ rows, $s = M-N+1$ columns, and entries bounded by

$$H = 4M^3 d_{2M} 4^M.$$

As $N \to \infty$, the prime number theorem gives

$$R = (2(1+w)+o(1))\frac{N}{\log N}, \quad s = wN + O(1), \quad \log H = (2+\log 4)(1+w)N + o(N).$$

In particular $s > R$ for large N. Apply Lemma 12.2 to obtain (12.6) and (12.7). The number of terms and their coefficient bound contribute only $\exp(o(N))$, so Proposition 12.1 proves (12.8). Omitting the last two rows makes the same argument apply to B alone. □

The coefficient $(1+w)^2/w = w + 2 + 1/w$ in this particular estimate is minimized at $w=1$. No finite onset, optimal window for primitive approximation, or necessary coefficient lower bound follows from this asymptotic observation.

Lemma 12.4 (Span of short relations)**.** *For an R-by-s integer matrix A with entries bounded by H, let*

$$V_K = \operatorname{span}_{\mathbb{Q}}\{u \in \mathbb{Z}^s : Au = 0,\ \|u\|_\infty \le K\}.$$

For every integer $K \ge 1$,

$$\dim V_K \ge \frac{s\log(K+1) - R\log(2sHK+1)}{\log(2K+1)}. \tag{12.9}$$

Proof. A fiber of A on $\{0,\dots,K\}^s$ contains at least $(K+1)^s/(2sHK+1)^R$ points. Subtract one point of that fiber. The resulting distinct vectors lie in $V_K \cap [-K,K]^s$. If $d = \dim V_K$, some projection onto d coordinate positions is injective on V_K: choose d independent rows of a matrix whose columns are a basis of V_K. There are at most $(2K+1)^d$ projected integer vectors. Taking logarithms gives (12.9). □

Corollary 12.5. *With the window and notation of Theorem 12.3, there are at least* $s - O(N/\sqrt{\log N})$ *linearly independent integer vectors satisfying* (12.6)*, each with*

$$\|u\|_\infty \le \left\lceil \exp\left(\frac{N}{\sqrt{\log N}}\right)\right\rceil, \qquad \sum_{n=N}^{M} |u_n| e_n \le \exp(-cN + o(N)).$$

Proof. Apply Lemma 12.4 to $[B; a; b]$ with the displayed K. In its right side, $R = O(N/\log N)$, $\log H = O(N)$, and $\log K \sim N/\sqrt{\log N}$. Hence its deficit from s is $O(N/\sqrt{\log N})$. Choose an independent subset from the vectors spanning V_K. Proposition 12.1 provides their stated absolute-sum bounds. □

Even a subspace of nearly full dimension generated by short relations can therefore consist entirely of zero-target relations. The next bound addresses the target map explicitly.

Theorem 12.6 (A quantitative target-survival bound)**.** *Let* $B \in \mathbb{Z}^{r\times s}$ *have rank* r *and entries of absolute value at most* $H \ge 1$*. Suppose a rational-linear map* $\Phi : \ker_{\mathbb{Q}} B \to \mathbb{Q}^t$ *has rank* t*. Let* $1 \le j \le t$ *and* $K \ge 1$ *be integers. If*

$$(K+1)^{t-j+1} > (2sH+1)^r, \tag{12.10}$$

then the images under Φ *of the integer vectors* $u \in \ker B$ *with* $\|u\|_\infty \le K$ *span dimension at least* j*.*

Proof. Suppose those images lie in a subspace $T \subseteq \mathbb{Q}^t$ of dimension at most $j-1$. Two points in the same B-fiber of $\{0,\dots,K\}^s$ differ by one of the specified short vectors. Each such fiber consequently lies in a translate of $\ker B \cap \Phi^{-1}(T)$, of dimension at most $s-r-t+j-1$. An injective coordinate projection on that linear space bounds the size of the fiber by $(K+1)^{s-r-t+j-1}$. The number of B-images is at most $(2sHK+1)^r$. Thus

$$(K+1)^s \le (2sHK+1)^r (K+1)^{s-r-t+j-1} \le (2sH+1)^r (K+1)^{s-t+j-1},$$

contrary to (12.10). □

For a rank-two target, this guarantees one surviving direction when $K+1 > (2sH+1)^{r/2}$, and two independent directions when $K+1 > (2sH+1)^r$. If its rank assumptions hold for a Sondow window, the coefficient estimates used above yield a sufficient bound of order $\log K = O(N^2/\log N)$. That sufficient bound is too large to combine with the exponential error estimate; it is not a lower bound on the size of actual Sondow target vectors.

Proposition 12.7 (Limits of rank-and-height information)**.** *There are integer matrices of rank* r*, coefficient bound* H*, and rank-two target maps for which every nonzero target requires an integer lift of sup norm at least* $H^{r/2}$ *when* r *is even. There are also such examples in which obtaining two independent target images requires a lift of sup norm at least* H^r*. Arbitrarily many zero-target kernel coordinates can be appended without changing these conclusions.*

Proof. Let $H \ge 2$ be an integer and first put $r = 2m$. Use the equations

$$x_i = Hx_{i+1}, \qquad y_i = Hy_{i+1} \qquad (0 \le i < m), \qquad \Phi = (x_m, y_m).$$

They have rank $2m$, and an integer solution has $x_i = H^{m-i}a$, $y_i = H^{m-i}b$, with $a, b \in \mathbb{Z}$. Any nonzero target has norm at least H^m in its lift; this bound is attained with $(a,b) = (1,0)$ or $(0,1)$.

For the second assertion use a single chain $x_i = Hx_{i+1}$, $0 \le i < r$, an unconstrained coordinate y, and $\Phi = (x_r, y)$. The y direction has a lift of norm one. Two independent images, however, require a vector with $x_r \ne 0$, whose lift has norm at least H^r. This is also attained. In both constructions further unconstrained coordinates may be added and mapped to zero. □

These examples concern general integer matrices, not the particular Sondow matrix. They show why additional information about the prime exponents and the primitive target coordinates is required: an abundance of short relations, by itself, does not guarantee short relations that survive in the target.

13 Primitive targets and an exact quotient norm

Fix a finite block $\mathcal{I}$ of s indices. Retain the integer coefficient identity from the preceding section in the vector form

$$e = a\gamma - b + B^T\lambda_\gamma, \qquad (\lambda_\gamma)_p = \log p, \qquad \Phi w = (a^T w, b^T w). \tag{13.1}$$

Here B has r rows, indexed by the relevant primes, and a, b, e are column vectors of length s. Put

$$\mathcal{C} = \begin{pmatrix} B \\ a^T \\ b^T \end{pmatrix}.$$

Throughout this section we assume

$$\operatorname{rank} \mathcal{C} = r + 2. \tag{13.2}$$

This assumption implies that $\Phi : \ker B \longrightarrow \mathbb{R}^2$ is surjective. We do not assert (13.2) for every moving block; it is checked exactly for each block used in the finite computations below.

Let $\varepsilon_n \in \mathbb{Q}_{>0}$ be proved upper bounds for e_n. The usual triangle inequality assigns to an integer relation $Bu = 0$ the bound $\sum_n \varepsilon_n |u_n|$. The size of this expression before dividing the target by its greatest common divisor can be misleading. The following quotient construction performs the division exactly.

Theorem 13.1 (Primitive quotient optimization)**.** *For $v \in \mathbb{R}^2$, define*

$$F(v) = \min_{Bw=0,\ \Phi w = v} \sum_{n \in \mathcal{I}} \varepsilon_n |w_n|. \tag{13.3}$$

Then F is a norm on $\mathbb{R}^2$ whose unit ball is a centrally symmetric rational polygon. If $v = (q, p) \in \mathbb{Z}^2$ is primitive, then

$$F(v) = \min_{\substack{u \in \mathbb{Z}^s,\ Bu=0,\ \Phi u = gv \\ g \in \mathbb{Z}_{>0}}} \frac{\sum_n \varepsilon_n |u_n|}{g}. \tag{13.4}$$

Both minima are attained. For a rational target, a rational optimal lift in (13.3) can be chosen with at most $r + 2$ nonzero coordinates. Moreover,

$$|q\gamma - p| \le F(q, p). \tag{13.5}$$

Proof. Surjectivity gives a feasible lift for every target. Positivity of all weights makes the intersections of a fiber with sublevel sets of the objective compact. Consequently the minimum exists. For a rational target this is a rational linear program, after writing $w = w^+ - w^-$ with $w^+, w^- \ge 0$, and it has a rational optimum.

Every integer lift on the right of (13.4) gives a feasible rational lift u/g on the left. Conversely, let w be a rational optimum and choose a positive common denominator g of its coordinates. Then $u = gw$ is integral, $Bu = 0$, and $\Phi u = g(q, p)$. Since $\gcd(q, p) = 1$,

$$\gcd(a^T u, b^T u) = g. \tag{13.6}$$

Thus the denominator used to clear the lift is exactly the greatest common divisor removed from the target. This proves the equality and attainment in (13.4).

The quotient of the weighted ℓ^1 norm is symmetric, homogeneous, and subadditive. If $F(v) = 0$, an attained minimum has zero lift, hence $v = 0$. The unit ball is the image under Φ of the intersection of $\ker B$ with the rational weighted cross-polytope $\sum \varepsilon_n |w_n| \le 1$. Surjectivity makes this image two-dimensional; it is a centrally symmetric rational polygon.

For the support bound, choose a rational optimum. If its support has more than $r+2$ entries, there is a nonzero rational vector in $\ker \mathcal{C}$ supported on those entries. In a sufficiently small interval about the optimum, the objective is affine along this vector. Its slope must be zero, or one direction would improve the objective. Move to an endpoint of the interval on which the coordinate signs are unchanged. At least one coordinate vanishes, and the objective remains optimal. Repetition proves the assertion.

Finally, (13.1) gives $e^T w = q\gamma - p$ whenever $Bw = 0$ and $\Phi w = (q, p)$. Taking the triangle inequality and then the minimum proves (13.5). □

The lattice $\Phi(\ker B \cap \mathbb{Z}^s)$ can have a nontrivial index in $\mathbb{Z}^2$. The theorem does not identify it with $\mathbb{Z}^2$. Instead, it allows a multiple gv of a primitive target and then divides by its actual target gcd. An obstruction formulated solely in terms of the unnormalized lift cannot be carried over to F without accounting for (13.6).

Corollary 13.2 (Invariance under positive column normalization)**.** *Let $D = \operatorname{diag}(t_n)$, where $t_n \in \mathbb{Q}_{>0}$. Replace (B, a^T, b^T) by $(BD, a^T D, b^T D)$ and replace ε_n by $t_n \varepsilon_n$. The function F is unchanged.*

Proof. The substitution $x = Dw$ identifies both feasible sets and their objectives. All data remain rational, so denominator clearing remains available even if the normalized presentation is not integral. □

In particular, the multipliers d_{2n} may be included in the columns or removed from them together with the corresponding error weights. Raw coefficient norms and raw dominance tests need not be invariant under this operation. The optimized primitive bound is invariant.

Proposition 13.3 (Dual certificate)**.** *For every $(q, p) \in \mathbb{R}^2$,*

$$F(q,p) = \max\left\{ \eta_1 q + \eta_2 p : \left| B^T \lambda + a\eta_1 + b\eta_2 \right| \le \varepsilon \right\}, \tag{13.7}$$

where inequalities and absolute values are coordinatewise. For a rational target, rational primal and dual optima exist.

Proof. The linear program with matrix $[\mathcal{C}, -\mathcal{C}]$ and nonnegative variables (w^+, w^-) has costs $(\varepsilon, \varepsilon)$. Its dual is precisely (13.7). The primal is feasible and has finite attained minimum by Theorem 13.1, so finite-dimensional linear programming duality applies. □

In particular a finite optimum can be checked without trusting the optimization algorithm. It is enough to supply a rational feasible lift w and rational $(\lambda, \eta_1, \eta_2)$ satisfying the dual inequalities and equality of the two objectives. Equivalently, if $z \in \operatorname{row}(\mathcal{C})$ satisfies $|z_n| \le \varepsilon_n$ and $z_n w_n = \varepsilon_n |w_n|$ for every n, then z certifies the global optimality of w on its fiber. These are rational equalities and inequalities.

14 Nonvanishing from polynomial positivity

The conditions $u \neq 0$ and $\Phi u \neq 0$ do not by themselves imply $e^T u \neq 0$. A separate analytic sign certificate is possible for many Sondow combinations. Put

$$T(x,y) = x(1-x)y(1-y), \qquad h(x,y) = \frac{1}{(1-xy)(-\log(xy))}.$$

Define a measure on $[0, 1/16]$ by

$$\nu(E) = \int_{\{(x,y)\in(0,1)^2 : T(x,y)\in E\}} T(x,y)h(x,y)\,dx\,dy. \tag{14.1}$$

Its mass is $I_1 < 1/4$, as follows also from Lemma 9.1. Every nonempty open subinterval of $(0, 1/16)$ has positive ν-measure: the continuous map T takes every such value in the open square, and its inverse image of the subinterval is nonempty and open. Moreover,

$$I_n = \int_0^{1/16} t^{n-1}\,d\nu(t), \qquad n \geq 1. \tag{14.2}$$

The factor T in (14.1) ensures that the measure is finite; no finite zeroth integral I_0 is used.

For a consecutive block $\mathcal{I} = \{N, \ldots, N+m\}$ with $N \geq 1$ and a real lift $w = (w_0, \ldots, w_m)$, define

$$R_w(X) = \sum_{j=0}^{m} w_j d_{2(N+j)} 16^{-j} X^j. \tag{14.3}$$

The notation w_j here refers to the coordinate at index $N + j$. Equation (14.2) gives the exact identity

$$\sum_{j=0}^{m} w_j d_{2(N+j)} I_{N+j} = \int_0^{1/16} t^{N-1} R_w(16t)\,d\nu(t). \tag{14.4}$$

Proposition 14.1 (A gamma-independent sign certificate)**.** *Suppose $Bw = 0$, $\Phi w = (q, p)$, and R_w is nonnegative on $[0, 1]$ and not identically zero. Then $q\gamma - p > 0$.*

Proof. A nonzero nonnegative polynomial is positive on a nonempty open subinterval of $(0, 1)$. Full support of ν, together with $t^{N-1} > 0$ for $t > 0$, makes the right side of (14.4) strictly positive. Its left side is $q\gamma - p$ by prime-log cancellation. □

14.1 Exact Bernstein coefficients

For $R(X) = \sum_{j=0}^{m} r_j X^j$, the degree-$m$ Bernstein expansion is

$$R(X) = \sum_{k=0}^{m} b_k \binom{m}{k} X^k (1-X)^{m-k}, \qquad b_k = \sum_{j=0}^{k} r_j \frac{\binom{k}{j}}{\binom{m}{j}}. \tag{14.5}$$

Indeed, the identity

$$X^j = \sum_{k=j}^{m} \frac{\binom{k}{j}}{\binom{m}{j}} \binom{m}{k} X^k (1-X)^{m-k}$$

follows from $\binom{m}{k}\binom{k}{j} = \binom{m}{j}\binom{m-j}{k-j}$ and the binomial theorem. Thus rational monomial coefficients give rational Bernstein coefficients by an explicit triangular transform. If all $b_k \geq 0$ and at least one is positive, then $R(X) > 0$ for $0 < X < 1$. If all $b_k > 0$, the partition-of-unity identity gives $R(X) \geq \min_k b_k > 0$ throughout $[0, 1]$.

The elementary sufficient condition

$$r_0 > \sum_{j=1}^{m} |r_j| \tag{14.6}$$

implies positivity of every b_k, since each binomial ratio in (14.5) belongs to $[0, 1]$. Its converse fails. Consequently Bernstein positivity supplies additional sign certificates beyond first-coefficient dominance; the finite Sondow example below exhibits this difference within an actual log-free kernel.

Proposition 14.2 (Eventual degree elevation)**.** *Let $R \in \mathbb{Q}[X]$ have degree at most m and be strictly positive on $[0,1]$. For all sufficiently large integers $M \geq m$, every coefficient in its degree-M Bernstein expansion is positive. More precisely, put*

$$C_R = \frac{1}{2}\sum_{j=2}^{m} |r_j| j(j-1).$$

For all $0 \leq k \leq M$,

$$\left| b_k^{(M)} - R(k/M) \right| \leq \frac{C_R}{M-m+1}, \qquad b_k^{(M)} = \sum_{j=0}^{\min(k,m)} r_j \frac{\binom{k}{j}}{\binom{M}{j}}. \tag{14.7}$$

In particular, if $R \geq \delta > 0$, it suffices that $C_R/(M-m+1) < \delta$.

Proof. The contributions $j = 0, 1$ agree exactly with evaluation at k/M. For $j \geq 2$ and $k \geq j$, compare

$$\frac{\binom{k}{j}}{\binom{M}{j}} = \prod_{i=0}^{j-1} \frac{k-i}{M-i} \quad \text{with} \quad (k/M)^j.$$

All factors belong to $[0,1]$, and

$$\left| \frac{k-i}{M-i} - \frac{k}{M} \right| = \frac{i(M-k)}{M(M-i)} \leq \frac{i}{M-m+1}.$$

Telescoping the difference between the two products gives the bound $j(j-1)/(2(M-m+1))$. If $k < j$, the binomial ratio is zero and

$$(k/M)^j \leq (j-1)/M \leq \frac{j(j-1)}{2(M-m+1)}.$$

Summing these bounds with weights $|r_j|$ proves (14.7). Compactness gives $\delta = \min_{[0,1]} R > 0$, proving eventual positivity. □

The proposition gives a rational semidecision procedure for strict positivity of a fixed polynomial. It makes no bound uniform over all lifts or all levels. A nonnegative polynomial with an interior zero cannot have a nonzero expansion with all Bernstein coefficients nonnegative, because every Bernstein basis function is positive in the interior. Such polynomials require a different exact nonnegativity certificate.

For an explicit lower bound, on $[1/4, 3/4]^2$ one has $T \geq 9/256$ and $h > 1/3$, using $-\log(xy) \leq \log 16 < 3$. The square has area $1/4$, so

$$I_N > \frac{1}{12}(9/256)^N. \tag{14.8}$$

If $s_0 \in \{1, -1\}$ orients R_w so that all its Bernstein coefficients are positive, and b_* is their minimum, then

$$s_0(q\gamma - p) > \frac{b_*}{12}(9/256)^N > 0. \tag{14.9}$$

Everything on the right is rational for a rational lift.

14.2 Positivity inside the primitive optimization

Let $\mathcal{B}$ denote the rational linear map from w to the Bernstein coefficients of R_w at a fixed degree. Define

$$F^+(v) = \min_{\substack{Bw=0,\ \Phi w=v \\ \mathcal{B}w\geq 0}} \sum_j \varepsilon_j |w_j|, \tag{14.10}$$

with value $+\infty$ if infeasible. This is an asymmetric, extended-valued optimization, not a norm. For a nonzero rational target, every feasible lift has a nonzero polynomial: the monomial coefficient map in (14.3) is invertible. Thus

$$0 < q\gamma - p \leq F^+(q,p) \tag{14.11}$$

whenever the minimum is finite. Compactness of weighted ℓ^1 sublevel sets and rational linear programming give an attained rational optimum. For a primitive integer target one has

$$F^+(v) = \min_{\substack{u\in\mathbb{Z}^s,\ Bu=0,\ \Phi u=gv,\ g\in\mathbb{Z}_{>0} \\ \mathcal{B}u\geq 0}} \frac{\sum_j \varepsilon_j |u_j|}{g}. \tag{14.12}$$

The proof is exactly denominator clearing as in Theorem 13.1: the cone inequalities are homogeneous and survive multiplication or division by g. This shows that an explicit sign condition can be imposed before optimization without losing the exact gcd cancellation. It does not establish feasibility or decay at arbitrarily large levels.

15 Two-sided bounds and an affine feasible interval

We use the strict bounds $\ell_n < e_n < h_n$ proved in Lemma 9.1.

Retain a fixed block satisfying (13.2) and write $w_n^+ = \max(w_n, 0)$, $w_n^- = \max(-w_n, 0)$. The two-sided information suggests the support function

$$U(v) = \min_{Bw=0,\ \Phi w=v} \{h^T w^+ - \ell^T w^-\}. \tag{15.1}$$

Theorem 15.1 (Asymmetric primitive optimization)**.** *For every target the minimum in (15.1) is attained; a rational optimum exists for every rational target. For $v = (q,p) \neq 0$,*

$$-U(-v) < q\gamma - p < U(v). \tag{15.2}$$

For a primitive integer target,

$$U(v) = \min_{\substack{u\in\mathbb{Z}^s,\ Bu=0,\ \Phi u=gv \\ g\in\mathbb{Z}_{>0}}} \frac{h^T u^+ - \ell^T u^-}{g}. \tag{15.3}$$

Its dual is

$$U(q,p) = \max\left\{\eta_1 q + \eta_2 p : \ell \leq B^T\lambda + a\eta_1 + b\eta_2 \leq h\right\}. \tag{15.4}$$

Proof. For a lift in the fiber of v, subtracting its exact value from the objective gives

$$h^T w^+ - \ell^T w^- - (q\gamma - p) = (h-e)^T w^+ + (e-\ell)^T w^-.$$

All coefficients on the right are strictly positive. On this fixed finite block their minimum is positive, so the objective has compact sublevel sets on each fiber. This proves attainment and, for every nonzero lift, strict domination of $q\gamma - p$. A nonzero target has a nonzero optimal lift. Applying the same argument to $-v$ proves (15.2). Rational linear programming gives a rational optimum for rational v. Clearing its denominators and using (13.6) proves (15.3).

To compute the dual, use costs $(h, -\ell)$ for nonnegative variables (w^+, w^-). The dual inequalities are $B^T\lambda + a\eta_1 + b\eta_2 \leq h$ and its negative at most $-\ell$, which gives (15.4). Strong duality applies because the primal optimum is finite and attained. □

If the only lower information is $e_n \geq 0$, the origin belongs to the dual feasible set and $U(v) \geq 0$ in every direction. Such a support function alone cannot give a strictly negative upper bound or a strictly positive lower bound. The quantitative lower bounds in (9.8) contain additional sign information.

15.1 Fixing the constant coordinate

Equation (13.1) specifies the coefficient of b exactly: it is -1. Preserving this coordinate gives the following relaxation of the gamma identities:

$$G = \left\{\theta \in \mathbb{R} : \exists y \in \mathbb{R}^r \quad \ell \leq a\theta - b + B^T y \leq h\right\}. \tag{15.5}$$

The variables y are arbitrary real numbers. In particular, this model does not impose that y consists of actual prime logarithms.

Theorem 15.2 (Exact interval of the finite boxed model)**.** *The feasible set G is a compact rational interval $[\alpha, \beta]$ with*

$$\alpha < \gamma < \beta. \tag{15.6}$$

Its exact endpoints are

$$\beta = \min_{Bw=0,\ a^T w=1} \{b^T w + h^T w^+ - \ell^T w^-\}, \tag{15.7}$$

$$-\alpha = \min_{Bw=0,\ a^T w=-1} \{b^T w + h^T w^+ - \ell^T w^-\}. \tag{15.8}$$

Both linear programs have rational primal and dual optima. For every real target (q, p), the best worst-case absolute error permitted by the model (15.5) is

$$J(q,p) := \max_{\theta \in G} |q\theta - p| = \max\{|q\alpha - p|, |q\beta - p|\}. \tag{15.9}$$

For $q > 0$, this gives the strict signed bounds

$$q\alpha - p < q\gamma - p < q\beta - p, \tag{15.10}$$

which are at least as strong as (15.2).

Proof. The point $(\theta, y) = (\gamma, \lambda_\gamma)$ satisfies every inequality strictly. It therefore has a neighborhood within the feasible set. The map $(y, \theta) \mapsto B^T y + a\theta$ is injective because $[B; a^T]$ has full row rank. Its inverse on its image is continuous, so the inverse image of the bounded box $b + [\ell, h]$ is bounded. It is a nonempty compact rational polytope. Its projection onto the θ-axis is a compact rational interval, and the neighborhood of the actual point places γ strictly between its endpoints.

For (15.7), split w into nonnegative parts. The costs become $(b+h, -b-\ell)$ and the constraint matrix is $[B; a^T]$. The dual maximizes θ under

$$\ell \leq B^T y + a\theta - b \leq h,$$

which is exactly the maximum in (15.5). Changing the target coordinate 1 to -1 changes the dual objective to $-\theta$ and proves (15.8). Feasibility, boundedness, and rationality give the asserted attained optima.

The interval formula (15.9) is immediate from linearity. The points $(y, \theta, -1)$ form a slice of the feasible set in (15.4). Taking the maximum of $q\theta - p$ on this slice cannot exceed $U(q, p)$; taking the minimum cannot be less than $-U(-q, -p)$. This proves the comparison of signed bounds. Strictness in (15.10) follows from (15.6) when $q > 0$. □

The qualification "of the finite boxed model" is essential. Endpoints are feasible in the relaxation, hence no smaller interval is implied solely by those linear inequalities. Additional valid constraints, for example the correlated positive-difference inequalities developed elsewhere in this part, can shrink this interval. The improvement over F also does not contradict the optimality of the fixed-target quotient norm: an endpoint certificate allows $b^T w$ to vary and subsequently adds an exactly known rational constant when estimating a prescribed (q, p).

For completeness, let $c = (\alpha + \beta)/2$ and $\rho = (\beta - \alpha)/2 > 0$. The elementary identity $\max(|u + v|, |u - v|) = |u| + |v|$ yields

$$J(q, p) = \rho|q| + |cq - p|. \tag{15.11}$$

Thus J is a norm whose unit ball is a rational parallelogram. The change $(q, p) \mapsto (q, cq - p)$ has determinant -1, so its area is $2/\rho = 4/(\beta - \alpha)$. This explicit geometry does not supply denominator growth or an infinite approximation theorem: the placement of the parallelogram relative to $\mathbb{Z}^2$ remains an arithmetic question.

16 Exact finite certificates at two levels

The computations use the blocks $\{N, \dots, 2N\}$ for $N = 32, 40$. All Sondow coefficients, bounds, linear programs, and comparisons are rational or integral. The targets are retained from the preceding quotient computation. That construction chose continued fractions of a rational center computed from a rational right inverse of $\mathcal{C}$ and a weighted-median minimization; it did not take continued fractions of a supplied decimal value of γ. Nothing about this finite selection establishes a uniform assertion as N varies.

Write (q_N, p_N) for the first retained target at level N. They are

$$\begin{aligned} q_{32} &= 754657555134112732808638400 27, \\ p_{32} &= 435600162459702074577366065 78, \end{aligned} \tag{16.1}$$

$$\begin{aligned} q_{40} &= 14660316118186948404524956764515505 25, \\ p_{40} &= 84621641158259386268764501033152289 9. \end{aligned} \tag{16.2}$$

Their numerator–denominator gcds are 1. In the following table, α_N, β_N are the exact rational endpoints of Theorem 15.2, $\mathsf{L}_N = q_N\alpha_N - p_N$, and $\mathsf{R}_N = q_N\beta_N - p_N$. Decimal logarithms are descriptive rounded values; they are not used to prove any displayed rational inequality.

N	$\log(\beta_N - \alpha_N)$	$\log\lvert\mathsf{L}_N\rvert$	$\log\lvert\mathsf{R}_N\rvert$	signs of $(\mathsf{L}_N, \mathsf{R}_N)$
32	−132.84450590	−68.20232457	−66.20516987	$(+, +)$
40	−174.74952280	−86.07078916	−86.07530188	$(-, -)$

Table 1: Exact endpoint certificates summarized on the target-form scale. The large rational endpoints and their primal–dual witnesses are stored in the accompanying certificate file.

The corresponding finite changes of error bound are shown in Table 2. Here the symmetric bound uses $\varepsilon = h$ in (13.3), the two-sided bound uses (15.2), and the projected bound uses (15.9).

N	symmetric log bound	two-sided log bound	projected log bound
32	−65.730623	−65.782655	−66.205170
40	−84.632505	−84.633619	−86.070789

Table 2: Three certified upper bounds for the same two primitive targets.

There is also a direct polynomial sign certificate for each of these targets, using the optimal lift for the symmetric bound. For $N = 32$ the polynomial fails (14.6), but all its 33 degree-32

Bernstein coefficients are positive. For $N = 40$ the negative of the lift polynomial has all 41 degree-40 Bernstein coefficients positive. Formula (14.9) gives positive rational lower bounds whose logarithms are approximately -85.084754 and -107.596244, respectively. These establish the signs directly from the integral representation at the same level. The affine interval bounds in Table 1 are stronger for these targets.

Proposition 16.1 (A finite exponent certificate)**.** *For the integers $q = q_{40}$ and $p = p_{40}$ in (16.2), the exact certificates prove*

$$0 < p - q\gamma < q^{-103/100}. \tag{16.3}$$

Proof. The certificate supplies exact rational endpoints α_{40}, β_{40} satisfying

$$q\beta_{40} - p < 0, \qquad E := p - q\alpha_{40} > 0, \qquad E^{100}q^{103} < 1.$$

Theorem 15.2 therefore gives $0 < p - q\gamma < E < q^{-103/100}$. All three certificate comparisons are rational or integral; no decimal evaluation of γ enters the proof. □

To make the role of computation precise, the supplied verifier reconstructs B, a, b, ℓ, h from the coefficient formulas. For each linear program it checks a rational lift, the exact target equations, a rational dual vector, every dual inequality, and equality of the objectives. In the endpoint programs the constraint matrix is $[B; a^T]$ and the extra objective offset is $b^T w$. For an arbitrary rational offset o and target t, a certificate for the problem

$$\min_{Aw=t} \{o^T w + h^T w^+ - \ell^T w^-\}$$

consists of w, y satisfying

$$Aw = t, \qquad \ell \le A^T y - o \le h, \qquad o^T w + h^T w^+ - \ell^T w^- = t^T y.$$

Weak duality proves optimality directly from these relations. The verifier separately checks denominator clearing and (13.6), the signs in Table 1, and $E^{100}q^{103} < 1$. The positivity verifier also reconstructs and inverts the Bernstein transform rather than sampling a grid of real points.

The principal data files are `gamma_push_quotient_sharp_certificate.json`, `gamma_sondow_intervals_certificate.json`, and `gamma_sondow_positivity_certificate.json`. The interval certificate can be checked without reoptimizing by running

```
python3 gamma_sondow_intervals.py --verify
```

the corresponding positivity checks are implemented in `gamma_sondow_positivity.py`. The accompanying sources specify the dependencies and preserve the exact integers and rational witnesses omitted from the printed tables.

Proposition 16.1 concerns one pair, not an infinite sequence. In particular, the exponent $103/100$ in the integer linear form, or $203/100$ after division by q, supplies no irrationality or transcendence conclusion from this finite certificate. The results of these sections are the exact normalization of primitive bounds, explicit nonvanishing criteria, the optimal interval in a specified finite relaxation, and verified finite applications of those constructions.

17 Positive differences of moving order

The absolute value of each coefficient in an alternating combination is often an unnecessarily expensive way to estimate its error. A common positive moment representation can retain the correlation between the integrals. Here that observation gives an explicit construction at every pair of indices, together with its exact exponential rate.

Lemma 17.1 (A finite moment carrier)**.** *Let σ be the pushforward of $th(x,y)\,dx\,dy$ under $s = 16t$. It is a finite positive measure on $[0,1]$, gives positive mass to every nonempty open subinterval of $(0,1)$, and satisfies*

$$\sigma([0,1]) = I_1 < \frac{1}{4}, \qquad I_r = 16^{-(r-1)} \int_0^1 s^{r-1}\, d\sigma(s) \quad (r \geq 1). \tag{17.1}$$

Proof. The upper comparison in Lemma 9.1 gives $th(x,y) \leq xy$, whose integral is $1/4$. The strict mass bound follows. For any interior value of s, an open subinterval around it has a nonempty open preimage inside the square; on that preimage the density is positive. This proves full interior support. The moment identity is the definition of pushforward applied to $(16t)^{r-1}$. □

Theorem 17.2 (All-index positive differences and rational bounds)**.** *For integers $n \geq 1$ and $m \geq 0$, put*

$$D_{n,m} = \sum_{k=0}^{m} (-16)^k \binom{m}{k} I_{n+1+k}.$$

Then

$$D_{n,m} = 16^{-n} \int_0^1 s^n (1-s)^m\, d\sigma(s) > 0 \tag{17.2}$$

and, with $0^0 = 1$,

$$D_{n,m} < \frac{16^{-n}}{4} \frac{n^n m^m}{(n+m)^{n+m}}, \tag{17.3}$$

$$D_{n,m} > \frac{1}{1024} \left(\frac{9}{256}\right)^{n+1} \left(\frac{31}{256}\right)^{m}. \tag{17.4}$$

Proof. The binomial theorem applied inside (17.1) proves (17.2); the integrand is positive in the interior. Its maximum is $n^n m^m/(n+m)^{n+m}$, at $s = n/(n+m)$ when $m > 0$; the case $m = 0$ is immediate. The mass bound in Lemma 17.1 proves the upper bound. For the lower bound, return to the square and write

$$D_{n,m} = \int_{(0,1)^2} t^{n+1}(1-16t)^m h(x,y)\, dx\, dy.$$

On $[1/4, 3/8]^2$ one has $t \geq 9/256$, $1 - 16t \geq 31/256$, and $h > xy \geq 1/16$. Its area is $1/64$, so restriction to this square proves (17.4). □

The bound obtained instead from $I_r < 4\,16^{-r}$ by taking absolute values term by term is $(16^{-n}/4)2^m$. Its ratio to (17.3) is

$$2^m \frac{(n+m)^{n+m}}{n^n m^m},$$

equal to 8^n when $m = n$. This is a comparison of certified upper bounds, not a claim that the actual errors change under a change of representation.

Theorem 17.3 (Exact moving-order rate)**.** *Suppose $m_n \geq 0$ are integers and $m_n/n \to \lambda \in [0,\infty)$. Define*

$$\mathcal{H}(\lambda) = (1+\lambda)\log(1+\lambda) - \lambda \log \lambda, \qquad \mathcal{H}(0) = 0.$$

Then

$$\lim_{n\to\infty} \frac{\log D_{n,m_n}}{n} = -\log 16 - \mathcal{H}(\lambda). \tag{17.5}$$

For $E_{n,m} = d_{2(n+m+1)} D_{n,m}$, one consequently has

$$\lim_{n\to\infty} \frac{\log E_{n,m_n}}{n} = -\kappa(\lambda), \quad \kappa(\lambda) = \log 16 - 2 - 2\lambda + \mathcal{H}(\lambda). \tag{17.6}$$

The maximum over $\lambda \geq 0$ is

$$\kappa_{\max} = \log 16 - 2 - \log(1 - e^{-2}) = 0.9180021801\ldots, \qquad \lambda_* = (e^2 - 1)^{-1}. \tag{17.7}$$

Proof. The upper limit in (17.5) follows from (17.3). If $\lambda > 0$, the function $\log s + \lambda \log(1-s)$ has its unique maximum at $s_* = (1+\lambda)^{-1}$. On every closed interval contained in $(0,1)$, the exponents with m_n/n converge uniformly to this function. Choose a small interval around s_* and restrict (17.2) to it. Its σ-mass is positive. The logarithm of that fixed mass, divided by n, tends to zero, which gives the matching lower limit as the interval shrinks. When $\lambda = 0$, use a fixed interval arbitrarily close to 1; on each such interval the term $(m_n/n)\log(1-s)$ tends uniformly to zero. This gives the same conclusion.

The prime number theorem in the form $\log d_j = j + o(j)$ proves (17.6). Finally

$$\kappa'(\lambda) = -2 + \log(1 + 1/\lambda), \qquad \kappa''(\lambda) = -\frac{1}{\lambda(1+\lambda)} < 0.$$

The unique critical point is λ_*, and substitution gives (17.7). The endpoint $\lambda = 0$ has rate $\log 16 - 2$. □

A construction with rational index rules is already available: $m_n = \lfloor n/6 \rfloor$ gives $\lambda = 1/6$ and a rate larger than $\log 16 - 2$. There is no need to know the decimals of γ to choose these orders. Nevertheless $E_{n,m}$ is an integer linear form in $1, \gamma$, and prime logarithms, not yet in $1, \gamma$ alone.

Proposition 17.4 (Growth of the displayed gamma coefficient)**.** *Let*

$$\mathcal{A}_{n,m} = \sum_{k=0}^{m} (-16)^k \binom{m}{k} C_{n+1+k}.$$

For every sequence in Theorem 17.3, this coefficient is eventually nonzero with sign $(-1)^{m_n}$ *and*

$$\lim_{n\to\infty} \frac{\log |\mathcal{A}_{n,m_n}|}{n} = \log 4 + \lambda \log 63. \tag{17.8}$$

Thus the gamma coefficient in E_{n,m_n} *has exponential rate* $2(1+\lambda) + \log 4 + \lambda \log 63$.

Proof. The arcsine moment formula for the central binomial coefficients gives

$$\mathcal{A}_{n,m} = \int_0^1 (4y)^{n+1} (1-64y)^m \frac{dy}{\pi\sqrt{y(1-y)}}.$$

On $y > 1/64$ the integrand has constant sign $(-1)^m$. Its logarithmic absolute value divided by n has supremum tending to $\log 4 + \lambda \log 63$, approached as $y \uparrow 1$. Restriction to a fixed interval near 1 gives the matching lower exponential bound, just as in the proof of Theorem 17.3. The absolute contribution of $0 < y < 1/64$ is at most 16^{-n-1}, and hence cannot cancel the first contribution for large n. This proves the sign and the limit; the integer normalization contributes $2(1+\lambda)$ to the rate. □

This coefficient estimate is not a statement about the final primitive height after another elimination. The analytic decay rate, the coefficient growth rate, and the primitive height must remain separate.

17.1 Correlated constraints in a fixed window

Consider a block $\mathcal{J} = \{N, \dots, M\}$ and choose $i \geq N$, $m \geq 0$ with $i + m \leq M$ and $i \geq 2$. Put $D = d_{2M}$ and define the integer vector

$$u_{i+k} = \frac{D}{d_{2(i+k)}} (-16)^k \binom{m}{k} \quad (0 \leq k \leq m), \qquad u_j = 0 \text{ otherwise.} \tag{17.9}$$

Then $u \cdot e = DD_{i-1,m}$. The rational lower and upper bounds in Theorem 17.2, multiplied by D, give valid inequalities on this linear combination. Append them to the individual interval constraints in the same variables θ and y used for the gamma projection. Equivalently, append the column $(Bu, a \cdot u)$ and the constant offset $b \cdot u$ in the dual optimization. No new prime-logarithm variables are introduced.

The feasible set can only decrease, so its projected gamma interval can only contract. Strict contraction is a finite statement requiring a certificate. For $\mathcal{J} = \{N, \dots, 2N\}$, the stored computations take starts

$$i \in \{N, N + \lfloor N/4 \rfloor, N + \lfloor N/2 \rfloor\}$$

and orders $m \in \{1, 2, 4, 8, 16\}$ at $N = 32, 40$. All fifteen pairs are contained in their respective windows. The exact primal-dual certificates give Table 3; its decimals are only summaries of rational endpoint comparisons.

Table 3: Finite contraction after fifteen correlated constraints. Widths are positive; all logarithms are natural.

N	log(box width)	log(correlated width)	width ratio
32	−132.8445058965	−135.2714119062	11.32379212
40	−174.7495228030	−176.9827640464	9.33005811

Both new endpoints lie strictly inside the old interval. Since additional constraints were imposed, this improvement is compatible with optimality of the earlier box relaxation. These two contractions do not prove convergence of moving-window widths, much less a persistent improvement in the Diophantine exponent.

18 Rational moment approximants and graded denominator bounds

This section concerns a rational approximation family distinct from the logarithm-elimination construction. Its analytic error has a fixed polynomial scale, whereas the remaining arithmetic question concerns the denominator after complete cancellation. The positive-moment representation and the Padé construction are classical mechanisms; the statements below make the normalization and the remaining denominator question explicit.

18.1 A positive rational moment problem

Write

$$\mathscr{B}(z) = \frac{z}{-\log(1-z)} = \sum_{m \ge 0} \mathfrak{b}_m z^m, \qquad c_j = -\frac{\mathfrak{b}_{j+1}}{j+1} = \frac{1}{(j+1)(j+1)!} \int_0^1 x \prod_{k=1}^{j} (k - x)\, dx. \tag{18.1}$$

In particular, every c_j is a positive rational number, $c_0 = 1/2$, and $c_1 = 1/24$. The coefficient identity follows by integrating $(1-z)^x$ over $0 < x < 1$.

For $0 < t < 1$, put

$$\rho(t) = \frac{1}{t\{\log^2((1-t)/t) + \pi^2\}}, \qquad v(t) = \int_t^1 \frac{\rho(s)}{s}\, ds, \qquad d\nu(t) = v(t)\, dt. \tag{18.2}$$

Schröder's classical representation of the Gregory coefficients gives $-\mathfrak{b}_{j+1} = \int_0^1 t^j \rho(t)\, dt$; a contour derivation and its history are given in [11]. One can check the density directly: the function

$(1-\mathscr{B}(z))/z$ is analytic off $[1,\infty)$, tends to zero at infinity, and its upper-bank jump at $z=1/t$ is $2\pi i/(\log^2((1-t)/t)+\pi^2)$. Cauchy's formula around the slit therefore yields the stated moment representation. Tonelli's theorem now gives

$$c_j=\int_0^1 t^j\,d\nu(t),\qquad \gamma=\sum_{j\ge 0}c_j=\int_0^1\frac{d\nu(t)}{1-t}. \tag{18.3}$$

For completeness, the middle identity follows without an arithmetic assumption on γ: integrating $(1-\mathscr{B}(z))/z$ over $(0,1)$ and substituting $t=1-z$ gives

$$\sum_{j\ge 0}c_j=\int_0^1\left(\frac{1}{1-t}+\frac{1}{\log t}\right)dt=\lim_{m\to\infty}(H_m-\log(m+1))=\gamma.$$

The last equality follows by multiplying the integrand by $1-t^m$, using the integral formulas for H_m and $\log(m+1)$, and applying monotone convergence. All displayed positive measures have positive density on every open subinterval of $(0,1)$.

Define

$$K_N=(c_{i+j}-c_{i+j+1})_{0\le i,j<N},\qquad \boldsymbol{c}_N=(c_0,\dots,c_{N-1})^{\mathsf{T}},\qquad \widehat{\gamma}_N=\boldsymbol{c}_N^{\mathsf{T}}K_N^{-1}\boldsymbol{c}_N. \tag{18.4}$$

Theorem 18.1 (Positive remainder and nonsingularity)**.** *For every $N\ge 1$, K_N is positive definite, $\widehat{\gamma}_N\in\mathbb{Q}$, and*

$$R_N:=\gamma-\widehat{\gamma}_N=\min_{\substack{\deg P\le N\\ P(1)=1}}\int_0^1\frac{P(t)^2}{1-t}\,d\nu(t)>0. \tag{18.5}$$

The sequence $\widehat{\gamma}_N$ is strictly increasing and converges to γ. Moreover,

$$R_N=\Theta(N^{-2}\log^{-2}N),\qquad 0<R_N<\frac{1}{9N(N+1)}\quad(N\ge 1). \tag{18.6}$$

There is no exceptional singular case at $N=1$: $K_1=(11/24)$ and $\widehat{\gamma}_1=6/11$.

Proof. For a nonzero polynomial r of degree below N, $\int(1-t)r(t)^2d\nu(t)>0$, proving positive definiteness. Every admissible P is uniquely $1-(1-t)r$. If $\boldsymbol{r}$ is the coefficient vector of r, the objective in (18.5) equals

$$\gamma-2\boldsymbol{r}^{\mathsf{T}}\boldsymbol{c}_N+\boldsymbol{r}^{\mathsf{T}}K_N\boldsymbol{r}.$$

Its unique minimum is attained at $\boldsymbol{r}=K_N^{-1}\boldsymbol{c}_N$, which proves the identity and rationality. The minimum is positive because $P(1)=1$ and the measure has full support. The minimizing polynomial is orthogonal to every polynomial of degree below N for ν, by the first variation. If two successive minima were equal, uniqueness would give a nonzero polynomial of degree at most N orthogonal to itself for ν. Thus the decrease of R_N is strict.

Set $w(t)=v(t)/(1-t)$. From (18.2),

$$0<w(t)<\frac{1}{\pi^2 t},\qquad w(t)\ge\frac{1}{2\{\log^2(2/(1-t))+\pi^2\}}\quad(t\ge 1/2). \tag{18.7}$$

The first bound discards the logarithmic square in the denominator and integrates s^{-2}. For the second, restrict the integral to $t\le s\le(1+t)/2$. On that interval, $s^{-2}\ge 1$ and $|\log((1-s)/s)|\le\log(2/(1-t))$.

Let J_j be the shifted Jacobi polynomials normalized by $J_j(1)=1$ for the weight t on $[0,1]$. Rodrigues' formula and integration by parts give

$$J_j(t)=\frac{1}{j!\,t}\frac{d^j}{dt^j}\{t^{j+1}(t-1)^j\},\qquad \int_0^1 tJ_j(t)J_k(t)\,dt=\frac{\mathbf{1}_{j=k}}{2j+2}.$$

Consequently $r = \sum_{j=0}^{N-1}(2j+2)J_j/[N(N+1)]$ satisfies $r(1) = 1$ and $\int tr^2 = 1/[N(N+1)]$. The trial $P = tr$ in (18.5) proves $R_N < 1/[\pi^2 N(N+1)] < 1/[9N(N+1)]$, and in particular convergence.

Here are details of the sharp order, since it determines the denominator target. If $t \geq 1-\epsilon$, $0 < \epsilon < 1/2$, then

$$w(t) \leq \frac{1}{t \log^2((1-\epsilon)/\epsilon)}.$$

Let r_d be the normalized endpoint reproducing kernel of degree d for Lebesgue measure on $[0,1]$. Thus $r_d(1) = 1$ and $\int_0^1 r_d^2 = 1/(d+1)^2$. With $k + d = N$ and $k \geq 1$, use $P = t^k r_d$ and split the integral at $1-\epsilon$ to get

$$R_N \leq \frac{1}{(d+1)^2}\left\{\frac{1}{\log^2((1-\epsilon)/\epsilon)} + \frac{(1-\epsilon)^{2k-1}}{\pi^2}\right\}. \tag{18.8}$$

The choices $k = \lfloor N/\log N \rfloor$, $d = N-k$, $\epsilon = (\log N)^3/N$ for sufficiently large N show $\limsup N^2 \log^2 N\, R_N \leq 1$.

For the lower bound take $I = [1/2, 1-N^{-2}]$, $N \geq 2$, of length $L \geq 1/4$. The second estimate in (18.7) reduces the problem to the endpoint evaluation norm for polynomials on I. After mapping I to $[-1,1]$, the evaluation point is $x_0 = 1+2/(N^2 L) \leq 1+8/N^2$. The ordinary Legendre polynomials satisfy $|P_j(x_0)| \leq \exp(4j/N) \leq e^4$ for $j \leq N$, by their integral representation and $\operatorname{arcosh}(1+y) \leq \sqrt{2y}$. The reproducing kernel is therefore at most $L^{-1}\sum_{j=0}^{N}(2j+1)P_j(x_0)^2 \leq 9e^8 N^2$. Cauchy–Schwarz for evaluation implies

$$R_N \geq \frac{1}{18e^8 N^2\{\log^2(2N^2) + \pi^2\}} \quad (N \geq 2). \tag{18.9}$$

Together with (18.8), this proves the asserted order. □

18.2 Degree-dependent integral normalization

Write $\widehat{\gamma}_N = p_N/q_N$ in lowest terms with $q_N > 0$. The following integers are the Todd numbers:

$$T_m = \prod_{p \text{ prime}} p^{\lfloor m/(p-1)\rfloor}, \qquad \mathsf{T}_N = \operatorname{diag}(T_1, \dots, T_N), \qquad \Delta_N = \operatorname{lcm}(1, \dots, 2N) \prod_{\substack{3 \leq p \leq 2N+1 \\ p \text{ prime}}} p. \tag{18.10}$$

Their classical topological interpretation is discussed in [12]. Only the explicit product and the elementary valuation bounds below are used here.

Lemma 18.2 (Coefficient denominators)**.** *For every prime p and $m \geq 0$,*

$$v_p(\mathfrak{b}_m) \geq -\left\lfloor \frac{m}{p-1} \right\rfloor, \qquad v_p(c_j) \geq -\left\lfloor \frac{j+1}{p-1} \right\rfloor - v_p(j+1).$$

Proof. Expand the reciprocal of $F(z) = \sum_{m \geq 0} z^m/(m+1)$. Every coefficient of $\mathscr{B} = F^{-1}$ is a finite sum over compositions $m = m_1 + \cdots + m_k$ of products $(-1)^k \prod_i (m_i+1)^{-1}$. If $p^a \mid m_i + 1$, then $(p-1)a \leq p^a - 1 \leq m_i$. Summing these bounds proves the first inequality. The second follows from (18.1). □

Theorem 18.3 (Graded integral Gram matrix)**.** *For every $N \geq 1$, the matrix and vector*

$$\mathsf{A}_N = \Delta_N \mathsf{T}_N K_N \mathsf{T}_N \in M_N(\mathbb{Z}), \qquad \boldsymbol{v}_N = \Delta_N \mathsf{T}_N \boldsymbol{c}_N \in \mathbb{Z}^N$$

are integral, A_N is positive definite, and

$$\widehat{\gamma}_N = \frac{1}{\Delta_N} \boldsymbol{v}_N^{\mathsf{T}} \mathsf{A}_N^{-1} \boldsymbol{v}_N, \quad q_N = \frac{\mathcal{Z}_N}{\gcd(\mathcal{Y}_N, \mathcal{Z}_N)}, \quad \begin{cases} \mathcal{Y}_N = \boldsymbol{v}_N^{\mathsf{T}} \operatorname{adj}(\mathsf{A}_N) \boldsymbol{v}_N, \\ \mathcal{Z}_N = \Delta_N \det \mathsf{A}_N > 0. \end{cases} \tag{18.11}$$

Proof. Put $a = i + 1$, $b = j + 1$ for a matrix entry. The relevant moments are c_{a+b-2} and c_{a+b-1}. For $h = p - 1$ and $m \in \{a + b - 1, a + b\}$,

$$\left\lfloor \frac{m}{h} \right\rfloor - \left\lfloor \frac{a}{h} \right\rfloor - \left\lfloor \frac{b}{h} \right\rfloor \le 1.$$

For odd p, the possible carry is cleared by the prime product in Δ_N, and $v_p(m)$ is cleared by its least common multiple. For $p = 2$ the floor difference is at most zero, so that the least common multiple suffices. No prime larger than $2N + 1$ occurs. For $T_a c_{a-1}$ the only remaining denominator exponent is at most $v_p(a)$, also cleared by Δ_N. This proves integrality. Positive definiteness follows by congruence. Direct inversion gives the scalar identity, and reduction of the fraction $\mathcal{Y}_N/\mathcal{Z}_N$ gives the exact denominator formula. □

Theorem 18.4 (Primitive denominator upper bound)**.** *For every $N \ge 1$,*

$$q_N \le \Delta_N^{N+1} \left(\prod_{j=1}^{N} T_j \right)^2 \left(\frac{11}{24} \right)^N 4^{-(N-1)^2}. \tag{18.12}$$

In particular,

$$\log q_N \le N^2 \log N + \left(\frac{5}{2} - \log 4 \right) N^2 + o(N^2). \tag{18.13}$$

The coefficient in this upper bound is not an asserted asymptotic constant for the actual reduced denominator.

Proof. Theorem 18.3 gives $q_N \le \mathcal{Z}_N = \Delta_N^{N+1} (\prod T_j)^2 \det K_N$. The matrix K_N is the Gram matrix of monomials for the measure $(1 - t)d\nu(t)$, of total mass $c_0 - c_1 = 11/24$. Replace the monomials by the monic shifted Chebyshev polynomials 1 and $2^{1-2j}\mathcal{T}_j(2t - 1)$, $1 \le j < N$. This triangular change has determinant one and preserves the Gram determinant. Hadamard's inequality and their uniform norms give

$$\det K_N \le \left(\frac{11}{24} \right)^N \prod_{j=1}^{N-1} 2^{2-4j} = \left(\frac{11}{24} \right)^N 4^{-(N-1)^2}.$$

This proves (18.12).

We give the arithmetic estimate needed for the second term. Legendre's formula, with $s_p(m)$ the sum of the base-p digits, implies

$$v_p(T_m/m!) = \left\lfloor \frac{s_p(m)}{p - 1} \right\rfloor \ge 0.$$

The primes $p \le \sqrt{m} + 1$ contribute $O(\sqrt{m} \log(m + 1))$ to $\log(T_m/m!)$, since the displayed exponent is at most $1 + \lfloor \log_p m \rfloor$. If $p > \sqrt{m} + 1$, then $p^2 > m$, and that exponent equals $\lfloor m/(p - 1) \rfloor - \lfloor m/p \rfloor \in \{0, 1\}$. When it is one there is an integer k with $m/p < k \le m/(p - 1)$, hence $p = \lfloor m/k \rfloor + 1$ and $k \le \sqrt{m} + 1$. There are at most $O(\sqrt{m})$ such primes. Consequently

$$\log T_m = \log(m!) + O(\sqrt{m} \log(m + 1)), \qquad 2 \sum_{j=1}^{N} \log T_j = N^2 \log N - \frac{3}{2} N^2 + O(N^{3/2} \log(N + 1)).$$

Finally the prime number theorem gives $\log \Delta_N = \psi(2N) + \vartheta(2N + 1) - \log 2 = 4N + o(N)$. Taking logarithms in (18.12) proves (18.13). □

Proposition 18.5 (An exact forced common factor)**.** *Put $T = T_{2N}$ and $W = \Delta_N T$. Define the uniform normalization $\mathsf{A}_N^{\mathrm{u}} = WK_N$, $\boldsymbol{v}_N^{\mathrm{u}} = W\boldsymbol{c}_N$, and $\mathcal{Y}_N^{\mathrm{u}}, \mathcal{Z}_N^{\mathrm{u}}$ by (18.11), with W in place of Δ_N. Then*

$$\mathcal{R}_N = \frac{T_{2N}^{N+1}}{(\prod_{j=1}^{N} T_j)^2} \in \mathbb{Z}_{>0}, \qquad \mathcal{Y}_N^{\mathrm{u}} = \mathcal{R}_N \mathcal{Y}_N, \qquad \mathcal{Z}_N^{\mathrm{u}} = \mathcal{R}_N \mathcal{Z}_N,$$

and therefore

$$\gcd(\mathcal{Y}_N^{\mathrm{u}}, \mathcal{Z}_N^{\mathrm{u}}) = \mathcal{R}_N \gcd(\mathcal{Y}_N, \mathcal{Z}_N), \qquad \log \mathcal{R}_N = N^2 \log N + \left(\log 4 - \frac{1}{2}\right) N^2 + o(N^2).$$

Proof. Integrality of the uniform normalization follows from Lemma 18.2. For each prime p, the exponent of p in $\mathcal{R}_N$ is nonnegative because

$$(N+1)\left\lfloor \frac{2N}{p-1} \right\rfloor - 2\sum_{j=1}^{N} \left\lfloor \frac{j}{p-1} \right\rfloor \geq 0.$$

The determinant scaling proves the equality for $\mathcal{Z}_N$; both scalar ratios equal $\widehat{\gamma}_N$, which proves the equality for $\mathcal{Y}_N$ and the gcd identity. The estimate for T_m in the preceding proof gives the stated asymptotic expansion. □

18.3 What a determinant does not determine

Let D be any common denominator of $c_0, \dots, c_{2N-1}$, and put $M = D^2 K_N$, $\boldsymbol{u} = D\boldsymbol{c}_N$. The finite group $G = \mathbb{Z}^N / M\mathbb{Z}^N$ has a nondegenerate pairing

$$\lambda([x],[y]) = x^{\mathsf{T}} M^{-1} y \pmod{\mathbb{Z}}.$$

Its well-definedness and nondegeneracy follow immediately by changing representatives and testing against the coordinate vectors. The exact denominator invariant is

$$q_N = \operatorname{ord}_{\mathbb{Q}/\mathbb{Z}} \lambda([\boldsymbol{u}],[\boldsymbol{u}]), \tag{18.14}$$

since $\boldsymbol{u}^{\mathsf{T}} M^{-1} \boldsymbol{u} = \widehat{\gamma}_N$. Neither $|G|$ nor the order of $[\boldsymbol{u}]$ alone determines this self-pairing. For example, $M = 5I_2$ and $\boldsymbol{u} = (1,2)^{\mathsf{T}}$ give a class of order five with integral self-pairing. Real positive definiteness therefore does not exclude denominator cancellation.

There is nevertheless a uniform local statement at prime indices. Write $\mathbb{Z}_p$ for the p-adic integers.

Proposition 18.6 (Prime entry after primitive reduction)**.** *For every prime $p \geq 5$,*

$$\widehat{\gamma}_{p-1} \in \mathbb{Z}_p, \qquad p\widehat{\gamma}_p \equiv 4 \pmod{p}, \qquad p \nmid q_{p-1}, \qquad v_p(q_p) = 1.$$

Proof. Up to degree $2p$, write

$$F(z) = \mathscr{B}(z)^{-1} = F_0(z) + \frac{z^{p-1}}{p} + \frac{z^{2p-1}}{2p}, \qquad F_0 \in \mathbb{Z}_p[[z]].$$

Since $3p - 3 > 2p$, reciprocal expansion gives, modulo z^{2p+1},

$$\mathscr{B} = F_0^{-1} - \left(\frac{z^{p-1}}{p} + \frac{z^{2p-1}}{2p}\right) F_0^{-2} + \frac{z^{2p-2}}{p^2} F_0^{-3}.$$

The coefficients $[z]F_0^{-2} = -1$ and $[z^2]F_0^{-3} = 1/2$ imply

$$p^2 c_{p-1} \equiv -1, \qquad p^3 c_{2p-1} \equiv -\frac{1}{4} \pmod{p}.$$

The same expansion shows that c_j is integral for $j \le p-3$, $v_p(c_{p-2}) \ge -1$, and $v_p(c_j) \ge -2$ for $j \le 2p-2$. Partition

$$K_p = \begin{pmatrix} A & v \\ v^{\mathsf{T}} & d \end{pmatrix}, \qquad \boldsymbol{c}_p = \begin{pmatrix} b \\ \beta \end{pmatrix}, \qquad A = K_{p-1}.$$

The reduction of $p^2 A$ vanishes strictly before its anti-diagonal and has anti-diagonal entries one. Thus $A^{-1} \in p^2 M_{p-1}(\mathbb{Z}_p)$. Here $b \in p^{-1}\mathbb{Z}_p^{p-1}$, $v \in p^{-2}\mathbb{Z}_p^{p-1}$, $\beta = c_{p-1}$, and $d = c_{2p-2} - c_{2p-1}$. Hence

$$b^{\mathsf{T}} A^{-1} b \in \mathbb{Z}_p, \quad p^2(\beta - v^{\mathsf{T}} A^{-1} b) \equiv -1, \quad p^3(d - v^{\mathsf{T}} A^{-1} v) \equiv \tfrac{1}{4} \pmod{p}.$$

The Schur-complement formula for $\widehat{\gamma}_p$ proves the result. □

Remark 18.7 (The remaining arithmetic requirement). Theorem 18.1 gives the exact equivalence

$$q_N \gamma - p_N \longrightarrow 0 \quad \Longleftrightarrow \quad q_N = o(N^2 \log^2 N),$$

also along any specified infinite subsequence. No such denominator estimate is proved here. The upper bound (18.13) is far above the required scale and does not show that the integer forms tend to zero. Conversely, Proposition 18.6 gives only a local lower constraint at different indices; multiplying its prime factors as though they belonged to a single q_N would be invalid.

There is an elementary collective lower bound. If $M_N = \max_{N \le j \le 2N} q_j$, strict monotonicity gives N successive rational gaps, each at least M_N^{-2}. Since their sum is smaller than $R_N = O(N^{-2}\log^{-2} N)$,

$$\max_{N \le j \le 2N} q_j = \Omega(N^{3/2} \log N).$$

This does not exclude exceptional sparse subsequences. The results of this section concern rational approximants and their exact normalization, and establish neither irrationality nor transcendence of γ.

19 Companion constructions and the specialization problem

The companion manuscripts [15, 16] develop constructions based on exponential integrals, moving levels, and arithmetic Gevrey series. This section records the parts needed to interpret the present arithmetic approximation results. The claims proved below concern functions and exact identities; their specialization at Euler's constant is identified separately. In particular, no assertion concerning joint algebraic independence at arbitrarily many sampling points is needed.

19.1 Conventions and a balanced approach to Euler's constant

On the positive real axis put

$$\mathrm{Ein}(x) = \sum_{k \ge 1} \frac{(-1)^{k-1} x^k}{k\, k!}, \qquad E_1(x) = \int_x^\infty \frac{e^{-t}}{t}\, dt, \qquad \delta = \int_0^\infty \frac{e^{-t}}{1+t}\, dt.$$

The connection identity, with the principal logarithm, is

$$\mathrm{Ein}(x) = \gamma + \log x + E_1(x), \qquad \delta = eE_1(1). \tag{19.1}$$

The derivative of the difference in the first identity is zero, and its limit at zero is γ; the second follows by translating the integral. Thus $\mathrm{Ein}(1) = \gamma + \delta/e$.

The dyadic level in [15, 16] is

$$Y_N = (N+1)\,\mathrm{Ein}(2^N) - N\,\mathrm{Ein}(2^{N+1}), \qquad N \ge 1. \tag{19.2}$$

Its definition uses values of an explicitly specified entire function, without using a decimal expansion of γ.

Proposition 19.1 (Dyadic error and polynomial detector)**.** *The numbers Y_N decrease strictly to γ. More precisely,*

$$\varepsilon_N := Y_N - \gamma = (N+1)E_1(2^N) - NE_1(2^{N+1}) > 0, \qquad \varepsilon_N \sim \frac{N+1}{2^N} e^{-2^N}.$$

For every nonzero $P \in \mathbb{C}[X]$, the following limit exists:

$$\lim_{N\to\infty} \frac{-\log|P(Y_N)|}{2^N} = \operatorname{ord}_{X=\gamma} P. \tag{19.3}$$

The expression on the left is defined for all sufficiently large N. Consequently γ is algebraic if and only if there exists a nonzero $P \in \mathbb{Q}[X]$ for which the limit in (19.3) equals one.

Proof. Substitution of (19.1) into (19.2) cancels the logarithms. Positivity follows because E_1 is positive and strictly decreasing. For $g(t) = E_1(2^t)$ one has

$$g''(t) = (\log 2)^2 2^t e^{-2^t} > 0, \qquad \varepsilon_N - \varepsilon_{N+1} = (N+1)\{g(N) - 2g(N+1) + g(N+2)\} > 0.$$

Integration by parts gives $E_1(x) = e^{-x}x^{-1}(1 + O(x^{-1}))$, proving the asymptotic formula and $-\log\varepsilon_N \sim 2^N$. Write $P(X) = (X-\gamma)^m R(X)$, where $R(\gamma) \neq 0$. Then $P(Y_N) = \varepsilon_N^m R(Y_N)$, which proves (19.3). If γ is algebraic, its minimal polynomial has a simple root there, since the characteristic is zero. Conversely, a positive value of the limit implies $P(\gamma) = 0$. □

This is an exact classification of a detector, not a lower bound excluding its critical slope. The same proof works whenever a specified sequence approaches a limit at a known rate. It does not turn the transcendental approximants Y_N into rational approximants of controlled height. The specialized quadratic detector of [16] is covered by (19.3); excluding its critical value still requires arithmetic information not furnished by the rapid convergence alone.

19.2 A normalized mixed family and the exact zero-removal bridge

An E-function is a holonomic series $\sum a_n z^n/n!$ with algebraic coefficients whose conjugates and common denominators grow at most exponentially. An arithmetic Gevrey series of order one is a series $\sum n! b_n w^n$ with $\sum b_n w^n$ a G-function. Let $\mathcal{M}$ denote the mixed space of formal sums

$$F(z) + \widehat{f}(1/z), \qquad \widehat{f}(0) = 0,$$

with these two respective types. We use direction-zero sums when that direction is admissible. Summation is injective on the mixed space, and differentiation and multiplication by z preserve it [14, Definition 1 and Lemma 1].

Consider the normalized inverse series

$$\widehat{D}(z) = \sum_{n\geq 0} (-1)^n n! z^{-n-1}, \qquad \widehat{W}(z) = e^z \operatorname{Ein}(z) - \widehat{D}(z), \qquad U(z) = e^z. \tag{19.4}$$

Its Borel transform is $(1+\xi)^{-1}$, so that

$$D_0(z) = \int_0^\infty \frac{e^{-zt}}{1+t}\, dt = e^z E_1(z), \qquad W_0(z) = e^z(\gamma + \log z). \tag{19.5}$$

Both the formal family and its sectorial sum satisfy

$$U' = U, \qquad \widehat{W}' = \widehat{W} + U/z. \tag{19.6}$$

The inverse tail used in the specific bridge of [15] is $z\widehat{D}(z)$, rather than $\widehat{D}(z)$. Its sectorial formula therefore has the extra term $(1-z)e^z E_1(z)$. The two normalizations have the same value at $z = 1$, but their functional identities must not be interchanged.

Lemma 19.2. *The numbers* $1, e, e\,\mathrm{Ein}(1)$ *are linearly independent over* $\overline{\mathbb{Q}}$.

Proof. The E-functions $1, e^z, e^z\,\mathrm{Ein}(z)$ form a differential system over $\mathbb{Q}(z)$ with no finite singularity except zero. They are linearly independent over $\mathbb{C}(z)$. Indeed, a relation involving the third would imply $\mathrm{Ein}(z) = A(z) + B(z)e^{-z}$ for rational functions A, B. Differentiation and comparison of the independent functions $1, e^{-z}$ give $A'(z) = 1/z$, impossible for a rational function. The remaining two functions are also independent. The linear specialization theorem for E-functions at the ordinary algebraic point 1 now gives the assertion [10]. □

Proposition 19.3 (Exact status of the restricted bridge)**.** *For every* $\beta \in \overline{\mathbb{Q}}$, *put* $\widehat{R}_\beta = \widehat{W} - \beta U$. *Then*

$$(R_\beta)_0(1) = e(\gamma - \beta), \qquad \widehat{R}_\beta \notin (z-1)\mathcal{M}.$$

Hence the implication

$$(R_\beta)_0(1) = 0 \quad \Longrightarrow \quad \widehat{R}_\beta \in (z-1)\mathcal{M} \tag{19.7}$$

holds if and only if $\gamma \neq \beta$. *Requiring* (19.7) *for every algebraic* β *is equivalent to the transcendence of* γ.

Proof. The value identity follows from (19.5). If $\widehat{R}_\beta = (z-1)(F(z) + \widehat{f}(1/z))$, comparison of the nonnegative and nonpositive powers in

$$e^z(\mathrm{Ein}(z) - \beta) - (z-1)F(z) = \widehat{D}(z) + (z-1)\widehat{f}(1/z)$$

forces both sides to be an algebraic constant c. Evaluation at one gives $e\,\mathrm{Ein}(1) - \beta e = c$, contradicting Lemma 19.2. The consequent of (19.7) is therefore always false, so the implication holds exactly when its antecedent is false. □

This recovers the logical content of the bridge theorem in [15] with the normalization used in [16]. It does not prove the required zero-removal assertion. Analytic removability of a zero at one is distinct from membership of the resulting quotient in an arithmetic class.

Let $\mathbf{E}$ be the ring of values of E-functions at algebraic points, and $\mathbf{D}$ the ring of directionally summed Gevrey-one values with the convention of [14]. The standard conditional conclusion can be stated without an uncertain closure claim.

Corollary 19.4 (Conditional intersection consequence)**.** *If* γ *is algebraic, then*

$$\delta \in (\mathbf{E} \cap \mathbf{D}) \setminus \overline{\mathbb{Q}}.$$

In particular, $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$ *implies that* γ *is transcendental.*

Proof. Under the stated assumption, $e^z(\mathrm{Ein}(z) - \gamma)$ is an E-function and its value at one is δ. The equality $D_0(1) = \delta$ gives membership in $\mathbf{D}$. If δ were algebraic, the identity $e\,\mathrm{Ein}(1) - \gamma e - \delta = 0$ would contradict Lemma 19.2. □

This implication is already contained in [14, Theorem 2]; it is included to fix the normalization. In particular, writing $E_1(1) = e^{-1}\delta$ does not by itself prove $E_1(1) \in \mathbf{D}$, since the needed closure under multiplication by e^{-1} has not been established here. Nor should γ be identified with an unframed Stokes jump multiplier: W_0 changes by $2\pi i e^z$ around zero, while adding ce^z changes its constant term without changing that jump. The arithmetic study of such connection and Stokes coefficients requires specified normalizations [13].

19.3 Mixed Gamma jets and the cost of polynomial targets

The all-order family in [16] permits a precise classification at the function level. For $r \geq 1$ set

$$\mathrm{Ein}_r(z) = \sum_{n\geq 1} \frac{(-1)^{n-1} z^n}{n^r n!}, \qquad \widehat{D}_r(z) = \sum_{n\geq 0} (-1)^n \begin{bmatrix} n+1 \\ r \end{bmatrix} z^{-n-1},$$

where brackets denote unsigned Stirling numbers of the first kind. Put $\widehat{W}_0 = e^z$ and $\widehat{W}_r = e^z \,\mathrm{Ein}_r(z) - \widehat{D}_r(z)$. These are mixed functions. For example, the normalized Borel transform of the inverse tail is the G-function

$$\mathcal{B}\widehat{D}_r(\xi) = \frac{(-1)^{r-1}}{(r-1)!} \frac{\log^{r-1}(1+\xi)}{1+\xi},$$

which has no positive-ray singularity. Write W_r for the direction-zero sum, and $\mathcal{M}_0$ for such sectorial mixed sums.

Lemma 19.5 (Generating function and arithmetic normal form)**.** *Coefficientwise near $t = 0$,*

$$\sum_{r\geq 0} W_r(z) t^r = e^z \Gamma(1-t) z^t. \tag{19.8}$$

Moreover, as analytic functions on a common sector,

$$\mathcal{M}_0 \cap e^z \mathbb{C}[\log z] = \bigoplus_{r\geq 0} \overline{\mathbb{Q}}\, W_r. \tag{19.9}$$

All sums on the right are finite.

Proof. The incomplete Gamma decomposition gives

$$\Gamma(1-t) z^t = 1 + \sum_{r\geq 1} \mathrm{Ein}_r(z) t^r - t z^t \Gamma(-t, z).$$

The expansion $e^z z^t \Gamma(-t,z) \sim \sum_{n\geq 0} (-1)^n (t+1)_n z^{-n-1}$, together with $[t^{r-1}](t+1)_n = \left[\begin{smallmatrix} n+1 \\ r \end{smallmatrix}\right]$, proves (19.8). Equivalently, one can sum the displayed Borel transforms before integrating for sufficiently small $|t|$.

Set $T = z(\partial_z - 1)$. Then $TW_0 = 0$ and $TW_r = W_{r-1}$. To compute its mixed kernel, write $\Psi = F + \sum_{n\geq 1} a_n z^{-n}$ and suppose $T\Psi_0 = 0$. Injectivity of summation gives $T\Psi = 0$ formally. Its constant coefficient is $-a_1$, and its negative coefficients give $-na_n - a_{n+1} = 0$. Thus all a_n vanish and $F' = F$, whence $F = ce^z$ with c algebraic. Applying T lowers the degree of a logarithmic polynomial. Induction on that degree, followed by subtraction of the corresponding W_r, proves (19.9). Uniqueness follows from their distinct logarithmic degrees, whose leading coefficients are $1/r!$. □

Put $X(z) = \gamma + \log z$. The Gamma expansion gives

$$e^{-z} W_r(z) = [t^r] \exp\left(X(z) t + \sum_{k\geq 2} \frac{\zeta(k)}{k} t^k \right).$$

The next proposition identifies the exact extra coefficients introduced by polynomial targets.

Proposition 19.6 (Integral coefficient conversion)**.** *Let $P(X) = \sum_{k=0}^{d} a_k X^k \in \mathbb{Z}[X]$, $H = \max_k |a_k|$, and define the formal polynomials*

$$c_n(Z) = [t^n] \exp\left(-\sum_{k\geq 2} \frac{Z_k}{k} t^k \right), \qquad B_r(Z) = \sum_{k=r}^{d} a_k k! c_{k-r}(Z).$$

Then $B_r \in \mathbb{Z}[Z_2, \dots, Z_d]$, with weighted degree at most $d - r$ when $\mathrm{wt}(Z_k) = k$, and

$$\|B_r\|_1 \le H(d+1)d!, \qquad e^z P(X(z)) = \sum_{r=0}^{d} B_r(\zeta(2), \dots, \zeta(d)) W_r(z).$$

Here $\|\cdot\|_1$ is the sum of absolute polynomial coefficients. For every $P \in \overline{\mathbb{Q}}[X]$, including zero,

$$e^z P(\gamma + \log z) \in \mathcal{M}_0 \quad \Longleftrightarrow \quad \deg P \le 1, \tag{19.10}$$

where zero is included on the right by convention.

Proof. The signed cycle expansion is

$$n! c_n(Z) = \sum_{\sum k m_k = n} (-1)^{\sum m_k} \frac{n!}{\prod_{k\ge 2} k^{m_k} m_k!} \prod_{k \ge 2} Z_k^{m_k}.$$

Each unsigned coefficient counts permutations with prescribed cycle lengths and no fixed points. It is an integer, and their sum is the derangement number, at most $n!$. Hence the stated integrality, weighted degree, and bound follow by summing over k. Multiplying the two formal generating functions with opposite zeta exponents gives $X^k = k! \sum_{r=0}^{k} c_{k-r}(\zeta) e^{-z} W_r$, proving the identity.

For the last assertion, degree at most one follows from $W_0 = e^z$, $W_1 = e^z X$. For degree $d \ge 2$, the unique expansion has coefficient

$$B_{d-2} = (d-2)! \left(a_{d-2} - \binom{d}{2} a_d \zeta(2) \right).$$

This is transcendental since $a_d \ne 0$ and $\zeta(2) = \pi^2/6$ is transcendental. The normal form (19.9) therefore excludes mixed membership. □

For instance, $e^z X^2 = 2W_2 - \zeta(2) W_0$. The coefficient bound is for integer polynomials in *formal* period variables; it is not a height bound for algebraic numbers after substituting $\zeta(k)$. At $z = 1$ the conversion represents $eP(\gamma)$, but no nonvanishing theorem for that value has been proved here. This obstruction also does not rule out the degree-one bridge: under any hypothetical algebraic equality $\gamma = \beta$, the function $W_1 - \beta W_0$ is already mixed, irrespective of the degree of β. The unresolved step remains arithmetic specialization, rather than the existence of a functional encoding.

20 What an infinite construction must establish

The finite arithmetic data above support exact theorems about particular forms. Passing to an infinite sequence requires a separate statement about the primitive targets and their nonzero values. The following criteria make that distinction explicit.

Proposition 20.1 (A nonzero small-form criterion)**.** *If there are integers q_j, p_j with*

$$0 < |q_j \gamma - p_j| \longrightarrow 0,$$

then γ is irrational.

Proof. If $\gamma = a/b \in \mathbb{Q}$ with $b > 0$, every nonzero value of $q_j \gamma - p_j$ has absolute value at least $1/b$. □

An alternative is to construct two linearly independent integer targets at each of infinitely many levels, with both error bounds tending to zero. If γ were rational, both errors would eventually be zero; their targets would then lie on the same one-dimensional rational line, contradicting independence. This version avoids certifying the sign of each individual form, but still requires control of the second independent target, not only one short kernel vector.

Theorem 20.2 (Fixed positive margin and transcendence)**.** *Suppose there are a fixed $\varepsilon > 0$ and primitive integer pairs (q_j, p_j), with $q_j > 0$ and $q_j \to \infty$, such that*

$$0 < |q_j\gamma - p_j| < q_j^{-1-\varepsilon}. \tag{20.1}$$

Then γ is transcendental.

Proof. Proposition 20.1 first excludes rationality. Division by q_j gives

$$0 < \left|\gamma - \frac{p_j}{q_j}\right| < q_j^{-2-\varepsilon}.$$

Roth's theorem [18] permits only finitely many such reduced rational approximations to an algebraic irrational number for each fixed $\varepsilon > 0$. The unbounded denominators give a contradiction. □

The theorem requires only an infinite subsequence of levels. The constant 0.03 in the finite certificate is not special: every fixed positive margin would suffice. Conversely, a sequence of exponents decreasing to 1 in the linear form, or a fixed excellent exponent at finitely many levels, does not supply the hypothesis of Theorem 20.2. The criterion is sufficient and is not necessary for transcendence.

20.1 An exact interval formulation of the missing estimate

Suppose a gamma-free exact procedure produces rational intervals (α_j, β_j) containing γ. Put

$$m_j = \frac{\alpha_j + \beta_j}{2}, \qquad r_j = \frac{\beta_j - \alpha_j}{2}.$$

For a target (q, p) with $q > 0$, every point of the interval gives

$$|q\gamma - p| \le |qm_j - p| + qr_j. \tag{20.2}$$

If $p/q \notin (\alpha_j, \beta_j)$, the strict inclusion of γ also proves $q\gamma - p \neq 0$. Therefore one concrete sufficient target for the present method is to prove, at infinitely many levels, that primitive pairs with $q_j \to \infty$ satisfy

$$\frac{p_j}{q_j} \notin (\alpha_j, \beta_j), \qquad |q_jm_j - p_j| + q_jr_j < q_j^{-1-\varepsilon} \tag{20.3}$$

for a fixed $\varepsilon > 0$. The first condition may instead be replaced by any independent positive-integral certificate.

Since the terms in the right side of (20.2) are nonnegative, meeting this particular budget necessarily requires

$$r_j < q_j^{-2-\varepsilon}, \qquad |m_j - p_j/q_j| < q_j^{-2-\varepsilon}.$$

Thus small interval width alone is insufficient: the arithmetic position of its midpoint and the primitive denominator chosen near it matter at the same time. These necessary inequalities refer to this interval budget, not to every possible proof of transcendence.

Problem 20.3 (Uniform primitive control)**.** For the Sondow interval models, possibly augmented by positive moving differences or stronger moment constraints, construct an infinite sequence of primitive targets satisfying (20.3), or prove another nonvanishing estimate and the bound (20.1).

No solution of Problem 20.3 is given here. Neither a large-dimensional kernel, a scalar denominator bound, nor numerical improvement in a fixed window proves it. In particular, an exact relation in $\ker B \cap \ker \Phi$ can be arbitrarily inexpensive while carrying no information about γ.

20.2 Scope of the methodological tests

The construction rule, effective target dimension, error-to-height ratio, and nonvanishing certificate are useful tests for an explicit small-form method. They should not be promoted to universal restrictions on all transcendence arguments. A gamma-independent construction rule prevents an unsupported numerical selection from replacing a proof, but access to a rational enclosing interval is not by itself circular. Additional variables may legitimately occur if they are eventually eliminated or treated by a theorem whose hypotheses control their joint arithmetic. The distinction is whether that elimination or arithmetic theorem has actually been proved.

Likewise, Dirichlet's approximation theorem is an existence guarantee, not a universal upper bound on how good approximations can be for arbitrary real numbers. It cannot supply an unconditional ceiling for an exponent of an unknown transcendental constant. Nor does a large coefficient in one representation of an algebraic approximant prove a lower bound for its minimal-polynomial height. These logical distinctions exclude several premature impossibility conclusions as well as premature positive proofs.

21 Reproducibility and the theorem boundary

The original package `KEIO_Gamma_Paper_v1_Complete.zip`, retained in the cumulative archive, contains the exact certificates and Python verifiers for the finite statements of this part. The main finite claims can be checked without evaluating γ, by running the following commands from its unpacked `certificates` directory:

```
python3 gamma_sondow_intervals.py --verify
python3 gamma_sondow_damping.py --verify
python3 gamma_sondow_relations.py
python3 gamma_sondow_positivity.py
python3 gamma_sondow_threshold_verify.py
```

The first two check rational primal-dual equalities and inequalities, the endpoint identities, exact target common factors, and exponent comparisons by integer powers. The next two verify prime-logarithm cancellation and polynomial-positivity certificates. The last checks the elementary finite constants accompanying the general kernel estimates. The all-index theorems rest on the written proofs, not on these finite checks. Package versions and execution results are recorded with the archive; some JSON entries exceed Python's default decimal-digit limit, which the supplied scripts adjust explicitly.

The results organize an explicit interaction between analytic positivity, prime arithmetic, and primitive lattice optimization. The remaining question is quantitative and arithmetic: whether the surviving primitive targets can be controlled along an unbounded sequence while the certified error retains a fixed improvement over the Roth threshold.

Part III
Additional early constructions and arithmetic audits

22 Early constructions not contained in the first gamma paper

The first gamma paper concentrated on Sondow windows, positive moments, primitive quotient optimization and arithmetic transfer. The research records of 6 September also contain several different constructions. This part retains their principal mathematical content. A statement called a source calculation below is retained with its exact proof-file reference; it is not silently promoted to a new theorem of this edition. The complete source files, including their finite certificates and internal reviews, are included in the accompanying source archive.

Throughout this part, $H_0 = 0$, $H_n = \sum_{j=1}^{n} j^{-1}$ and $L_n = \operatorname{lcm}(1, \dots, n)$, with $L_0 = 1$. We use $\delta = \int_0^\infty e^{-t}(1+t)^{-1}\,dt$ and $\operatorname{Ein}(z) = \sum_{k\geq 1}(-1)^{k+1}z^k/(k\,k!)$. Thus $\operatorname{Ein}(1) = \gamma + e^{-1}\delta$. Integral positivity, a nonzero coefficient vector, a nonzero determinant and a nonzero value at γ are four different statements.

23 A full-rank logarithmic Laguerre compression

For $N \geq 1$ define the rational symmetric matrix

$$M_N = (m_{ij})_{0\leq i,j<N}, \qquad m_{ii} = H_i, \qquad m_{ij} = -\frac{1}{|i-j|} \quad (i \neq j).$$

The standard Laguerre polynomials are normalized by $\int_0^\infty e^{-t}L_i(t)L_j(t)\,dt = \delta_{ij}$.

Proposition 23.1 (The logarithmic compression)**.** *For every $i, j \geq 0$,*

$$\int_0^\infty e^{-t}L_i(t)L_j(t)\log t\,dt = m_{ij} - \gamma\delta_{ij}.$$

Every M_N has simple real eigenvalues, and the eigenvalues of M_N and M_{N+1} strictly interlace.

Proof. Differentiate the generalized Laguerre orthogonality relation

$$\int_0^\infty t^s e^{-t}L_i^{(s)}(t)L_j^{(s)}(t)\,dt = \frac{\Gamma(i+s+1)}{i!}\delta_{ij}$$

at $s = 0$, using $\partial_s L_n^{(s)}|_{s=0} = \sum_{k<n} L_k/(n-k)$. The diagonal gives $H_i - \gamma$, and an off-diagonal entry gives $-1/|i-j|$.

Let x be an eigenvalue and let p be the corresponding polynomial of degree at most $N-1$. Set $a = e^{x-\gamma}$. Then

$$\int_0^\infty e^{-t}\log(t/a)p(t)q(t)\,dt = 0 \qquad (\deg q < N).$$

If $\deg p \leq N-2$, the admissible choice $q = (t-a)p$ gives a strictly positive integral, a contradiction. Hence every eigenvector has degree $N-1$. An eigenspace of dimension at least two would contain a nonzero combination of smaller degree, so the eigenvalues are simple. If two consecutive compressions shared an eigenvalue, the equality case of symmetric-matrix interlacing would give an eigenvector of the larger compression supported in the smaller space. Its degree would be at most $N-1$, contradicting the same argument for the size $N+1$ compression. Cauchy's interlacing theorem now gives strict interlacing. □

Let P_N be the positive-leading primitive integer multiple of $\det(XI_N - M_N)$. For example,

$$P_1 = X, \quad P_2 = X^2 - X - 1, \quad P_3 = 4X^3 - 10X^2 - 3X + 11.$$

Clearing matrix entries by L_{N-1} and bounding the row sums gives

$$\log H(P_N) = O(N^2).$$

For $S_N = P_N^2 + P_{N+1}^2$, strict interlacing implies $S_N(x) > 0$ for every real x, while $\deg S_N = 2N+2$ and $\log H(S_N) = O(N^2)$.

Corollary 23.2 (An unfulfilled determinant criterion)**.** *If*

$$\limsup_{N\to\infty} \frac{-\log S_N(\gamma)}{N^2} = +\infty,$$

then γ is transcendental.

Proof. For a fixed algebraic number α, the nonzero resultant of its primitive minimal polynomial with S_N gives $|S_N(\alpha)| \geq \exp(-C_\alpha N^2)$. Nonvanishing is automatic here because S_N is positive on the real line. This contradicts the stated limsup. □

The small-value hypothesis is not proved. The finite tests in `gamma_spectral_attempt.md` instead give large integer values in their tested range. This construction changes the rank-one occurrence of γ into a full-rank shift; it does not solve the height comparison.

24 Exact logarithmic denominators and the limits of a fixed transform

Theorem 24.1 (Prime residues and cumulative denominators)**.** *Let K be a number field, $a \in K$ and $P(z, Y) \in K[z, Y]$ have Y-degree $d \geq 1$. Write*

$$P(z, a + \log(1 - z)) = \sum_{n\geq 0} c_n z^n.$$

If Δ_N is the least positive integer such that $\Delta_N c_0, \ldots, \Delta_N c_N$ are algebraic integers, then

$$\log \Delta_N = H_d N + o(N).$$

More explicitly, write the first nonzero coefficient of $T_r(z) = \partial_Y^r P(z, a + \log(1 - z))/r!$ as $\beta_r z^{k_r}$. For every sufficiently large prime p outside a fixed finite exceptional set, and every prime ideal $\mathfrak{p}$ above p,

$$p^r c_{rp+k_r} \equiv (-1)^r \beta_r \pmod{\mathfrak{p}} \qquad (1 \leq r \leq d).$$

In particular the local denominator exponent is exactly r.

Proof. The logarithm is transcendental over $K(z)$, so each T_r is nonzero. Remove the finitely many primes at which the coefficients or the β_r are nonintegral or at which a β_r is not a unit. Choose $p > k_r$ with $rp + k_r < p^2$. In a contribution to the coefficient of degree $rp + k_r$, at most r logarithmic factors can have a denominator divisible by p. After multiplying by p^r, only contributions with exactly r such factors survive modulo $\mathfrak{p}$. Each must be $-z^p/p$: replacing one by a term of degree $2p$ would exceed the degree budget. Their remaining coefficient is exactly β_r by Taylor's formula in Y. This proves the congruence and the exact valuation.

Set $K_0 = \max k_r$. The congruences force

$$\log \Delta_N \geq \sum_{r=1}^{d} \vartheta((N - K_0)/r) - O(1), \qquad \vartheta(x) = \sum_{p \leq x} \log p.$$

For $p > \sqrt{N}$ a term of degree at most N contains at most $\min(d, \lfloor N/p \rfloor)$ factors with a p in their denominator. For $p \leq \sqrt{N}$ the bound $d\lfloor \log_p N \rfloor$ suffices, apart from a fixed denominator contribution. These small primes contribute $o(N)$. Thus

$$\log \Delta_N \leq \sum_{r=1}^{d} \vartheta(N/r) + o(N).$$

The prime number theorem proves the result. □

Corollary 24.2 (A delimited failure of global boundedness)**.** *If $P(z, Y) \in \mathbb{Q}(z)[Y]$ has positive Y-degree and a is algebraic, then a Taylor germ of $P(z, a + \log(1 - z))$, after removal of a fixed pole at zero, is not globally bounded. Here globally bounded means that a fixed nonzero algebraic change $z \mapsto cz$, followed by multiplication by a fixed nonzero algebraic constant, makes every coefficient integral.*

Proof. Multiply by a fixed polynomial to clear the rational functions in z. Global boundedness would be preserved. Fixed algebraic scalings affect only finitely many finite places, whereas the theorem forces a denominator at every sufficiently large good prime. □

For the harmonic bridge

$$B_a(z) = \frac{-a - \log(1 - z)}{1 - z},$$

every fixed differential polynomial with coefficients in $\mathbb{Q}(z)$ is a polynomial in $a + \log(1 - z)$ with rational-function coefficients. Such a transform can become globally bounded only after losing that variable, and therefore losing the shift a. For example,

$$((1 - z)\partial_z - 1)B_a = \frac{1}{1 - z}.$$

The exact constant H_d is asserted for polynomial coefficients in z; rational coefficients may add geometric denominators at finitely many primes. Growing-degree transforms and other coupled functions are outside this obstruction.

25 The Vacca generator and a rational Markov realization

Put $\ell(n) = 1 + \lfloor \log_2 n \rfloor$ and

$$A(z) = \sum_{n \geq 1} \frac{\ell(n)}{2n(2n + 1)} z^n, \qquad f(z) = \sum_{j \geq 0} z^{2^j}.$$

The classical Vacca–Sondow identity gives $A(1) = \gamma$, and direct coefficient comparison gives

$$2\theta(2\theta + 1)A(z) = \frac{f(z)}{1 - z}, \qquad f(z) = z + f(z^2), \qquad \theta = z\frac{d}{dz}.$$

Proposition 25.1 (Loss of the Mahler class under this integration)**.** *The series $A(z)$ is not a rational-coefficient k-Mahler function for any integer $k \geq 2$.*

Proof. Suppose a rational series $C(z)$ satisfies

$$P_0(z)C(z) + \sum_{i=1}^{q} P_i(z)C(z^{k^i}) = 0, \qquad P_i \in \mathbb{Z}[z], \quad P_0 \neq 0.$$

Let cz^v be the first nonzero term of P_0. For all sufficiently large n, comparison of the coefficient of z^{n+v} expresses the nth coefficient of C as $1/c$ times an integer combination of earlier coefficients. Consequently only finitely many primes occur in the coefficient denominators. For every odd prime p, the coefficient of A at $n = (p-1)/2$ has p-valuation -1, because $0 < \ell(n) < p$. This is a contradiction. □

Set $a_n = \ell(n)/(2n(2n+1))$. A birth-and-death chain on the positive integers is specified by

$$P_{n,n+1} = \frac{a_{n+1}}{4a_n}, \qquad P_{n,n-1} = \frac{1}{4} \quad (n \geq 2),$$

and the remaining row mass is assigned to $P_{n,n}$. Since $a_{n+1}/a_n < 2$, these are nonnegative rational transition probabilities. Detailed balance gives the stationary probability $\pi_n = a_n/\gamma$. The chain is irreducible and positive recurrent, so the mean positive return time to state 1 is

$$\mathbb{E}_1\tau_1^+ = \frac{1}{\pi_1} = 6\gamma.$$

Its spectral gap is zero. Indeed, for the indicator of the tail $\{n \geq N\}$, the Dirichlet energy is $O(\log N/N^2)$ and the variance is comparable to $\log N/N$. Thus the Rayleigh quotient tends to zero. All transition probabilities are rational; no arithmetic theorem about this infinite-chain return time has been established here.

26 The full logarithm-free floor-integral space

For $m \geq 2$ put $Q_m(u) = \prod_{j=1}^{m}(u+j)$ and let

$$V_m = \{P/Q_m^2 : P \in \mathbb{Q}[u],\ \deg P \leq 2m-3\}.$$

Write a member $g \in V_m$ in partial fractions as

$$g(u) = \sum_{j=1}^{m} \left(\frac{\alpha_j}{u+j} + \frac{\beta_j}{(u+j)^2} \right).$$

The decay at infinity is equivalent to $\sum \alpha_j = 0$ and $\sum \beta_j = \sum j\alpha_j$. For

$$A = \sum \beta_j, \qquad B = \sum \beta_j H_{j-1}, \qquad \lambda_p = \sum \alpha_j v_p((j-1)!),$$

integration over successive unit intervals yields

$$\mathcal{T}(g) := \int_0^\infty (\lfloor u \rfloor + 1)g(u)\,du = A\gamma - B + \sum_{p \leq m-1} \lambda_p \log p.$$

Absolute convergence follows from $g(u) = O(u^{-3})$. The two decay identities cancel the divergent terms and the common $\log(2\pi)$ term in the partial-fraction calculation.

Theorem 26.1 (Exact dimension and target freedom)**.** *Let $W_m \subset V_m$ be the subspace where every λ_p vanishes. Then*

$$\dim W_m = 2m - 2 - \pi(m-1).$$

The target map $W_m \to \mathbb{Q}^2$, $g \mapsto (A, B)$ is onto, and its kernel has dimension $2m - 4 - \pi(m-1)$.

Proof. Put $T_j = \sum_{r>j} \alpha_r$ for $1 \le j < m$. These coordinates are free and

$$\sum_r \alpha_r \log((r-1)!) = \sum_{j=1}^{m-1} T_j \log j, \qquad A = \sum_{j=1}^{m-1} T_j.$$

The prime-exponent matrix of the numbers $1, \dots, m-1$ has full row rank: its columns at prime indices form an identity matrix. It imposes $\pi(m-1)$ independent conditions. Choosing only T_1 nonzero gives arbitrary A without introducing logarithms. With that A fixed, opposite changes of β_1 and β_2 vary B arbitrarily, since $H_0 = 0$ and $H_1 = 1$. The dimension statements follow. □

This theorem concerns the whole numerator space, not the more restricted Sondow numerator. Its surjectivity is a reason to keep the primitive-height question explicit: the existence of a target pair is no smallness theorem.

Theorem 26.2 (Completeness for strict rational linear inequalities)**.** *For $A, B \in \mathbb{Q}$, the inequality $A\gamma - B > 0$ holds if and only if some m and some nonzero $g = P/Q_m^2 \in W_m$ satisfy $P(u) \ge 0$ for $u \ge 0$ and have target pair (A, B).*

Proof. One direction is immediate from the positive integral. For the converse, use

$$\mathcal{K}[R](u) = \int_0^1 x^u (-\log x)(1-x) R(x)\, dx.$$

For a polynomial R this is a rational function with double poles at negative integers. Direct calculation and Tonelli's theorem give

$$\mathcal{K}[x^{j-1}] = \frac{1}{(u+j)^2} - \frac{1}{(u+j+1)^2}, \qquad \mathcal{T}(\mathcal{K}[R]) = \int_0^1 R(x)\, dx.$$

Let

$$F(x) = \frac{1}{1-x} + \frac{1}{\log x} = \sum_{n\ge 0} a_n (1-x)^n,$$

where the positive rational Gregory coefficients satisfy

$$a_n = \frac{1}{(n+1)!} \int_0^1 t \prod_{j=1}^{n} (j-t)\, dt, \qquad \sum_{n\ge 0} a_n = 1.$$

Then $\mathcal{K}[F] = ((u+1)^2(u+2))^{-1}$ has integral γ. Set $L_N(x) = \sum_{n<N} a_n (1-x)^n$, $S_N = \int_0^1 L_N$, $E_N = \mathcal{K}[F - L_N]$, and $b_N = \mathcal{K}[(1-x)^N]$. Both E_N and b_N are positive rational functions. The ratio $E_N(u)/b_N(u)$ is the average of the decreasing function $J_N(x) = \sum_{k\ge 0} a_{N+k}(1-x)^k$ under a density proportional to $x^u(-\log x)(1-x)^{N+1}$. Its derivative is $\mathrm{Cov}(\log x, J_N(x)) \le 0$. Consequently

$$E_N^+(u) := \delta_N b_N(u) - E_N(u) \ge 0, \qquad \delta_N = E_N(0)/b_N(0) \in \mathbb{Q}.$$

Here $E_N(0)$ is rational because both terms defining E_N are rational functions. Also $0 \le J_N \le \sum_{n\ge N} a_n \to 0$, so $\delta_N \to 0$. Therefore

$$S_N < \gamma < U_N := S_N + \frac{\delta_N}{N+1}, \qquad S_N, U_N \longrightarrow \gamma.$$

Let $h_1 = (u+1)^{-2} - (u+2)^{-2}$, so $h_1 > 0$ and $\mathcal{T}(h_1) = 1$. If $A > 0$, choose N with $AS_N > B$ and put $g = AE_N + (AS_N - B)h_1$. If $A < 0$, choose N with $AU_N > B$ and put $g = (-A)E_N^+ + (AU_N - B)h_1$. If $A = 0$, take $g = -Bh_1$. These are nonnegative rational functions with the required integral. Putting their finitely many poles over one Q_m^2 gives the asserted numerator. All their logarithmic coefficients vanish by construction. □

Completeness here is a terminating certificate search for a true strict inequality. It cannot reject equality. The quantitative interval bound, proved in `gamma_bridge_cones.md`, is

$$U_N - S_N < \frac{N+2}{(N+1)^2 H_N^2}, \qquad U_N - S_N \sim \frac{1}{N \log^2 N}.$$

This is an explicit family of feasible intervals, not the optimal rate of the full positive cone. The first upper endpoint is $U_1 = 53/88$.

27 Gamma filters: positivity, integer sampling and algebraic elimination

For $N \geq 2$ and $q \geq 1$ define

$$\mathcal{E}_{N,q} = \gamma + q \sum_{j=1}^{N} \frac{(-1)^{j+1}}{j} \binom{N}{j} \log \Gamma(1 + j/q).$$

The binomial finite-difference identities remove the powers of degrees $2, \ldots, N$ in the local Gamma expansion and retain its linear term. More usefully, the Gamma integral and the beta identity give the positive remainder

$$\mathcal{E}_{N,q} = q \sum_{m \geq 1} \int_{mq}^{\infty} \frac{N!\, du}{u^{N+2} \prod_{j=1}^{N} (1 + j/u)} > 0. \tag{27.1}$$

One way to verify this formula is to differentiate in $1/q$, apply $\sum_{j=0}^{N} (-1)^j \binom{N}{j}/(u+j) = N!/[u(u+1)\cdots(u+N)]$, and match the vanishing limit at $1/q = 0$; the absolutely convergent Gamma integral justifies the interchange.

Theorem 27.1 (All-parameter error and a unit gamma coefficient)**.** *With $S = N(N+1)/2$ and $A_{N,q} = N!\zeta(N+1)/((N+1)q^N)$, one has*

$$e^{-S/q} A_{N,q} \leq \mathcal{E}_{N,q} \leq A_{N,q}.$$

For $q = L_N$, every Gamma logarithm has an integer coefficient and

$$\mathcal{E}_{N,L_N} \sim \frac{N!}{(N+1)L_N^N}, \qquad \frac{\log \mathcal{E}_{N,L_N}}{N^2} \longrightarrow -1.$$

Proof. In (27.1), $u \geq q$ and $1 \leq \prod_{j=1}^{N}(1 + j/u) \leq e^{S/q}$. Integrating the comparison power $N!u^{-N-2}$ and summing over m gives the two bounds. The prime number theorem gives $\log L_N = N + o(N)$ and $N^2/L_N \to 0$; Stirling's formula proves the limit. Integer coefficients follow from $j \mid L_N$. □

Proposition 27.2 (A necessary sampling denominator)**.** *Suppose q, M are positive integers, $c_i, n_i \in \mathbb{Z}$, and*

$$\sum_i c_i (n_i/q) = M, \qquad \sum_i c_i (n_i/q)^r = 0 \quad (2 \leq r \leq N).$$

Then $L_N \mid Mq$.

Proof. The integer-valued polynomial $\binom{x}{r}$ has zero constant term and linear coefficient $(-1)^{r-1}/r$. Hence the assumptions give

$$\sum_i c_i \binom{n_i}{r} = (-1)^{r-1} Mq/r \in \mathbb{Z}.$$

Thus every $1 \leq r \leq N$ divides Mq. □

This applies to unequal nodes and arbitrarily many nodes; at coefficient $M = 1$, the denominator L_N is minimal within this exact moment filter. It is not a universal denominator theorem for all gamma approximants.

An earlier odd-cumulant filter has the distinct form

$$A_r = \sum_{i=1}^{r} \frac{2r(-1)^{i+1}(r!)^2}{i(r-i)!(r+i)!} \log \frac{\Gamma(1-i/(2r))}{\Gamma(1+i/(2r))}.$$

The source calculation in `gamma60_probability.md`, using the partial-fraction expansion at the nodes $i^2/(4r^2)$, proves

$$0 < (-1)^{r+1}(A_r - \gamma) < \frac{(r!)^3}{r(3r)!} \sim \frac{2\pi}{\sqrt{3}} 27^{-r}.$$

This is an alternating family of upper and lower bounds; monotonicity of the two subfamilies is not asserted. The surviving Gamma logarithms cannot be assumed to be logarithms of algebraic numbers.

27.1 The full coefficient denominator and uniform phase law

For integer $q \geq 1$, the least positive integer making every coefficient $q(-1)^{j+1}\binom{N}{j}/j$ integral is exactly

$$D_{N,q} = \frac{L_N}{\gcd(L_N, q)}.$$

To see the lower bound, for a prime $p \leq N$ let p^e be its largest power at most N, and write $N = bp^e + c$, $1 \leq b < p$, $0 \leq c < p^e$. The congruence $(1+X)^N \equiv (1+X^{p^e})^b(1+X)^c \pmod p$ gives $\binom{N}{p^e} \equiv b \pmod p$. The coefficient at $j = p^e$ therefore attains the required denominator exponent. The upper bound is immediate. In particular $D_{N,N} = L_N/N$, not L_N.

Put $\lambda_N(u) = 2/u + \sum_{j=1}^{N} 1/(u+j)$ and $\epsilon_N(q) = 1/(q\lambda_N(q))$. The stronger uniform estimate from `gamma_bridge_phase.md` is

$$\frac{B(q, N+1)}{\lambda_N(q)} \leq \mathcal{E}_{N,q} \leq \frac{1+1/N}{1-\epsilon_N(q)} \frac{B(q, N+1)}{\lambda_N(q)},$$

where $B(u, N+1) = N!/[u(u+1)\cdots(u+N)]$. Here is the short argument retaining its uniform scope. For $f_N(u) = B(u, N+1)/u$, $-f_N'/f_N = \lambda_N$, and $-\lambda_N' \leq \lambda_N/u$. Integration by parts gives

$$\frac{f_N(v)}{\lambda_N(v)} \leq \int_v^\infty f_N(u)\, du \leq \frac{f_N(v)}{(1 - 1/(v\lambda_N(v)))\lambda_N(v)}.$$

Also $u\lambda_N(u)$ increases, $m\lambda_N(mq) \geq \lambda_N(q)$, and

$$\sum_{m \geq 2} \frac{B(mq, N+1)}{B(q, N+1)} \leq \sum_{m \geq 2} \frac{B(m, N+1)}{B(1, N+1)} = \frac{1}{N}.$$

Apply these estimates to the beta-tail formula. Since $\epsilon_N(q) \leq 1/(1+H_{N+1})$, the equivalent $\mathcal{E}_{N,q} \sim B(q, N+1)/\lambda_N(q)$ is uniform over all $q \geq 1$. In particular

$$\mathcal{E}_{N,q} \sim \frac{\Gamma(q)}{N^q \log N} \quad (q \text{ fixed}),$$

with constant $\Gamma(q)$, not $\Gamma(q+1)$. If $q/N \to \rho > 0$, then

$$-N^{-1} \log \mathcal{E}_{N,q} \to (1+\rho)\log(1+\rho) - \rho \log \rho.$$

The exact error comparison in Theorem 27.1 is more convenient at $q = L_N$; this uniform phase formula covers the smaller sampling regimes as well.

There is also a useful formal classical-Gamma obstruction. At conductor q, let e_r denote $\log\Gamma(r/q)$ modulo logarithms of algebraic numbers and $\log\pi$, with $e_q = 0$, and impose the stated reflection and multiplication relations. The integer functional

$$\Psi_q(e_r) = 2r - q \quad (1 \le r < q)$$

annihilates them: both reflection and each multiplication relation are checked by summing an arithmetic progression. For the filter vector $V_{N,q} = q\sum_{j=1}^{N}(-1)^{j+1}\binom{N}{j}e_{j \bmod q}/j$ it gives

$$\Psi_q(V_{N,q}) = q(2 - qH_N) \quad (q > N),$$

and $N(2 - NH_N - (-1)^{N+1})$ when $q = N$. These are nonzero for $q \ge N$, $q \ge 3$. Thus an individual filter does not lose every Gamma symbol under the specified classical relations. The statement concerns that formal quotient and does not assert completeness of all relations between actual Gamma values. Combinations of different filters, including the following algebraic-valued construction, are not excluded.

27.2 Removing the Gamma and pi coordinates

There is a different, algebraic-valued family. For $N \ge 2$, let

$$r_i = \frac{1}{8} + \frac{i}{24N}, \quad 0 \le i < N, \qquad I_k(r) = r^k - (1/2 - r)^k - (2r)^k,$$

and solve

$$\sum_i w_i = 0, \qquad \sum_i w_i I_k(r_i) = -1 \quad (2 \le k \le N).$$

Define the positive algebraic number

$$\alpha_N = \prod_{i=0}^{N-1}\left(\frac{2^{1-2r_i}\cos(\pi r_i)}{1 - 2r_i}\right)^{w_i}.$$

The weights are rational; the powers denote their positive real branches. The source theorem in `gamma_next_relations.md` states and proves that the system is nonsingular for every $N \ge 2$ and

$$|\gamma + \log\alpha_N| \le 150(N+1)^2(90/139)^N. \tag{27.2}$$

For clarity, the all-order mechanism is recorded here. The operator

$$(Tu)(r) = u(r) - u(1/2 - r) - u(2r)$$

maps polynomials of degree at most N onto the codimension-one space $p(0) = p(1/4)$. Its kernel consists of multiples of $r - 1/2$: the coefficient of the highest term of degree $k \ge 2$ is $1 - (-1)^k - 2^k \ne 0$, and the degree-one calculation gives that kernel. For $W_N(z) = \prod_i(z - r_i)$, the source's elementary node-product bound $|W_N(1/4)/W_N(0)| \le 5/7$ proves the constrained interpolation is nonsingular. Differentiating $Tu = E$ twice gives

$$u''(x) = \tfrac{1}{4}u''(x/2) - \tfrac{1}{4}u''((1 - x)/2) - \tfrac{1}{4}E''(x/2).$$

This is a contraction in the second-derivative norm. Reflection and duplication for Γ then reduce the error to at most $\frac{1}{4}\|(f - p_N)''\|_{[0,1/2]}$, where $f(r) = \log(2^{1-2r}\cos(\pi r)/(1 - 2r))$ and p_N is the constrained interpolant. Cauchy's formula on the circle with center $7/48$ and radius $3/5$, with the source bound $|f| < 7$, yields (27.2). The full contour-distance and constant audit is retained in that exact proof file.

Under the assumption that γ is algebraic, $\gamma + \log \alpha_N$ is nonzero for every N: otherwise the Hermite–Lindemann theorem would make the positive algebraic number $\alpha_N = e^{-\gamma}$ transcendental. This resolves nonvanishing under the contradiction hypothesis, not the degree-height comparison. The source's Cijsouw-based sufficient condition $D_N^2(D_N + \log H_N) = o(N)$ on an infinite subsequence, where D_N, H_N are the actual degree and primitive polynomial height of α_N, remains unproved. Large displayed powers do not by themselves determine those primitive invariants.

27.3 The E-function replacement and what the constant three quarters measures

For rational $a > 0$ put

$$E_a(z) = \sum_{k \geq 0} \frac{a}{a+k} \frac{(-z)^k}{k!}, \qquad G(a, X) = \frac{X^a}{a} E_a(X) = \int_0^X t^{a-1} e^{-t}\, dt.$$

The functions E_a are rational-coefficient E-functions, with $zE_a' + aE_a = ae^{-z}$. Take

$$q = L_N, \qquad d_j = q(-1)^{j+1} \binom{N}{j} / j, \qquad a_j = 1 + j/q,$$

and let X be the least power of two at least $8(N+1)(\lceil \log_2 q \rceil + 1)$. Replacing $\Gamma(a_j)$ by $G(a_j, X)$ gives a remainder $\widetilde{\varepsilon}_N$ satisfying

$$\tfrac{3}{4}\mathcal{E}_{N,q} < \widetilde{\varepsilon}_N < \tfrac{5}{4}\mathcal{E}_{N,q}.$$

Indeed $0 < \log \Gamma(a) - \log G(a, X) \leq 4(X+1)e^{-X}$ for $1 \leq a \leq 2$, $X \geq 4$, and $\sum |d_j| \leq q2^N$; comparison with the lower bound of Theorem 27.1 gives the displayed one-quarter budget.

The identities $\sum d_j a_j = qH_N + 1 \in \mathbb{Z}$ give a rational prefactor

$$C_N = X^{qH_N+1} \prod_j a_j^{-d_j} > 0$$

and E-functions

$$P_N(z) = \prod_{j \text{ odd}} E_{a_j}(Xz)^{d_j}, \qquad Q_N(z) = \prod_{j \text{ even}} E_{a_j}(Xz)^{-d_j}$$

such that

$$e^{\widetilde{\varepsilon}_N} = C_N e^{\gamma} P_N(1)/Q_N(1).$$

If γ is algebraic, $C_N e^{\gamma z} P_N(z) - Q_N(z)$ is an E-function with a positive small value at 1. Its defining family, orders and heights still vary with N.

The proof files `gamma_next_rescaling.md` and `gamma_next_efunction.md` provide two precise cost statements:

$$\frac{\log h(C_N)}{N} \longrightarrow 1 + \log 2, \qquad \frac{\log \operatorname{ord} P_N}{N^2}, \ \frac{\log \operatorname{ord} Q_N}{N^2} \longrightarrow \frac{3}{4}.$$

Here h is the logarithmic height of the displayed reduced rational prefactor, and ord is the minimal order of a scalar linear differential equation over $\mathbb{C}(z)$. Thus the 3/4 is a constant for the logarithm of a differential order, not a transcendence exponent of γ or a denominator exponent for a rational gamma approximation. The minimum-order input, after the ramification $z = w^q$, is algebraic independence of the primitives $\int_0^w v^{j-1} e^{-Xv^q}\, dv$ and their translation orbits. It gives

$$\operatorname{ord} \prod_{j \in J} E_{1+j/q}(Xz)^{n_j} = \prod_{j \in J} (n_j + 1), \qquad J \subseteq \{1, \ldots, q-1\}, \quad n_j \geq 0.$$

The exceptional endpoint $j = q$, when present at the first small indices, is treated separately; it does not affect the asymptotic statement. These costs do not establish a contradiction with a value lower bound.

28 Primitive exterior elimination in the Euler–Gompertz construction

Let $s = e^{-1}$ and $E(z) = \sum_{k\geq 1} z^k/(k\,k!)$, so $E(-1) = -\gamma - s\delta$. For integer rows $r_i = (A_i, B_i, C_i)$, set

$$R_i = A_i s + B_i E(-1) + C_i = s(A_i - B_i\delta) + C_i - B_i\gamma.$$

Define

$$U_{ij} = A_iC_j - A_jC_i, \quad V_{ij} = A_iB_j - A_jB_i, \quad W_{ij} = C_iB_j - C_jB_i,$$

and $L_{ij} = U_{ij} - V_{ij}\gamma + W_{ij}\delta$.

Proposition 28.1 (The surviving primitive pair)**.** *For $D = \det(r_1, r_2, r_3)$,*

$$W_{23}L_{12} - W_{12}L_{23} = D(B_2\gamma - C_2).$$

If $D \neq 0$ and $(B_2, C_2) \neq (0, 0)$, removal of the full coefficient content leaves exactly $(q, p) = (B_2/g, C_2/g)$, up to common sign, where $g = \gcd(|B_2|, |C_2|)$.

Proof. Expansion makes the δ coefficient zero, the γ coefficient DB_2, and the constant coefficient $-DC_2$. The common content is therefore exactly $|D|g$. □

The cofactor matrix satisfies

$$\operatorname{cof}(\operatorname{cof} M) = \det(M)M$$

for invertible 3 by 3 matrices. This follows from $\operatorname{cof} M = \det(M)M^{-T}$. Consequently repeated full cofactor formation, followed by primitive normalization, has projective period two. Moving the two outside rows while retaining the common row does not change its surviving rational approximation. This corrects a crude product-of-heights estimate; it does not prove the remaining form small or nonzero.

28.1 The actual Rivoal rows and an all-degree determinant

The published Hermite–Padé input is

$$R_n(z) = \frac{z^{3n+1}}{n!^2}\int_0^1\int_0^1 e^{zuv}u^{2n}(1-u)^n v^{2n}(1-v)^n\,du\,dv = A_n(z)e^z + B_n(z)E(z) + C_n(z),$$

with $\deg A_n \leq n-1$ and $\deg B_n, \deg C_n \leq n$. Writing

$$f_{n,k} = \frac{(2n+k)!}{k!^2(n-k)!^2}, \qquad h_{n,k} = H_{2n+k} - 2H_k + 2H_{n-k},$$

integration by parts, or the partial-fraction formulas in `gamma_cont_rivoal.md`, gives

$$B_n(z) = \sum_{k=0}^{n}(-1)^k f_{n,k}z^{n-k}, \qquad C_n(z) = \sum_{k=0}^{n}(-1)^k f_{n,k}h_{n,k}z^{n-k}.$$

In particular $B_n(-1) = (-1)^n\sum_k f_{n,k} \neq 0$; the primitive shared-row output is the rational number $C_n(-1)/B_n(-1)$.

For completeness, define

$$D_n(z) = \det\begin{pmatrix} A_n & B_n & C_n \\ A_{n+1} & B_{n+1} & C_{n+1} \\ A_{n+2} & B_{n+2} & C_{n+2}\end{pmatrix}(z).$$

Put $m = n + 1$ and

$$\beta_t = (-1)^t (3t)!/(t!^2), \qquad d_m = \frac{27m^2 + 18m + 1}{3(m+1)(3m+1)(3m+2)},$$

$$K_n = \frac{(2n)!^2}{(3n+1)!^2}, \quad M_{0,n} = -\beta_m \beta_{m+1} d_m > 0,$$

$$t_n = \frac{(2m-1)^2}{(3m-1)^2} - \frac{4(9m^2+2m-2)}{3(27m^2+18m+1)}.$$

Proposition 28.2 (Consecutive-row determinant)**.** *For all $n \geq 0$,*

$$D_n(z) = K_n M_{0,n} z^{3n+1}(1 + t_n z), \qquad (-1)^{n+1} D_n(-1) > 0.$$

Proof. The degree bounds give $\deg D_n \leq 3n + 2$. Replacing the final column by the remainders shows $z^{3n+1} \mid D_n$. Only the first two Taylor coefficients of R_n and of $M = A_m B_{m+1} - A_{m+1} B_m$ are needed. Beta integration gives

$$R_n = K_n z^{3n+1} \left(1 + \frac{(2n+1)^2}{(3n+2)^2} z + O(z^2)\right).$$

Put $a_t = 2H_t - H_{3t}$. The displayed coefficient formulas and the vanishing of the first coefficients of R_t give, for $t \geq 1$,

$$B_t(0) = \beta_t, \quad B_t'(0) = -t\beta_t/3, \quad A_t(0) = \beta_t a_t,$$

$$A_t'(0) = \beta_t \left[-(t/3+1)a_t + 2t/3 - 4/9\right].$$

Since $a_m - a_{m+1} = -d_m$, one obtains

$$M(0) = M_{0,n}, \qquad M'(0)/M(0) = -\frac{4(9m^2+2m-2)}{3(27m^2+18m+1)}.$$

Multiplication proves the formula. The subtracted term in t_n is positive and $(2m-1)/(3m-1) < 2/3$, hence $t_n < 4/9 < 1$. □

This determinant nonvanishing holds in every degree. Nonvanishing of the evaluated primitive form was certified only at the finitely tested indices in the source; these two scopes must not be conflated.

29 Gregory arithmetic: exact cancellation before integration

Let

$$F(z) = \frac{-\log(1-z)}{z} = \sum_{n \geq 0} \frac{z^n}{n+1}, \quad B(z) = F(z)^{-1} = \sum_{n \geq 0} b_n z^n,$$

and let $g_n = -b_n > 0$ for $n \geq 1$. Then

$$G(z) = \sum_{n \geq 1} \frac{g_n}{n} z^n, \qquad G'(z) = \frac{1 - B(z)}{z}, \qquad G(1) = \gamma.$$

The moments used in the first gamma paper are $c_n = g_{n+1}/(n+1)$.

Theorem 29.1 (Todd denominators)**.** *For every $N \geq 0$, the least integer clearing $b_0, \ldots, b_N$ is*

$$T_N = \prod_{p \leq N+1} p^{\lfloor N/(p-1) \rfloor}.$$

For every prime p and every $k \geq 1$,

$$v_p(b_n) \geq -\lfloor n/(p-1) \rfloor, \qquad p^k b_{k(p-1)} \equiv (-1)^k \pmod{p}.$$

Moreover $\log T_N = N \log N + O(N)$.

Proof. For $j \geq 1$, $(p-1)v_p(j+1) \leq j$, with equality only when $j = p-1$. Expand $B = (1 + (F-1))^{-1}$ geometrically. Each contribution to b_n is a signed product of $1/(j_i+1)$ for an ordered composition $\sum j_i = n$. Its denominator exponent is at most $\lfloor n/(p-1) \rfloor$. At $n = k(p-1)$ only the composition with k entries all equal to $p-1$ attains exponent k. This proves the congruence and the exact cumulative denominator. Finally

$$\log T_N = \sum_{p \leq N+1} \lfloor N/(p-1) \rfloor \log p = N \log N + O(N)$$

by the weighted prime estimate $\sum_{p \leq x} \log p/(p-1) = \log x + O(1)$. □

Theorem 29.2 (Reciprocal Hankel cancellation)**.** *For every $N \geq 1$,*

$$\det(b_{i+j})_{0 \leq i,j < N} = \frac{(-1)^{N-1}}{\displaystyle\prod_{j=0}^{N-2} (2j+3) \binom{2j+2}{j}^2}.$$

The empty product is 1*. The absolute determinant has N^2-root limit* $1/4$*; its exact reduced denominator has N^2-root limit* 4*.*

Proof. For a general reciprocal pair $F = \sum f_n z^n$, $B = 1/F$, $f_0 = 1$, let $T = (b_{i-j})_{0 \leq j \leq i < N-1}$ and $H = (f_{i+j+2})_{0 \leq i,j < N-1}$. Coefficient comparison gives

$$(b_{i+j+2} - b_{i+1} b_{j+1}) = -THT^{\mathsf{T}}.$$

It follows either by multiplying the bivariate generating function

$$\frac{(F(x)-1)/x - (F(y)-1)/y}{x - y}$$

by $-B(x)B(y)$, or directly by convolution. Taking the Schur complement of the initial entry $b_0 = 1$ gives the determinant $(-1)^{N-1} \det H$. Here $H_{ij} = 1/(i+j+3)$; the Cauchy determinant formula gives the displayed product. Stirling's formula proves the two limits. □

Thus coefficientwise factorial-scale denominators can collapse in a full Hankel determinant. The exact integer after clearing this determinant is nevertheless always ± 1: the capacity and denominator exponents balance. This identity applies before the division by n defining G, not automatically to the gamma Padé output.

29.1 Formal-group Frobenius and an exact Lyapunov equation

For $\psi_p(z) = 1-(1-z)^p$, the multiplicative formal-group logarithm gives

$$B(\psi_p(z)) = \frac{\psi_p(z)}{pz}B(z), \qquad \psi_p(z) \equiv z^p \pmod p.$$

This is not the exact Mahler substitution $z \mapsto z^p$ over $\mathbb{Q}$. Theorem 29.1 also gives the exact p-adic radius

$$R_p(B) = R_p(G) = p^{-1/(p-1)}.$$

Division of the nth coefficient by n changes its nth-root growth by a factor tending to one. The coordinate $x = -\log(1-z)$ gives

$$B(1-e^{-x}) = \frac{1-e^{-x}}{x}, \qquad G(1-e^{-x}) = \mathrm{Ein}(x) + \log\frac{1-e^{-x}}{x}.$$

The first transformed function is an E-function. In the second formula, γ still occurs at the limit $x \to +\infty$, not as a regular algebraic-argument E-value.

Let

$$T = (b_{i-j})_{0\le j\le i<N}, \quad J = (1/(i+j+2))_{0\le i,j<N}, \quad M = (c_{i+j})_{0\le i,j<N},$$

and $\Lambda = \mathrm{diag}(1/2, 3/2, \dots, N-1/2)$.

Proposition 29.3 (The integrated moment matrix as a Gramian)**.** *The matrix M is the unique solution of*

$$\Lambda M + M\Lambda = TJT^{\mathsf{T}}, \qquad M = \int_0^\infty e^{-t\Lambda}TJT^{\mathsf{T}}e^{-t\Lambda}\,dt.$$

If $D = \mathrm{diag}(0, 1, \dots, N-1)$, U is lower triangular with every entry on and below the diagonal equal to 1, and $X = T^{-1}MT^{-\mathsf{T}}$, then

$$\mathcal{A}X + X\mathcal{A}^{\mathsf{T}} = J, \qquad \mathcal{A} = \Lambda + I - UT.$$

Both M and X are positive definite.

Proof. The reciprocal-series identity

$$(b_{i+j+1}) = -T(f_{i+j+1})T^{\mathsf{T}}$$

and $(i+j+1)c_{i+j} = -b_{i+j+1}$ give the first equation. The Lyapunov operator has positive eigenvalues $i+j+1$ on matrix units, so its inverse is the stated convergent integral. Positivity follows because J is positive definite and T is invertible. Differentiating $B = z/[-\log(1-z)]$ gives

$$zB' = B - B^2/(1-z).$$

Using truncated Toeplitz multiplication and $[D, T(B)] = T(zB')$ gives $T^{-1}DT = D + I - UT$. Congruence of the first equation proves the second. □

The inversion of the factors $i+j+1$ is the exact new arithmetic cost introduced by integration. Positive stability alone does not control the primitive denominator of a Schur-complement value.

For the elementary partial sums $S_N = \sum_{n=1}^N g_n/n = A_N/Q_N$ in lowest terms, the same valuation calculation proves

$$v_2(Q_{2^k}) = 2^k + k.$$

Indeed, at $N = 2^k$ the last summand has valuation $-(N+k)$ and every preceding summand has larger valuation. The coefficient integral gives $g_{N+1} \ge 1/(6N(N+1))$, hence

$$Q_N(\gamma - S_N) \ge \frac{2^N}{6(N+1)^2} \longrightarrow \infty \qquad (N = 2^k).$$

This is a failure theorem for these dyadic truncations, not for every subsequence or for the Padé family.

30 Moving factorial parameters and the exact target they approach

For integers $n \geq 1$, $m \geq 0$, define

$$f_{n,m,k} = \frac{(m+k)!}{k!^2(n-k)!^2}, \quad h_{n,m,k} = H_{m+k} - 2H_k + 2H_{n-k}, \quad r_{n,m} = \frac{\sum_{k=0}^{n} f_{n,m,k} h_{n,m,k}}{\sum_{k=0}^{n} f_{n,m,k}}.$$

Up to common factors, $m = 0$, $m = n$ and $m = 2n$ correspond respectively to the Laguerre, Aptekarev and surviving Rivoal families.

Theorem 30.1 (Complete convergence classification)**.** *For every sequence of nonnegative integers m_n,*

$$r_{n,m_n} \to \gamma \quad \Longleftrightarrow \quad m_n = o(n^2).$$

If $m_n/n^2 \to a \in (0,\infty)$, then

$$r_{n,m_n} \longrightarrow \gamma + \frac{2K_0(2/\sqrt{a})}{I_0(2/\sqrt{a})} > \gamma.$$

If $m_n/n^2 \to \infty$, then

$$r_{n,m_n} = \log(m_n/n^2) - \gamma + O(1/n + n^2/m_n) \longrightarrow +\infty.$$

Proof. Put $j = n - k$ and normalize by the endpoint term:

$$w_j = \frac{(n^{\underline{j}})^2}{(m+n)^{\underline{j}}(j!)^2}, \qquad g_j = H_{m+n-j} - 2H_{n-j} + 2H_j.$$

Then $r_{n,m}$ is the weighted average of g_j and $0 \leq g_j - g_0 \leq 4H_j$. At quadratic scale $m/n^2 \to a > 0$, for each fixed j, $w_j \to a^{-j}/(j!)^2$ and $g_j \to \log a - \gamma + 2H_j$. The bounds $w_j \leq (2/a)^j/(j!)^2$ and the displayed harmonic inequality permit dominated convergence. The classical series

$$K_0(2\sqrt{x}) = -(\tfrac{1}{2}\log x + \gamma) I_0(2\sqrt{x}) + \sum_{j \geq 0} \frac{H_j x^j}{(j!)^2}$$

gives the stated limit. Both Bessel functions are positive at positive real arguments. If $m/n^2 \to \infty$, then $w_j \leq \varepsilon^j/(j!)^2$ with $\varepsilon = n^2/(m+1) \to 0$, so $r_{n,m} = H_{m+n} - 2H_n + O(\varepsilon)$, proving the last formula.

Suppose now that $m = o(n^2)$ and set $\rho = n/\sqrt{m+n} \to \infty$. Compare the normalized w_j with the Bessel distribution

$$\mathbb{P}_\rho(J = j) = \frac{\rho^{2j}}{(j!)^2 I_0(2\rho)}.$$

Their unnormalized ratio is

$$R_j = \prod_{s<j} \frac{(1 - s/n)^2}{1 - s/(m+n)} \leq 1.$$

For $j \leq 2\rho$ it is at least e^{-8} for large n. Since $\mathbb{E}_\rho J^2 = \rho^2$, this region has probability at least $3/4$, and the actual distribution $\mathbb{Q}$ satisfies $\mathbb{Q}(J = j) \leq (4e^8/3)\mathbb{P}_\rho(J = j)$. The Bessel moments

$$\mathbb{E}_\rho J = \rho I_1(2\rho)/I_0(2\rho), \qquad \mathbb{E}_\rho J^4 = \rho^4 + 2\rho^3 I_1(2\rho)/I_0(2\rho) + \rho^2$$

imply $J/\rho \to 1$ in probability under both distributions. Moreover

$$\mathbb{E}_\rho(J+1)/\rho \leq 2, \qquad \mathbb{E}_\rho \rho/(J+1) = I_1(2\rho)/I_0(2\rho) \leq 1.$$

The inequality $(\log y)^2 \le C(y + y^{-1})$ gives uniform integrability, and therefore $\mathbb{E}_{\mathbb{Q}} H_J = \log \rho + \gamma + o(1)$. For $j \le n/2$, $H_n - H_{n-j} \le 2j/n$; for the remaining tail use H_n and the fourth-moment bound. Thus both endpoint harmonic shifts have expectation $o(1)$, and

$$r_{n,m} = H_{m+n} - 2H_n + 2\log\rho + 2\gamma + o(1) = \gamma + o(1).$$

Finally, if m_n/n^2 does not tend to zero, some subsequence has a further ratio limit in $(0, \infty]$. The already proved alternatives exclude convergence to γ on that subsequence. □

The manifest integer denominator $Q^0_{n,m} = \sum_k \binom{n}{k}^2 (m+k)!/m!$ satisfies

$$\frac{(m+n)!}{m!} \le Q^0_{n,m} \le \frac{(m+n)!}{m!} I_0(2n/\sqrt{m+n}).$$

This is not a lower bound on the denominator of the reduced fraction $r_{n,m}$. The finite source test of $5 \le n \le 100$, $0 \le m \le 2n$ contains 10,176 pairs and certifies $|Q\gamma - P| > 1$ for every tested primitive pair. It does not prove failure at all indices. The quadratic movement changes the limiting target; below that scale the primitive arithmetic problem remains.

31 Sign variation, normal indices and growing prime windows

An additional local result preceding the first paper is not implied by its first-prime-entry theorem. Its real component has a useful abstract form.

Theorem 31.1 (A quadratic sign change leaves no three-index gap)**.** *Let ω be a finite positive measure with infinite support on a compact interval, and let s_2 be a real monic quadratic polynomial. Write $h_j = \int t^j s_2(t)\, d\omega(t)$ and $D_r = \det(h_{i+j})_{0 \le i,j < r}$, with $D_0 = 1$. Then $(D_r, D_{r+1}, D_{r+2}) \ne (0,0,0)$ for every $r \ge 0$. Each Hankel matrix has nullity at most two.*

Proof. If $D_n \ne 0$, let P_n be the monic polynomial orthogonal to degrees less than n for the signed functional $\mathcal{L}(f) = \int f s_2\, d\omega$. The first κ with $\mathcal{L}(t^\kappa P_n) \ne 0$ is at most $n+2$, because otherwise

$$\mathcal{L}(s_2 P_n^2) = \int s_2^2 P_n^2\, d\omega > 0$$

would vanish. Write $\kappa = n + k - 1$, $1 \le k \le 3$. In the monic basis $1, \dots, t^{n-1}, P_n, tP_n, \dots, t^{k-1}P_n$, the cross block vanishes and the last block has nonzero constant anti-diagonal and zeros on the side $i + j < k - 1$. Therefore

$$D_{n+k} = D_n (-1)^{k(k-1)/2} \mathcal{L}(t^\kappa P_n)^k \ne 0,$$

while the intervening determinants vanish. Start with $D_0 = 1$ and iterate. For nullity, a kernel of dimension at least three would contain a nonzero polynomial f of degree at most $r - 3$. Testing against $s_2 f$ contradicts $\int s_2^2 f^2\, d\omega > 0$. □

The gap bound is sharp in this class: with $d\omega = dt$ on $[0,1]$ and $s_2 = t^2 - t + 1/6$, one has $D_1 = D_2 = 0$ and $D_3 = -1/180^3$.

For the gamma Padé local model, define from the Gregory series B

$$\lambda_s = \frac{[z^{s+2}]B^2}{s+1}, \quad \mu_s = -\frac{[z^{s+3}]B^3}{s+1}, \quad d_s = c_s - 8\lambda_s + 16\mu_s, \quad h_s = d_s - d_{s+1}.$$

The source `gamma_push_local.md` identifies this h_s with the preceding signed-moment form. The identification is explicit: with $x = \log(t/(1-t))$ put

$$\rho(t) = \frac{x^4 + 16x^3 + (2\pi^2 + 48)x^2 + 16\pi^2 x + \pi^4 - 16\pi^2}{t(x^2 + \pi^2)^3}, \quad v(t) = \int_t^1 \rho(s)\, \frac{ds}{s}.$$

Then $h_j = \int_0^1 t^j(1-t)v(t)\,dt$. The quartic numerator has two real roots: after $y = x + 4$, its derivative is four times $y^3 + (\pi^2 - 24)y + 32$, which has negative discriminant and one real root, and its minimum is negative. Consequently ρ has signs $+, -, +$. The derivative $v' = -\rho/t$, the endpoint positivity and $\int v = d_0 = -1/6$ show that v has precisely two simple zeros $\alpha < \beta$ with the same sign pattern. Therefore $d\omega = (1-t)v(t)\,dt/((t-\alpha)(t-\beta))$ is positive and finite, which verifies the sign-change hypothesis.

The arithmetic transfer uses the fixed-offset congruence from `gamma_cont_pade.md`: if $p > 2r+3$ and D_r is a p-adic unit, then $p\ell_{p+r} \equiv 4 \pmod{p}$, hence the reduced denominator q_{p+r} has valuation $v_p(q_{p+r}) = 1$. The rational moments above give an explicit increasing bound $\mathcal{B}_r$ on the reduced numerator of D_r with

$$\log \mathcal{B}_r = 6r^2 \log r + O(r^2).$$

Indeed, through degree $M = 2r+2$, $M!L_{M+1}$ clears B, its cube clears B^3, and an extra L_M clears the divisions by $s+1$. Expanding the determinant gives the asserted leading growth. The exact bound retained in the source is

$$\mathcal{B}_r = r!(C_M Q_M)^r, \quad Q_M = (M!L_{M+1})^3 L_M, \quad C_M = 2\{1 + 8(M+1)(M+3)\}.$$

Combining this with Theorem 31.1 proves: for every $c < 1/\sqrt{3}$, every sufficiently large prime p, and simultaneously every integer

$$1 \le r \le c\sqrt{\log p / \log\log p},$$

at least one of $q_{p+r}, q_{p+r+1}, q_{p+r+2}$ has p-valuation one. The growing window is too narrow to imply a uniform prime-count lower bound. In particular this is not an exponential lower bound on every q_N. The previously suggested stronger assertion that every prime $N/2 < p \le N$ survives is false at $(N, p) = (51, 43)$.

32 A raw quotient index and why primitive cancellation remains separate

Let $B \in \mathbb{Z}^{r\times s}$ have full row rank, and suppose the matrix C formed by adjoining the two target rows a, b has rank $r+2$. Let $\Lambda = \ker_{\mathbb{Z}} B$ and $\Omega = \{(a\cdot u, b\cdot u) : u \in \Lambda\} \subset \mathbb{Z}^2$. For a full-row-rank integer matrix M, write $\Delta(M)$ for the gcd of its maximal minors.

Proposition 32.1 (Exact target-lattice index)**.** *The lattice Ω has rank two and*

$$[\mathbb{Z}^2 : \Omega] = \frac{\Delta(C)}{\Delta(B)}.$$

Equivalently, for a saturated integer-kernel basis $u_1, \dots, u_k$, the index is the gcd of all determinants $\det(\Phi(u_i), \Phi(u_j))$, where $\Phi(u) = (a\cdot u, b\cdot u)$.

Proof. An integer column Hermite reduction gives a unimodular V with $BV = [H\ 0]$ and H invertible. The last columns of V form a basis of the entire integer kernel. In this basis

$$CV = \begin{pmatrix} H & 0 \\ * & F \end{pmatrix},$$

where the columns of F generate Ω. Every nonzero maximal minor contains all first r columns, so $\Delta(C) = |\det H|\Delta(F)$, while $\Delta(B) = |\det H|$. The rank-two image index is $\Delta(F)$. □

For a Sondow window, write the positive integer-normalized integrals as $e_n = a_n\gamma - b_n + \sum_p \beta_{p,n} \log p$ and choose rational upper bounds $e_n \le E_n$. Set $S_a = \sum |a_n|$ and $S_e = \sum E_n$. If $u, v \in \Lambda$ have independent target images and $D = [\mathbb{Z}^2 : \Omega]$, then

$$\|u\|_\infty \|v\|_\infty \ge \frac{D}{2S_aS_e}.$$

Indeed, for $L_u = Q_u\gamma - P_u$, the nonzero target determinant is a multiple of D, whereas

$$|Q_uP_v - Q_vP_u| = |Q_vL_u - Q_uL_v| \le 2\|u\|_\infty \|v\|_\infty S_aS_e.$$

This holds without any assumption on γ. It bounds raw representatives of two independent targets. Dividing target pairs by their gcd changes that cost: the lattice $M\mathbb{Z}^2$ has index M^2, but its two coordinate generators reduce to primitive pairs of determinant one. The primitive quotient norm in the first paper is precisely the later correction needed to account for this operation.

The source `gamma_cont_quotient.md` checks a proposed prime divisibility law at finitely many windows $N = 32, 40, 48, 64$. The all-N divisibility law remains a conjecture. Neither the raw index nor those finite checks give a no-go theorem for primitive output forms.

33 Retained auxiliary routes and their exact status

The following items are part of the cumulative record, but their long special-purpose calculations are retained in the source archive rather than duplicated in the main proof development.

1. **Restricted positive cones and square certificates.** `gamma60_operators.md` solves the full nonnegative numerator cone at $m = 3$, rather than only a chosen finite sample, and gives its exact algebraic endpoints and a dual certificate. The resulting bound $|q\gamma - p| > q/200$ belongs to that fixed cone. The larger-pole completeness theorem above supersedes an interpretation of this finite obstruction as a global one. The distinct Laguerre square-certificate calculation in `gamma60_arithmetic.md` obtains $|a - b\gamma| \ge |b|/(\sqrt{80}N^{3/2})$ in its specified square family, with tensor extensions. It does not cover arbitrary signed integrals or the whole floor-integral space.

2. **Growing-degree exponential transforms.** `gamma60_geometry.md` constructs increasing-degree transforms outside the fixed-degree logarithmic denominator obstruction of Theorem 24.1. Their limit coefficients retain e^γ. This is a genuine scope distinction; no arithmetic classification of that remaining constant is derived from the transform.

3. **Positive tensors and finite support.** `gamma_next_polynomial.md` proves finite positive tensor certificates for strict polynomial signs and constructs positive rational atomic measures realizing alternative values in a fixed-pole feasible interval. Consequently increasing tensor degree at fixed poles does not isolate γ. The particular $m = 3$ alternative is $4/7$. `gamma_frontier_arithmetic_cones.md` also constructs totally real algebraic alternatives of arbitrary degree under finitely many moment conditions. Their positivity is a real condition, not an integrality theorem for all divided powers.

4. **Divided powers.** The same arithmetic-cones note proves that an algebraic number x for which every $\binom{x}{n}$ is an algebraic integer must be an integer. No such factorial divisibility is established for γ or for a fixed multiple of it. This implication is a source theorem with unverified target hypothesis, not an irrationality proof.

5. **Geometric nodes and actual reduced denominators.** `gamma_frontier_scaling.md` gives a Gaussian-binomial Gamma filter with positive error $\log R_N = -\frac{1}{2}(\log B)N^3 + O_B(N)$, but with a displayed rational prefactor of logarithmic height $\Theta_B(q \log q)$, $\log q \sim \frac{1}{2}(\log B)N^2$.

`gamma_frontier_spectral.md` constructs a distinct purely rational family with $e^{-n}/6 < \gamma - r_n < 5e^{-n}$ for $n \geq 4$ and proves for its actual reduced denominator d_n that $\log d_n = \Theta(n^4 \log n)$. Thus $d_n(\gamma - r_n) \to +\infty$ for that family, even after cancellation.

6. **Mahler means and fixed kernels.** `gamma_frontier_mahler.md` isolates a Fourier mean $1/2 - \gamma/\log b$; the arithmetic status of that mean does not follow from a Mahler special-value theorem. The fixed-kernel analysis in `gamma_next_renormalization.md` uses

$$W(x) = \kappa(1-x) + (1-x^2)A(x), \qquad \kappa = W(-1)/2,$$

for $W(1) = 0$, to distinguish retention and loss of the gamma coefficient. The denominator exponents proved there concern its specified truncations and do not apply to every generating function for γ.

7. **Information and diffusion.** `gamma60_physics.md` records the classical normalized identity $D_{\mathrm{KL}}(\mathrm{Exp}(1) \| \,\mathrm{Gamma}(2,1)) = \gamma$. The integer moments $n!$ determine the exponential distribution, but the associated Ser–Sondow reconstruction has error asymptotic to $1/(N \log N)$ and a prime-log coefficient denominator with logarithm at least $(1/2 + o(1))N$. Uniqueness of the distribution supplies no small integer form. For the diffusion generator $L = x\partial_x^2 + (1-x)\partial_x$, the formal identity $L \log x = -1$ cannot be integrated against stationarity without the nonzero boundary term. The source computes that term; the purported contradiction from dropping it is withdrawn.

8. **Multiscale quadrature and spectral realizations.** `gamma60_analysis.md` combines two Gamma quadratures and proves an additional factor N^{-3} in its error, with a retained algebraic-logarithmic target. It does not remove those coordinates. `gamma_next_spectral.md` constructs transcendental finite approximants in $\overline{\mathbb{Q}}(\pi)$ converging to γ with error $\Theta(n^{-2})$. Transcendence of each approximant is not preserved under a limit. The earlier spectral note also realizes any prescribed positive total trace by rapidly decreasing, finitely algebraically independent positive eigenvalues. Those qualitative spectral properties alone therefore cannot decide the arithmetic of a total trace.

9. **Mixed Gamma jets and local discriminant pairings.** The first gamma paper incorporates the integral coefficient conversion and graded denominator bound from the continuation notes. The additional closed multiplication law over a larger period ring is retained in `gamma_cont_mixed.md`; enlarging the coefficient ring does not prove a zero-lifting theorem. The local Padé pairing $[v]^T M^{-1}[v] \pmod{\mathbb{Z}}$ can be isotropic even for a positive definite real Gram matrix. Its order, not the order of $[v]$ or the size of the discriminant group, is the reduced output denominator. The source example $(N, p) = (7, 5)$ exhibits this distinction.

The source register is part of completeness: it identifies which claims have been re-proved here, which are inputs with retained detailed proofs, which are finite computations and which remain conditional. No item in this part proves that γ is irrational or transcendental.

Part IV
Analytic variations and imported methods

34 Automatic series, periodic identities, and normalization

The following investigations concern explicit representations of Euler's constant and the information preserved by changes of representation. A functional transcendence theorem is not, without an additional specialization argument, a theorem about a particular value. The counterexamples below are kept alongside the positive constructions because they delimit precisely which information is insufficient.

34.1 A Mahler derivative whose primitive leaves every Mahler class

For an integer $B \geq 2$, put

$$\epsilon_B(n) = B\mathbf{1}_{B|n} - 1, \qquad f_B(z) = \sum_{n\geq 1} \epsilon_B(n) \frac{\lfloor \log_B n \rfloor}{n} z^n, \qquad G_B(z) = \sum_{k\geq 1} z^{B^k}.$$

The generalized Vacca identity [26] and direct differentiation give

$$f_B(1) = \gamma, \qquad z f_B'(z) = G_B(z)\left(\frac{B}{1-z^B} - \frac{1}{1-z}\right), \qquad G_B(z) - G_B(z^B) = z^B. \tag{34.1}$$

Here the boundary series converges. Indeed, its sum through $B^M - 1$ is

$$MH_{B^{M-1}-1} - (M-1)H_{B^M-1},$$

which tends to γ; a partial last block is $O_B(MB^{-M})$. In particular f_B' is B-Mahler. The primitive has a different arithmetic property.

Lemma 34.1 (Finite denominator support for a Mahler series)**.** *If $H(z) = \sum h_n z^n \in \overline{\mathbb{Q}}[[z]]$ satisfies $\sum_{i=0}^{r} a_i(z) H(z^{b^i}) = 0$, where $b \geq 2$, $a_i \in \overline{\mathbb{Q}}(z)$, and $a_0 \neq 0$, then all h_n belong to the ring of S-integers of one number field for one finite set S of finite places.*

Proof. Clear rational-function denominators, and write $d = \operatorname{ord}_0 a_0$, $\alpha = [z^d]a_0 \neq 0$. The coefficient of z^{n+d} solves for h_n by division by α. All other indices in $a_0 H$ are smaller than n, and those in $a_i H(z^{b^i})$, $i > 0$, are of the form $(n + d - k)/b^i < n$ for sufficiently large n. Thus a finite initial segment and the finitely many coefficients of the a_i, together with α^{-1}, generate all subsequent coefficients by ring operations. □

Proposition 34.2 (Persistent new-prime denominators)**.** *The series f_B is not Mahler in any integer base. More generally, let $c \in \mathbb{Q}^\times$, $R \in z\mathbb{Q}[[z]]$ be algebraic, and $r = R'(0)$. If $\exp(cf_B + R) = \sum e_n z^n$, then at every sufficiently large prime p,*

$$pe_p \equiv -c\lfloor \log_B p \rfloor - r \pmod{p}, \qquad v_p(e_p) = -1.$$

Consequently this exponential is neither algebraic nor Mahler in any integer base. Multiplication by an algebraic unit in $\mathbb{Q}[[z]]$, or a fixed nonzero rational rescaling of the variable, preserves the conclusion.

Proof. For sufficiently large $p \nmid B$, $p[z^p]f_B = -\lfloor \log_B p \rfloor$ is nonzero modulo p, while all earlier coefficients are p-integral. Eisenstein's denominator theorem makes all coefficients of R simultaneously p-integral outside a fixed finite set. In the coefficient of degree p of $\exp h$, $h = cf_B + R$, only

h_p and $h_1^p/p!$ can have a p-denominator. Wilson's and Fermat's congruences give the displayed residue. After a fixed common denominator is cleared, its negative is a nonzero integer of size $O(\log p) < p$ for all sufficiently large p. Lemma 34.1 and Eisenstein's theorem apply. For an algebraic unit U, only $U(0)e_p$ contributes a nonintegral summand to $[z^p]U\exp h$; the rescaling claim follows because a fixed nonzero rational number is a unit at almost all primes. □

The same conclusion holds for a nontrivial finite combination $F = \sum_{d\in\mathcal{D}} c_d f_{b^d}$, including $\exp(F+R)$. For $b^n < p < b^{n+1}$ the detecting residue is $-A(n)$, respectively $-A(n)-R'(0)$, where $A(n) = \sum_d c_d\lfloor n/d\rfloor$. This function is not eventually constant: otherwise its periodic difference $\sum_{d|n} c_d$ would vanish identically, and triangular induction would give every $c_d = 0$. Bertrand's theorem supplies a prime in $(b^n, 2b^n)$ for each sufficiently large detecting index. This proves persistence for powers of one fixed base, not for arbitrary independent bases. Nevertheless

$$(f_b - f_{b^2})(1) = 0, \qquad \exp\big((f_b - f_{b^2})(1)\big) = 1. \tag{34.2}$$

Thus the function-level obstructions just proved cannot themselves decide the arithmetic of the boundary value.

34.2 Finite Gamma data and global rigidity

Proposition 34.3 (Finite-jet normalization counterexamples)**.** *Fix finitely many positive rational evaluation points, finitely many rational test points for Gamma multiplication formulas, and a finite jet order $R \geq 1$. For every real η there is a meromorphic function $\widetilde{\Gamma}$ satisfying the Gamma recurrence and reflection formula globally, with the same poles, residues, zeros and positive integer values as Γ, the prescribed rational function values, and the multiplication identities through jet order R, such that $\widetilde{\Gamma}'(1) = \eta$ and $\widetilde{\Gamma}(x) > 0$ for $x > 0$. It is differentially transcendental over $\mathbb{C}(z)$.*

Proof. Choose N clearing the denominators of all test arguments, including their multiplication translates. Put $M = \lceil R/2\rceil$ and

$$\phi(z) = \frac{1}{2\pi N}\sum_{k=0}^{M}\frac{\binom{2k}{k}}{4^k(2k+1)}\sin^{2k+1}(2\pi N z), \qquad \widetilde{\Gamma}(z) = \Gamma(z)e^{(\eta+\gamma)\phi(z)}.$$

The truncated arcsine identity gives $\phi(t+h) = h + O(h^{2M+3})$ at every test argument t. Moreover ϕ is entire, 1-periodic, zero at the integers, and $\phi(1-z) = -\phi(z)$. These facts prove recurrence, reflection, divisor, residue, value and derivative assertions. In a multiplication identity the added exponents have difference $\sum_{j=0}^{m-1}\phi(z+j/m) - \phi(mz)$; its linear jets cancel and all jets through R vanish. Positivity follows on the positive axis. The difference-Galois form of Hölder's theorem applies to every nonzero meromorphic solution of $F(z+1) = zF(z)$ [27]. This is an existence counterexample, explicitly using γ in its definition, not a proposed gamma-independent approximation algorithm. □

This freedom disappears if the finite multiplication tests are replaced by a global duplication identity and $\widetilde{\Gamma}/\Gamma$ is entire and nowhere zero. In that case

$$\widetilde{\Gamma}(z) = e^{4\pi i k z}\Gamma(z) \quad (k\in\mathbb{Z}),$$

and positivity selects $k = 0$. To see this, the periodic logarithmic derivative h of the quotient satisfies $2h(2z) = h(z) + h(z+1/2)$. Its Fourier coefficients obey $a_{2n} = a_n$; exponential decay on horizontal lines forces $a_n = 0$ for $n \neq 0$. Periodicity, the value at 1, and duplication then give

the displayed exponential. This is analytic uniqueness, not arithmetic specialization. An even simpler counterexample to unrestricted difference-value principles is

$$F(z)=\psi(z)+\gamma=\sum_{k\ge1}\left(\frac{1}{k}-\frac{1}{k+z-1}\right),\qquad F(z+1)-F(z)=z^{-1},\qquad F(n)=H_{n-1}\in\mathbb{Q}.$$

It is differentially transcendental yet has an entire algebraic shift orbit of rational values. Shift, reflection and multiplication identities for ψ are invariant under adding a constant [28].

34.3 Periodic constants and cyclotomic averaging

If f is algebraic-valued and q-periodic, let $m=q^{-1}\sum_{a=1}^{q}f(a)$ and $\widehat{f}(r)=q^{-1}\sum_{a=1}^{q}f(a)\zeta_q^{-ra}$. Fourier expansion and Dirichlet convergence give

$$\lim_{x\to\infty}\left(\sum_{n\le x}\frac{f(n)}{n}-m\log x\right)=m\gamma-\sum_{r=1}^{q-1}\widehat{f}(r)\log(1-\zeta_q^r).\tag{34.3}$$

Thus the finite periodic family adds algebraic logarithmic directions to one unresolved γ coordinate. For example,

$$\gamma(\Omega)=\prod_{p\in\Omega}(1-p^{-1})\left(\gamma+\sum_{p\in\Omega}\frac{\log p}{p-1}\right).$$

Baker's theorem controls a nonzero logarithmic part but does not by itself exclude algebraicity of the remaining γ coordinate.

For $L(q)=\sum_{n\ge1}d(n)q^n$, an integer $p>1$ and prime m, the root-of-unity filter gives

$$\frac{1}{m}\sum_{j=0}^{m-1}L(\zeta_m^j p^{-1/m})=2L(p^{-1})-L(p^{-m}).\tag{34.4}$$

Indeed only indices divisible by m survive, and $d(mn)=2d(n)-\mathbf{1}_{m|n}d(n/m)$. Weighting instead by $1+X+\cdots+X^{m-1}$, with m now arbitrary, gives

$$W_m=(1-p^{-1})\sum_{\ell\ge1}p^{-\ell}D(m\ell),\quad D(x)=\sum_{n\le x}d(n).$$

With $D(x)=x\log x+(2\gamma-1)x+\Delta(x)$, set

$$C_p=\frac{(p-1)^2}{p}\sum_{\ell\ge1}\frac{\ell\log\ell}{p^\ell},\quad T_m=\frac{1}{2}\left(\frac{p-1}{m}W_m-\log m+1-C_p\right).$$

Then the exact identity

$$T_m-\gamma=\sum_{\ell\ge1}\frac{(p-1)^2\ell}{p^{\ell+1}}\frac{\Delta(m\ell)}{2m\ell}\tag{34.5}$$

has positive weights summing to one. It retains γ, but introduces the logarithmic moment C_p and is not a rational approximation. Infinite analytic averages in these formulas are not field traces of known algebraic numbers; only their finite polynomial truncations have that interpretation.

A separate truncation makes the height cost explicit. For prime $M\ge101$, $q=1-1/M$, $K=\lceil 4M\log M\rceil$, the reduced denominator of $\sum_{n\le K}d(n)q^n=A_M/D_M$ is $D_M=M^K$: its numerator modulo M is $(-1)^K d(K)\neq0$, since $d(K)\le2\sqrt{K}<M$. The normalized logarithmic approximant $G_M=-\log M-\log(q)A_M/D_M$ satisfies $\gamma-G_M=(4M)^{-1}+O(M^{-2})$, by Euler–Maclaurin and the geometric tail estimate. Its primitive integer logarithmic form is

$$D_M\gamma-(A_M-D_M)\log M+A_M\log(M-1),$$

of size asymptotic to $M^K/(4M)$. The approximation is neither rational nor a small linear form in $1,\gamma$.

35 Divisor sums, exact dyadic denominators, and logarithmic jets

Throughout this section $\Delta(x) = D(x) - x\log x - (2\gamma-1)x$. The elementary hyperbola estimate $|\Delta(x)| \le 8\sqrt{x}$ suffices for all explicitly stated elementary error bounds.

Theorem 35.1 (A gamma-independent reduced dyadic family)**.** *For odd $j \ge 3$, define*

$$P_j = 2^{j+1} + 4jD(2^{j-2}) - (j-2)D(2^j), \quad r_j = P_j/2^{j+2}, \quad \mathcal{J} = \{j : \lfloor\sqrt{2^j}\rfloor \text{ is odd}\}.$$

The set $\mathcal{J}$ is infinite. For $j \in \mathcal{J}$, r_j is reduced, its denominator is exactly $q_j = 2^{j+2}$, these fractions are distinct, and

$$|r_j - \gamma| \le (6j-4)2^{-j/2}, \qquad P_j - q_j\gamma = 4j\Delta(2^{j-2}) - (j-2)\Delta(2^j). \tag{35.1}$$

At most one fraction equals γ. Consecutive selected indices satisfy $j' \le 2j+3$, and $q_{j'} \le 2q_j^2$.

Proof. For $g(n) = \frac{1}{2}(D(n)/n - \log n + 1)$, $g(n) = \gamma + \Delta(n)/(2n)$. The combination $((k+h)g(b^k) - kg(b^{k+h}))/h$ cancels the logarithm and preserves the coefficient of γ. Taking $b=2$, $k=j-2$, $h=2$ proves the formula and error estimate. Since $d(n)$ is odd exactly at squares, $D(n) \equiv \lfloor\sqrt{n}\rfloor \pmod 2$, so P_j is odd on $\mathcal{J}$. Writing $j = 2h+1$, selection means that the hth binary digit of $\sqrt{2}$ is 1; there are infinitely many such digits. If the next 1 occurs at $H > h$ and $A = \lfloor 2^h\sqrt{2}\rfloor$, then

$$(3\cdot 2^{2h})^{-1} < \sqrt{2} - A/2^h < 2^{1-H}.$$

The lower bound follows by rationalizing and using the positive integer $2\cdot 2^{2h} - A^2$. Hence $H \le 2h+2$. Distinct reduced denominators imply distinct fractions and therefore eventual nonvanishing of their errors. □

The error bound for the integer form in (35.1) grows like $j2^{j/2}$. Even an individual bound $\Delta(x) = O(x^\theta)$ yields only $O(j2^{\theta j})$ for that integer form. This is a limitation of the available upper bound, not a lower bound on actual cancellation. Ridout's theorem [29] gives a precise sufficient condition: if, for some fixed $\varepsilon > 0$, infinitely many selected j satisfy

$$0 < |4j\Delta(2^{j-2}) - (j-2)\Delta(2^j)| < q_j^{-\varepsilon}, \tag{35.2}$$

then γ is transcendental. Rational γ is excluded by a fixed integer-lattice lower bound; algebraic irrational γ is excluded by Ridout after division by q_j. Condition (35.2) remains unproved and is not asserted equivalent to transcendence.

35.1 An interior value and its divided logarithmic jet

Put $A_h = P_{2h+1}$, $C_h = A_{h+1} - 4A_h$, and

$$F(z) = \sum_{h\ge1} A_h z^h, \qquad G(z) = \sum_{h\ge1} C_h z^h.$$

The bound $A_h = 8\gamma 4^h + O(h2^h)$ and a telescoping sum imply

$$G \text{ is analytic for } |z| < \tfrac{1}{2}, \qquad \gamma = 1 + G(1/4)/32. \tag{35.3}$$

A finite partial sum is exactly the old dyadic approximation, so the identity does not accelerate it.

Proposition 35.2 (Functional transcendence and a stronger countermodel)**.** *The series F, G are transcendental over $\mathbb{Q}(z)$ and have no finite linear 2^m-Mahler relation for any $m \geq 1$. Nevertheless, the initial value $A_1 = 32$, coefficient parities, divisibility $C_h \in (2h+1)\mathbb{Z}$, and the displayed value identity do not determine the arithmetic of its value: they admit models with any prescribed real value α in place of γ.*

Proof. Let $b_h = \lfloor 2^h\sqrt{2} \rfloor \bmod 2$. Then $A_h \equiv b_h$ and $C_h \equiv b_{h+1} \pmod 2$. Adamczewski–Bugeaud [30] implies that these binary digits are not 2-automatic, while Christol's theorem [31] makes the corresponding reduced series transcendental over $\mathbb{F}_2(z)$. A primitive integer polynomial relation would reduce to a nonzero relation modulo 2. A primitive 2^m-Mahler relation reduces, by Frobenius, to $\sum_i \overline{Q}_i(z)\overline{F}(z)^{2^{mi}} = 0$, also a nonzero polynomial relation. The same proof applies to G.

For the countermodel start with $\widetilde{A}_1 = 32$ and let $e_h = \widetilde{A}_h - 8\alpha 4^h$. Choose $v_h \in b_{h+1} + 2\mathbb{Z}$ nearest to $-4e_h/(2h+1)$, and put $\widetilde{A}_{h+1} = 4\widetilde{A}_h + (2h+1)v_h$. Then $|e_h| \leq 2h-1$ for $h \geq 2$, and $|v_h| \leq 4$ for $h \geq 2$. The same telescoping argument gives $\alpha = 1 + \widetilde{G}(1/4)/32$. The countermodel series has polynomially bounded integer coefficients and the same nonautomatic parity. Its radius is 1, it is not rational, and Pólya–Carlson gives the unit circle as natural boundary [32]. In particular, even this additional analytic property does not preclude an algebraic prescribed value. □

The exact division $C_h = (2h+1)V_h$ has further content. Set

$$B_h = D(2^{2h-1}), \qquad V_h = 8B_{h+1} - B_{h+2} - 16B_h, \qquad V(z) = \sum_{h\geq 1} V_h z^h.$$

Then

$$V(1/4) = 8 - 16\log 2, \qquad V'(1/4) = 64\gamma + 32\log 2 - 80. \tag{35.4}$$

To prove this, for $J(z) = \sum B_h z^h$ use

$$J(z) = \frac{4\log 2\, z}{(1-4z)^2} + \frac{4(\gamma - (1+\log 2)/2)z}{1-4z} + E(z), \qquad V(z) = -\frac{(1-4z)^2}{z^2}J(z) + \frac{3}{z} - 4,$$

where E is analytic on $|z| < 1/2$. Its contribution to both the value and first derivative vanishes. For general $b \geq 2$, the analogous values are

$$\begin{aligned} X_b &= D(b^3) - b^2 D(b) - 2b^3 \log b, \\ Y_b &= b^5(2\gamma - 1 + \log b) - b^4 D(b), \qquad \overline{\mathbb{Q}}(X_b, Y_b) = \overline{\mathbb{Q}}(\gamma, \log b). \end{aligned}$$

Every X_b is transcendental by Hermite–Lindemann; at most one Y_b is algebraic because $Y_b/b^5 - Y_c/c^5$ is a rational translate of $\log(b/c)$. Neither statement excludes algebraicity of γ.

Even the fixed value in (35.4) admits an arbitrary derivative parameter. For $h \geq 3$ round

$$4^h\big(h\log 2 + \alpha - (1+\log 2)/2\big)$$

down to the nearest integer of the parity of B_h, retaining $B_1 = 3$, $B_2 = 20$. The error lies in $(-2, 0]$, so the resulting $\widetilde{V}_h$ are bounded, with $-16 < \widetilde{V}_h < 34$ for $h \geq 3$. The preceding double-zero calculation preserves $V(1/4)$ but replaces γ by α in $V'(1/4)$. This countermodel preserves the finite-difference structure and all indicated parities, not the actual identity $B_h = D(2^{2h-1})$ or the full multiplicativity of d.

35.2 Local Euler factors and Riesz smoothing

For a prime p, interpreting $d(x) = 0$ at nonintegral x, one has

$$d(n) - 2d(n/p) + d(n/p^2) = d(n)\mathbf{1}_{p\nmid n}.$$

The local identity $(1-X)^2(1-X)^{-2} = 1$ proves it. Accordingly $O_k = D(2^k) - 2D(2^{k-1}) + D(2^{k-2})$ is the odd-part divisor sum, and

$$R_j = \frac{2^{j-1} + 4(j+2)O_{j-2} - jO_j}{2^j} \longrightarrow \gamma.$$

For odd $j = 2h + 1 \geq 5$, the numerator is odd whenever $b_h \neq b_{h-1}$. This gives infinitely many distinct reduced denominators 2^j, with

$$|R_j - \gamma| \leq 2(2 + \sqrt{2})^2(3j + 4)2^{-j/2}.$$

Indeed the parity of O_j counts odd squares, and the main-term expansion proves convergence and the bound. Removing finitely many Euler factors changes constants, not the exponent guaranteed by individual error bounds. For a fixed filter $\sum_i c_i D(\lambda_i x)$ the main term is exactly

$$x\{a \log x + (2\gamma - 1)a + b\}, \qquad a = \sum_i c_i\lambda_i, \quad b = \sum_i c_i\lambda_i \log \lambda_i.$$

Thus cancelling $x \log x$ by a fixed filter also cancels its γ coefficient. A shift polynomial at geometric nodes b_0^h cancels both main terms exactly when it has a double root at b_0.

For comparison, let

$$M(x) = \sum_{n \leq x} d(n)(x - n), \qquad A(x) = M(x)/x^2 + 3/4 - 1/(4x) - \tfrac{1}{2}\log x.$$

The integrated Voronoi estimate gives $A(x) - \gamma = O(x^{-5/4})$ [33]. For odd $j \geq 5$ and $x = 2^j$,

$$s_j = \frac{jA(x/4) - (j-2)A(x)}{2} = \frac{16jM(x/4) - (j-2)M(x) + 3x^2/2 - (3j+2)x/4}{2x^2}.$$

If $\lfloor \sqrt{2^j} \rfloor \bmod 4 \in \{1, 2\}$, the numerator is odd. Counting square contributions to $M(x)$ modulo 2 proves this; the condition occurs infinitely often because it detects changes in consecutive binary digits of $\sqrt{2}$. Hence there is an infinite reduced family with $q_j = 2^{2j+1}$ and

$$|s_j - \gamma| = O((\log q_j)q_j^{-5/8}).$$

The stronger real-variable smoothing has squared the denominator. Even the classical unsmoothed bound $\Delta(x) = O(x^{1/3} \log x)$ gives exponent $2/3$ in the original family, whereas the Ridout sufficient threshold is $1 + \varepsilon$. These are guaranteed upper-bound exponents, not proved optimal errors.

35.3 The first singular local layer of the moment denominator

For the previously defined moment approximation $\ell_N = \boldsymbol{c}_N^T K_N^{-1} \boldsymbol{c}_N$, retain its local Schur representation at $N = p + r$, $p > 2r + 3$:

$$\ell_{p+r} = \beta + z^T T^{-1} z, \quad T \in \mathrm{Sym}_r(\mathbb{Z}_p), \quad z \in \mathbb{Z}_p^r, \quad p\beta \equiv 4 \pmod{p}.$$

Here $p\beta$ is integral; the displayed representation is the input from the earlier moment calculation, not an assertion for arbitrary matrices. If Z lifts a basis of $\ker(T \bmod p)$ and

$$\kappa = p^{-1} Z^T T Z \bmod p, \qquad h = Z^T z \bmod p,$$

then, provided κ is nonsingular,

$$p\ell_{p+r} \equiv 4 + h^T \kappa^{-1} h \pmod{p}. \tag{35.5}$$

To prove this, choose a complementary basis and write $T = \left(\begin{smallmatrix} A & pC \\ pC^T & pD \end{smallmatrix}\right)$, where A is invertible and $D \bmod p = \kappa$. Its Schur complement is $p(D - pC^T A^{-1} C)$, so the only negative valuation layer is p^{-1} and has the residue in (35.5). The formula is invariant under changing lifts and bases. A nonzero residue means valuation 1 in the reduced denominator; a zero residue means valuation 0. In dimension one, $(-\kappa/p) = -1$ excludes cancellation. The verified example $(N, p, \kappa, h) = (51, 43, 19, 28)$ has residue 0, unlike $(15, 13, 10, 11)$, $(49, 47, 3, 39)$ and $(24, 19, 6, 15)$, whose residues are $7, 41, 13$. If κ is itself singular, a deeper layer is required.

36 Exact filter optimization and continuous divisor correlations

36.1 What individual error intervals can certify

Proposition 36.1 (Two-support minimax principle)**.** *For $N < K$ and positive ϵ_k,*

$$\min_{\sum c_k = 1,\ \sum k c_k = 0} \sum_{k=N}^{K} \epsilon_k |c_k| = \min_{N \le i < j \le K} \frac{j\epsilon_i + i\epsilon_j}{j - i}.$$

A minimizer has $c_i = j/(j-i)$, $c_j = -i/(j-i)$ and all other coefficients zero. If $\epsilon_k = C2^{-ak}$, $a > 0$, the minimum is $\Theta_a(CK2^{-aK})$, uniformly in the left endpoint $N < K$.

Proof. Write $c_k = x_k - y_k$, $x_k, y_k \ge 0$. An optimal basic feasible solution of this positive-cost linear program with two equations has at most two positive variables. One variable cannot meet both equations; two distinct indices with opposite signs give the stated solution. For the asymptotic bound the pair $K-1, K$ is sufficient above. Below, put $s = j - i$; when $j \ge K/2$, the first cost term is at least $(CK/2)2^{-aK}2^{as}/s$, bounded below by a positive constant times $CK2^{-aK}$. When $j < K/2$, the exponential bound is stronger. □

There is an exact optimum for arbitrary variable length in the elementary divisor bound. For $4 \le L \le k$, impose $\sum_{\ell=0}^{L} w_\ell = 1$, $\sum \ell w_\ell = k$. Then

$$\min \sum_{\ell=0}^{L} |w_\ell| 2^{\ell/2} = 5k/4 - 1, \qquad w_0 = 1 - k/4, \quad w_4 = k/4. \tag{36.1}$$

The dual certificate is $|5\ell/4 - 1| \le 2^{\ell/2}$, with equality only at $\ell = 0, 4$. Its signs also prove uniqueness. This yields

$$R_{k,4} = \frac{2^{k+2} + (4-k)D(2^k) + 16kD(2^{k-4})}{2^{k+3}}, \quad |R_{k,4} - \gamma| \le (5k-4)2^{-k/2}.$$

The denominator is exactly 2^{k+3} on the selected odd indices of Theorem 35.1. Optimizing the error is different from optimizing the integerized error: among two-point formulas with displayed denominator $2m2^k$, that latter elementary budget is minimized at $m = 1$, not $m = 4$. Actual primitive-height optimization over all filters is not solved here.

Allowing continuous nodes Xe^{-s_i} with $0 \le s_i \le \log X$ leads to the analogous exact relaxation. If $u = W(e^{-1}) > 0$ and $\log X \ge (1+u)/\beta$, then

$$\min_{\sum w_i = 1,\ \sum w_i s_i = \log X} \sum_i |w_i| e^{\beta s_i} = \frac{\beta}{u} \log X - 1.$$

The line $(\beta/u)s - 1$ is bounded in absolute value by $e^{\beta s}$ and tangent at $s_* = (1+u)/\beta$; equality uses $s = 0, s_*$. The optimal continuous nodes need not be rational. These optimizations concern independent error intervals, leaving actual correlations untouched.

36.2 Covariance, uniform growing filters, and exact prediction

Let

$$X_r(x) = 2^{-r/2}\Delta(4^r x), \quad t = 2^{-3/2}, \quad c = \frac{2(1-t)}{1+t}, \quad \rho_h = t^h(1+ch), \quad S = \sum_{n\geq 1} d(n)^2 n^{-3/2}.$$

For fixed r, s, the continuous covariance is

$$\lim_{T\to\infty} T^{-3/2}\int_1^T X_r(x)X_s(x)\,dx = \frac{S}{6\pi^2}\rho_{|r-s|}. \tag{36.2}$$

Its normalized spectral density is

$$f(\theta) = \frac{(1-t)^3}{1+t}\frac{1+4t+t^2+2t\cos\theta}{(1-2t\cos\theta+t^2)^2}, \quad f_{\min} = \left(\frac{1-t}{1+t}\right)^3 > 0. \tag{36.3}$$

These identities use the actual divisor error, not an assumed stochastic model.

Theorem 36.2 (Uniform continuous mean for growing dyadic filters)**.** *Fix $0 < \eta < 1/2$. Uniformly for all distinct nonnegative integers $r_1, \dots, r_m$, all real vectors w, and $m \leq T^{1/2-\eta}$,*

$$T^{-3/2}\int_T^{2T}\left(\sum_{i=1}^m w_i X_{r_i}(x)\right)^2 dx = C_*\sum_{i,j} w_i w_j \rho_{|r_i-r_j|} + O_\eta(T^{-\eta/8})\|w\|_2^2, \quad C_* = (2^{3/2}-1)S/(6\pi^2).$$

There is no upper bound on the r_i. They and w may depend on T but are held fixed during integration in x.

Proof. Use the truncated Voronoi formula [34]

$$V_M(u) = \frac{u^{1/4}}{\pi\sqrt{2}}\sum_{n\leq M} d(n)n^{-3/4}\cos(4\pi\sqrt{nu} - \pi/4).$$

Its standard truncation, followed by integration of distinct frequencies, gives, for $2 \leq M \leq Y$,

$$Y^{-3/2}\int_Y^{2Y} |\Delta(u) - V_M(u)|^2 du \ll_\varepsilon M^{-1/2+2\varepsilon} + Y^{-1/2+2\varepsilon}\log(2Y).$$

For completeness, first truncate at $\lfloor Y\rfloor$; the pointwise remainder contributes $O_\varepsilon(Y^{1+2\varepsilon})$ to the square integral. The diagonal tail is $O(Y^{3/2}\sum_{n>M} d(n)^2 n^{-3/2})$. The off-diagonal integral is $O(Y/|\sqrt{a}-\sqrt{b}|)$; splitting $a < b$ at $a = b/2$ bounds its sum by $O_\varepsilon(Y^{1+2\varepsilon}\log(2Y))$. Rescaling $u = 4^r x$ preserves this normalized estimate uniformly for $r \geq 0$.

Use a Hilbert space with equal frequencies $2^r\sqrt{n}$ identified. The vector v_r of coefficients $d(n)n^{-3/4}$ has inner products

$$\langle v_r, v_s\rangle = 4^{-3h/4}\sum_{n\geq 1} d(n)d(4^h n)n^{-3/2} = S\rho_h, \qquad h = |r-s|.$$

The last equality follows by writing $n = 2^\nu m$, m odd, and summing $(\nu+1)(\nu+2h+1)t^\nu$. Summing the Poisson kernel and its derivative gives (36.3). Parseval bounds the limiting quadratic form above and below by fixed positive multiples of $\|w\|_2^2$.

For $a, b \leq M$, unequal frequencies satisfy $|2^r\sqrt{a} - 2^s\sqrt{b}| \geq (3\sqrt{M})^{-1}$: after assuming $r \leq s$, either $4^{s-r}b \leq 4M$ and rationalization applies, or the gap exceeds $\sqrt{M}$. Group equal frequencies before squaring. If the distinct frequencies are ordered, their successive gaps give row sums

bounded by $O(\sqrt{M}\log(2mM))$ for the reciprocal-gap matrix, using $2|A_iA_j| \le A_i^2+A_j^2$. Therefore the nonconstant terms contribute

$$O(T^{-1/2}\sqrt{M}\log(2mM))\left\|\sum_i w_i v_{r_i,M}\right\|^2.$$

The Hilbert-space tail has norm $O_\varepsilon(\sqrt{m}M^{-1/4+\varepsilon})\|w\|_2$. The replacement of the actual functions by their truncated sums has normalized norm at most

$$O_\varepsilon\left(\sqrt{m}M^{-1/4+\varepsilon}+\sqrt{m}T^{-1/4+\varepsilon}\sqrt{\log(2T)}\right)\|w\|_2.$$

Take $M=\lfloor T^{1-\eta}\rfloor$, $\varepsilon=\eta/16$. All these perturbations are $O_\eta(T^{-\eta/8})\|w\|_2$, and their quadratic effects have the stated bound. The constant term of the cosine-square integral is exactly $(2^{3/2}-1)/(6\pi^2)$ times the Hilbert-space squared norm. □

The exact prediction problem also has a closed solution. Set

$$A=1+4t+t^2,\quad \kappa=\frac{A+\sqrt{A^2-4t^2}}{2},\quad s=t/\kappa,\quad \sigma^2=\frac{(1-t)^3}{1+t}\kappa.$$

Then $f(\theta)=\sigma^2|(1+se^{i\theta})/(1-te^{i\theta})^2|^2$. For every polynomial W with $W(0)=1$, Parseval gives

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} f(\theta)|W(e^{i\theta})|^2d\theta \ge \sigma^2.$$

The infimum is attained among infinite absolutely summable filters by $W_\infty(z)=(1-tz)^2/(1+sz)$; its degree-$H$ Taylor truncations have excess $O(s^{2H})$. Here $\sigma^2\simeq 0.4967594891$. Under the different constraint $\sum w_i=1$, the exact finite optimum is $(\mathbf{1}^TR_m^{-1}\mathbf{1})^{-1}=\Theta(1/m)$, where $R_m=(\rho_{|i-j|})$. None of these continuous mean statements supplies a lower bound at any selected dyadic point.

36.3 Exact resonance removal and its normalization

Let $h_\ell=\frac{1}{2}(D(4^\ell)/4^\ell+1)$ and $P(Z)=\sum_j c_jZ^j\in\mathbb{Z}[Z]$, $q=P(1)\neq 0$. The combination $q^{-1}\sum_j c_jh_{k+j}$ cancels its logarithm precisely when $P'(1)=-kP(1)$. In a common finite Voronoi truncation, its square-index resonance vanishes precisely when $P(2^{-3/2})=0$, or $8Z^2-1\mid P$.

Proposition 36.3. *These two conditions force $49\mid q$. The least positive value 49 is attained by*

$$P_k(Z)=(8Z^2-1)\{7-(7k+16)(Z-1)\}.$$

More generally, for primitive $F\in\mathbb{Z}[Z]$, $A=F(1)\neq 0$, $B=F'(1)$, the least positive $|P(1)|$ under $F\mid P$ and $P'(1)=-kP(1)$ is $A^2/\gcd(|A|,|B|)$.

Proof. Write $P=FQ$ by Gauss's lemma. Then $AQ'(1)=-(kA+B)Q(1)$, so $A/\gcd(A,B)$ divides $Q(1)$. Conversely choose Q linear with that constant value at 1 and the required integer derivative. For $F=8Z^2-1$ use $A=7$, $B=16$. □

The concrete approximation has elementary error at most $(126k+372)2^{-k}/49$. The nonsquare frequencies remain. The number 49 is a filter normalization, not necessarily a surviving factor of the reduced rational denominator.

37 Reciprocity, special-function conversion, and arithmetic transfer tests

37.1 A positive exponentially convergent reciprocity formula

Define $S(y) = \sum_{n\geq 1} d(n)K_0(2\pi ny)$. Tonelli's theorem applied to the positive Bessel integral gives $S(y) = \int_0^\infty L(e^{-2\pi y\cosh u})\,du$. The classical Ramanujan–Koshliakov relation [35] is

$$\sqrt{y}\{\gamma - \log(4\pi) + \log y + 4S(y)\} = \frac{\gamma - \log(4\pi) - \log y + 4S(1/y)}{\sqrt{y}}.$$

One derivation writes $4S(y) = (2\pi i)^{-1}\int_{(\sigma)} \Omega(s)^2 y^{-s}\,ds$, where $\Omega(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$, and uses $\Omega(s) = \Omega(1-s)$. The residues at $1, 0$ are, respectively, $(c - \log y)/y$ and $-c - \log y$, $c = \gamma - \log(4\pi)$. Differentiation at the self-dual point gives

$$\gamma = \log(4\pi) - 2 + \sum_{n\geq 1} d(n)\{16\pi n K_1(2\pi n) - 4K_0(2\pi n)\}. \tag{37.1}$$

Every summand is positive. In fact, with $x = 2\pi n$,

$$8xK_1(x) - 4K_0(x) = 4\int_0^\infty (2x\cosh u - 1)e^{-x\cosh u}\,du > 0.$$

From $K_1(x) \leq e^{-x}(\sqrt{\pi/(2x)} + x^{-1})$ and $d(n) \leq 2\sqrt{n}$, the tail after N is at most

$$16(\pi+1)\rho^{N+1}\left(\frac{N+1}{1-\rho} + \frac{\rho}{(1-\rho)^2}\right), \qquad \rho = e^{-2\pi}. \tag{37.2}$$

Thus this route proves one-sided nonvanishing and exponential analytic convergence. Its partial sums contain Bessel values and $\log(4\pi)$; there is no associated fixed integer lattice. The identity

$$K_0(z) = \sum_{k\geq 1} \frac{H_k(z^2/4)^k}{(k!)^2} - \{\gamma + \log(z/2)\}I_0(z)$$

reintroduces γ and does not split (37.1) into two convergent sums, since the separate I_0 contribution grows. The dual arguments $2\pi y$ and $2\pi/y$ cannot both be algebraic because their product is $4\pi^2$.

For the parameter family $F_a(y) = 4\sum_{n\geq 1} n^{-a}\sigma_{2a}(n)K_a(2\pi ny)$, the Ramanujan–Guinand relation reads

$$F_a(y) - y^{-1}F_a(1/y) = \Omega(1+2a)(y^{-1-a} - y^a) + \Omega(2a)(y^{-1+a} - y^{-a}),$$

interpreted by removable limits at $a = 0, \pm 1/2$. At $a = 1/2$ the expansions $\Omega(1+t) = t^{-1} + c/2 + O(t)$ and $\Omega(t) = -t^{-1} + c/2 + O(t)$ cancel the simple-pole coefficient carrying γ. Since $K_{1/2}(x) = \sqrt{\pi/(2x)}e^{-x}$, the resulting identity is the Dedekind eta transformation $\log\eta(i/y) = \frac{1}{2}\log y + \log\eta(iy)$, not a rational formula for γ. Recovering the order derivative gives instead

$$\partial_\nu K_\nu(x)|_{\nu=1/2} = \sqrt{\frac{\pi}{2x}}e^x E_1(2x) \tag{37.3}$$

[36]. The differentiated full reciprocity relation is preserved in the companion source note; it contains additionally $\zeta'(2)/\zeta(2)$ and an E_1 series. Its termwise E_1 coefficient cannot be removed by a fixed finite same-scale half-integer order combination. This last assertion concerns the displayed functions, not linear independence of their specialized values.

37.2 Fixed rational maps amplify existing denominator costs

Proposition 37.1 (Fixed-map height and error)**.** *Let $R = A/B \in \mathbb{Q}(X)$ be nonconstant with coprime integer polynomials, $D = \max(\deg A, \deg B)$. Suppose $B(\alpha) \neq 0$ and R has a zero of order $m \geq 1$ at $\alpha \in \mathbb{R}$. If $x_n = a_n/q_n \to \alpha$ are reduced, $q_n \to \infty$, $x_n \neq \alpha$, then*

$$\operatorname{den}(R(x_n)) \asymp q_n^D, \qquad |\operatorname{num}(R(x_n))| \asymp q_n^D |x_n - \alpha|^m.$$

If α is algebraic of degree d, then $D \geq dm$. In particular $q_n|x_n - \alpha| \to \infty$ implies divergence of the reduced numerators after every such fixed transformation.

Proof. Homogenize A, B to degree D. Their common divisor at a primitive pair (a, q) is uniformly bounded: Bezout identities in the original coordinate bound it by Cq^L, and identities for the reciprocal polynomials bound it by $C'a^{L'}$. Coprimality of a, q gives a fixed bound $|CC'|$. Moreover $B_h(a_n, q_n) = q_n^D B(x_n) \sim q_n^D B(\alpha)$. Taylor expansion gives $|R(x_n)| \asymp |x_n - \alpha|^m$. If α is algebraic, its minimal polynomial divides A with multiplicity at least m. Finally $q_n^D|x_n - \alpha|^m = q_n^{D-m}(q_n|x_n - \alpha|)^m$. □

For a polynomial P with leading coefficient A_d, every denominator prime $p \nmid A_d$ survives with valuation multiplied by d; uniformly, $q^d/|A_d|^d \leq \operatorname{den}(P(a/q)) \leq q^d$ when the value is nonzero. A hypothetical minimal polynomial has a simple root at γ, so $P(x_n)$ gains only one power of the approximation error, not d. Algebraic norms have the same local denominator amplification. These statements apply after an endpoint is formed, not to increasing-degree maps or simultaneous coefficient design before integration.

Positivity also cannot be transported through arbitrary conjugations. For $\alpha = \sqrt{2}$, $\int_0^1 t^n(1 + \alpha t)dt = (n+1)^{-1} + \alpha(n+2)^{-1}$ is a positive Hausdorff sequence, whereas its conjugated second moment is $1/3 - \sqrt{2}/4 < 0$. An argument transporting the real positive model to all algebraic conjugates therefore needs an additional arithmetic theorem.

37.3 Prime defects of the polyexponential trace tower

Let

$$F_s(z) = \sum_{n \geq 1} \frac{(-1)^{n+1} z^n}{n^s n!}, \qquad P_{s,n}(X) = -n[z^n] \log(1 - XF_s(z)), \quad s \geq 1.$$

For every prime p,

$$p^s P_{s,p}(X) \in \mathbb{Z}_{(p)}[X], \qquad p^s P_{s,p}(X) \equiv -X \pmod{p}. \tag{37.4}$$

Indeed expand $p \sum_{j=1}^{p} X^j [z^p] F_s^j / j$. Only $j = 1$ survives multiplication by p^s and reduction; its residue is $-X$ by Wilson's theorem. Thus at every nonzero algebraic level c, the trace $P_{s,p}(c^{-1})$ has valuation $-s$ at every sufficiently large good prime (normalizing $v_{\mathfrak{p}}(p) = 1$). Fixed multipliers and rescalings cannot make the ordinary trace sequence globally bounded.

More strongly, let a globally bounded algebraic-coefficient series have first nonconstant term $R(w) = a_0 + a_r w^r + \cdots$, $a_r \neq 0$. If $R(F_s(z)) = \sum b_n z^n$, then at all sufficiently large good primes

$$p^{s+1} b_{p+r-1} \equiv -r a_r \pmod{\mathfrak{p}}, \qquad v_{\mathfrak{p}}(b_{p+r-1}) = -(s+1). \tag{37.5}$$

The only composition involving an index at least p is one of the r permutations of $(p, 1, \ldots, 1)$ in F_s^r; all larger outer powers use indices below p. This proves (37.5). Removing the first $R \geq 1$ outer logarithmic terms merely moves the defect: for $U_{s,n}^{[R]} = n \sum_{j=R+1}^{n} X^j [z^n] F_s^j / j$ and $p > R + 1$,

$$p^{s+1} U_{s,p+R}^{[R]}(X) \equiv -RX^{R+1} \pmod{p}.$$

These are obstructions to global boundedness and ordinary integral realizations, not impossibility theorems for capacity criteria that allow explicit denominator costs.

Order of operations matters for Borel conversion. Multiplication of the coefficients of F_s by $n!$ gives $-\operatorname{Li}_s(-z)$, a G-function. But the ordinary trace generating function at $s = 1$ is

$$H_X(z) = \sum_{n\geq 1} P_{1,n}(X) z^n = \frac{X z \operatorname{Ein}'(z)}{1 - X \operatorname{Ein}(z)}, \qquad X \neq 0,$$

and is not D-finite. The entire order-one function $\operatorname{Ein} -X^{-1}$ has infinitely many zeros: otherwise Hadamard factorization would make it $e^{az+b}P(z)$, whose derivative has finitely many zeros, contrary to $\operatorname{Ein}'(z) = (1 - e^{-z})/z$. Its logarithmic derivative has a pole at every such zero. A D-finite function has only finitely many finite singularities. Multiplying or dividing a coefficient sequence by $n!$ preserves P-recursiveness in both directions [37]; therefore neither factorial transform of this full trace series is D-finite.

37.4 Finite algebraic specializations and filtered local data

Proposition 37.2 (No exceptional Euler shift at a finite algebraic point)**.** *For every $c \in \mathbb{C}$, $F_c = \operatorname{Ein} -c$ has minimal inhomogeneous differential order 2, with $zF_c'' + (z+1)F_c' = 1$, and minimal homogeneous order 3. For every algebraic c and nonzero algebraic α, the numbers $e^{-\alpha}, F_c(\alpha)$ are algebraically independent over $\overline{\mathbb{Q}}$. Thus the algebraic-value set of F_c among algebraic arguments is exactly $\{0\}$; it has an algebraic zero only if $c = 0$, at zero.*

Proof. The derivative proves the equation. Functional independence of e^{-z} and $\operatorname{Ein}(z)$ follows along the positive axis: in a polynomial relation $\sum_j e^{-jz} P_j(z, \operatorname{Ein}(z)) = 0$, select the smallest nonzero j. Since $\operatorname{Ein}(x) = \log x + \gamma + O(e^{-x}/x)$, that term has a nonzero leading power-logarithmic asymptotic, while every subsequent exponential is smaller. Translation by c preserves independence. Hence $1, F_c, F_c'$ are linearly independent over $\mathbb{C}(z)$, giving the asserted minimal orders by the displayed equation and its derivative. For algebraic c, the vector $(1, e^{-z}, F_c)$ consists of E-functions satisfying a system with only finite pole 0. Siegel–Shidlovskii specialization at $\alpha \neq 0$ preserves transcendence degree two [10]. At zero $F_c(0) = -c$. □

This theorem applies to every algebraic shift, including a hypothetical algebraic γ; it does not distinguish the finite part at infinity. The exact Laplace formula

$$s \int_0^\infty e^{-sx} \operatorname{Ein}(bx)\, dx = \log(1 + b/s), \qquad b, s > 0,$$

shows why a simple regularization loses that coordinate: the contribution γ cancels the universal correction $s \int_0^\infty e^{-sx} \log x\, dx = -\log s - \gamma$.

A local filtration retains an extension parameter but does not force it nonzero. For

$$\phi e_0 = e_0, \quad \phi e_1 = c e_0 + p e_1, \qquad \operatorname{Fil}^1 D = \mathbb{Q}_p(e_1 + a e_0), \quad \operatorname{Fil}^0 D = D, \quad \operatorname{Fil}^2 D = 0,$$

every $(a, c) \in \mathbb{Q}_p^2$ is weakly admissible. The invariant $\kappa = a - c/(p-1)$ is zero exactly for the filtered split extension. Indeed the eigenlines are $\mathbb{Q}_p e_0$ and $\mathbb{Q}_p(e_1 + c e_0/(p-1))$, with Newton numbers $0, 1$ and Hodge numbers 0 and either 0 or 1; total numbers are both 1. Under $e_1' = e_1 + t e_0$, one has $c' = c + (1-p)t$, $a' = a - t$, leaving κ unchanged. Arbitrary local choices have not been identified with a canonical Euler comparison realization. The historical source's suggested rapid-decay realization is retained as a research proposal, not as an established relation-preserving map.

38 Residue-first positivity, convex tensor tests, and two-pole escape

Write $T(R) = \int_1^\infty \lfloor t \rfloor R(t)\,dt$ and

$$r_j(t) = (t+j)^{-2} - [t(t+1)]^{-1}, \qquad H_0 = 0.$$

For rational $A_0, \dots, A_M$ and $q = \sum_j A_j$, the exact log-free identity is

$$R_A = \sum_j A_j r_j = O(t^{-3}), \qquad T(R_A) = q\gamma - \sum_j A_j H_j. \tag{38.1}$$

To prove it, expand $\lfloor t \rfloor = \sum_{m\ge 1} \mathbf{1}_{t\ge m}$ and integrate first. The first L primitives sum to $\sum_j A_j(H_{L+j} - H_j) - q\log(L+1)$ and tend to the stated limit. Absolute convergence follows from $O(t^{-3})$.

For $q = 1$, put $z = 1/t$ and

$$K_A(z) = \frac{1}{1+z} - \sum_{j=1}^{M} \frac{A_j j(2+jz)}{(1+jz)^2}, \qquad D_M(z) = \prod_{j=1}^{M} (1+jz)^2.$$

Then $R_A = z^3 K_A$, and $N_A = D_M K_A$ is a polynomial of degree at most $2M-1$. Since $D_M > 0$ on $[0,1]$, positivity is exactly a polynomial interval question; nonnegative Bernstein coefficients are a sufficient, not necessary, certificate. For example

$$A = (17/48, 5/6, -3/16), \quad \sum_j A_j H_j = 53/96, \quad N_A(z) = \tfrac{1}{12}(1-z)^3 + \tfrac{3}{2}z^3 > 0,$$

proves $\gamma > 53/96$ with signed coefficients. The upper vector $(13/44, 10/11, -9/44)$ gives $\gamma < 53/88$. The error is exactly $\int_0^1 z\lfloor 1/z \rfloor N_A(z)/D_M(z)\,dz$, which admits rational upper bounds by interval subdivision.

The archived finite optimization proposed 16 bounds and certified 14; the degree-$M = 10$ upper and $M = 12$ lower proposals failed exact sign checks. For $M = 2, 4, 6, 8$, the certified lower endpoints had denominator bit lengths $7, 14, 53, 85$ and integer errors exceeding $2, 86, 10^{13}, 10^{22}$, respectively. These are feasibility certificates, not global optimum certificates. No asymptotic impossibility is inferred from them.

38.1 Why the positive-semidefinite coefficient cone is narrower

Suppose a symmetric matrix $X \succeq 0$ has $X\mathbf{1} = 0$, trace $Q > 0$, and all logarithms in its floor-integral transform cancel. Its remaining value is $Q\gamma - A$, $A = \sum_j H_j X_{jj}$. Then

$$\gamma - A/Q > 2^{-25}. \tag{38.2}$$

Indeed normalize $Q = 1$ and write $X_{jk} = \langle v_j, v_k \rangle$, $h(t) = \sum_j v_j/(t+j)$. The integral is $\int_1^\infty \lfloor t \rfloor \|h(t)\|^2 dt$. Since $A < \gamma < 3/5$, $\|v_0\|^2 > 2/5$. Cauchy–Schwarz and $\zeta(2) - 1 < 13/20$ imply

$$\|h(1)\| > \sqrt{2/5} - \sqrt{39/100} > 1/128, \qquad \|h'(t)\| < 2.$$

Thus $\|h(t)\| > 1/256$ on $[1, 1 + 1/512]$, giving (38.2). This excludes convergence in that PSD subclass; signed residue kernels are outside its hypotheses.

The natural Lagrange-quadrature subclass is classical. With $G_n = \int_0^1 \binom{x}{n}\,dx$ and $C_n = \int_0^1 x(x+1)\cdots(x+n-1)\,dx/n!$, $C_0 = 1$, it gives

$$\ell_M = \sum_{n=1}^{M} \frac{|G_n|}{n} \uparrow \gamma, \qquad u_M = 1 - \sum_{n=1}^{M} \frac{C_n}{n(n+1)} \downarrow \gamma, \qquad u_M - \ell_M = \frac{C_M}{M+1}.$$

These are Fontana–Mascheroni/Cauchy formulas [38]; their rediscovery is not a new irrationality result.

38.2 The complete acceptance set of every fixed residue basis

Let $h = r_0 - r_1 > 0$, $\nu(dt) = \lfloor t \rfloor h(t)dt$ and $v_j = r_j/h$. Then ν is a full-support probability measure on the compactification $T = [1, \infty]$, and

$$v_0 - v_1 = 1, \quad v_0(t) = \frac{t+1}{2t+1}, \quad v_j(\infty) = \tfrac{1}{2} - j, \quad \int v_j \, d\nu = \gamma - H_j.$$

For finite $M \geq 1$, define

$$C_M = \operatorname{conv}\{(v_0(t), \ldots, v_M(t)) : t \in T\}, \quad I_M = \{\alpha : (\alpha - H_j)_{j=0}^{M} \in C_M\}.$$

Theorem 38.1 (All tensor degrees inherit the linear interval)**.** *Each I_M is a compact nondegenerate interval contained in $[1/2, 2/3]$, with γ in its interior. Every nonnegative rational-coefficient residue tensor*

$$K_C(\boldsymbol{t}) = \sum_{j_1, \ldots, j_d} C_{j_1 \cdots j_d} \prod_{i=1}^{d} r_{j_i}(t_i)$$

has polynomial integral $P_C(\gamma)$, where $P_C(X) = \sum C_{j_1 \cdots j_d} \prod_i (X - H_{j_i})$. It satisfies $P_C(\alpha) \geq 0$ for every $\alpha \in I_M$, and strictly > 0 on the interior if the kernel is nonzero. Conversely every $\alpha \notin I_M$ fails a rational-coefficient positive linear test. Thus the acceptance sets for all tensor degrees and for all linear tests are exactly the same set I_M.

Proof. The rational functions $r_0, \ldots, r_M$ are linearly independent: their unique double poles first force each coefficient for $j \geq 2$ to vanish, and then the double poles at $0, -1$ force the remaining two. Consequently the affine span of C_M is exactly $x_0 - x_1 = 1$. The full-support barycenter $y(\gamma) = \int v \, d\nu$ lies in its relative interior. Otherwise a proper supporting affine functional would be nonnegative on the curve and have integral zero, forcing it to vanish on the entire affine span. Intersecting with the line $y(\alpha)$ proves nondegeneracy and interiority of I_M.

Every interior point has a full-support representing measure: mix ν with a measure representing a farther point on the same interval. For an arbitrary point of I_M a representing probability measure suffices. Integrating the normalized tensor $K_C / \prod_i h(t_i)$ against its product measure gives $P_C(\alpha)$; full support makes a nonzero nonnegative continuous kernel strictly positive in integral.

For a point outside C_M, strict convex separation gives an affine functional nonnegative on C_M and negative at the point. Use $v_0 - v_1 = 1$ to absorb its constant. Adding a sufficiently small positive multiple of that constant function makes positivity uniform, while negativity at the point persists. Rational approximation of the finite coefficient vector then produces the required rational linear test. □

For $M = 1$, the theorem can be sharpened to an exact coefficient criterion: a polynomial P of degree at most d has a nonnegative tensor representation precisely when its degree-d Bernstein coefficients on $[1/2, 2/3]$ are nonnegative. Symmetrization makes the multiaffine lift unique, and its vertex values are those coefficients. In particular the lift of $(X - a)^2$, $a \in (1/2, 2/3)$, is $(x_1 - a)(x_2 - a)$, which changes sign despite its nonnegative diagonal.

For $M = 2$ one obtains explicitly

$$I_2 = [\beta, 53/88], \quad 8\beta^3 - 27\beta^2 + 18\beta - 3 = 0, \quad \beta \in (1/2, 2/3).$$

To compute it, use $x = v_0$ and $v_2 = x(x-1)(5x-1)/(3x-1)^2$, a strictly concave function on this interval. Intersect its graph and endpoint chord with $y = x - 3/2$. The intervals are nested and the classical quadrature bounds show $\bigcap_M I_M = \{\gamma\}$, an analytic identification with no arithmetic classification. If $I_M = [\ell, u]$ and $\epsilon_M = \min(\gamma - \ell, u - \gamma) > 0$, any positive nonzero integer linear form represented in this fixed basis obeys

$$q\gamma - p \geq \epsilon_M |q| \quad (q \neq 0), \qquad -p \geq 1 \quad (q = 0).$$

There is no claim of a uniform positive ϵ_M as $M \to \infty$.

38.3 Two fixed poles with increasing order

Setting tensor variables equal changes the integral. Direct partial fractions give

$$T(r_0^2) = 4\gamma - 1 - \zeta(2) + \tfrac{1}{3}\zeta(3), \quad T(r_0 r_1) = 6\gamma - \tfrac{7}{2},$$

$$T(r_1^2) = 4\gamma - \tfrac{13}{3} + \zeta(2) + \tfrac{1}{3}\zeta(3).$$

Cancelling the displayed $\zeta(2), \zeta(3)$ coefficients in their span leaves only the cross product. This is formal coefficient cancellation, not an assumption about independence of the actual constants.

For $k \geq 2$, put $d_k(t) = t^{-k} - (t+1)^{-k}$. Telescoping primitives give $T(d_k) = 1/(k-1)$. Consequently

$$R_{q,B} = qr_0 - \sum_{k=2}^{L} B_k d_k, \qquad T(R_{q,B}) = q\gamma - \sum_{k=2}^{L} \frac{B_k}{k-1}. \tag{38.3}$$

This is the complete formally zeta-free space of rational functions with poles only at $0, -1$, orders at most L, and decay $O(t^{-3})$. Indeed writing its coefficients as $a_k/t^k + b_k/(t+1)^k$ gives $a_1 + b_1 = 0$, $a_2 + b_2 = b_1 = q$ and

$$T(R) = q\gamma - b_2 + \sum_{k=3}^{L} \frac{(a_k + b_k)\zeta(k-1) - b_k}{k-1}.$$

The formal zeta coefficients vanish exactly when $a_k + b_k = 0$ for $k \geq 3$, which is precisely (38.3). The null directions $(k-1)d_k - (\ell-1)d_\ell$ have zero integral and necessarily change sign unless zero. They can alter shape, but do not supply positive zero integrals.

For $u = 1/t$, positivity is equivalent to nonnegativity on $[0, 1]$ of

$$N_{q,B}(u) = q(1+u)^{L-1} - \sum_{k=2}^{L} B_k u^{k-3}\{(1+u)^L - (1+u)^{L-k}\}. \tag{38.4}$$

The apparent pole at $k = 2$ cancels. Its degree is at most $2L - 3$ for $L \geq 3$ and at most one for $L = 2$.

The arithmetic-first sublattice $B_k = (k-1)c_k$, $c_k \in \mathbb{Z}$, has $p = \sum c_k \in \mathbb{Z}$. It produced finitely many positive forms: $(L, q, p) = (3, 14, 8), (5, 14, 8), (8, 208, 120), (10, 532, 307)$. The first has the explicit certificate

$$14r_0 - 7d_2 - 2d_3 = \frac{u^4(1-u)(1+2u)}{(1+u)^3} \geq 0.$$

For $L = 8$ the coefficients are $(c_2, \ldots, c_8) = (104, 17, 2, -8, 10, -7, 2)$; the exact Bernstein certificate for (38.4) is nonnegative and nonzero. After reducing by 8 it gives $0 < 26\gamma - 15 < 0.007608$. These are finite certificates, with duplicate rational endpoints among runs, not an infinite small-integer-form sequence.

Theorem 38.2 (Explicit analytic escape with no arithmetic rate)**.** *The two-pole space has gamma-independent rational endpoints $a_m^- < \gamma < a_m^+$ for every $m \geq 1$, using pole order $L = m^2 + 2$, such that $a_m^+ - a_m^- \leq 3/(8m)$.*

Proof. Let $f(t) = \psi_1(t) - 1/t = \sum_{n\geq 0} r_0(t+n)$, $F(u) = f(1/u)$, and $g(u) = F'(u)/u$, with $g(0) = 1$. Writing $v = u/(1+u)$ gives $F(u) - F(v) = r_0(t)$. For $u > 0$,

$$g(u) = 2\sum_{n\geq 0}(1+nu)^{-3} - u^{-1} = \int_0^\infty s^2 e^{-s}\left(\frac{1}{1-e^{-us}} - \frac{1}{us}\right) ds.$$

If $h(x) = (1-e^{-x})^{-1} - x^{-1}$, then $0 < h'(x) = x^{-2} - (4\sinh^2(x/2))^{-1} \le 1/12$. The upper bound follows by differentiating the partial fraction expansion of coth: each term $2(4\pi^2a^2 - x^2)/(4\pi^2a^2 + x^2)^2$ is at most $2/(4\pi^2a^2)$. Thus $0 \le g' \le 1/2$. The same telescoping integral gives

$$\gamma = \int_0^1 (1-u)g(u)\,du.$$

Put $n = m^2$, $N = m^3$, $\delta_m = 1/(4m)$ and $u_j = j/n$. For $j > 0$, define the rational quantities

$$s_j = 2\sum_{k=0}^{N}(1+ku_j)^{-3} - u_j^{-1}, \quad \ell_j = s_j + \frac{1}{u_j(1+(N+1)u_j)^2}, \quad v_j = s_j + \frac{1}{u_j(1+Nu_j)^2},$$

with $\ell_0 = v_0 = 1$. The integral test gives $\ell_j \le g(u_j) \le v_j$ and $v_j - \ell_j \le 2/(1+m)^3 \le \delta_m$. Let $b_{n,j}(u) = \binom{n}{j}u^j(1-u)^{n-j}$ and set

$$p_- = \sum_j \ell_j b_{n,j} - \delta_m, \qquad p_+ = \sum_j v_j b_{n,j} + \delta_m.$$

The Lipschitz bound and binomial variance give $|g(u) - \sum_j g(j/n)b_{n,j}(u)| \le 1/(4m)$, so $p_- \le g \le p_+$ and $p_+ - p_- \le 3\delta_m$. For $P_\pm(u) = \int_0^u sp_\pm(s)ds = \sum B_{\pm,k}u^k$, the kernels $r_0 - [P_\pm(u) - P_\pm(v)]$ have the required opposite signs. Their rational endpoints are explicitly

$$a_m^- = \sum_{j=0}^{n} \frac{n-j+1}{(n+1)(n+2)}\ell_j - \delta_m/2, \quad a_m^+ = \sum_{j=0}^{n} \frac{n-j+1}{(n+1)(n+2)}v_j + \delta_m/2.$$

These follow from $\sum_k B_k/(k-1) = \int_0^1 (1-u)p(u)du$. The interval width is at most $3\delta_m/2$. Strictness holds because r_0 has nonzero simple-pole residues and cannot be a finite combination of the d_k. □

The denominator bit lengths for $m = 1, 2, 3$ in this explicit rule are $(7, 5), (89, 79), (642, 627)$ for the two endpoints. No estimate proves that the denominator times the error tends to zero. Maximal cancellation at infinity in the same space instead returns the classical divergent trigamma expansion

$$f(t) \sim \frac{1}{2t^2} + \sum_{n\ge1} \frac{\mathsf{B}_{2n}}{t^{2n+1}}, \quad |\mathsf{B}_{2n}| = \frac{2(2n)!\zeta(2n)}{(2\pi)^{2n}}.$$

Its first signed examples give $23/40 < \gamma < 7/12$; adding more terms at fixed $t = 1$ eventually worsens that particular rule. This is not an obstruction to every global positivity optimization.

38.4 The exact Todd cost of integer-residue flatness

Proposition 38.3 (Integer quadrature flatness)**.** *Let $A_j \in \mathbb{Z}$, $q = \sum_j A_j$, and suppose $R_A(t) = O(t^{-L-3})$. Then*

$$T_L \mid q, \qquad T_L = \prod_{p\ prime} p^{\lfloor L/(p-1)\rfloor}.$$

For $M \ge L$ this is necessary and sufficient for the existence of integer residues with that flatness, already supported on $0, \dots, L$. The condition $\sum A_j H_j \in \mathbb{Z}$ is separate.

Proof. Writing $\mu_r = \sum_j A_j j^r$, expansion gives coefficients $(-1)^r((r+1)\mu_r - q)t^{-r-2}$; hence flatness means $\mu_r = q/(r+1)$ for $0 \le r \le L$. In the integer-valued binomial basis this is $\sum_j A_j\binom{j}{k} = qG_k$, where $G_k = \int_0^1 \binom{x}{k}dx$. The matrix $(\binom{j}{k})_{0\le k,j\le L}$ is unimodular triangular, so integer residues exist exactly when all qG_k are integral.

For clarity the required denominator identity is proved here. The inverse of $\sum_{j\ge0}(-1)^j z^j/(j+1)$ is $\sum G_n z^n$. For each prime p, its coefficient recurrence gives $v_p(G_n) \ge -\lfloor n/(p-1)\rfloor$ by induction, since $v_p(j+1) \le \lfloor j/(p-1)\rfloor$. When $n = m(p-1)$, the recurrence term with $j = p-1$ has valuation $-m$; all other terms have valuation at least $-m+1$. For $j = k(p-1) > p-1$ this uses $v_p(j+1) \le k-1$, because $p^k - 1 > k(p-1)$; the remaining indices lose a floor unit. Thus $v_p(G_{m(p-1)}) = -m$, proving $\operatorname{lcm}_{k\le L} \operatorname{den}(G_k) = T_L$. □

These are the classical Todd/Hirzebruch denominators [12]. Legendre's formula gives $L! \mid T_L$. For minimal support $M = L$, the unique normalized rule is the classical Lagrange rule with endpoint $\ell_L = \sum_{k=1}^{L} |G_k|/k$. Keeping integer residues and $q > 0$ gives the actual lower bound

$$q\gamma - p \ge \frac{q}{16(L+1)^2 H_L^2} \ge \frac{L!}{16(L+1)^2 H_L^2} \longrightarrow \infty.$$

Indeed on $0 \le x \le (2H_L)^{-1}$, $\prod_{j=1}^{L}(1-x/j) \ge 1 - xH_L \ge 1/2$, yielding $|G_{L+1}| \ge 1/[16(L+1)H_L^2]$; use the next positive series term. Subsequent primitive reduction can destroy the integer-residue hypothesis, so this last assertion is not a lower bound for every reduced endpoint. Nor is high flatness necessary for every small positive kernel.

39 Ten further routes: completed algebra and the remaining value problem

This section preserves the unique results of the ten-route audit. Route R02, classification of G-function solutions to a rationally forced first-order exponential equation, is incorporated in the later Euler zero-division analysis. It is not counted a second time here. The other routes address different transformations and do not combine automatically into one arithmetic proof.

39.1 R01: signed factorial-tail acceleration

Put $X = 2^N$, $Y_N = (N+1)\operatorname{Ein}(X) - N\operatorname{Ein}(2X)$, and $A_m(x) = \sum_{k<m}(-1)^k k! x^{-k-1}$. Then

$$Z_N = Y_N - (N+1)e^{-X}A_X(X) > \gamma$$

eventually, with

$$Z_N - \gamma \sim (N+1)\sqrt{\frac{\pi}{2X}}e^{-2X}. \tag{39.1}$$

Indeed integration by parts gives $E_1(x) = e^{-x}A_m(x) + (-1)^m r_m(x)$, $r_m(x) = m!\int_x^\infty e^{-t}t^{-m-1}dt > 0$. Since X is even, $Z_N - \gamma = (N+1)r_X(X) - NE_1(2X)$. The substitution $t = X+s$ and Stirling's formula give $r_X(X) \sim \sqrt{\pi/(2X)}e^{-2X}$, while $E_1(2X) \sim e^{-2X}/(2X)$ is smaller. If $m/X \to \theta > 0$, the exponential rate is $1 + \theta - \theta\log\theta$, uniquely maximized at $\theta = 1$.

The primitive coefficient height in the specified coordinates $(\operatorname{Ein}(X), \operatorname{Ein}(2X), e^{-X})$ is exactly

$$H_N = (N+1)2^{(N-1)X+N+1-v_2(N+1)}. \tag{39.2}$$

For the consecutive summands of $A_X(X)$ the valuation differences are $v_2(k+1) - N < 0$, so its last summand uniquely determines its denominator, $2^{(N-1)X+N+1}$. Multiplication by $N+1$ gives the displayed clearing factor; consecutive integers $N, N+1$ and the odd reduced final numerator make the resulting vector primitive. Since $0 < A_X(X) < 1/X$, its first coefficient is largest. Hence the analytic height ratio tends to $2/\log X = 0$. These coordinates are E-values, so clearing their coefficients does not give an integer linear form in $1, \gamma$. For fixed nonzero $P \in \mathbb{C}[T]$, the new slope is

$$\lim_N \frac{-\log|P(Z_N)|}{2^N} = 2\operatorname{ord}_{T=\gamma} P,$$

by Taylor expansion. This doubles an analytic slope without proving that no rational polynomial vanishes at γ.

39.2 R03: a quantitative all-prime budget

The classical Euler–Gompertz Padé sequences [39] satisfy

$$q_n = \sum_{k=0}^{n} k! \binom{n}{k}^2, \quad u_{n+1} = 2(n+1)u_n - n^2 u_{n-1}, \quad (q_0, q_1) = (1, 2), \quad (s_0, s_1) = (0, 1),$$

$$q_n s_{n+1} - q_{n+1} s_n = (n!)^2, \quad q_n \delta - s_n = n! \int_0^\infty \frac{t^n e^{-t}}{(1+t)^{n+1}} dt > 0,$$

$$n! \le q_n \le n! e^{2\sqrt{n}}, \quad 0 \le s_n < q_n, \qquad q_n \delta_p - s_n \in (n!)^2 \mathbb{Z}_p, \quad \delta_p = \sum_{k \ge 0} (-1)^k k!.$$

The height bound follows from $q_n/n! = \sum_{j=0}^{n} \binom{n}{j}/j! \le \sum_{j \ge 0} n^j/(j!)^2 \le e^{2\sqrt{n}}$.

Proposition 39.1 (Unmatched factorial mass)**.** *For reduced $\alpha = a/b \in \mathbb{Q}$, $b > 0$, set $\mathcal{B}_\alpha = \{p : \delta_p \ne \alpha\}$. For every n except at most one,*

$$\sum_{\substack{p \le n \\ p \in \mathcal{B}_\alpha}} v_p(n!) \log p \ge \tfrac{1}{2} \log(n!) - \sqrt{n} - \tfrac{1}{2} \log(|a| + b).$$

Consequently, for sufficiently large n,

$$\sum_{\substack{p \le n \\ p \in \mathcal{B}_\alpha}} \frac{\log p}{p-1} \ge \tfrac{1}{2} \log n - \tfrac{1}{2} - \frac{1}{\sqrt{n}} - \frac{\log(|a| + b)}{2n}.$$

Proof. The determinant identity makes s_n/q_n strictly increasing. Thus $L_n = aq_n - bs_n$ vanishes at most once. At a matched prime, $p \nmid b$ and $v_p(L_n) \ge 2v_p(n!)$. The product of those prime powers divides the nonzero integer L_n, while $|L_n| \le (|a| + b) n! e^{2\sqrt{n}}$. Taking logarithms proves the first bound. Use $v_p(n!) \le n/(p-1)$ and $\log(n!) \ge n \log n - n$ for the second. □

This quantitative consequence of the classical Padé method is stronger than mere divergence of the unmatched prime weight; it does not assert natural density or identify the real value with finite-place values. For $P \in \mathbb{Z}[T]$ of degree d and height H, the integer $q_n^d P(s_n/q_n)$ vanishes at most d times. At a prime satisfying $P(\delta_p) = 0$ it lies in $(n!)^2 \mathbb{Z}_p$, by subtracting homogeneous polynomial evaluations, even if q_n is not a p-adic unit. Its archimedean upper bound is $(d+1)Hq_n^d$. The resulting budget, after division by $(n!)^2$, tends to zero for $d = 1$ but is not small for $d \ge 2$; already for $d = 2$ it is at least $3H(n+1)^2$. This diagnoses those estimates, not all possible refinements.

39.3 R04: minimal universal Bell–Gould filters

Let $B_0 = 1$, $B_{n+1} = \sum_{k=0}^{n} \binom{n}{k} B_k$, and $G_0 = 0, G_1 = 1$, with the same recurrence for G_{n+1} when $n \ge 1$. A common monic filter modulo M is $C \in \mathbb{Z}[X]$ with $C(E)B_n = C(E)G_n = 0 \pmod M$ for every $n \ge 0$, where E is forward shift. Write $\mu(M)$ for its least degree.

Proposition 39.2. *Over $\mathbb{F}_p[X]$, the common annihilator ideal is $((X^p - X)(X^p - X - 1))$. Thus $\mu(p) = 2p$, and*

$$\mu(M) = \max_{p^r \| M} \mu(p^r).$$

If a known monic common recurrence modulo p^r has degree D, a proposed filter is universal exactly when it kills the first D shifts of both sequences.

Proof. The Bell–Gould recurrence identities, retaining the initial boundary term, give $T(E)B = 0$, $T(E)G = a$ over $\mathbb{F}_p$, where

$$T = X^p - X - 1, \qquad a_n = \sum_{j=0}^{p-1} (-1)^{p-1-j}(p-1-j)!\, j^n,$$

with $0^0 = 1$; these follow by applying the binomial recurrence [40]. All weights are nonzero. Vandermonde independence of j^n, beginning at $n = 0$, makes the annihilator of a equal to $(X^p - X)$. The polynomial T is irreducible because a root has $\theta^{p^k} = \theta + k$, a Frobenius orbit of length p. Since B is nonzero its annihilator is (T). Every common filter thus contains both coprime factors, whose product works. Reduction proves the lower CRT bound; equalize local degrees by multiplying by powers of X and apply coefficientwise CRT for the upper bound. Finally shift operators commute, so the filtered sequence retains the known degree-D monic recurrence, and its first D zero values propagate. □

The finite exact systems retained in the source certificates give

M	8	64	81	125	12!	20!
previous degree	10	20	21	30	32	56
$\mu(M)$	6	12	15	20	22	38

For example $X^6 + X^5 - 3X^4 + 3X^3 - X^2 + 3X - 2$ works modulo 8 and gives $28\delta - 17$ after the integral normalization. These finite certified minima are not a formula for every $\mu(p^r)$. If p_N is the largest prime at most N, reduction gives $\mu(N!) \geq 2p_N$. Combined with the previously constructed degree-$3N$ upper bound and the prime number theorem, this gives $2 \leq \liminf \mu(N!)/N \leq \limsup \mu(N!)/N \leq 3$. No existence of that limit or small real values follows. The 19 retained forms at $N = 2, \ldots, 20$ were certified nonzero, but the final value was approximately 1.25×10^{16}.

39.4 R05: a rational spectral edge with an actual localization cost

For $0 \leq i, j < N$, let

$$A_N(i,j) = (i+j)!, \quad B_N(i,j) = (i+j)!\{H_{i+j} - i - j - 1\}, \quad \lambda_N = \max_{v \neq 0} \frac{v^T B_N v}{v^T A_N v}.$$

Both matrices are integral and A_N is positive definite.

Theorem 39.3. *One has $\lambda_N < \gamma - 1$, $\lambda_N \leq \lambda_{N+1}$ and $\lambda_N \to \gamma - 1$. Equivalently, $cA_N - B_N \succeq 0$ for every N exactly when $c \geq \gamma - 1$; every finite pencil is positive definite at the threshold. In the standard Laguerre basis the same eigenvalues are those of*

$$T_N(i,j) = \begin{cases} H_i - 2i - 1, & i = j, \\ \min(i,j), & |i-j| = 1, \\ -1/|i-j|, & |i-j| \geq 2. \end{cases}$$

The infinite matrix on finitely supported sequences is essentially self-adjoint, with purely absolutely continuous spectrum $(-\infty, \gamma - 1]$ of multiplicity two. Moreover, for $\gamma_N = 1 + \lambda_N$,

$$\gamma - \gamma_N \geq \frac{1}{64N^2(1 + 1/(4N))}. \tag{39.3}$$

Proof. Gamma integrals and their derivatives identify the Rayleigh quotient with the average of $f(x) = \gamma + \log x - x$ against $p(x)^2 e^{-x} dx$. Its unique maximum is $\gamma - 1$ at $x = 1$; a nonzero

polynomial cannot concentrate its mass at that single point. Nested spaces give monotonicity. Polynomials are dense in $L^2((1+|f|)e^{-x}dx)$: for a function orthogonal to them, the measure $g(1+|f|)e^{-x}dx$ has a Laplace transform analytic on $\Re z < 1/2$ with all derivatives zero at zero; Fourier uniqueness gives the zero measure. Approximation near $x = 1$ proves the limit.

The Laguerre generating function yields

$$\int_0^\infty x^\alpha e^{-x} \sum_i L_i(x)t^i \sum_j L_j(x)s^j dx = \Gamma(\alpha+1)\frac{(1-t)^\alpha(1-s)^\alpha}{(1-ts)^{\alpha+1}}.$$

Differentiation at $\alpha = 0$ gives diagonal log moments $H_i - \gamma$ and off-diagonal moments $-1/|i-j|$. Combining the Laguerre recurrence for multiplication by x gives T_N. The same density proof for weight $(1+f^2)e^{-x}$ shows the polynomial domain is a core. Each interval $(0,1)$ and $(1,\infty)$ maps bijectively onto $(-\infty, \gamma-1)$ under f, giving two positive absolutely continuous spectral measures.

The classical inequality $|e^{-x/2}L_j(x)| \le 1$ [41] follows also from the normalized Hermite-function identity $\langle h_j(\cdot + a), h_j\rangle = e^{-a^2/4}L_j(a^2/2)$ and Cauchy–Schwarz. Hence every normalized polynomial of degree below N satisfies $e^{-x}|p(x)|^2 \le N$. At most half its mass lies in $[1-\epsilon, 1+\epsilon]$, $\epsilon = 1/(4N)$. Elsewhere

$$x - 1 - \log x \ge \epsilon - \log(1+\epsilon) \ge \epsilon^2/[2(1+\epsilon)].$$

Integrating over at least half the mass proves (39.3). □

For $\ell_N = \operatorname{lcm}(1,\dots,N-1)$, $\ell_1 = 1$, $\det\{\ell_N(X-1)I_N - \ell_N T_N\}$ is an integer defining polynomial of γ_N, with height at most $[\ell_N(1+5N)]^N$. This is an upper bound for one defining polynomial, not a sharp minimal height. The first values are $\gamma_1 = \gamma_2 = 0$, and γ_3 is the largest root of $4X^3 + 14X^2 + 5X - 1$. The actual error bound (39.3) excludes exponential accuracy of raw finite sections in N, without excluding acceleration or other operators.

39.5 R06: reciprocal levels and one-zero excision

The reciprocal construction in companion [001] supplies a real entire function H with

$$H(0) = 2\,\mathrm{Cin}(1), \qquad H'(w) = \int_0^1 e^{-sw}k(s)ds, \quad k(s) = J_0(2\sqrt{s(1-s)}),$$

where $\mathrm{Cin}(1) = \int_0^1 (1-\cos t)t^{-1}dt$. Its inherited derivative-zero set is $\{\pm 2i\sqrt{\pi^2n^2-1} : n \ge 1\}$, and its real restriction is an increasing bijection. These are inputs from the reciprocal construction. Let w_c denote the unique real solution $H(w_c) = 2c$.

Proposition 39.4. *For every real c, all zeros of $H - 2c$ are simple and every nonreal zero lies in $\Re w < 0$. A right-half-plane zero occurs exactly when $c > \mathrm{Cin}(1)$, and is then $w_c > 0$. A zero on the imaginary axis occurs only for $c = \mathrm{Cin}(1)$, at zero.*

Proof. For $u = 2\sqrt{s(1-s)} \le 1$, the Bessel series give $J_0(u) \ge 1 - s + s^2$ and $0 < J_1(u) \le u/2$ for $u > 0$. For $s \le 1/2$ one has $k'(s) \le 0$; for $s > 1/2$, $k'(s) \le 2s - 1$, so $sk'(s) - k(s) \le s^2 - 1 < 0$. Thus $k(s)/s$ is strictly decreasing. For $\sigma \ge 0$, $y > 0$,

$$\Im H(\sigma + iy) = \int_0^1 e^{-\sigma s}\frac{k(s)}{s}\sin(ys)ds > 0.$$

Extend the decreasing weight by zero after 1 and pair positive and negative sine half-periods; all pairs are nonnegative and the first is strictly positive. The $1/s$ singularity is canceled by the sine. Conjugation and the real bijection prove zero location. The inherited critical lattice is on the imaginary axis away from zero, where this same inequality makes the H-value nonreal; real levels therefore have no multiple zeros. □

For $c > \operatorname{Cin}(1)$ and $a = w_c$, removing the single upper zero of the rotated function gives

$$\widehat{E}_c(z) = \frac{e^{-iz/2}(H(-iz) - 2c)}{z - ia}, \qquad \widehat{\Theta}_c(z) = \frac{z - ia}{z + ia}\, e^{iz}\, \frac{H(iz) - 2c}{H(-iz) - 2c}.$$

The inherited Cartwright bounds are $h_{\widehat{E}_c}(\theta) = \frac{1}{2}|\sin\theta|$ and $|\widehat{E}_c(x)| = O(\log(2+|x|)/(1+|x|))$. All remaining zeros lie below the real axis. Cartwright factorization, together with $H(y) = \log y + O(1)$, $H(-y) \sim -e^y/y$, gives $\widehat{\Theta}_c(iy) \sim -1/(y \log y)$ and zero exponential mean type. Thus its reflection quotient is nonconstant inner and $\widehat{E}_c$ is Hermite–Biehler of type $1/2$. Logarithmic differentiation at zero gives the convergent positive trace

$$\sum_{\substack{H(\rho)=2c \\ \rho \neq w_c}} \frac{-2\Re\rho}{|\rho|^2} = 1 + \frac{\sin 1}{\operatorname{Cin}(1) - c} + \frac{2}{w_c}.$$

The last term records the excised zero. At the Euler level, $\operatorname{Cin}(1) = \gamma - \operatorname{Ci}(1)$ gives $1 - \sin 1/\operatorname{Ci}(1) + 2/w_*$. The same structure holds at algebraic levels such as $c = 1$; recovering w_* spectrally does not establish its algebraicity.

39.6 R07: exact evaluation images of integer divisor filters

For $k \geq 1$, $d \geq 2$, put $M = 4^{k+d}$, $a_j = 4^{d-j} D(4^{k+j})/2 \in \mathbb{Z}$. The constraints $\sum_{j=0}^{d} c_j = q$, $\sum j c_j = -kq$ give a rational value $N/(Mq)$, where $N = Mq/2 + \sum a_j c_j$. Define

$$s_j = a_j - 2a_{j+1} + a_{j+2}, \quad g = \gcd(|s_0|, \ldots, |s_{d-2}|), \quad P = M/2 + (k+1)a_0 - ka_1.$$

Then the image for fixed integer q is exactly

$$N \in Pq + g\mathbb{Z}, \qquad \operatorname{den}(N/(Mq)) = M|q|/\gcd(|N|, M|q|). \tag{39.4}$$

For $g = 0$ the image is the singleton $\{Pq\}$. Indeed $q(k+1, -k, 0, \ldots)$ is a particular solution and $e_j - 2e_{j+1} + e_{j+2}$ form an integer basis of the homogeneous kernel, by successive coordinate elimination. Bezout gives (39.4); for $g > 0$ the three-row map has Smith invariants $1, 1, g$. For $(k, d) = (1, 3)$ the actual values are $(D(4), D(16), D(64), D(256)) = (8, 50, 280, 1466)$, $(s_0, s_1) = (16, 13)$ and $P = 240$. Every $p/256$ is realized by

$$c = (2, -1, 0, 0) + (p - 240)(-4, 13, -14, 5).$$

Such surjectivity supplies no privileged approximation of γ; smallness, primitive height, and nonvanishing remain separate.

39.7 R08: exact Sondow prefix rank and a larger positive cone

Retain $I_n = \binom{2n}{n}\gamma - A_n + L_n$ and put

$$w_{n,t} = 2\binom{n}{t}^2 (H_t - H_{n-t}), \qquad \lambda_{p,n} = \sum_{t=0}^{n} w_{n,t} v_p((n+t)!).$$

An integer vector u cancels every logarithm exactly when $\sum_n u_n \lambda_{p,n} = 0$ for all relevant primes, by unique factorization after clearing rational exponents.

Proposition 39.5. *For every $M \geq 1$, the logarithm matrix on columns $1, \ldots, M$ has rank $\pi(2M)$ and nullity $M - \pi(2M)$. On a general window $N, \ldots, M$ its rank is at least the number of odd primes in $[2N - 1, 2M - 1]$. If a nonzero kernel vector has last supported index $m \geq 2$, then $2m - 1$ is composite.*

Proof. The weights are antisymmetric under $t \mapsto n-t$, negative before the midpoint and positive afterwards. Every proper tail $\sum_{t=r-n}^{n} w_{n,t}$, $n+1 \le r \le 2n$, is strictly positive. For $n < p \le 2n$, this tail at $r = p$ equals $\lambda_{p,n}$; for $p > 2n$ the entry is zero. For each odd prime choose column $(p+1)/2$. In increasing order this is a triangular minor with positive diagonal. For a prefix add row 2 and column 1, whose entry is 2; all prime rows are covered. The same observation at $p = 2m-1$ gives the final claim. □

Let $U(z) = \sum_{j=0}^{M-N} u_{N+j} z^j$. Since $r = x(1-x)y(1-y) \in [0, 1/16]$,

$$\sum_{n=N}^{M} u_n I_n = \iint_{(0,1)^2} \frac{r^N U(r)}{(1-xy)(-\log(xy))}\, dx\, dy.$$

Positivity of U on that interval proves nonvanishing. If $d = M-N$, the Bernstein coefficients of $U(t/16)$ are exactly

$$b_k = \sum_{j=0}^{k} \frac{u_{N+j}}{16^j} \frac{\binom{k}{j}}{\binom{d}{j}}.$$

Their strict positivity is less restrictive than first-coefficient dominance. The retained certificate on $1, \dots, 8$ has $\min b_k = 17363115$, fails first-coefficient dominance, and gives

$$57429190525504380\gamma - \frac{426310477993846627277}{12870} > 50000.$$

This enlarges the certified positive cone, not the approximation quality. The lower bound follows from $I_1 > 3/1024$, obtained by restricting its integral to $[1/4, 3/4]^2$. That prefix's two-dimensional logarithm kernel also maps with rank two to the (Q, A)-plane; rank is different from shortness of vectors in that quotient.

An unconstructed sufficient family for irrationality would combine exact logarithm cancellation, strict positivity, and

$$d_{2M} \sum_{n=N}^{M} |u_n| 16^{-(n-1)} \longrightarrow 0, \qquad d_{2M} = \operatorname{lcm}(1, \dots, 2M).$$

Indeed $I_n \le 16^{-(n-1)} I_1$ and denominators of A_n divide d_{2M}. This condition has not been proved for an infinite family and addresses irrationality rather than all algebraic degrees.

39.8 R09: exact primitive cost of two-trace recovery

For $q \ge 1$, $x = c^{-1}$, let $U = T_2^{(q)}(c)$ and $V = T_4^{(q)}(c)$ be the universal polyexponential inverse traces. Their explicit expressions are

$$U = x^2 - 2^{-q}x, \quad V = x^4 - 2^{1-q}x^3 + \left(\frac{2}{3^{q+1}} + \frac{1}{2^{2q+1}}\right)x^2 - \frac{x}{6 \cdot 4^q}.$$

Set $D_q = 6 \cdot 24^q$, $N_q = 4^{q+1} - 6^q - 3^{q+1}$ and $B_q = 4 \cdot 8^q - 3 \cdot 6^q$. Then

$$N_q x = D_q V - D_q U^2 - B_q U \tag{39.5}$$

is primitive of coefficient height exactly D_q and, after division by N_q, the unique total-degree-at-most-two expression in U, V identically equal to x. No affine expression works. Indeed $V - U^2 = A_q U + C_q x$, with

$$A_q = \frac{2}{3^{q+1}} - \frac{1}{2^{2q+1}}, \qquad C_q = -\frac{1}{6 \cdot 4^q} + \frac{2}{3 \cdot 6^q} - \frac{1}{2^{3q+1}}.$$

Clearing gives $D_qA_q = B_q$, $D_qC_q = N_q$. The number N_q is odd and congruent to 1 modulo 3, hence coprime to D_q. The strict inequalities $0 < A_q < 1$, $|C_q| < 1$ give the height. Comparison of degrees eight, six, four, two and one in x successively fixes all coefficients of any quadratic recovery expression. The unnormalized residual is asymptotic to $-4^{-q}/(6c)$, but after integerization it is N_q/c, of size asymptotic to $6^q/|c|$. This is optimality only in the stated low-degree coordinate problem. At the algebraic level $c = 2$, for example, $T_2^{(1)}(2) = 0$, $T_4^{(1)}(2) = 1/288$, and all universal traces are rational. Inverse-branch complexity therefore does not imply arithmetic complexity of these values.

39.9 R10: raw-moment recovery and determinant information loss

For independent formal variables g, d over a characteristic-zero field K, let

$$a = (g+1)d, \quad b = (g+1)(1-d), \quad c = (g+\tfrac{3}{2})d.$$

Then $g = a + b - 1$, $d = 2(c - a)$ and $K[a, b, c] = K[g, d]$. The complete relation ideal is generated by $2(A+B)(C-A)-A$, as the same formulas define an inverse in the quotient. At the actual moments this recovers γ and δ with coefficients of height at most two, but the moments themselves are not rational approximants.

The first two determinant coordinates are instead

$$U = (g+1)d, \qquad V = (2g+3)^2(2d-1) - \frac{2g^2+5g+5}{2}d^2.$$

They satisfy

$$[K(g,d) : K(U,V)] = 4, \qquad K[U,V] \cap K[g] = K. \tag{39.6}$$

For the first assertion substitute $g = U/d - 1$ to obtain

$$2d^4 + (U-4)d^3 + (2U^2 - 16U + 2V + 2)d^2 + (-16U^2 + 8U)d + 8U^2 = 0.$$

Over $K(U)$ this is $A(d) + 2Vd^2$, with coprime coefficients since $A(0) = 8U^2 \neq 0$. It is irreducible as a primitive linear polynomial in V; Gauss's lemma, using its constant leading coefficient as a polynomial in d, proves irreducibility over $K(U, V)$. Its degree is four. For the second assertion, give U, V weights $1, 2$. If a nonconstant $Q(U, V)$ has highest weighted degree M, its coefficient of d^M after substitution is

$$(g+1)^M \sum_{i+2j=M} q_{ij} \left(-\frac{2g^2+5g+5}{2(g+1)^2} \right)^j \neq 0.$$

The rational function in parentheses is nonconstant and hence transcendental over K. Thus no nonconstant such Q belongs to $K[g]$. These are universal identities, not restrictions on accidental specialized relations or on larger coordinate sets.

Finally all mixed positivity tests admit an algebraic-coordinate model: for $\alpha \geq \gamma - 1$,

$$k!(\alpha + k + 1 - H_k) = \int_0^\infty x^k e^{-x}(x - \log x + \alpha - \gamma)dx$$

has a positive almost-everywhere density. Keeping the actual positive Gompertz measure preserves the mixed Hankel positivity at $\alpha = 0$. The measure has changed; it does not preserve the defining unshifted Euler integral. This establishes insufficiency of those positivity tests alone, not a contradiction in either arithmetic world.

Part V
Infinite polynomial families and exact targets

40 Infinite polynomial families and exact arithmetic targets

This section integrates the polynomial-family records of 9–10 September 2026. Its principal distinction is between a proved criterion, a terminating finite certificate, and a still unproved assertion that certificates succeed at unbounded complexity. Throughout, $H(P)$ is the maximum absolute coefficient of an integer polynomial. No criterion below is asserted to be satisfied by γ at unbounded complexity.

40.1 The degree-sensitive target is an equivalence

For a nonzero polynomial P with $P(x) \neq 0$, put

$$B(P) = \log \max(2, H(P)) + \deg P, \qquad \mathcal{R}_x(P) = \frac{-\log|P(x)|}{B(P)}.$$

Lemma 40.1 (Resultant lower bound)**.** *Let α be algebraic of degree d, with primitive minimal polynomial f, leading coefficient $a \neq 0$, and conjugates $\alpha = \alpha_1, \ldots, \alpha_d$. Set*

$$C_\alpha = |a| \prod_{i=2}^{d} \max(1, |\alpha_i|).$$

For every $P \in \mathbb{Z}[X]$ of degree r with $P(\alpha) \neq 0$,

$$|P(\alpha)| \geq C_\alpha^{-r} \big((r+1)H(P)\big)^{-(d-1)}. \tag{40.1}$$

If $0 < \alpha < 1$, then $C_\alpha = M(f)$, where M denotes Mahler measure.

Proof. The integer $\operatorname{Res}(f, P) = a^r \prod_i P(\alpha_i)$ is nonzero. For $i \geq 2$, $|P(\alpha_i)| \leq (r+1)H(P)\max(1, |\alpha_i|)^r$. Substitution into $|\operatorname{Res}(f, P)| \geq 1$ proves the assertion. □

Theorem 40.2 (Equivalent infinite-family target)**.** *A real number x is transcendental if and only if there are integer polynomials P_n, with $P_n(x) \neq 0$, for which*

$$\limsup_n \mathcal{R}_x(P_n) = \infty.$$

For $x \in [0, 1]$ this is also equivalent to existence, at unbounded integers $D \geq 2$, of polynomials satisfying

$$\deg P_D, H(P_D) \leq D, \qquad 0 < |P_D(x)| < 2U_D, \quad U_D = \frac{D(D+1)}{(D+1)^{D+1} - 1}. \tag{40.2}$$

Proof. For algebraic x, Lemma 40.1 and $\log(r+1) \leq r$ bound $\mathcal{R}_x(P)$ by a constant depending only on x.

Conversely, evaluate the $(D+1)^{D+1}$ polynomials with coefficient vectors in $\{0, \ldots, D\}^{D+1}$ at a transcendental $x \in [0, 1]$. All values are distinct and lie in an interval of length at most $D(D+1)$. Two adjacent values therefore differ by a nonzero polynomial value of absolute value at most U_D. Since

$$\frac{-\log(2U_D)}{D + \log D} \sim \log D,$$

these differences give the required family. For an arbitrary real x, the interval length is at most $D(D+1)\max(1, |x|)^D$; its additional logarithmic cost is only $O_x(D)$ and does not change the conclusion. □

Remark 40.3 (Correction of an earlier formulation). The earlier statement that this varying-degree criterion is stronger than necessary is replaced by Theorem 40.2. Fixed-degree or fixed-exponent criteria can be stronger than necessary, but the precise varying-degree condition above is an equivalence. Its converse uses transcendence to exclude collisions; it cannot be used to assume those collisions absent for γ.

For linear forms, $0 < |q_n x - p_n| \to 0$, with integer p_n, q_n, implies irrationality. Roth's theorem gives transcendence if, for a fixed $\epsilon > 0$, infinitely many distinct reduced fractions satisfy $0 < |x - p_n/q_n| < q_n^{-2-\epsilon}$ [18]. A finite exponent improvement supplies neither infinitely many fractions nor the required uniform margin. In contrast, the varying-orbit criterion in Section 43 permits a specified vanishing margin; its normalization and hypotheses are different.

For completeness, a fixed-degree alternative uses the classical Nesterenko criterion. If, with D fixed first,

$$\deg P_{D,n} \le D, \quad H(P_{D,n}) \le e^{b_D n + o(n)}, \quad |P_{D,n}(x)| = e^{-a_D n + o(n)}, \qquad a_D, b_D > 0,$$

then

$$\dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}}(1, x, \dots, x^D) \ge 1 + a_D/b_D$$

by the nonoscillating case of [43, Theorem 1]. The value asymptotic includes a lower estimate: positivity alone is insufficient. An unbounded set of ratios a_D/b_D would prove transcendence, but no such family is obtained here.

40.2 A finite certificate and a total output rule

Suppose $[a, b] \subset [0, 1]$ is a rational interval containing x, with midpoint c and width w. For polynomials of degree at most D and height at most Q, define

$$M_{D,Q} = \frac{QD(D+1)}{2}, \qquad U_{D,Q} = \frac{Q(D+1)}{(Q+1)^{D+1} - 1}.$$

Every such polynomial satisfies

$$|P(x) - P(c)| \le M_{D,Q} w/2. \tag{40.3}$$

Evaluate the entire coefficient grid $\{0, \dots, Q\}^{D+1}$ at c, retaining repeated values. Let g be the least adjacent gap.

Proposition 40.4 (Finite separation and exact termination). *If $g > M_{D,Q} w/2$, every nonzero polynomial in the stated coefficient box is nonzero at x. If also $M_{D,Q} w/2 < U_{D,Q}$, the minimum-gap difference satisfies*

$$0 < |P(x)| < 2U_{D,Q}.$$

Given rational enclosures of widths tending to zero, refinement until both tests hold terminates exactly when the box contains no nonzero polynomial vanishing at x.

Proof. Each polynomial in the symmetric coefficient box is a difference of two grid vectors, so its midpoint value has absolute value at least g. Equation (40.3) proves the first assertion. The minimum gap is at most $U_{D,Q}$, giving the second. If a relation exists, the corresponding midpoint difference is at most $M_{D,Q} w/2$, and the test cannot pass. Otherwise the finite collection of distinct evaluations at x has a positive minimum gap δ. The midpoint minimum gap is at least $\delta - M_{D,Q} w/2$; a sufficiently narrow interval passes both tests. □

Here is a completely specified total rule for γ. Let

$$G_D = (D+1)^{D+1}, \quad M_D = \frac{D^2(D+1)}{2}, \quad 2^{k_D} > \frac{11 M_D}{8 U_D},$$

where k_D is the least positive integer satisfying the inequality. Put $N = 2^{k_D}$ and, with $k = k_D$,

$$s_k = 2\sum_{j=0}^{k-1} \frac{1}{(2j+1)3^{2j+1}}, \qquad r_k = \frac{3}{4(2k+1)9^k},$$

$$a_D = H_N - k(s_k + r_k) - N^{-1}, \quad b_D = H_N - ks_k, \quad c_D = (a_D + b_D)/2. \tag{40.4}$$

The atanh series gives $s_k < \log 2 < s_k + r_k$, and harmonic integral comparison gives $0 < H_N - \log N - \gamma < N^{-1}$. Consequently

$$0 < a_D < \gamma < b_D < 1, \qquad b_D - a_D < \frac{U_D}{M_D}, \qquad |c_D - \gamma| < \delta_D := \frac{1}{D(G_D - 1)}. \tag{40.5}$$

Indeed $b_D - a_D = 2^{-k} + kr_k < (11/8)2^{-k}$; the displayed choice of k gives the width estimate. The elementary bounds $1/2 < \gamma < 2/3$, together with $k \geq 6$, place these intervals inside $(0, 1)$.

Order the grid polynomials by their rational values at c_D, breaking ties lexicographically by their coefficient vectors, and choose the lexicographically first pair realizing the least adjacent gap. Let P_D be their difference, oriented to have nonnegative value at c_D. Then

$$|P_D(c_D)| = \min_{\substack{0 \neq P \in \mathbb{Z}[X] \\ \deg P, H(P) \leq D}} |P(c_D)|, \qquad 0 \leq |P_D(\gamma)| < \tfrac{3}{2}U_D. \tag{40.6}$$

This rule always terminates, since it consists of finitely many rational operations. The assertion $P_D \neq 0$ concerns its coefficient vector, not its value at γ.

Proposition 40.5 (Full degree and primitivity)**.** *For every $D \geq 2$, the canonical output P_D has degree D and integer content* 1*.*

Proof. Bertrand's theorem supplies a prime p with $N/2 < p < N$ [42]. The choice of k implies $N/2 > D$ and $N/2 > 2k + 1$. In H_N, only the term $1/p$ has denominator divisible by p. The corrections in

$$c_D = H_N - ks_k - (2N)^{-1} - kr_k/2$$

are p-integral, so $v_p(c_D) = -1$. Its reduced denominator exceeds D. The rational-root theorem therefore excludes a zero at c_D of every nonzero polynomial of height at most D. Dividing a minimizer by content greater than 1 would strictly decrease its nonzero value. If its degree were below D, multiplication by X would also strictly decrease its value because $0 < c_D < 1$. Both contradict minimality. □

40.3 Algebraic collapse, its effective onset, and the relation lattice

Theorem 40.6 (Exact dichotomy for the total family)**.** *If γ is transcendental, every $P_D(\gamma)$ is nonzero. If γ is algebraic, every sufficiently late $P_D(\gamma)$ is exactly zero. More generally, the latter conclusion holds at any algebraic $\alpha \in (0, 1)$ for any family with degree and height at most D and $|P_D(\alpha)| < 2U_D$.*

Proof. Let d and $M = M(f)$ be the degree and Mahler measure of a hypothetical minimal polynomial. A nonzero value satisfies

$$|P_D(\alpha)| \geq M^{-D}[D(D+1)]^{-(d-1)}. \tag{40.7}$$

This is incompatible with $2U_D = \exp(-D\log D + O(\log D))$ for sufficiently large D. The transcendental case is the definition of transcendence. □

An exact sufficient condition for collapse is

$$(D+1)^{D+1} - 1 > 2[D(D+1)]^d M^D.$$

Two explicit, nonoptimal sufficient onsets are

$$D \geq \max\{4d, \lceil 4M^2 \rceil\}, \tag{40.8}$$
$$D \geq \lceil \max\{4M, 16d \log(16dM)\} \rceil . \tag{40.9}$$

For the first, use $2U_D < 4(D+1)^{1-D}$; its ratio to the lower bound in (40.7) is at most $4M^D/(D+1)^{D-2d+1} < 4/2^D < 1$. For the second, put $t = 16dM$. At the prescribed onset D_1, one has $D_1 + 1 \leq t^2$ and $D_1 \geq 8d \log(D_1 + 1)$; monotonicity of $D/\log(D+1)$ extends this to larger D. The logarithm of the same comparison ratio is less than

$$\log 4 - D \log 4 + D/4 < 0.$$

Since $M(f) \leq \sqrt{d+1} H(f)$ by Jensen and Parseval, (40.8) also yields the coefficient-only onset $D \geq \max\{4d, 4(d+1)H(f)^2\}$. A finite nonvanishing certificate excludes a bounded region of degree *and* height, not every algebraic number of a given degree.

Proposition 40.7 (Saturated relation lattice and abundant collisions)**.** *For $D \geq d$, evaluation at algebraic α has kernel*

$$\{P \in \mathbb{Z}[X]_{\leq D} : P(\alpha) = 0\} = f\mathbb{Z}[X]_{\leq D-d}.$$

This kernel is saturated of rank $D + 1 - d$. All nonzero Smith invariant factors of the matrix with columns $f, Xf, \ldots, X^{D-d}f$ equal 1. Moreover, at most

$$V_D \leq 1 + [D(D+1)]^d M^D$$

distinct values occur on the coefficient grid $\{0, \ldots, D\}^{D+1}$.

Proof. Gauss's lemma proves the kernel formula. Saturation follows because $mP(\alpha) = 0$ implies $P(\alpha) = 0$. Alternatively, modulo every prime p, multiplication by the nonzero polynomial $f \bmod p$ is injective in $\mathbb{F}_p[X]$. Its columns remain independent, so no prime divides all maximal minors, proving the Smith assertion. The image has rank d, though for nonintegral α it need not have basis $1, \alpha, \ldots, \alpha^{d-1}$. For example, the image at $\alpha = 1/2$ is $2^{-D}\mathbb{Z}$.

Distinct grid values are separated by the lower bound (40.7) and lie in an interval of length $D(D+1)$. Counting separated points gives the bound for V_D. □

At least one fiber contains $\lceil G_D/V_D \rceil$ vectors. Subtracting one member produces at least $\lceil G_D/V_D \rceil - 1$ distinct exact relations. Thus the algebraic branch permits many short relations; an assertion that an exact zero is unlikely does not contradict it.

40.4 Canonical selection forces maximal multiplicity

For a primitive irreducible polynomial f, define

$$\nu_D(f) = \max_{\substack{0 \neq A \in \mathbb{Z}[X] \\ \deg A, H(A) \leq D}} v_f(A).$$

Here $v_f(A)$ is the multiplicity of f as a factor of A.

Theorem 40.8 (Maximum admissible multiplicity)**.** *If γ is algebraic with primitive minimal polynomial f of degree d, then the canonical outputs satisfy*

$$v_f(P_D) = \nu_D(f)$$

for every sufficiently large D. In particular, with $L = \|f\|_1$,

$$f^{m_D} \mid P_D, \qquad m_D = \min\left\{\left\lfloor \frac{\log D}{\log L}\right\rfloor, \left\lfloor \frac{D}{d}\right\rfloor\right\} = \frac{\log D}{\log L} + O_f(1).$$

Proof. Write $P_D = f^r B$ with $B(\gamma) \neq 0$. Integer quotients follow from Gauss's lemma. The elementary Mahler estimates

$$H(B) \le 2^{\deg B} M(B) \le 2^D M(P_D) \le T_D := 2^D D(D+1)$$

follow from expanding the product over roots and Jensen's unit-circle formula. They are uniform in r. Lemma 40.1 gives

$$|B(\gamma)| \ge \Lambda_D := M(f)^{-D}((D+1)T_D)^{-(d-1)} = \exp(-O_f(D + \log D)).$$

By (40.5),

$$|B(c_D) - B(\gamma)| \le \tfrac{1}{2} D(D+1) T_D \delta_D = \exp(-D\log D + O(D + \log D)),$$

so $|B(c_D)| \ge \Lambda_D/2$ eventually.

Choose an admissible competitor $A = f^s C$ with $s = \nu_D(f)$. Again $H(C) \le T_D$. If $r < s$, exact minimality and $f(c_D) \neq 0$ give

$$|B(c_D)| \le |f(c_D)|^{s-r} |C(c_D)| \le S(f)\delta_D(D+1)T_D, \qquad S(f) = \sum_{j\ge 1} j|f_j|.$$

For large D the last expression is less than $\Lambda_D/2$, a contradiction. Thus $r = s$. Finally $f^{m_D} X^{D - d m_D}$ is admissible, since its height is at most $L^{m_D} \le D$; hence $\nu_D(f) \ge m_D$. □

Let $J_D = \gcd_{\mathbb{Q}[X]}(P_D, P_D')$, normalized to be monic. Under algebraicity, $f^{m_D - 1} \mid J_D$ eventually. It follows that any one of the following assertions, if established on an unbounded set of canonical indices, proves transcendence:

- P_D is squarefree;
- a proved rational interval containing γ has no real root of J_D;
- $\deg J_D = o(\log D)$;
- the number of roots of J_D in such an interval, counted with multiplicity, is $o(\log D)$.

Sturm counts and squarefree factorization certify individual cases. Their unbounded success is unproved. If intervals may be freely refined, the second local criterion is equivalent in strength to transcendence: for a transcendental γ each fixed J_D has no zero at γ, whereas the algebraic branch forces the indicated repeated root.

The minimization rule is essential. At $x = 1/2$,

$$(2X - 1)(X^{D-1} - 1), \qquad D \ge 3,$$

is primitive, squarefree, of degree D and height 2, yet vanishes at x. Squarefreeness plus the size bounds alone therefore does not suffice.

40.5 Localized common factors, modular certificates, and infeasible jets

For any target-small family, coprime finite blocks whose least index tends to infinity exclude eventual divisibility by one minimal polynomial. The more precise certificate is absence of a common real root in a proved interval containing γ. A persistent factor away from this interval is harmless. For example, for a transcendental $x \in [1/2, 3/5]$, pigeonholing coefficient grids of degree $D-1$ gives R_D with

$$0 < |R_D(x)| < \frac{(5/2)D}{(D+1)^D - 1}.$$

Then $P_D = XR_D$ satisfies (40.2), yet every member contains X. Thus global coprimality is not a necessary consequence of transcendence for arbitrary bounded families.

For fixed N, the monic gcd of $P_N, \dots, P_M$ eventually stabilizes as M increases, because it is a descending sequence of divisors of P_N. Equality at several consecutive stages cannot certify permanent stabilization: a later output can remove a factor. Under a hypothetical fixed ideal (f), sums, products, Euclidean remainders and Bezout combinations remain in (f). A Wronskian obeys

$$(fB)'(fC) - (fB)(fC)' = f^2(B'C - BC'),$$

so it does not remove the relation either.

Proposition 40.9 (Degree-preserving modular escape)**.** *If reductions of integer polynomials $F_1, \dots, F_s$ have gcd 1 over $\mathbb{F}_p$, and at least one F_i retains its degree, then their gcd over $\mathbb{Q}$ is 1.*

Proof. A primitive common factor A divides every F_i integrally. Degree preservation in one product forces the leading coefficient of A to remain nonzero modulo p, so its reduction is a nonconstant common factor, a contradiction. □

The degree check cannot be omitted: $(pX+1)X$ and $(pX+1)(X+1)$ share $pX+1$ over $\mathbb{Q}$, but reduce to the coprime pair $X, X+1$. Alternatively, a nonzero reduction of the Sylvester determinant formed with the original degrees certifies coprimality.

Homogeneous congruences can be retained by pigeonholing. If a linear map $L : \mathbb{F}_p^{D+1} \to \mathbb{F}_p^r$ is prescribed, one residue fiber of $\{0, \dots, Q\}^{D+1}$ has at least $\lceil (Q+1)^{D+1}/p^r \rceil$ members. Their nearest difference satisfies $L(P \bmod p) = 0$ and

$$|P(x)| \le \frac{Q(D+1)}{\lceil (Q+1)^{D+1}/p^r \rceil - 1}.$$

This gives no prescribed nonzero residue. Selecting differences between distinct fibers requires a separate close-pair theorem. Moreover, a subset of $\mathbb{F}_p^{D+1}$ whose every nonzero difference has degree D has at most p members, since the leading-coefficient projection must be injective. Requiring every difference to have a full-degree irreducible reduction therefore destroys the large grid.

At the comparator $x = 1/2$, every nonzero degree-D integer polynomial value has size at least 2^{-D}, while $2U_D < 2^{-D}$ for $D \ge 3$. Consequently every target-small output must be divisible by $2X - 1$. Prescribing an odd leading coefficient, or irreducibility of degree at least 2, is incompatible with the universal target there. These counterexamples concern constructions uniform in x; they do not exclude a future use of special identities of γ.

Differentiation escapes an ideal but does not preserve a small-value estimate. The stronger difficulty is the following feasibility obstruction.

Proposition 40.10 (Identically stringent jets are impossible)**.** *Set $\epsilon_D = 2U_D$ and let r_D be the least positive integer such that $(5/3)^{r_D} \ge 8D(D+1)$. For all sufficiently large D and every $x \in [1/2, 3/5]$, no nonzero polynomial of degree and height at most D satisfies*

$$|P^{(j)}(x)| < \epsilon_D \qquad (0 \le j < r_D).$$

Proof. Write $P = X^s R$ with $R(0)$ a nonzero integer, and put $K = r_D - 1 = O(\log D)$. Expanding X^{-s} at x gives, for $j \le K$,

$$\left|\frac{R^{(j)}(x)}{j!}\right| \le \eta := \epsilon_D (K+1)(D+K)^K 2^{D+K}.$$

For $s = 0$ the same upper bound is immediate. Remove the Taylor polynomial $T = \sum_{j=0}^{K} R^{(j)}(x)(X-x)^j/j!$ and put $Q = R - T$. Its coefficient norm satisfies

$$\|T\|_1 \le \theta := \eta(K+1)(8/5)^K = \exp(-D\log D + O(D + (\log D)^2)) < 1/2.$$

Thus Q has a root of multiplicity at least r_D at x, $|Q(0)| \ge 1/2$, and $\|Q\|_1 \le D(D+1) + 1/2$. The root product and Jensen's formula imply

$$\|Q\|_1 \ge M(Q) \ge |Q(0)| x^{-r_D} \ge \tfrac{1}{2}(5/3)^{r_D} \ge 4D(D+1),$$

a contradiction. Although Q has real coefficients, these Mahler inequalities remain valid; integrality was needed only for $|R(0)| \ge 1$. □

A formally sufficient degree-d criterion demanding the same accuracy through derivative order $\lfloor D/d \rfloor$ is therefore empty on this interval for large D. Resultants would indeed force all these derivatives to vanish and hence a factor of degree exceeding D, but Proposition 40.10 shows that such a criterion cannot be used as an achievable universal construction. Unequal derivative budgets are outside this obstruction.

40.6 Multiplicative recycling and common-denominator alternatives

Define the additive cost $C(P) = \log M(P) + \deg P$ for nonconstant integer polynomials. The inequalities

$$H(P) \le 2^{\deg P} M(P), \qquad M(P) \le (\deg P + 1) H(P)$$

show $C(P) \le 2B(P) \le 6C(P)$, while $C(PQ) = C(P) + C(Q)$.

Proposition 40.11 (Primitive-factor reduction)**.** *If $0 < |P(x)| < 1$, some primitive irreducible positive-degree factor F of P satisfies*

$$\frac{-\log|F(x)|}{C(F)} \ge \frac{-\log|P(x)|}{C(P)}.$$

Consequently the unbounded-ratio target is unchanged if all outputs are required to be primitive and irreducible.

Proof. Write $P = c\prod_i F_i^{m_i}$. The numerator equals $\sum_i m_i[-\log|F_i(x)|] - \log|c|$ and the denominator equals $\sum_i m_i C(F_i) + \log|c|$. The ratio is no larger than the greatest positive factor ratio. The comparison between B and C completes the reduction. □

Products and powers of a bounded-quality collection cannot amplify its normalized quality. This statement allows different factors and includes coefficient cancellation after expansion; it does not exclude addition or determinants that create new factors.

A different sufficient condition approximates powers simultaneously. Fix D and suppose integer $q_n > 0, p_{1,n}, \dots, p_{D,n}$ have nonzero errors

$$e_{j,n} = q_n x^j - p_{j,n}, \qquad e_{1,n} \to 0, \qquad e_{j+1,n} = o(e_{j,n}).$$

Then $1, x, \dots, x^D$ are linearly independent over $\mathbb{Q}$. Indeed an integer relation, multiplied by q_n, becomes an integer tending to zero and hence eventually zero. Dividing by the earliest error with

a nonzero relation coefficient leaves that coefficient as a nonzero limit. Families for every fixed D would prove transcendence. This elementary criterion is related to the vector criteria of [44].

The rank-one moment curve imposes exact constraints. If $e_j = qx^j - p_j$, then

$$\Delta_{i,j} = qp_{i+j} - p_i p_j = qx^i e_j + qx^j e_i - qe_{i+j} - e_i e_j \in \mathbb{Z}. \tag{40.10}$$

Under the stated hierarchy and $x \neq 0$, $\Delta_{i,j}/(qe_i) \to x^j$ for $i < j$, and to $2x^i$ for $i = j$. These integers are eventually nonzero, implying

$$\liminf_n q_n|e_{i,n}| \geq |x|^{-j} \quad (i < j), \qquad \liminf_n q_n|e_{i,n}| \geq (2|x|^i)^{-1} \quad (i = j),$$

whenever the required coordinates exist. In particular the earliest errors cannot also be $o(1/q_n)$.

Taking powers of one approximation p_n/q_n with common denominator q_n^D gives errors

$$jx^{j-1}q_n^{D-1}(q_n x - p_n)(1 + o(1)),$$

all on the same scale, and makes every minor (40.10) vanish. A determinant descent is nevertheless an exact identity: for

$$q' = qp_2 - p_1^2, \qquad p' = 2(qp_3 - p_1 p_2),$$

one has

$$q'x - p' = -3qxe_2 + 2qe_3 - xe_1^2 + 2e_1 e_2.$$

Even when $q' \sim 2xqe_1$ is smaller than q, the term qe_2 remains uncontrolled. No improved approximation follows without additional cancellation or divisibility.

41 Positive realization and the complete two-pole integer lattice

This section uses the two-pole identities established in (38.3), restated here to fix normalization:

$$T(R) = \int_1^\infty \lfloor t \rfloor R(t)\, dt, \quad r_0(t) = \frac{1}{t^2(t+1)}, \quad d_k(t) = t^{-k} - (t+1)^{-k},$$
$$T(r_0) = \gamma, \qquad T(d_k) = \frac{1}{k-1}.$$

The new issue is the height of the *output polynomial*, which need not include the denominators of exact null-mean corrections.

41.1 The full lattice and its explicit basis

Put $\ell(b) = \sum_{j=1}^m b_j/j$ and $d_m = \operatorname{lcm}(1, \ldots, m)$. The full integer lattice is

$$\Lambda_m = \{b \in \mathbb{Z}^m : \ell(b) \in \mathbb{Z}\} = \ker\left(b \longmapsto \sum_{j=1}^m (d_m/j)b_j \pmod{d_m}\right).$$

The map is surjective, because the integers d_m/j have gcd 1: for each prime factor of d_m, choose j equal to its largest power at most m. Thus $[\mathbb{Z}^m : \Lambda_m] = d_m$, whereas the coordinatewise sufficient sublattice $\operatorname{diag}(1, \ldots, m)\mathbb{Z}^m$ has index $m!$.

An explicit triangular basis is $v_1 = e_1$ and, for $k \geq 2$,

$$v_k = \begin{cases} pe_k - e_{k/p}, & k = p^a, \\ e_k - ae_u - be_w, & k = uw,\ (u,w) = 1,\ u, w > 1,\ aw + bu = 1. \end{cases}$$

Each v_k, $k \ge 2$, has $\ell(v_k) = 0$. The determinant of the column matrix is $\prod_{p^a \le m} p = d_m$, so these vectors form a basis of the entire lattice. Consequently

$$b = pv_1 + \sum_{k=2}^{m} x_k v_k \quad (p, x_k \in \mathbb{Z}) \quad \Longrightarrow \quad T\left(qr_0 - \sum_j b_j d_{j+1}\right) = q\gamma - p.$$

The basis comparison is factorial versus exponential: d_{2n}/d_n divides $\binom{2n}{n}$ by prime-power valuations, and iteration gives the sufficient bound $d_m < 16^m$. This is a covolume fact, not a short-vector theorem for the intersection with a positivity cone.

For each prime $p \le m$, write $j = p^{s_j} u_j$ with $p \nmid u_j$ and $a = \lfloor \log_p m \rfloor$. The local integrality conditions are exactly

$$\sum_j b_j p^{a-s_j} u_j^{-1} \equiv 0 \pmod{p^a}.$$

If $m/2 < p \le m$, this forces $p \mid b_p$. For arbitrary integer b, with $S = \sum_j (d_m/j) b_j$,

$$\operatorname{den} \ell(b) = \frac{d_m}{\gcd(d_m, S)}.$$

For a rational shape $\beta = (\beta_j)$ normalized by $q = 1$, let $D_\beta = \operatorname{lcm}_j \operatorname{den} \beta_j$ and $p_0 = \sum_j \beta_j / j$. The least integer multiplier making all residues and the rational output term integral is

$$Q_{\text{full}} = \operatorname{lcm}(D_\beta, \operatorname{den} p_0).$$

The coordinatewise rule uses $Q_{\text{coord}} = \operatorname{lcm}_j \operatorname{den}(\beta_j / j)$. Primitive reduction of the final pair can be smaller still, so Q_{full} is not a lower bound on the primitive denominator.

41.2 Null-mean corrections preserve output height

The measure $d\eta(t) = \lfloor t \rfloor d_2(t)\, dt$ is a probability measure. Define

$$X(t) = \frac{r_0(t)}{d_2(t)} = \frac{t+1}{2t+1}, \qquad V_k(t) = \frac{(k-1) d_k(t)}{d_2(t)}.$$

Then $\mathbb{E}X = \gamma$ and $\mathbb{E}V_k = 1$. The quantitative Bernstein construction of Theorem 38.2 gives rational $B_{m,k}$, $2 \le k \le m^2 + 2$, for which

$$R_m = r_0 - \sum_k B_{m,k} d_k, \quad 0 \le R_m \le \epsilon_m d_2, \qquad \epsilon_m = \frac{3}{8m}.$$

With $a_m = \sum_k B_{m,k}/(k-1)$, put

$$F_m = X - \sum_k \frac{B_{m,k}}{k-1}(V_k - 1) = a_m + R_m/d_2.$$

It follows exactly that $\mathbb{E}F_m = \gamma$ and $F_m \in [a_m, a_m + \epsilon_m]$; for $m \ge 2$ this interval lies in $[0, 1]$.

For $P(Z) = \sum_{j=0}^{D} p_j Z^j$, define its symmetric multiaffine polarization

$$\mathcal{P}_D(x_1, \dots, x_D) = \sum_{j=0}^{D} \frac{p_j}{\binom{D}{j}} \sum_{|S|=j} \prod_{i \in S} x_i,$$

and the rational tensor kernel

$$K_{m,P}(t_1, \dots, t_D) = \prod_i d_2(t_i)\, \mathcal{P}_D(F_m(t_1), \dots, F_m(t_D)).$$

Independence yields the exact identity

$$T^{\otimes D}(K_{m,P}) = P(\gamma). \tag{41.1}$$

The final polynomial remains $P \in \mathbb{Z}[Z]$, with its original degree and height; clearing all internal kernel denominators would charge a cost that the exact output identity does not require.

If $S(P) = \sum_{j\geq 1} j|p_j|$, telescoping products give

$$|\mathcal{P}_D(x_1, \ldots, x_D) - P(a_m)| \leq \epsilon_m S(P)$$

throughout the range cube. Thus the rational test $P(a_m) > \epsilon_m S(P)$ proves strict positivity of the kernel. It eventually passes whenever $P(\gamma) > 0$, since $|P(a_m) - P(\gamma)| \leq \epsilon_m S(P)$. A sufficient conditional cost is $m > 3S(P)/(4P(\gamma))$.

Conversely, a continuous nonnegative nonzero kernel has positive integral because the integration measure has full support. Therefore every nonzero integer polynomial has a positive lift for one sign if and only if γ is transcendental. The construction is complete as a strict-sign semidecision procedure; proving termination for every nonzero P would already solve the arithmetic question.

The same correction mechanism shows that the image of the positive two-pole cone is exactly the set of strict positive rational forms $q\gamma - p$, $q, p \in \mathbb{Q}$. For $q > 0$, choose a certified rational lower endpoint $p/q < a < \gamma$ and a positive kernel with integral $\gamma - a$; multiplying it by q and adding $(qa - p)d_2$ realizes the prescribed form. For $q < 0$, use an upper endpoint and its positive error kernel; $q = 0$ is immediate. Unconstrained existence of a positive kernel therefore adds no approximation exponent.

41.3 A completed nonlinear family and a fixed-degree sum-of-squares bound

Let $A(Z) = 26Z - 15$ and $B(Z) = 7Z - 4$. The prior positive certificate gives $A(\gamma) > 0$. Their exact relation is $7A - 26B = -1$. The quadratic

$$Q(Z) = A(Z)^2 + B(Z)^2 = 725Z^2 - 836Z + 241$$

has discriminant -4, hence is irreducible. With $a = 15/26$ and a two-pole random variable $F \geq a$ of mean γ, its polarization is strictly positive:

$$725(a+u)(a+v) - 418(2a+u+v) + 241 = \frac{1}{676} + \frac{7}{26}(u+v) + 725uv > 0 \quad (u, v \geq 0).$$

Using $\gamma < 5773/10000$ gives

$$0 < Q(\gamma) < \frac{7141}{4000000} < 1.$$

Thus Q^n has degree $2n$, height at most 1802^n, and positive values bounded by $(7141/4000000)^n$. The same family is small at the rational point $577/1000$, where $Q = 61/40000$. It does not prove irrationality. Likewise $(26X - 15)^n$ has coefficient norm 41^n and equals 500^{-n} at that rational point. Proposition 40.11 explains why powering does not improve normalized quality.

Proposition 41.1 (Coprime integral sums of squares)**.** *If $A, B \in \mathbb{Z}[X]$ are coprime over $\mathbb{Q}[X]$, $m = \max(\deg A, \deg B) \geq 1$, and $P = A^2 + B^2$, then for $0 \leq x \leq 1$,*

$$P(x) \geq c_m H(P)^{-(2m-1)}, \quad c_m = \left((2m)!^2 C_m^{4m-2}(2m+1)^{2m-1}\right)^{-1}, \quad C_m = (m+1)(2m)^m.$$

Proof. A nonzero constant component gives $P \geq 1$. Otherwise put $r = \deg A$, $s = \deg B$, and $H_0 = \max(H(A), H(B))$. The adjugate of the Sylvester matrix gives $UA + VB = \mathrm{Res}(A, B)$ with degrees below s, r and coefficient bounds $(r+s-1)!H_0^{r+s-1}$. Evaluation on $[0, 1]$ yields

$$1 \leq (r+s)!H_0^{r+s-1}\sqrt{P(x)}, \qquad P(x) \geq ((2m)!)^{-2}H_0^{-(4m-2)}.$$

Interpolate A, B at j/m, $0 \le j \le m$. Every Lagrange basis polynomial has coefficient norm at most $(2m)^m$, giving

$$H_0 \le C_m \max_{[0,1]} \sqrt{P(x)} \le C_m \sqrt{(2m+1)H(P)}.$$

Substitution proves the claimed bound. □

This is a fixed-degree result for coprime integral components. Its degree-dependent constant cannot be ignored when m grows, and a common factor can retain the original small-value problem. For the displayed quadratic the sharper identity is $Q(x) = ((725x - 418)^2 + 1)/725$.

42 An all-order Bernoulli–Stieltjes family and its rate obstruction

42.1 Positive measure, exact Padé bounds, and endpoint arithmetic

The two-pole kernel of Theorem 38.2 admits the gamma-free expression

$$g(u) = 2\sum_{j\ge 0}(1+ju)^{-3} - u^{-1}, \quad g(0) = 1, \qquad \gamma = \int_0^1 (1-u)g(u)\,du.$$

Let $b(t) = (e^{2\pi t} - 1)^{-1}$ and $\rho(t) = 2t^3 b''(t) > 0$. Then

$$H(v) := \frac{g(\sqrt{v}) - 1}{\sqrt{v}} = \int_0^\infty \frac{\rho(t)}{1+vt^2}\,dt. \tag{42.1}$$

To derive it, differentiate Binet's digamma integral [45, 5.9.10] twice and use $\psi''(x) = -2\zeta(3, x)$. The resulting formula is

$$\frac{g(u) - 1}{u} = 4\int_0^\infty \frac{t(3 - u^2t^2)b(t)}{(1+u^2t^2)^3}\,dt.$$

The rational factor is the second derivative of $2t^3/(1+u^2t^2)$. Two integrations by parts give (42.1); boundary terms vanish using $b = O(t^{-1})$, $b' = O(t^{-2})$ at zero and exponential decay at infinity.

Push $\rho(t)dt$ forward by $x = t^2$ to a measure ν. Its moments are rational:

$$\mu_k = 2\int_0^\infty t^{2k+3} b''(t)\,dt = \frac{2(2k+3)(2k+2)!\zeta(2k+2)}{(2\pi)^{2k+2}} = (2k+3)|B_{2k+2}|.$$

Both ordinary and shifted Hankel matrices are positive definite because ν and $x\nu$ have positive density. The moment problem is determinate: for $0 < a < 2\pi$, the series $\sum a^{2k}\mu_k/(2k)!$ converges. Any measure with these moments has finite $\cosh(a\sqrt{x})$ integral. Symmetrizing its square-root pushforward gives a bilateral Laplace transform analytic near zero and fixed by the moments, hence fixes the measure. The same proof applies to $x\nu$. The formal moment series has radius zero; no convergence of that power series is asserted.

Let $L_n = [n-1/n]_H$ and $U_n = [n/n]_H$ for this formal moment series. Their coefficients are rational and all orders exist. If p_n is the monic degree-n orthogonal polynomial for ν, the Gaussian quadrature formula gives

$$L_n(v) = \sum_{j=1}^n \frac{w_j}{1+vx_j}, \qquad H(v) - L_n(v) = \int \frac{(p_n(x)/p_n(-1/v))^2}{1+vx}\,d\nu(x) > 0, \tag{42.2}$$

with simple positive nodes and positive weights. Applying Gaussian quadrature to $x\nu$, with lower approximant $\widetilde{L}_n$, gives

$$U_n(v) = \mu_0 - v\widetilde{L}_n(v) > H(v).$$

This is Gauss–Radau quadrature including the node zero; its zero-node weight is $\int \widetilde{p}_n(x)^2 d\nu(x)/\widetilde{p}_n(0)^2 > 0$, and other weights are the Gaussian weights divided by their positive nodes.

The quadrature measures have fixed mass, are tight by their first moments, and eventually reproduce each fixed moment. A higher fixed moment gives uniform integrability of lower powers. Every weak limit therefore has all the prescribed moments and equals the determinate measure. This proves $L_n(v), U_n(v) \to H(v)$ for $v > 0$. Their transforms are bounded by μ_0, so dominated convergence yields

$$\alpha_n^- := \int_0^1 (1-u)(1+uL_n(u^2))\,du < \gamma < \alpha_n^+ := \int_0^1 (1-u)(1+uU_n(u^2))\,du, \qquad \alpha_n^\pm \to \gamma. \tag{42.3}$$

At a positive algebraic node s the primitive integral is

$$\int_0^1 \frac{u(1-u)}{1+su^2}\,du = \frac{\log(1+s)}{2s} - \frac{1}{s} + \frac{\arctan\sqrt{s}}{s\sqrt{s}}.$$

All nodes and weights are algebraic. The logarithmic part is strictly positive, because the displayed logarithm and arctangent terms have positive weights. Writing the arctangent using logarithms of algebraic numbers makes it a nonzero algebraic-coefficient logarithmic linear form. Baker's theorem implies that this part, and hence each positive-order endpoint $\alpha_n^\pm$, is transcendental [47]. Their limit need not be transcendental. The first bounds are

$$-1 + \frac{3}{4}\log\frac{4}{3} + \frac{\sqrt{3}\pi}{4} < \gamma < \frac{7}{18} + \frac{\log 2}{12} + \frac{\pi}{24}.$$

No integer-polynomial height may be assigned to these endpoints merely from their Padé order.

42.2 A quantitative all-order lower bound on the error

Theorem 42.1 (Rate obstruction for the unchanged lower family)**.** *For every $n \geq 1$, writing $M = 2n+1$,*

$$\gamma - \alpha_n^- \geq \frac{3e^{-4/3}}{6400\pi^2 M^4 (\log M)^2}.$$

The same lower bound applies to the full bracket width, but is not claimed for the individual upper error.

Proof. The error in (42.2) is also the minimum of $\int q(x)^2(1+vx)^{-1}d\nu(x)$ over degree-n polynomials with $q(-1/v) = 1$. Indeed write a competitor as $p_n/p_n(-1/v) + (x+1/v)r$; the cross term vanishes by orthogonality.

Set $v = u^2$ and $Q(t) = q(t^2)$. It has degree at most $2n$ and $Q(i/u) = 1$. By even extension,

$$H(u^2) - L_n(u^2) = \int_{\mathbb{R}} Q(t)^2 w_u(t)\,dt, \qquad w_u(t) = \frac{\rho(|t|)}{2(1+u^2t^2)}.$$

For $0 < u \leq 1$,

$$w_u(t) \geq \frac{1}{2\pi}\operatorname{sech}^2(\pi t). \tag{42.4}$$

For $t > 0$, the ratio to $\operatorname{sech}^2(\pi t)$ is $\pi^2 t^3 \coth^3(\pi t)/(1+u^2t^2)$. If $t \leq 1$, use $t\coth(\pi t) \geq 1/\pi$ and denominator at most 2; if $t \geq 1$, use $\coth(\pi t) \geq 1$ and denominator at most $2t^2$.

The continuous Hahn normalization [46, 18.19.2–18.19.4; 18.20.9] gives an orthonormal basis for $\operatorname{sech}^2(\pi t)dt$:

$$\phi_k(t) = \sqrt{\frac{\pi(2k+1)}{2}}\,i^k F_k(2it), \qquad F_k(z) = {}_3F_2\left(\begin{matrix}-k, k+1, (1+z)/2 \\ 1, 1\end{matrix}; 1\right).$$

For $1 \le y \le 4/3$ we claim

$$|F_k(-2y)| \le 5(k+1)^{2y-1}. \tag{42.5}$$

Put $a = 1/2 - y \in [-5/6, -1/2]$. A finite beta-integral expansion of the Legendre polynomial P_k gives

$$F_k(-2y) = 1 + \frac{1}{\Gamma(a)\Gamma(1-a)} \int_0^1 s^{a-1}(1-s)^{-a}[\mathsf{P}_k(1-2s) - 1], ds.$$

The bracket vanishes to first order at zero, so the integral converges. Expand it termwise to recover the terminating hypergeometric series. The elementary Legendre estimate is $|\mathsf{P}_k(1-2s) - 1| \le \min(2, k(k+1)s)$; its derivative version follows by expressing P_k' as the sum of $(2j+1)\mathsf{P}_j$ of opposite parity and using $|\mathsf{P}_j| \le 1$. With $T = k(k+1)$, split at $1/T$ to bound the integral by

$$\left(\frac{1}{a+1} + \frac{2}{-a}\right) T^{-a} \le 10T^{-a}.$$

Reflection gives $|\Gamma(a)\Gamma(1-a)|^{-1} \le 1/\pi$, proving (42.5), including the immediate case $k = 0$.

It follows that $\sum_{k=0}^{2n} |\phi_k(iy)|^2 \le 25\pi M^{4y}$. Expanding Q in this basis and applying Cauchy–Schwarz at iy, together with (42.4), yields

$$H(u^2) - L_n(u^2) \ge \frac{1}{50\pi^2 M^{4/u}}, \qquad 3/4 \le u \le 1.$$

Let $h = 1/(4\log M)$; then $1 - h \ge 3/4$ and $M^{-4/u} \ge e^{-4/3}M^{-4}$ on $[1-h, 1]$. Integrating $u(1-u)$ there, whose integral is at least $3h^2/8$, proves the theorem. □

Thus $-\log(\gamma - \alpha_n^-) \le 4\log(2n+1) + 2\log\log(2n+1) + O(1)$. A fixed exponential upper error e^{-cn} cannot hold even on an unbounded subsequence. This is a lower bound, not a matching asymptotic, and does not exclude new cancellation between levels or moving evaluation points.

Two further restrictions retain their exact scope. Each positive-node integrand $w_j u(1-u)/(1+s_j u^2)$ has nonzero residue $w_j(1-z)/(2s_j)$ at $z = \pm i/\sqrt{s_j}$. Consecutive orthogonal polynomials have disjoint zero sets by their three-term recurrence. Therefore a nontrivial constant combination of two consecutive lower integrands still has a nonzero residue and has no rational antiderivative. This does not exclude a rational *definite* integral from global logarithmic identities. Also a quadrature scheme supported on one fixed finite set S cannot reproduce arbitrarily many moments: $\prod_{s\in S}(x-s)^2$ vanishes on that support but has positive ν-integral. A fixed number field is not a fixed finite support.

43 Bounded rational orbits and a vanishing approximation margin

The orbit $xj/(j+1)$ stays in a fixed compact interval. Its conditional arithmetic analysis differs from that of x/j: translating the latter by x would lose integer coefficients unless additional information about x were already available.

43.1 Exact-zero height has the sharp leading exponent

Let $\alpha \in (0,1)$ be algebraic, with primitive irreducible polynomial

$$f(X) = a_d X^d + \cdots + a_0, \qquad a_d > 0,\ a_0 \ne 0.$$

Set

$$\beta_j = \alpha\frac{j}{j+1}, \quad F_j(X) = j^d f\left(\frac{j+1}{j}X\right), \quad f_j = F_j/g_j,$$

where g_j is the positive content of F_j.

Lemma 43.1 (Content and leading coefficients)**.** *The f_j are pairwise distinct primitive irreducible polynomials of degree d, and*

$$g_j \mid |a_0 a_d|, \qquad \operatorname{lc}(f_j) = \frac{a_d (j+1)^d}{g_j} \geq \frac{(j+1)^d}{|a_0|}.$$

Proof. Rational linear substitution preserves irreducibility. The norms $N(\beta_j) = (j/(j+1))^d N(\alpha)$ are nonzero and distinct, hence the minimal polynomials differ. If a prime p divides j, the leading coefficient of F_j has valuation $v_p(a_d)$; if $p \mid j+1$, its constant coefficient has valuation $v_p(a_0)$. If neither is divisible by p, primitivity of f implies that one coefficient of F_j is a p-unit. Taking minimum coefficient valuations proves the content divisibility and the leading-coefficient bound. □

Theorem 43.2 (Exact minimum-height exponent)**.** *For $D \geq dK$, let $h_{D,K}$ be the smallest coefficient height of a nonzero integer polynomial of degree at most D vanishing at all $\beta_1, \ldots, \beta_K$. Then*

$$\frac{((K+1)!)^d}{|a_0|^K} \leq h_{D,K} \leq \|f\|_1^K ((K+1)!)^d, \qquad \log h_{D,K} = dK \log K + O_f(K).$$

Proof. The product $S_K = \prod_{j=1}^{K} f_j$ divides every such polynomial over $\mathbb{Q}$, hence integrally by Gauss's lemma. Its leading coefficient must divide the leading coefficient of the polynomial, giving the lower bound. Conversely S_K itself has degree dK and

$$H(S_K) \leq \prod_j \|f_j\|_1 \leq \|f\|_1^K ((K+1)!)^d.$$

Stirling's elementary logarithmic estimate proves the last assertion. □

For $K = \lfloor c \log D / \log\log D \rfloor$, the height is $D^{dc+o(1)}$. Thus exact-zero polynomials within height D exist eventually if $c < 1/d$ and cannot exist if $c > 1/d$. At the endpoint $c = 1/d$ for this particular schedule, the upper construction still fits: writing $\ell = \log D$ and $L = \log\log D$,

$$\log H(S_K) \leq \ell - \frac{\ell}{L} \log L + O_f(\ell/L) < \ell.$$

This leading-coefficient argument improves an earlier comparison using $M(R) \leq D\sqrt{D+1}$, which lost an unnecessary exponent $1/2$.

43.2 A fixed gain above one is sufficient

If $\deg R, H(R) \leq D$ and $R(\beta_j) \neq 0$, the resultant and $M(f_j) \leq M(f)(j+1)^d$ imply

$$|R(\beta_j)| \geq [M(f)(j+1)^d]^{-D} [D(D+1)]^{-(d-1)}. \tag{43.1}$$

Theorem 43.3 (Conditional orbit criterion)**.** *Fix $\kappa > 1$. For every positive integer d, put*

$$c_d = \frac{1+\kappa}{2d}, \qquad K_{D,d} = \left\lfloor c_d \frac{\log D}{\log\log D} \right\rfloor.$$

Suppose that, for each d, at unbounded integers D there is a nonzero $R_D^{(d)} \in \mathbb{Z}[X]$ with degree and height at most D such that

$$\max_{1 \leq j \leq K_{D,d}} \left| R_D^{(d)} \left(\gamma \frac{j}{j+1} \right) \right| \leq \exp\left(-\kappa \frac{D \log D}{K_{D,d}} \right). \tag{43.2}$$

Then γ is transcendental.

Proof. Assume γ algebraic of degree d and use that degree's family. Uniformly for $j \le K_{D,d}$, the negative logarithm of the nonzero lower bound (43.1) is at most $(d+o(1))D\log\log D$. The requested exponent is $(\kappa/c_d+o(1))D\log\log D$, and $\kappa/c_d > d$. Thus every value is eventually exactly zero. Theorem 43.2 then forces

$$H(R_D^{(d)}) \ge D^{dc_d+o(1)} = D^{(1+\kappa)/2+o(1)} > D,$$

a contradiction. □

The theorem proves sufficiency, not existence or an equivalence. The $d=1$ instance would exclude rationality only. The unbounded sets of successful D need not be common to the different fixed degrees. The numerical choice $\kappa = 1.03$ is permitted here with $c_d = 1.015/d$; it is not the exponent of any earlier isolated linear-form certificate.

43.3 A margin tending to zero can also suffice

Let

$$\ell = \log D, \quad L = \log\log D, \quad T = \log\log\log D, \qquad w_D \to \infty, \quad w_D = o(L),$$

and define, for sufficiently large D,

$$\kappa_D = 1 + \frac{2w_D}{L}, \qquad K_{D,d} = \left\lfloor \frac{\ell}{d(L-T-w_D)} \right\rfloor. \tag{43.3}$$

Theorem 43.4 (Vanishing-margin version)**.** *The conclusion of Theorem 43.3 holds if its parameters are replaced by* (43.3)*.*

Proof. Under an algebraic hypothesis of degree d, put $B_D = L - T - w_D$ and $K = K_{D,d}$. Since $B_D \sim L$,

$$K = \frac{\ell}{dB_D} + O(1), \qquad \log K = L - T - \log d + o(1).$$

The negative logarithm of the nonzero-value lower bound is at most

$$A_D = D\log M(f) + dD\log(K+1) + (d-1)\log[D(D+1)] = dD(L-T) + O_f(D).$$

The demanded exponent, including the floor error, is

$$\begin{aligned} E_D &= \kappa_D D\ell/K \\ &= dD\left(L - T + w_D - \frac{2w_D(T+w_D)}{L}\right) + O_d(DL^2/\ell). \end{aligned}$$

Hence $E_D - A_D = (d+o_f(1))Dw_D > 0$, and all orbit values vanish. On the other hand,

$$\begin{aligned} d\log((K+1)!) - K\log|a_0| - \ell &= \frac{\ell}{B_D}\left(w_D - \log d - 1 - \frac{\log|a_0|}{d} + o(1)\right) + O_f(L) \\ &= (1+o_f(1))\frac{\ell w_D}{L} > 0. \end{aligned}$$

The leading-coefficient lower bound again contradicts height at most D. □

A prescribed margin $\epsilon_D = \kappa_D - 1$ can therefore be used when $\epsilon_D \to 0$ and $\epsilon_D \log\log D \to \infty$, by choosing $w_D = \epsilon_D L/2$. An explicit alternative schedule is

$$\widetilde{\kappa}_D = 1 + 4T/L, \qquad \widetilde{K}_{D,d} = \left\lfloor (1+2T/L)\frac{\ell}{dL} \right\rfloor.$$

Its exponent exceeds the resultant threshold by $(3d+o_f(1))DT$, while the exact-zero logarithmic height exceeds ℓ by $(1+o_f(1))\ell T/L$. Both inequalities survive integer rounding. Neither schedule has been achieved by a gamma-specific infinite family.

43.4 Universal counting and a finite success that does not extend

For any fixed $x \in (0,1)$, evaluate $\{0,\dots,D\}^{D+1}$ at K points $xj/(j+1)$. Its images lie in $[0, D/(1-x)]^K$. Partition into $t^K < G_D$ boxes, taking t maximal. Two coefficient vectors lie in one box, giving a nonzero difference with

$$\max_j |R(xj/(j+1))| \le \frac{D}{(1-x)t} = \exp\left(-(1+o(1))\frac{D\log D}{K}\right)$$

for the above scales of K. Zero values are allowed. The notation $o(1)$ supplies neither a positive sign nor the necessary rate for the margin in Theorem 43.4.

The archived finite search used the older schedule $\kappa = 2$, $d = 1$, $K = \lfloor (7/4)\log D/\log\log D\rfloor$. At the four selected degrees $K = 4$. The candidate

$$R_8 = X^6(2X-1)^2, \qquad H(R_8) = 4,$$

was certified by rational arithmetic to satisfy

$$\max_{j\le 4} |R_8(\gamma j/(j+1))| < 173/10^6 < 1/4096.$$

The selected candidates $X^9(2X-1)^3$, $X^{13}(2X-1)^3$ and $X^{21}(2X-1)^3$ at $D = 12, 16, 24$ failed that same target. Failure of selected candidates is not a nonexistence theorem.

There is, however, an all-order obstruction to extending a success just by powers of X. For fixed nonzero $Q \in \mathbb{R}[X]$ and $x \in (0,1)$, set $R_D = X^{D-\deg Q}Q$ and $K_D \sim c\log D/\log\log D$. Then

$$\frac{-K_D \log \max_{j\le K_D} |R_D(xj/(j+1))|}{D\log D} \longrightarrow 0.$$

Indeed, if Q has a root of multiplicity $r \ge 0$ at x, its value at $xK/(K+1)$ is comparable to K^{-r}. Boundedness on $[0,x]$ and this last point give

$$-\log \max_j |R_D(xj/(j+1))| = D\log(1/x) + O(D/K + \log K).$$

Normalization proves the claim. A construction whose factor structure changes with D is not covered.

44 Bernoulli convolutions: an explicitly conditional arithmetic branch

Let

$$\nu_x = \mathrm{Law}\left(\sum_{j\ge 0} \varepsilon_j x^j\right), \qquad \Pr(\varepsilon_j = \pm 1) = \tfrac{1}{2},$$

and denote by $\mathcal{F}$ the additional hypothesis $\dim_H \nu_\gamma < 1$. All entropies in this section use natural logarithms.

The imported facts are the following: transcendental parameters in $(1/2, 1)$ have full dimension; dimension drop produces height-one polynomials of arbitrarily strong exponential smallness; and for algebraic x the dimension is $\min(1, h_x/\log(1/x))$, where h_x is entropy per digit. These are recorded with their sources in [48, Theorems 3, 6, 8], using Hochman's theorem [49]. They are external theorems, not new assertions proved by the present program.

Proposition 44.1 (Arithmetic consequences of dimension drop)**.** *If $\mathcal{F}$ holds, then γ is an algebraic unit, is a root of a nonzero polynomial with coefficients in $\{-1,0,1\}$, and satisfies*

$$\frac{1}{2} < |\sigma(\gamma)| < 2 \quad \textit{for every conjugate}, \qquad |N(\gamma)| = 1,$$
$$[\mathbb{Q}(\gamma):\mathbb{Q}] \geq 4, \qquad \gamma^{-1} \leq M(\gamma) < 16.$$

In particular every non-Archimedean absolute value of γ equals 1.

Proof. Varjú's theorem first forces algebraicity. Let d and M be its hypothetical degree and Mahler measure. For every fixed $0 < \theta < 1/M$, dimension drop gives nonzero polynomials Q_n of degree at most n and coefficients in $\{-1,0,1\}$ with $|Q_n(\gamma)| < \theta^n$ for large n. Nonzero values would instead be at least $M^{-n}(n+1)^{1-d}$ by the resultant. Thus some $Q_n(\gamma)$ equals zero.

Remove any power of X and change sign to make its leading coefficient 1. Its constant coefficient is ± 1, hence γ and γ^{-1} are algebraic integers. If a root z of this polynomial had $|z| \geq 2$, its leading term would strictly exceed the sum of all lower terms in modulus. Applying the same argument to the reciprocal gives $|z| > 1/2$. All conjugates obey these bounds. The unit norm identity yields

$$M(\gamma) = \prod_{|\sigma(\gamma)|<1} |\sigma(\gamma)|^{-1} \geq \gamma^{-1}.$$

For the upper Mahler bound, the entropy estimate of Breuillard–Varjú [50, Proposition 13], specialized to algebraic units and a fair sign E, gives $h_x \geq \Phi(M(x))$, where for an independent standard Gaussian G,

$$\Phi(a) = \sup_{t>0}\{h_{\mathrm{diff}}(taE+G) - h_{\mathrm{diff}}(tE+G)\}.$$

Choose $t = a^{-1/2}$. The difference equals $I(E;\sqrt{a}E+G) - I(E;a^{-1/2}E+G)$. A sign decision in the first channel has error probability at most $e^{-a/2}$. Its binary conditional entropy is at most $2e^{-a/4}$, since $h_b(p) \leq 2\sqrt{p}$. The second information is at most $1/(2a)$ by comparison of its two shifted Gaussian components with $N(0,1)$ and nonnegativity of mixture relative entropy. Hence

$$\Phi(a) \geq \log 2 - 2e^{-a/4} - \frac{1}{2a}.$$

For $a \geq 16$, use $e^4 > 50$ and $\log(11/10) > 1/11 > 57/800$ to obtain

$$\Phi(a) > \log 2 - 57/800 > \log(20/11).$$

Since $\gamma > 11/20$, $M(\gamma) \geq 16$ would force $h_\gamma > \log(1/\gamma)$ and therefore full dimension, contradicting $\mathcal{F}$.

For the degree bound, the elementary harmonic/logarithm enclosure certifies $577/1000 < \gamma < 578/1000$. A monic unit polynomial of degree at most 3 whose roots all have modulus below 2 belongs to one of the following larger coefficient boxes:

$$X + c \ (c = \pm 1), \quad X^2 + aX + c \ (|a| \leq 3, c = \pm 1), \quad X^3 + aX^2 + bX + c \ (|a| \leq 5, |b| \leq 7, c = \pm 1).$$

These contain 2, 14, and 330 polynomials. At $m = 231/400$ and $r = 1/2000$, the rational lower bound

$$|P(x)| \geq |P(m)| - r\sum_j j|p_j|$$

is positive for all of them throughout the gamma interval. Direct finite evaluation gives the respective minima 211/500, 13999/160000, and 677991/64000000. This excludes all three degrees under the unit and conjugate restrictions. □

The final finite enumeration is reproducible in the archived `verify_fractal_subworlds.py`; it does not restrict the degree of γ without the hypothesis $\mathcal{F}$.

There are actual algebraic dimension-drop models in the wider interval $(1/2, 3/5)$. Let β be the real root of $X^3 - X^2 - X - 1$ and $\lambda = \beta^{-1}$. The discriminant is -44, and the other conjugates of β have modulus $\beta^{-1/2} < 1$, so β is a Pisot unit. Moreover $\lambda \in (1/2, 3/5)$, $M(\lambda) = \beta < 2$, and there is no unit-circle conjugate. The strict entropy bound of [50, Proposition 7] gives $h_\lambda < \log M(\lambda) = \log(1/\lambda)$, hence dimension drop. This parameter lies outside the narrower gamma interval. The inference uses the entropy theorem, not the invalid general implication from singularity to dimension drop.

Conversely, full dimension does not force transcendence: $4/7$ is algebraic in the same broad interval and cannot be a root of any nonzero $\{-1, 0, 1\}$ polynomial by the rational-root theorem, so the imported dimension theorem gives full dimension. Neither $\mathcal{F}$ nor its negation has been established for γ here. Proving $\mathcal{F}$ would support algebraicity, whereas excluding it alone would not prove transcendence.

44.1 The exact logical closure and the finite record

Let A mean algebraicity of γ, B its transcendence, and C the definitions and proved statements used above. The orbit result has the form

$$\mathrm{C} + \mathrm{A} + \text{orbit-family existence} \implies \bot.$$

The fractal result gives $\mathrm{C} + \mathrm{B} + \mathcal{F} \implies \bot$, and excludes combinations of $\mathcal{F}$ with nonunit behavior, a forbidden conjugate, degree at most 3, or Mahler measure at least 16. None is a contradiction from A alone or B alone. A finite implication registry can trace minimal sets of extra assumptions; it is not a general theorem-discovery engine, a consistency proof, or a proof of independence.

The archived finite records preserve several different experiments without identifying their outputs:

- Full-box separation at $D = 2, 3, 4, 5$ was certified using 27, 256, 3125, and 46656 grid values. The associated positive-gap polynomials are $2X^2 - X$, $3X^3 - 3X^2 - X + 1$, $3X^4 + 3X^3 - 4X^2 - X + 1$, and $-3X^5 + 3X^4 - 5X^3 + 2X^2 + 2X - 1$. These exclude only the respective simultaneous degree-and-height boxes.

- The stricter canonical midpoint rule was independently executed for $D = 2, 3, 4$, with $k_D = 6, 10, 14$. Its outputs are the first three polynomials just listed and are squarefree. Their midpoint denominators retain the primes 37, 521, and 8209. The $D = 5$ finite-box experiment is not asserted to be that canonical output.

- A separate primitive-irreducible search produced degrees $1, 2, 4, 6, 8, 10, 12$ and certified $\mathcal{R}_\gamma(P) > 0, 1, 3, 4, 5, 6, 6$, respectively. Irreducibility was checked at primes $2, 3, 17, 11, 2, 19, 101$. These do not satisfy the same height-at-most-D schedule. A certified upper value U and the rational inequality $U(3^{\deg P} \max(2, H(P)))^s < 1$ imply $\mathcal{R}_\gamma(P) > s$, since $e < 3$.

- The full-lattice positive candidate at pole order 8 had $(q, p) = (104, 60)$ and coefficient vector $(52, 17, 3, -16, 25, -21, 7)$. It is exactly half the previous $(208, 120)$ kernel and reduces to the same primitive form $26\gamma - 15$, so it is not counted as a new approximation. Other archived positive candidates reduce to $7\gamma - 4$ and $201\gamma - 116$; solver timeouts certify no optimum.

- The first six lower and upper Padé identities were checked exactly; the sixth bracket width is below $3.772 \cdot 10^{-6}$. The all-order existence and rate claims rest on the proofs above, not this sample.

The source packages retain exact coefficients, rational intervals and independent check scripts. Their reported finite verification counts are experimental provenance, not a substitute for any quantified theorem. The incomplete assertions remain unbounded nonvanishing of the canonical outputs, an infinite local multiplicity escape, a common-denominator separated-error family, and the orbit improvement required in (43.2) or (43.3).

Part VI
The two alternatives and local arithmetic

45 The two arithmetic cases and the obligation to preserve relations

This part consolidates the two-world audit, the algebraic-system notes, the one-prime investigation, and their cross-field extensions. Throughout, the actual real number under investigation is fixed. We use the labels $\mathrm{A}_x : x \in \overline{\mathbb{Q}}$ and $\mathrm{B}_x : x \notin \overline{\mathbb{Q}}$, and specialize to $x = \gamma$ only after proving the general statements. These labels do not describe newly constructed models of the full definition of Euler's constant. None of the conclusions below determines which case holds for γ.

The logical bookkeeping is simple but essential. If a body of established facts $\mathcal{T}$ proves both $\mathrm{A}_\gamma \Rightarrow C$ and $\neg C$, it proves transcendence. If instead the derivation uses an additional hypothesis H, then a contradiction proves $\neg \mathrm{A}_\gamma \vee \neg H$. One cannot discard H without a separate proof. Several unsuccessful transfers in this part fail precisely at this point.

45.1 The relation ideal, derivations, and formal motions

For a real number x, put

$$I_x = \{P \in \mathbb{Q}[X] : P(x) = 0\}, \qquad R_x = \mathbb{Q}[x], \qquad K_x = \mathbb{Q}(x).$$

The evaluation presentation $R_x \simeq \mathbb{Q}[X]/I_x$ always holds. In the algebraic case, let $f \in \mathbb{Z}[X]$ be the primitive irreducible minimal polynomial, with positive leading coefficient, and write $d = \deg f$. Then $I_x = (f)$, $R_x = K_x$, and $[K_x : \mathbb{Q}] = d$. In the transcendental case these objects are respectively $0, \mathbb{Q}[X], \mathbb{Q}(X)$. In particular,

$$\dim_{\mathbb{Q}} \langle 1, x, \dots, x^D \rangle = \begin{cases} \min(D+1, d), & \mathrm{A}_x, \\ D+1, & \mathrm{B}_x. \end{cases} \tag{45.1}$$

The ground field in this formula cannot be replaced by $\mathbb{R}$: real coefficient evaluation has rank one in either case.

Proposition 45.1 (Equivalent forms of algebraic freedom)**.** *For a real number x, the following conditions are equivalent:*

1. *x is transcendental;*
2. *a $\mathbb{Q}$-derivation $D : K_x \to K_x$ satisfies $Dx = 1$;*
3. *$\dim_{K_x} \Omega_{K_x/\mathbb{Q}} = 1$;*
4. *$P(x) = 0$ implies $P'(x) = 0$ for every $P \in \mathbb{Q}[X]$;*
5. *a $\mathbb{Q}$-algebra homomorphism $\Phi : K_x \to K_x[[t]]$ has constant term the identity and satisfies $\Phi(x) = x + t$;*
6. *a $\mathbb{Q}$-automorphism of K_x sends x to $x + 1$.*

In the algebraic case every $\mathbb{Q}$-derivation is zero, and every formal self-deformation with identity constant term is constant.

Proof. In the transcendental case use ordinary differentiation and substitution in $\mathbb{Q}(X)$. For example,

$$D\Big(\frac{P(x)}{Q(x)}\Big) = \frac{P'(x)Q(x) - P(x)Q'(x)}{Q(x)^2}, \qquad \Phi\Big(\frac{P(x)}{Q(x)}\Big) = \frac{P(x+t)}{Q(x+t)}.$$

The latter denominator is invertible as a formal power series. In the algebraic case, separability gives $f'(x) \neq 0$; differentiating the relation gives $f'(x)Dx = 0$, and the same equation for Kähler differentials gives $\Omega_{K_x/\mathbb{Q}} = 0$. If a nonzero ideal of relations were stable under formal differentiation, differentiating its element of least positive degree would contradict minimality. If $\Phi(x) - x = ct^m + O(t^{m+1})$, its first nonzero coefficient in $f(\Phi(x))$ would be $cf'(x)$. Finally, an automorphism sending x to $x+1$ would make all the distinct numbers $x+n$ conjugates of x, which is impossible in finite degree. These arguments also prove the asserted rigidity; the standard module formulation is recorded in [51, 52]. □

A derivation on a real subfield which is continuous for the usual topology must vanish: approximate every element by rational numbers. Likewise an order-preserving $\mathbb{Q}$-automorphism of a real subfield is the identity, because a rational number strictly between an element and its purported image contradicts order preservation. Consequently neither continuity nor order preservation may be silently added to the maps in Proposition 45.1.

There is a useful quantitative form of this warning. For

$$s_n = 2\sum_{j=0}^{n-1} \frac{1}{(2j+1)3^{2j+1}}, \quad \rho_n = \frac{3}{4(2n+1)9^n}, \quad u_n = H_{2^n} - ns_n, \tag{45.2}$$

the positive tail of $\log 2 = 2\operatorname{arctanh}(1/3)$ and integral comparison for harmonic numbers give

$$0 < u_n - \gamma < 2^{-n} + n\rho_n. \tag{45.3}$$

Under B_γ, the derivation with $D\gamma = 1$ has

$$\frac{|D(u_n - \gamma)|}{|u_n - \gamma|} > \frac{1}{2^{-n} + n\rho_n} = (1 + o(1))2^n.$$

This is the expected discontinuity, not a contradiction. More generally, for every fixed m,

$$\{(u, Du, \dots, D^m u) : u \in \mathbb{Q}[\gamma]\} \quad \text{is dense in } \mathbb{R}^{m+1}.$$

Indeed the map from the coefficients of a polynomial of degree at most m to its jet at γ is an invertible real linear map; apply it to the dense set of rational coefficient vectors. No integer height bound is present in this density statement.

One can also express the missing assertion as a finite-box estimate: B_γ is equivalent to the existence, for every D, of some finite C_D such that

$$|P'(\gamma)| \le C_D |P(\gamma)| \quad (P \in \mathbb{Z}[X],\ \deg P, H(P) \le D).$$

In case B take the maximum over the finite box; in case A a sufficiently large box contains f, for which the inequality fails. Merely renaming this existence assertion does not supply a bound or prove it.

45.2 Counting distinct values and the relation lattice

Define

$$W_D(x) = \#\left\{ \sum_{j=0}^{D} a_j x^j : (a_0, \dots, a_D) \in \{0, \dots, D\}^{D+1} \right\}, \qquad 0 < x < 1.$$

Proposition 45.2 (Exact counting dichotomy)**.** *If x is transcendental then $W_D(x) = (D+1)^{D+1}$. If x is algebraic with primitive minimal polynomial f of degree d, then*

$$W_D(x) \le 1 + [D(D+1)]^d M(f)^D. \tag{45.4}$$

Consequently

$$x \notin \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad \limsup_{D\to\infty} \frac{\log W_D(x)}{D} = \infty.$$

Proof. Equality of two grid values is a polynomial relation, proving the first assertion. In the algebraic case a nonzero value of an integer polynomial P of degree and height at most D obeys

$$|P(x)| \ge M(f)^{-D}[D(D+1)]^{-(d-1)}.$$

To see this, use the nonzero integer $\mathrm{Res}(f, P)$, and bound each conjugate value other than $P(x)$ by $D(D+1)\max(1, |x_i|)^D$. Since $0 < x < 1$, multiplication gives the stated bound. All grid values lie in an interval of length at most $D(D+1)$; divide that length by the separation bound. The last claim follows because the algebraic bound is exponential times a polynomial, whereas the transcendental count has logarithm asymptotic to $D \log D$. □

The normalized quantity $\log W_D(x)/((D+1)\log(D+1))$ therefore tends to zero in A and is identically one in B. It counts exact values, not coefficient vectors or cells at a prescribed resolution.

In case A and $D \ge d$, Gauss's lemma gives the exact integer relation lattice

$$\{P \in \mathbb{Z}[X]_{\le D} : P(x) = 0\} = f\mathbb{Z}[X]_{\le D-d}. \tag{45.5}$$

Its basis $f, Xf, \dots, X^{D-d}f$ has covolume, in its own real span, at most $\|f\|_2^{D+1-d}$ by Hadamard's inequality. It is saturated: $mP(x) = 0$ with $m \ne 0$ implies $P(x) = 0$. Equivalently its nonzero Smith invariant factors are one. Reduction of the multiplication matrix modulo any prime is injective, since $f \bmod p$ is a nonzero polynomial. Writing $L = \|f\|_1$, the height-D box contains at least

$$(2\lfloor D/L \rfloor + 1)^{D-d+1} - 1 \tag{45.6}$$

nonzero exact relations: multiply f by every coefficient vector in $[-\lfloor D/L \rfloor, \lfloor D/L \rfloor]^{D-d+1}$. Thus an abundance of short exact relations is compatible with A. For $x = 1/2$, the basis from $f = 2X-1$ has Gram matrix with diagonal 5 and adjacent entries -2; its covolume is $\sqrt{(4^{D+1}-1)/3}$, while $W_D(1/2) \le 2D\,2^D + 1$.

Certified interval packing can give lower bounds for W_D. If $|c - x| \le \varepsilon$, two grid polynomials change their difference by at most $B_D\varepsilon$, where $B_D = D^2(D+1)/2$. In A, if $2B_D\varepsilon$ is smaller than the resultant separation above, clustering sorted midpoint values across gaps at most $B_D\varepsilon$ recovers their exact value classes. In B, a sufficiently fine precision separates any fixed finite grid, but no uniform bound on that precision is obtained here. Such a uniform packing theorem at γ would be additional arithmetic input.

45.3 Fractal dimension is a different observable

Section 44 records the imported Varjú and Hochman theorems and the corresponding conditional arithmetic conclusions. For comparison with W_D, retain the rational example $x = 4/7$. Differences of distinct binary words cannot vanish by the rational-root theorem, and their length-n separation is at least $7^{-(n-1)}$. Thus Hochman's theorem [49] gives full Bernoulli-convolution dimension, while

$$W_D(4/7) \le 1 + \tfrac{7}{3}D\,7^D.$$

The latter follows by counting multiples of 7^{-D} in an interval of length less than $7D/3$. Full dimension and vanishing normalized exact counting growth therefore coexist at a rational parameter.

The scale mismatch can be seen without a dimension theorem. The law of $\sum_{j=0}^{D} A_j x^j$, with digits uniform in $\{0,\dots,D\}$, has support in $[0, D/(1-x)]$. At interval resolution e^{-cD} its Shannon entropy is at most

$$\log\left(2+\frac{De^{cD}}{1-x}\right) = cD + O_x(\log D)$$

for every $x \in (0,1)$. Fixed exponential resolution cannot read the $D\log D$ exact atomic count. A dimension-zero exceptional set also need not exclude a specified computable parameter.

46 One-prime character counts and optimal finite camouflage

All homomorphisms in this section preserve the identity and fix the integers. Set

$$n_p(x) = \#\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}[x], \mathbb{F}_p), \qquad N_{p,k}(x) = \#\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}[x], \mathbb{Z}/p^k\mathbb{Z}).$$

For algebraic x, primitivity and Gauss's lemma give $\mathbb{Z}[x] \simeq \mathbb{Z}[X]/(f)$, even when f is not monic. Thus these counts are precisely the respective root counts of f. For transcendental x the source is the free polynomial ring, so

$$n_p(x) = p, \qquad N_{p,k}(x) = p^k. \tag{46.1}$$

In either case, a proposed image a in a target ring S defines an actual homomorphism exactly when

$$P(x) = 0 \quad \Longrightarrow \quad P(a) = 0 \text{ in } S \qquad \text{for every } P \in \mathbb{Z}[X]. \tag{46.2}$$

This implication is the unresolved input at $x = \gamma$.

46.1 All primes and exact eventual counts at one prime

Proposition 46.1 (Prime characters)**.** *For every real x,*

$$\left[\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}[x], \mathbb{F}_p) \neq \varnothing \text{ for every prime } p\right] \Longleftrightarrow x \in \mathbb{Z} \text{ or } x \notin \overline{\mathbb{Q}}.$$

If the condition is imposed only at all sufficiently large primes, the right side becomes $x \in \mathbb{Q}$ or $x \notin \overline{\mathbb{Q}}$. Requiring only infinitely many primes does not distinguish the cases. For algebraic x,

$$\sum_{p \le X} n_p(x) \sim \operatorname{Li}(X),$$

whereas for transcendental x the sum is asymptotic to $X^2/(2\log X)$.

Proof. For $x = a/b$ in lowest terms, $b > 0$, a prime dividing $b > 1$ prevents any map: the relation $bx = a$ would become $0 = \bar{a}$. At all other primes the rational case has a unique image. In degree $d > 1$, the Galois group of the splitting field acts transitively on the roots of f. Its average number of fixed points is one, while the identity has $d > 1$ fixed points. It therefore contains a fixed-point-free element. Chebotarev supplies a positive-density set of primes with this Frobenius class, at which f has no root. Exclude the finitely many leading-coefficient and discriminant primes in making the Frobenius identification. The same theorem applied to the identity class supplies infinitely many primes with roots. Averaging the fixed point character gives the algebraic asymptotic; in the transcendental case use (46.1) and partial summation of the prime number theorem. The needed Frobenius correspondence and density theorem are recorded in [53, 54]. □

More generally, if m_r is the number of Galois orbits on ordered r-tuples of roots, then $\pi(X)^{-1}\sum_{p\le X} n_p(x)^r \to m_r$. In particular $m_1 = 1$ and the limiting variance is $m_2 - 1$. These are conditional fingerprints, not measured values of $n_p(\gamma)$.

Theorem 46.2 (Exact eventual one-prime count)**.** *Let x be algebraic with primitive minimal polynomial f, and choose $A, B \in \mathbb{Z}[X]$, $M \in \mathbb{Z}\setminus\{0\}$, such that $Af + Bf' = M$. For a fixed prime p, put $s = v_p(M)$. Then*

$$k > 2s \quad\Longrightarrow\quad N_{p,k}(x) = \sum_{\substack{\alpha\in\mathbb{Z}_p\\ f(\alpha)=0}} p^{v_p(f'(\alpha))} \le dp^s. \tag{46.3}$$

The sum is zero if f has no integral p-adic root. In particular $N_{p,k}(x)$ is eventually constant.

Proof. The Bézout identity and $f(a) \equiv 0 \pmod{p^k}$ imply $t = v_p(f'(a)) \le s$. Since $k > 2t$, the strong Hensel lemma [55, Theorem 4.1] gives a root $\alpha \in \mathbb{Z}_p$ with $v_p(a-\alpha) \ge k - t > s$. As f' has integral coefficients, $v_p(f'(\alpha)) = t$. Conversely, the residues in

$$a \equiv \alpha \pmod{p^{k-s_\alpha}}, \qquad s_\alpha = v_p(f'(\alpha)),$$

all solve the congruence, by Taylor expansion; each such ball contains p^{s_α} residues modulo p^k. The balls are disjoint. If two overlapped, distinct roots α, β would satisfy $v_p(\alpha-\beta) > s$; division by both monic linear factors in $\mathbb{Z}_p[X]$ would then force $v_p(f'(\alpha)) > s$. Summing the disjoint contributions proves the formula. □

Corollary 46.3 (One-prime dichotomy)**.** *For any fixed prime p and every real x,*

$$x \notin \overline{\mathbb{Q}} \iff \sup_k N_{p,k}(x) = \infty, \qquad \lim_{k\to\infty} \frac{\log_p(1 + N_{p,k}(x))}{k} = \begin{cases} 0, & x \in \overline{\mathbb{Q}},\\ 1, & x \notin \overline{\mathbb{Q}}. \end{cases}$$

Also

$$\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[x], \mathbb{Z}_p) \simeq \varprojlim_k \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[x], \mathbb{Z}/p^k\mathbb{Z}),$$

and this set is infinite if and only if x is transcendental.

Proof. Combine (46.1) and Theorem 46.2. The inverse-limit statement follows by taking componentwise ring operations. Its left side has at most d elements in degree d and is in bijection with $\mathbb{Z}_p$ in the transcendental case. □

Finite-level cardinalities need not count infinite branches. For $x = \sqrt{17}$, $N_{2,k}(x) = 4$ for $k \ge 3$, although only two maps to $\mathbb{Z}_2$ exist. Indeed $-2(X^2 - 17) + X(2X) = 34$, and both 2-adic roots have derivative valuation one. The roots modulo 8 are $1, 3, 5, 7$, whereas the roots modulo 16 are $1, 7, 9, 15$; only two of the former four branches lift at that step.

There is a second counting trap. If J consists of finitely many verified relations and $J \subset I = \ker(\mathbb{Z}[X] \to \mathbb{Z}[x])$, then

$$\mathrm{Hom}(\mathbb{Z}[X]/I, S) \hookrightarrow \mathrm{Hom}(\mathbb{Z}[X]/J, S). \tag{46.4}$$

Thus the number of not-yet-excluded images is an *upper* bound on actual maps. It cannot provide the lower bound required by Corollary 46.3.

46.2 A height-free bound valid at every depth

Theorem 46.4 (Primitive polynomial root bound)**.** *Let $f \in \mathbb{Z}_p[X]$ have positive degree d, with at least one unit coefficient. For every $k \geq 1$,*

$$\#\{a \bmod p^k : f(a) \equiv 0 \pmod{p^k}\} \leq \min\{p^k, dp^{k-\lceil k/d \rceil}\}. \tag{46.5}$$

It follows that $N_{p,k}(x)$ satisfies this bound for every algebraic x of degree d, including nonintegral numbers and bad primes.

Proof. Factor $f(X) = c\prod_{i=1}^{d}(X - \alpha_i)$ in $\mathbb{C}_p$. Multiplicativity of the Gauss norm gives $1 = |c|_p \prod_i \max(1, |\alpha_i|_p)$. For $a \in \mathbb{Z}_p$, the factors from roots outside the unit disc cancel, leaving

$$|f(a)|_p = \prod_{|\alpha_i|_p \leq 1} |a - \alpha_i|_p.$$

If this product is at most p^{-k}, one of its at most d factors has valuation at least k/d. Any two integral points in the ball around that root differ by an element of valuation at least k/d, hence are congruent modulo $p^{\lceil k/d \rceil}$. Such a ball contains at most $p^{k-\lceil k/d \rceil}$ residues modulo p^k. Take the union over at most d roots. Primitivity supplies the unit coefficient for a minimal polynomial at every prime. □

If M_0 distinct genuine maps at depth k satisfy

$$M_0 > Dp^{k-\lceil k/D \rceil},$$

then algebraic degree at most D is excluded. This is a criterion without a height parameter, but no such lower bound for the actual γ-ring is established here.

46.3 The exact minimum degree of finite camouflage

For $M \geq 2$, define $\mu(M) = \min\{m \geq 0 : M \mid m!\}$. The relevant classical mechanism is fixed divisors, or the elementary $\mathbb{Z}$-case of generalized factorials [56].

Lemma 46.5 (Fixed-divisor bound)**.** *For $0 \neq f \in \mathbb{Z}[X]$ of degree d, let $c(f)$ be its coefficient content and let $\mathrm{fd}(f) = \gcd\{f(a) : a \in \mathbb{Z}\}$. Then $\mathrm{fd}(f) \mid d!c(f)$.*

Proof. The Newton expansion

$$f(X) = \sum_{j=0}^{d} \Delta^j f(0) \binom{X}{j} = \sum_{j=0}^{d} \frac{\Delta^j f(0)}{j!} (X)_j$$

has every $\Delta^j f(0)$ divisible by the fixed divisor. Multiplication by $d!$ shows that this divisor divides every coefficient of $d!f$, whose gcd is $d!c(f)$. □

Theorem 46.6 (Optimal finite camouflage)**.** *Fix finitely many primes $S \neq \varnothing$, depths $L_p \geq 1$, and a nonempty open real interval J. Put $M = \prod_{p \in S} p^{L_p}$. The least possible degree of an algebraic $x \in J$ satisfying*

$$N_{p,k}(x) = p^k \qquad (p \in S,\ 1 \leq k \leq L_p) \tag{46.6}$$

is exactly $\mu(M)$. This minimum is attained by an algebraic integer. Allowing nonmonic minimal polynomials cannot reduce it.

Proof. If f is the primitive minimal polynomial of any such x, then (46.6) says that all integer values of f are multiples of M. Lemma 46.5 gives $M \mid d!$, so $d \geq \mu(M)$.

For existence let $m = \mu(M) \geq 2$, $Q(X) = (X)_m$, and choose a prime $\ell \notin S$. Coefficientwise Chinese remaindering produces a monic degree-m polynomial f_0 congruent to Q modulo M, with every lower coefficient divisible by ℓ, and constant coefficient congruent to ℓ modulo ℓ^2. Choose a rational subinterval $(u, v) \subset J$ and $a/b \in (u, v)$, $b > 0$. Set

$$f_A(X) = f_0(X) + M\ell^2 A(bX - a), \qquad A \in \mathbb{Z}_{>0}.$$

This preserves the degree, monicity, congruences, and Eisenstein irreducibility at ℓ. For large A, its signs at u, v are opposite, so it has a real root in J. Since $(a)_m = m!\binom{a}{m}$ is divisible by M for every integer a, all prescribed residue classes are roots. The presentation of $\mathbb{Z}[x]$ proves (46.6). □

The construction also reveals a separating measurement outside the prescribed list: $N_{\ell,1}(x) = 1$ and $N_{\ell,2}(x) = 0$. Modulo ℓ its polynomial is X^m; a root modulo ℓ^2 would be divisible by ℓ, which leaves the nonzero constant term modulo ℓ^2.

Legendre's formula gives

$$\mu(p^L) = (p-1)L + O_p(\log(L+1)), \qquad \mu\Big(\prod p^{L_p}\Big) = \max_p \mu(p^{L_p}).$$

For example, the optimal degrees at $p = 2$ and depths $3, 7, 10, 100$ are $4, 8, 12, 104$, respectively. The earlier construction of degree p^L was valid but is superseded as an optimization claim. An exact degree-four example in $(577215/10^6, 577216/10^6)$ is the root of

$$X^4 + 18X^3 + 27X^2 + 144000018X - 83119056.$$

It is Eisenstein at 3 and is a null polynomial modulo 8. The earlier fourth-degree example $X^4 - 6X^3 + 3X^2 + 28794X - 16620$, with a root in $(577/1000, 578/1000)$, instead has counts $2, 4, 0, 0, \ldots$. Both are finite-data comparisons, not alternate realizations of γ's full definition.

Corollary 46.7 (Exact full-residue depth)**.** *The first depth at which full counts exclude every algebraic degree $\leq D$ is*

$$k_D = v_p(D!) + 1.$$

For $p = 2$, this is $D - s_2(D) + 1$, where s_2 is binary digit sum. In particular, $N_{2,9}(\gamma) = 512$, if proved by genuine relation-preserving maps, would exclude degree at most 10.

Proof. Full counts in degree $d \leq D$ force $p^k \mid d! \mid D!$. Conversely, for every $k \leq v_p(D!)$, Theorem 46.6 supplies a witness of degree at most D in any desired interval. □

No assertion about $N_{2,9}(\gamma)$ is proved by this corollary.

47 Real limits, arithmetic places, and invariant analytic data

47.1 The original approximants and denominator support

The real convergence in (45.3) has a different 2-adic behavior:

$$v_2(u_n) = -n. \tag{47.1}$$

Indeed $1/2^n$ is the unique term of H_{2^n} with valuation $-n$; every denominator in s_n is odd. Thus the original upper approximants cannot be reduced directly to integral residue rings at 2. The later simultaneous real/2-adic construction repairs this particular defect but still requires relation

preservation. Equation (47.1) is retained as the exact scope of the older failed construction, not as an obstruction to all approximations.

For a nonzero reduced rational $r = a/b$, $|r| \le B$, and prescribed $v_p(r) \ge k_p \ge 0$ at finitely many primes, one necessarily has

$$b \ge B^{-1} \prod_p p^{k_p}. \tag{47.2}$$

The numerator is divisible by that product, and $|a| \le Bb$. Moreover a sequence with denominator primes in a fixed finite set S, which converges in $\mathbb{R}$ and at every place of S, is eventually constant. Its differences are small at all these places, whereas a nonzero element of $\mathbb{Z}[S^{-1}]$ satisfies $|r|_\infty \prod_{p\in S} |r|_p \ge 1$, by the product formula and integrality outside S.

Uniform support is indispensable. The sequence

$$r_n = \frac{(n!)^2}{2(n!)^2 + 1} \tag{47.3}$$

converges to $1/2$ in $\mathbb{R}$ and to 0 in every fixed $\mathbb{Q}_p$. For large n, the denominator is a p-adic unit and the numerator valuation is $2v_p(n!) \to \infty$. Nevertheless no unital map $\mathbb{Z}[1/2] \to \mathbb{F}_2$ exists. Even convergence at every finite place does not imply descent of the real limit's relations. The assertion concerns separate convergence at each place, not convergence in the adelic restricted-product topology. More generally, finite Chinese remaindering with sufficiently large denominators, followed by a diagonal construction, realizes arbitrary prescribed p-adic integer limits together with an arbitrary real limit.

47.2 Exact valuations of the canonical rational midpoint

Use the midpoint c_D of (40.4), put $N = 2^k$, and write $c_D = A_D/B_D$ in lowest terms, $B_D > 0$. To avoid confusing the small logarithmic tail with a rational approximation, denote that tail bound here by ρ_k, as in (45.2). Then

$$c_D = H_N - ks_k - \frac{1}{2N} - \frac{k\rho_k}{2}.$$

The canonical schedule has $k \ge 6$ and $N/2 > \max(D, 2k+1)$. Let $\Pi_N = \prod_{N/2<p<N} p$, over primes in the indicated interval. Only the harmonic term $1/p$ is nonintegral at such a prime, whereas $-1/(2N)$ is the unique term with smallest 2-adic valuation. Consequently

$$v_2(c_D) = -(k+1), \qquad v_p(c_D) = -1 \quad (N/2 < p < N). \tag{47.4}$$

If $\ell_N = \operatorname{lcm}(1, \dots, N)$, inspection of all explicit denominators gives

$$2N\Pi_N \mid B_D, \qquad B_D \mid 8 \cdot 9^k \ell_N, \qquad N^2 < B_D \le 8 \cdot 9^k 4^{N-1}. \tag{47.5}$$

The strict lower bound uses Bertrand's postulate; the last elementary upper bound follows by iterating $\ell_{2n}/\ell_n \mid \binom{2n}{n} \le 4^n$.

For every polynomial $P(X) = \sum_{j=0}^{D} h_j X^j$ of degree exactly D, with $H(P) \le D$, the leading term is uniquely smallest at these places. Hence

$$v_p(P(c_D)) = -D \ (N/2 < p < N), \qquad v_2(P(c_D)) = v_2(h_D) - D(k+1). \tag{47.6}$$

Equivalently the homogenized integer numerator

$$T_P(A_D, B_D) = \sum_{j=0}^{D} h_j A_D^j B_D^{D-j}$$

is coprime to Π_N and has 2-adic valuation $v_2(h_D)$. In particular

$$|P(c_D)| \geq B_D^{-D} \geq (8 \cdot 9^k 4^{N-1})^{-D}.$$

This lower bound is compatible with the canonical small real value. Its guaranteed logarithmic scale is $-O(DN)$, whereas the target has scale $-D\log D + O(\log D)$. Large denominator valuations permit small real values in the product formula; they are not numerator divisibilities.

Under a hypothetical factorization $P_D = f^r C$, the large-prime valuations satisfy

$$-r\deg f - (D - r\deg f) = -D.$$

They impose no inconsistent constraint on r. A concrete comparison makes this compatibility explicit: for $D \geq 3$, put

$$c^* = \frac{1}{2} + \frac{1}{2N\Pi_N}, \quad r = \left\lfloor \frac{\log D}{\log 3} \right\rfloor, \quad Q_D(X) = (2X-1)^r X^{D-r}.$$

Then $\deg Q_D = D$, $H(Q_D) \leq 3^r \leq D$, $Q_D(1/2) = 0$, while

$$0 < Q_D(c^*) = (N\Pi_N)^{-r}(c^*)^{D-r} < 2U_D.$$

Its valuations coincide with (47.4) and (47.6), with leading coefficient 2^r. The comparator is not the actual harmonic midpoint or the canonical minimizer. It shows that these particular real and local properties can coexist with a repeated algebraic factor; an additional estimate using the actual harmonic numerator would be required to separate them.

47.3 All Hankel tests and a freely movable mean

Let E_0 have the standard exponential distribution and define

$$G_a = -\log E_0 + a - \gamma \qquad (a \in \mathbb{R}).$$

The Gamma integral and its Taylor expansion [57] give

$$M_a(z) = \mathbb{E}(e^{zG_a}) = e^{(a-\gamma)z}\Gamma(1-z) = \exp\left(az + \sum_{n\geq 2} \frac{\zeta(n)}{n} z^n\right), \qquad |z| < 1. \tag{47.7}$$

Thus $\mathbb{E}G_a = a$. If $m_n(a) = \mathbb{E}(G_a^n)$,

$$m_0(a) = 1, \qquad m_{n+1}(a) = a m_n(a) + \sum_{j=1}^{n} \binom{n}{j} j!\zeta(j+1) m_{n-j}(a). \tag{47.8}$$

Proposition 47.1 (Shift invariance of all principal Hankel determinants)**.** *For every $N \geq 1$ and every real a,*

$$\Delta_N(a) = \det(m_{i+j}(a))_{0\leq i,j<N} = Delta_N(0) > 0.$$

In particular,

$$\Delta_2 = \zeta(2), \qquad \Delta_3 = 2\zeta(2)^3 + 6\zeta(2)\zeta(4) - 4\zeta(3)^2.$$

Proof. Translation of the polynomial basis $1, X, \ldots, X^{N-1}$ is a triangular change of basis with diagonal entries one. The corresponding moment matrices are congruent by this matrix, preserving determinant. The density of G_a is positive on all of $\mathbb{R}$, so the integral of the square of a nonzero polynomial is strictly positive. Its moment Gram matrix is therefore positive definite. □

The entire infinite family is compatible with every mean. For example, when $a = 1/2$, $P(X) = 2X - 1$ satisfies $P(a) = 0$, while $\mathbb{E}(P(G_a)^2) = 4\zeta(2) > 0$. Positivity of a random polynomial's square is not nonvanishing at its mean. The later tensor-statistic construction refines this distinction by computing the exact variance terms and their signed cancellation; it does not turn this older Hankel family into an arithmetic detector.

47.4 Renormalization, multiplication, and the full Lévy measure

Proposition 47.2 (Complete maximum fixed-point family)**.** *Let F be a continuous distribution function with $0 < F(t) < 1$ for every real t. Then*

$$F(t + \log m)^m = F(t) \quad (m = 2, 3, \ldots)$$

holds if and only if $F(t) = \exp(-Ce^{-t})$ for some $C > 0$.

Proof. Put $h = -\log F > 0$. The equations give $h(t + \log(m/n)) = (n/m)h(t)$ for positive integers m, n. Density of the positive rationals and continuity extend this to $h(t + s) = e^{-s}h(t)$ for all real s. Take $t = 0$. Direct substitution proves the converse. □

The mean in this family is $\gamma + \log C$, so every real mean occurs. Equivalently, for independent copies, $\max(G_{a,1}, \ldots, G_{a,m}) - \log m$ has the law of G_a for all a and all m.

Gauss multiplication applied to (47.7) gives

$$\prod_{r=0}^{m-1} M_a\Big(\frac{z+r}{m}\Big) = (2\pi)^{(m-1)/2} m^{z-1/2} e^{(a-\gamma)(m-1)/2} M_a(z), \tag{47.9}$$

$$\prod_{r=0}^{m-1} \frac{M_a((z+r)/m)}{M_a(r/m)} = m^z M_a(z). \tag{47.10}$$

These identities hold initially where the factors are finite and then meromorphically. Normalization at $z = 0$ removes the entire drift factor; all the normalized identities therefore hold at every real mean.

There is also a construction which never uses a decimal value of γ. For independent standard exponentials E_n, the sums

$$S_{N,a} = a + \sum_{n=1}^{N} \frac{E_n - 1}{n}$$

are Cauchy in L^2, and their characteristic functions converge to $M_a(it)$ by the Gamma product. Equivalently, $\max(E'_1, \ldots, E'_N) - H_N + a$ has the same finite law: both uncentered moment generating functions are $\prod_{n=1}^{N} n/(n - z)$. This construction works at rational a just as well as at $a = \gamma$.

Proposition 47.3 (Infinite divisibility with fixed Lévy measure)**.** *Every G_a is infinitely divisible, and for $\Re z < 1$,*

$$\log M_a(z) = az + \int_0^\infty \frac{e^{zx} - 1 - zx}{x(e^x - 1)}\, dx. \tag{47.11}$$

Only the drift a varies; the positive Lévy measure is $\mathbf{1}_{x>0}\, dx/[x(e^x - 1)]$.

Proof. Near zero the compensated integrand is bounded; at infinity it is integrable in the stated half-plane. Expanding for $|z| < 1$ and using

$$\int_0^\infty \frac{x^{n-1}}{e^x - 1}\, dx = (n-1)!\zeta(n) \qquad (n \geq 2)$$

proves the identity, followed by analytic continuation. To construct convolution roots directly, take independent Gamma variables $V_{n,q}$ of shape $1/q$ and scale $1/n$. The centered sum

$$a/q + \sum_{n \geq 1} \Big(V_{n,q} - \frac{1}{qn} \Big)$$

converges in L^2, with variance $\zeta(2)/q$. The sum of q independent copies has the law of the limit of $S_{N,a}$, because Gamma shapes add. □

The exponentiated variable is $e^{-G_a} = \lambda_a E_0$, where $\lambda_a = e^{\gamma - a}$. Its Laplace transform and moments satisfy

$$L_a(s) = \frac{1}{1+\lambda_a s}, \qquad (-1)^j L_a^{(j)}(s) = \frac{j!\lambda_a^j}{(1+\lambda_a s)^{j+1}} > 0, \qquad \mathbb{E}(e^{-nG_a}) = n!\lambda_a^n.$$

Thus complete monotonicity and the positive moment recursion also allow every mean. Gamma reflection gives two further shift-invariant identities:

$$M_a(z)M_a(-z) = \frac{\pi z}{\sin \pi z}, \qquad |M_a(it)|^2 = \frac{\pi t}{\sinh \pi t}.$$

Their modulus discards the phase which carries the location parameter.

Within this family, the normalization selecting the actual constant is exactly

$$a = \gamma \iff M_a(-1) = 1 \iff \lambda_a = 1 \iff F_a(0) = e^{-1} \iff e^{(a-\gamma)(m-1)/2} = 1 \tag{47.12}$$

for one fixed integer $m \geq 2$. These are several forms of one normalization, not independent arithmetic constraints. Subtracting $\log N$ from the finite maximum selects γ; subtracting $H_N - a$ selects a. Prescribing the divergent centering term alone leaves a finite location freedom.

Hermite–Lindemann implies that among algebraic real a, at most one can have λ_a algebraic: two such values would give the algebraic quotient e^{b-a} for nonzero algebraic $b-a$. Under A, that unique value is precisely $a = \gamma$. This is compatible with A. Proving e^γ algebraic would instead prove γ transcendental, since $\gamma \neq 0$, but none of the preceding identities proves that assertion.

Finally, none of the functions M_a is itself an arithmetic E-function or G-function. Its first two Taylor coefficients are

$$[z]M_a = a, \qquad [z^2]M_a = \frac{a^2 + \zeta(2)}{2}.$$

If both were algebraic, $\zeta(2) = \pi^2/6$ would be algebraic. The same argument survives any nonzero algebraic rescaling of z. This excludes a direct invocation of arithmetic E/G-function theorems on this exact family, not every possible transformation of it.

47.5 The earlier curvature and capacity entry tests

The subsequent tensor-category classification supersedes the narrow rank-one twist test, but the original all-prime formula remains useful. With $\Phi = \mathrm{Ein}$ and

$$Y_a = (1, e^{-z}, \Phi - a)^T, \qquad A(z) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ z^{-1} & -z^{-1} & 0 \end{pmatrix}, \qquad \nabla = \partial - A,$$

the p-curvature in characteristic p, including $p = 2$, is

$$\psi_p = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ z^{-p} & -z^{-p}\displaystyle\sum_{j=0}^{p-1} z^j/j! & 0 \end{pmatrix}. \tag{47.13}$$

For completeness, $\nabla e_1 = -z^{-1}e_3$, $\nabla e_2 = e_2 + z^{-1}e_3$, and $\nabla e_3 = 0$. Wilson's congruence gives the first column; iteration gives the second lower entry as $\sum_{j=0}^{p-1}(-1)^j j! z^{-j-1}$, which is the displayed truncated exponential. The matrix has rank two, kernel generated by e_3, and both characteristic and minimal polynomials $T^2(T-1)$.

Tensoring with any rank-one connection shifts the eigenvalues to $c_p, c_p, c_p + 1$. A fixed nonconstant rational pullback $z = h(t)$ multiplies their difference by $h'(t)^p$, nonzero at all good primes; a rational gauge conjugates curvature. Therefore these operations cannot make the entire module globally nilpotent. The connections and transformations must be defined over one fixed number field, with only finitely many primes excluded. These are the standard tensor and pullback rules [62]; no general p-curvature conjecture is assumed. The pure logarithmic subconnection spanned by e_1, e_3 does have nonzero nilpotent curvature, and its arbitrary integration constant does not acquire an arithmetic classification from that fact. The scalar a does not occur in $A(z)$; changing it changes a framed solution, not the differential module.

For the capacity route, the direct fixed-scaling candidate also fails an explicit input test. If B, C are nonzero integers and $p \nmid BC$, then the z^p coefficient of $B\,\mathrm{Ein}(Cz)$ has valuation -2. Thus no fixed integral scaling makes its coefficients all integral. Multiplying coefficient n by $n!$, the inverse Borel convention used here, gives $\log(1+z)$; the same coefficient test now has valuation -1. Differentiation gives a rational function but removes the constant $-a$ under investigation. These facts test only this specific sequence of transformations. They do not prohibit an adelic, nonlinear, multivariable, or moving-denominator capacity construction.

The analytic relationship to the A/B split remains conditional. With $\eta = \mathrm{Ein}(1)$ and $\delta = \int_0^\infty e^{-t}/(1+t)\,dt$, $\eta = \gamma + \delta/e$. The known algebraic independence of e, η [63] implies, under A, the algebraic independence of e, δ, because $\overline{\mathbb{Q}}(e, \eta) = \overline{\mathbb{Q}}(e, \delta)$. It also forces

$$\delta \in (\mathbf{E} \cap \mathbf{D}) \setminus \overline{\mathbb{Q}},$$

where the two rings have the specified meanings of [14]. This contradicts the *conjectural* intersection equality, not a proved equality. Here it is $\delta \in \mathbf{D}$ that is used; division by e is not automatically allowed merely because $\mathbf{D}$ is a ring. The earlier displayed Stokes invariance and the later curvature computation do not add the missing arithmetic lifting theorem.

48 A feasibility obstruction for replication near zero

This section concerns new auxiliary polynomials, not the canonical minimum-gap outputs. Ordinary simultaneous pigeonholing on the grid $\{0, \dots, D\}^{D+1}$, at K points in $(0, 1)$, supplies a nonzero polynomial of degree and height at most D with error

$$\exp(-(1/c + o(1))D \log\log D), \qquad K = \left\lfloor c \frac{\log D}{\log\log D} \right\rfloor.$$

For the rational dilation orbit of a hypothetical degree-d algebraic number, the resultant comparison forces exact zeros when $c < 1/d$, whereas the product of the different minimal-polynomial copies exceeds the Mahler budget only when $c > 3/(2d)$. These two windows do not intersect. Trying to bridge the gap by an arbitrary stronger error requirement can make the proposed construction impossible even without assuming algebraicity.

Theorem 48.1 (Universal obstruction for the orbit tending to zero)**.** *Fix $x \in (0, 1)$, $c > 3/2$, and $\kappa > c$, and put $K = \lfloor c \log D / \log\log D \rfloor$. For all sufficiently large D, no nonzero $P \in \mathbb{Z}[X]$ with $\deg P, H(P) \le D$ satisfies*

$$\max_{1 \le j \le K} |P(x/j)| \le \exp\left(-\kappa \frac{D \log D}{K}\right). \tag{48.1}$$

Proof. Write $P = X^s R$, where $R(0)$ is a nonzero integer. If the right side of (48.1) is ε, then $|R(x/j)| \le \varepsilon (K/x)^D =: \eta$. Interpolate these values by T of degree at most $K-1$. Node separation is at least x/K^2, so the Lagrange basis yields

$$\|T\|_1 \le K\eta (2K^2/x)^{K-1} =: \theta_D.$$

Consequently

$$\log\theta_D \le -\kappa\frac{D\log D}{K} + D\log(K/x) + O_x(K\log K) = -(\kappa/c - 1 + o(1))D\log\log D \to -\infty.$$

The real polynomial $Q = R - T$ has all K roots x/j, with $|Q(0)| \ge 1/2$ for large D, while

$$M(Q) \le \|Q\|_2 \le D\sqrt{D+1} + 1/2.$$

The constant-term form of the Mahler product gives independently

$$M(Q) \ge |Q(0)| \prod_{j=1}^{K} \frac{j}{x} \ge \tfrac{1}{2}K!x^{-K} = D^{c+o(1)}.$$

This contradicts the upper bound $D^{3/2+o(1)}$. Integrality is used before interpolation, to bound the nonzero constant coefficient below; the Mahler inequalities themselves also hold for real or complex coefficients. □

The contradiction eliminates the specified demand, not either arithmetic case. Replacing the nodes by $xj/(j+1)$ removes this particular geometric obstruction: their inverse product is only $(K+1)x^{-K}$. A real-coefficient polynomial with these roots and constant coefficient one has coefficient norm at most $(1+2/x)^K = D^{o(1)}$. Its coefficients need not be integers. Thus this observation neither constructs the desired integer family nor proves its feasibility. The bounded-orbit criterion of Section 43 retains the arithmetic factor cost and the missing simultaneous approximation assumption.

There is also no free recovery of the obstruction by the translation $Y = x - X$. For transcendental x, translating an integer polynomial introduces coefficients involving x; the new constant term is uncontrolled. For instance, if $B_K(X) = \prod_{j=1}^{K}(X - xj/(j+1))$, then $B_K(x) = x^K/(K+1)!$, exactly compensating the apparent factorial from the translated roots.

A related audit of the canonical multiplicity theorem is worth retaining. If $f^{r_D} \mid P_D$, $H(P_D) \le D$, and $\deg P_D \le D$, then

$$r_D \le \frac{\log D + \frac{1}{2}\log(D+1)}{\log M(f)}.$$

For $0 < x < 1$ algebraic, $M(f) > 1$; the usual available competitors $f^m X^{D-md}$, with $\|f\|_1^m \le D$, already allow m of order $\log D$. Hence the lower logarithmic multiplicity and this upper logarithmic budget are compatible. Similarly, after removing roots at zero from an integer polynomial R, the number of its roots in $|z| \le b < 1$, counted with multiplicity, is at most

$$\frac{\log D + \frac{1}{2}\log(D+1) - \log|R(0)|}{\log(1/b)}.$$

This is $O(\log D)$, not the smaller bound that would contradict the canonical algebraic branch.

49 Effective nonvanishing tests and the limits of generic world search

49.1 What can be certified by exact rational computation

A computable real x is supplied with rational approximants q_n and a guaranteed bound $|q_n - x| \le 2^{-n}$. Choose a rational R containing these values in $[-R, R]$. For $P(X) = \sum_{j=0}^{d} a_j X^j$, put $L_P = \sum_{j=1}^{d} j|a_j|R^{j-1}$. Then

$$|P(q_n)| > L_P 2^{-n} \quad\Longrightarrow\quad P(x) \ne 0. \tag{49.1}$$

The test is exact and terminates precisely when $P(x) \neq 0$: once $L_P 2^{-n} < |P(x)|/3$, the inequality holds. A zero value can never pass. If $[a_n, b_n] \subset [0,1]$ encloses γ, its midpoint r_n gives the common finite-box certificate

$$|P(r_n)| > \frac{HD(D+1)}{4}(b_n - a_n) \quad \Longrightarrow \quad P(\gamma) \neq 0 \quad (\deg P \leq D,\ H(P) \leq H). \tag{49.2}$$

Proposition 49.1 (Finite-box termination and least survivor)**.** *The parallel test of all nonzero integer polynomials of degree at most D and height at most H terminates if and only if none vanishes at x. Consequently termination for every pair (D, H) is equivalent to transcendence of x.*

Effectively enumerate the primitive irreducible integer polynomials, with positive leading coefficient, as $f_1, f_2, \ldots$, and delete candidates whenever (49.1) certifies nonvanishing. Let $m_s(x)$ be the least index not deleted after stage s, with all candidates eventually tested. Then m_s is computable and nondecreasing, and

$$\begin{aligned} x \in \overline{\mathbb{Q}}: &\quad m_s(x) \text{ eventually equals the minimal-polynomial index,} \\ x \notin \overline{\mathbb{Q}}: &\quad m_s(x) \longrightarrow \infty. \end{aligned}$$

Proof. There are finitely many tests in a box, so if all its values are nonzero their finite termination times have a finite maximum. Conversely a zero value cannot be certified. For the survivor claim, all candidates preceding a true minimal polynomial are eventually deleted and that polynomial never is. In the transcendental case each fixed finite initial list is eventually deleted. Permanent deletion makes the least surviving index nondecreasing. □

Long observed stability does not certify eventual stability. If algebraicity and explicit degree/height bounds are supplied independently, exact isolation of the finitely many possible roots does identify the number from sufficiently narrow intervals. This is a result under a finite-candidate promise, not a proof of that promise for γ. The archived degree-3, height-4 calculation checked 2,927 primitive polynomials and certified $|P(\gamma)| > 10^{-4}$ throughout that box. It is an independent reproduction of a subset of the earlier degree-4, height-4 verification, not a new exclusion range. The exact scripts and certificates are retained in the accompanying archive.

49.2 Two explicit reductions from the halting problem

Theorem 49.2 (No uniform arithmetic semidecision for computable reals)**.** *On inputs consisting of a program for a fast rational Cauchy name of a real number, neither algebraicity nor transcendence has a semidecision procedure valid for every such input. This remains true under the promise that the input is either rational or transcendental.*

Proof. Fix a computable transcendental $\tau \in (0,1)$, and let $H_e = t$ mean that a machine e first halts at step $t \geq 1$. Define

$$x_e = \begin{cases} 0, & e \text{ never halts}, \\ 2^{-t}\tau, & H_e = t. \end{cases}$$

These reals have fast names uniformly in e: to precision 2^{-n} simulate n steps; if the machine has not halted, any future value is smaller than 2^{-n}, and otherwise compute the indicated multiple of τ. Here x_e is algebraic exactly when e never halts. A semidecision procedure for algebraicity would enumerate nonhalting machines, contradicting the undecidability of halting.

For the other direction choose rational q_t with $|q_t - \tau| \leq 2^{-t-2}$, and put

$$y_e = \begin{cases} \tau, & e \text{ never halts}, \\ q_t, & H_e = t. \end{cases}$$

To precision 2^{-n}, simulate $n+2$ steps. If no halt has occurred, output q_{n+2}; even a later halt changes the value by less than the error budget. If a halt has occurred, output the corresponding rational exactly. Thus y_e is uniformly computable and is transcendental exactly when e never halts. This excludes a uniform semidecision for transcendence. Both constructions have the stated rational/transcendental promise. Nonzero rational affine changes put them in any prescribed rational open interval. □

One may take $\tau = \sum_{m\geq 1} 10^{-m!}$. Its partial sums p_n/q_n, with $q_n = 10^{n!}$, satisfy $0 < \tau - p_n/q_n < 2q_n^{-n-1}$. If a fixed nonzero integer polynomial P of degree d vanished at τ, sufficiently late partial sums would not be its roots, so $|P(p_n/q_n)| \geq q_n^{-d}$, contradicting the derivative bound times the displayed tail. This also makes the auxiliary transcendental used in the reduction explicit. The general undecidability background is [58]; the fast names and reductions above are supplied directly.

Theorem 49.2 does not exclude a proof for the specific constant γ, nor establish independence of its arithmetic nature from any axiom system.

49.3 Finite observations, infinitesimals, and proof search

Any finite set of strict polynomial inequalities true at a real point persists in a neighborhood. That neighborhood contains rationals, algebraic numbers of arbitrarily high degree, and transcendental numbers. Hence these finite tests alone cannot classify the point. This does not apply to an exact algebraic equality or assert that the comparison points satisfy γ's full analytic definition.

An explicit non-Archimedean example locates a different possible confusion. In the ordered Laurent series field $\mathbb{R}((t))$, with t a positive infinitesimal, $c = \gamma + t$ belongs to every strict rational interval containing γ, but $c \neq \gamma$ and is transcendental over $\mathbb{R}$. In the actual complete real field the same shrinking intervals determine one element. Passing to an infinitesimal field changes that uniqueness property; it does not construct another Euler constant satisfying all the original definitions. Likewise the formal map of Proposition 45.1 may preserve the inherited order into $K_x((t))$, although a nontrivial real-field self-automorphism cannot preserve order.

Finally, fix a consistent effectively axiomatized classical theory T in which the algebraicity assertion A is expressible. Enumerating contradiction proofs in $T + A$ and $T + \neg A$ in parallel terminates exactly when T decides A, because the deduction theorem gives

$$T + A \vdash \bot \iff T \vdash \neg A, \qquad T + \neg A \vdash \bot \iff T \vdash A.$$

The excluded-middle theorem alone does not guarantee termination. This is conditional proof-search bookkeeping, not a claim that either extension is inconsistent or that Euler's constant is independent of ZFC.

Remark 49.3 (What this part supplies). The completed general results are the exact local-count dichotomy, the height-free bound, the optimal finite-camouflage degree, the formal-motion classification, the analytic invariant families, the specific replication obstruction, and the effective-test theorems. Their applications to γ are equivalences, sufficient conditions, or explicitly limited failures. The missing positive input for the local route remains an unbounded family of *actual* relation-preserving maps. The missing input for value counting is a superexponential lower bound on *distinct evaluated values*. Neither follows from a collection of surviving candidates, from the number of coefficient vectors, or from shift-invariant positivity.

Part VII
Arithmetic transfer and its obstructions

50 Arithmetic transfer: notation and scope

This part preserves the arithmetic-transfer revision as a coherent collection of four mechanisms with explicit hypotheses and proofs. The arithmetic question is whether a nonzero polynomial $P \in \mathbb{Q}[X]$ can satisfy $P(\gamma) = 0$. Calling the algebraic and transcendental alternatives two "worlds" introduces no additional axiom: to exclude one alternative one must derive a contradiction from it and independently established facts. None of the constructions below produces that contradiction.

Throughout,

$$\begin{aligned} \gamma &= \lim_{n\to\infty}(H_n - \log n), & H_n &= \sum_{j=1}^{n} \frac{1}{j}, \\ \mathrm{Ein}(z) &= \sum_{n\geq 1} \frac{(-1)^{n+1} z^n}{n\, n!}, & E_1(x) &= \int_x^\infty \frac{e^{-t}}{t}\, dt \quad (x > 0), \\ \delta &= \int_0^\infty \frac{e^{-t}}{1+t}\, dt = eE_1(1), & \eta &= \mathrm{Ein}(1) = \gamma + \delta/e. \end{aligned} \tag{50.1}$$

On the positive real branch, and by continuation on compatible branches, $\mathrm{Ein}(z) = \gamma + \log z + E_1(z)$; see [59]. We use $|p|_p = p^{-1}$. Congruences for rational numbers with denominator prime to p are interpreted in $\mathbb{Z}_p$. A common denominator for algebraic numbers is the least positive rational integer making them all algebraic integers. Differential modules have derivation $\partial = d/dz$ over $K = \overline{\mathbb{Q}}(z)$.

There are four separate assertions in this part. The approximation and grid results concern the preservation of polynomial relations under different completions. The differential classification concerns the finite tensor category of one specified module, not arbitrary transformations of functions. The statistical result concerns kernels valid for an entire location family, not every possible identity special to γ. The denominator result concerns an explicitly defined formal quotient; no closure under division at a zero of a Borel sum is assumed.

Attribution and scope. The approximation argument is an elementary instance of simultaneous approximation in different completions. The statistical constructions use classical U-statistics and finite differences [61]; their identities are proved directly. The differential argument uses the primitive-dependence theorem and differential Galois theory [65, 62]. The factorial-denominator argument uses the regular-singularity theorem for G-operators [13]. Its prime-discrepancy statement is also a consequence of [60, Theorem 1], as proved below. We make no claim of research priority for these auxiliary results. The mixed-series comparison uses the arithmetic framework of [14]; it does not prove the missing lifting assertion. This part is a research note documenting constructions and obstructions, not a claimed solution of the constant problems.

51 Explicit real and 2-adic approximation

Theorem 51.1 (Prescribed simultaneous limits)**.** *Let $x \in \mathbb{R}$, $a \in \mathbb{Z}_2$, and let $[l_n, u_n]$ be rational intervals containing x with width at most $(2/9)^n$. Write $c_n = (l_n + u_n)/2$, $t_n = 2^n$, $q_n = 9^n$, and let a_n be the residue of a modulo t_n. Choose $0 \leq b_n < t_n$ with $b_n \equiv q_n a_n \pmod{t_n}$ and set*

$$r_n = \frac{b_n + t_n \,\mathrm{round}((q_n c_n - b_n)/t_n)}{q_n}. \tag{51.1}$$

Then $r_n \in \mathbb{Z}[1/3]$, and

$$|r_n - x| \le (2/9)^n, \qquad r_n \equiv a \pmod{2^n}. \tag{51.2}$$

In particular, $r_n \to x$ in $\mathbb{R}$ and $r_n \to a$ in $\mathbb{Q}_2$.

Proof. Rounding gives $|r_n - c_n| \le t_n/(2q_n)$, while the interval gives $|c_n - x| \le t_n/(2q_n)$. Their sum is the real estimate. Since q_n is odd, it is a unit in $\mathbb{Z}_2$, and the numerator congruence gives the 2-adic estimate. The residues are compatible because they are residues of one fixed a. Either tie convention for rounding is valid. □

For computable x and a the construction is an algorithm. A noncomputable a requires its residues as oracle input. Thus no computability assertion about arbitrary 2-adic integers is implicit.

Lemma 51.2 (Rational intervals for γ)**.** *For $m \ge 1$, define*

$$s_m = 2\sum_{j=0}^{m-1} \frac{1}{(2j+1)3^{2j+1}}, \quad R_m = \frac{3}{4(2m+1)9^m}, \quad U_m = H_{2^m} - ms_m, \quad \beta_m = 2^{-m} + mR_m. \tag{51.3}$$

Then $U_m - \beta_m < \gamma < U_m$, and $\beta_{3n} < (2/9)^n$ for $n \ge 1$.

Proof. The expansion $\log 2 = 2\sum_{j\ge0}((2j+1)3^{2j+1})^{-1}$ has a positive tail less than R_m, by a geometric-series bound. The integral comparison for the harmonic sum gives $0 < H_N - \log N - \gamma < 1/N$. Combining the two bounds proves the interval. Finally

$$\frac{\beta_{3n}}{(2/9)^n} = \left(\frac{9}{16}\right)^n + \frac{9n}{4(6n+1)162^n} \le \frac{9}{16} + \frac{3}{8\cdot 162} < 1.$$

□

Consequently, Theorem 51.1 gives explicit rational approximants to γ without reading its decimal expansion or assuming its arithmetic nature. The harmonic sum at $m = 3n$ has 2^{3n} terms; the construction is not asserted to be computationally efficient.

Proposition 51.3 (Failure of relation preservation)**.** *The simultaneous limits in Theorem 51.1, including compatibility at every level, do not imply $P(x) = 0 \Rightarrow P(a) = 0$ for $P \in \mathbb{Z}[X]$.*

Proof. Take $x = c_n = 1/3$ and $a = 0$. Then

$$r_n = \frac{2^n}{9^n}\operatorname{round}\left(\frac{9^n}{3\cdot 2^n}\right) \longrightarrow \frac{1}{3} \text{ in } \mathbb{R}, \qquad r_n \longrightarrow 0 \text{ in } \mathbb{Q}_2.$$

For $P(X) = 3X - 1$ we have $P(x) = 0$, but $P(r_n) \equiv -1 \pmod{2^n}$ and $P(a) = -1$. Evaluation therefore defines a homomorphism on $\mathbb{Z}[X]$, but it does not descend to $\mathbb{Z}[x] = \mathbb{Z}[X]/\ker(P \mapsto P(x))$ with $x \mapsto a$. The choice $x = 1/2$, $a = 0$, $P = 2X - 1$ gives another example. □

Proposition 51.4 (Denominator escape)**.** *Suppose $r_n \in \mathbb{Z}[1/3]$ converges to x in $\mathbb{R}$ and to a in $\mathbb{Q}_2$. Unless $x \in \mathbb{Z}[1/3]$ and a is its natural 2-adic image, the exponents of 3 in the reduced denominators of r_n tend to infinity. For the construction at $x = \gamma$, $a = 0$, one has $|r_n|_3 > 2^n$ for all sufficiently large n.*

Proof. If infinitely many reduced denominators have exponent at most e, that subsequence lies in the discrete real set $3^{-e}\mathbb{Z}$. Real convergence makes it bounded, so a further subsequence is constant. Uniqueness of both limits yields precisely the stated exception. At $x = \gamma$ and $a = 0$, eventually $0 < r_n < 1$. Write $r_n = A'_n/3^{e_n}$ in lowest terms. The numerator is a positive multiple of 2^n. Hence $|r_n|_3 = 3^{e_n} > A'_n \ge 2^n$. □

52 What is sufficient to transfer a relation?

Proposition 52.1 (An integer-zero sufficient condition)**.** *Let $r_n = A_n/q_n$ be reduced rationals with $q_n > 0$ and $p \nmid q_n$, $r_n \to x$ in $\mathbb{R}$, and $r_n \to a$ in $\mathbb{Q}_p$. If, for an integer $d \geq 1$, $q_n^d |r_n - x| \to 0$, then every $P \in \mathbb{Z}[X]$ of degree at most d satisfies*

$$P(x) = 0 \quad \Longrightarrow \quad P(r_n) = 0 \text{ eventually} \quad \Longrightarrow \quad P(a) = 0.$$

Proof. For $e = \deg P \leq d$, the number $q_n^e P(r_n)$ is an integer. On a fixed compact real interval containing x and the tail of the sequence, bound $|P'|$ by C_P. Under $P(x) = 0$ its absolute value is at most $C_P q_n^d |r_n - x| \to 0$, so it eventually vanishes. Polynomial continuity in $\mathbb{Q}_p$ proves the last implication. Constant polynomials cause no difficulty. □

This is a sufficient condition for this direct procedure, not a necessary condition for arbitrary proofs. If infinitely many r_n are distinct and the condition holds for every fixed d, then x is already transcendental: a nonzero polynomial has finitely many roots.

Theorem 52.2 (Common-denominator grid bound)**.** *For $M \geq 2$ distinct rationals $r_j = A_j/q$ with integer numerators and the same positive denominator, every $x \in \mathbb{R}$ satisfies*

$$\max_{1 \leq j \leq M} |r_j - x| \geq \frac{M-1}{2q}. \tag{52.1}$$

For the 2^n distinct residues in construction (51.1), at each n this gives

$$9^{nd} \max_{a_n \bmod 2^n} |r_{n,a_n} - \gamma| \geq \frac{2^n - 1}{2}\, 9^{n(d-1)} \qquad (d \geq 1). \tag{52.2}$$

Proof. The largest and smallest of M distinct integer numerators differ by at least $M - 1$. Their rational values have that difference divided by q, and at least one lies at distance half the difference or more from x. Distinct residues in the construction force distinct numerators. □

The theorem excludes a uniformly vanishing integer-zero budget for the entire common-denominator grid. It does not exclude a favorable single sequence, estimates using individual reduced denominators, or a different relation-preserving construction. For one prescribed residue b modulo t and a fixed denominator q, the exact minimax identity is

$$\sup_{x \in \mathbb{R}} \inf_{m \in \mathbb{Z}} \left| x - \frac{b + tm}{q} \right| = \frac{t}{2q}.$$

Indeed the admissible points are an equally spaced lattice, and its midpoints attain the worst distance.

Proposition 52.3 (Finite denominator support)**.** *Suppose a rational sequence converges in $\mathbb{R}$ and in every $\mathbb{Q}_p$, and the prime factors of all sufficiently late reduced denominators belong to a fixed finite set S. Then the sequence is eventually constant.*

Proof. Convergence in each $\mathbb{Q}_p$, $p \in S$, bounds the denominator exponent at that prime. A fixed positive integer B therefore clears all late denominators. The sequence Br_n is a convergent real sequence of integers and is eventually constant. □

The support condition cannot be omitted. The rationals $r_n = (n!)^2/(2(n!)^2 + 1)$ converge to $1/2$ in $\mathbb{R}$ and to 0 in every fixed $\mathbb{Q}_p$. Controlling each fixed place separately does not control primes that enter the denominators with n.

53 Globally nilpotent objects in the tensor category of the Ein connection

Put $k = \overline{\mathbb{Q}}$, $K = k(z)$, and $\partial = d/dz$. Let M be the differential module represented by

$$Y' = A(z)Y, \qquad A(z) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ z^{-1} & -z^{-1} & 0 \end{pmatrix}. \tag{53.1}$$

We write $\langle M \rangle_{\otimes}$ for the category obtained from M by finitely many tensor products, duals, finite direct sums, subobjects, and quotients over K. Every object and every finite collection of morphisms in this category descend to a number field. An object is *globally nilpotent* if, for a number-field model, its p-curvature is nilpotent at all but finitely many finite places of good reduction. This property is unchanged by finite extension of the number field or by a rational change of basis.

We give a concrete application of standard Picard–Vessiot and tensor-category theory. No priority claim is made for the resulting classification. The classification concerns the differential module; the additional analytic normalization containing γ is discussed separately at the end of the section.

53.1 Primitive independence and the Picard–Vessiot field

On a simply connected domain in $\mathbb{C} \setminus \{0\}$ containing a positive real interval, choose the branches

$$E = e^{-z}, \qquad L = \log z, \qquad J = E_1(z).$$

Here E_1 is continued from its usual positive-real definition. They satisfy

$$E' = -E, \qquad L' = z^{-1}, \qquad J' = -E/z, \qquad \mathrm{Ein}(z) = \gamma + L + J; \tag{53.2}$$

the last normalization agrees with [59, Eq. 6.2.4].

Lemma 53.1. *If $b, c \in \mathbb{C}$ and $R \in \mathbb{C}(z, E)$ satisfy*

$$R' = b/z - cE/z,$$

then $b = c = 0$.

Proof. There is no nonzero algebraic function r over $\mathbb{C}(z)$ satisfying $r' = \lambda r$ for a nonzero constant λ. Indeed, in a finite extension containing r, the norm q of r would satisfy $q'/q = n\lambda$, where n is the extension degree. The logarithmic derivative of a nonzero rational function tends to zero at infinity, which is impossible. In particular, E is transcendental over $\mathbb{C}(z)$.

View R as a rational function of the independent variable E, and factor its denominator over an algebraic closure of $\mathbb{C}(z)$. At a nonzero finite pole $E = r(z)$, let $a(z)(E - r)^{-s}$ be its leading principal part. Its derivative has coefficient $sa(r + r')$ at pole order $s + 1$. The prescribed derivative has no such pole, so $r' = -r$, contrary to the preceding paragraph. Consequently the only possible finite E-pole is zero, and

$$R = \sum_{n=u}^{v} r_n(z) E^n, \qquad r_n(z) \in \mathbb{C}(z).$$

For $n \neq 0, 1$, comparison of coefficients gives $r_n' - n r_n = 0$, hence $r_n = 0$. At $n = 0$ it gives $r_0' = b/z$; the residue of a rational derivative is zero, so $b = 0$. At $n = 1$ it gives $r_1' - r_1 = -c/z$. A finite pole of r_1 would produce, upon differentiation, a pole of strictly higher order than the right-hand side permits, including at zero. Thus r_1 is a polynomial, which forces $c = 0$. □

Lemma 53.2. *The functions E, L, J are algebraically independent over $\mathbb{C}(z)$. Moreover, the constant field of*

$$F = K(E, L, J)$$

is exactly k.

Proof. The transcendence of E was proved above. Both L and J are antiderivatives over $\mathbb{C}(z, E)$. Their meromorphic-function extension has constant field $\mathbb{C}$, so the no-new-constants hypothesis in the Kolchin–Ostrowski theorem is satisfied. If L, J were algebraically dependent over $\mathbb{C}(z, E)$, that theorem would give $bL + cJ \in \mathbb{C}(z, E)$ for some $(b, c) \neq (0, 0)$; see [65, Theorem 2.5]. Differentiating contradicts Lemma 53.1.

Now an element of F with zero derivative is a complex constant d. Write it as a quotient of polynomials in E, L, J with coefficients in $k(z)$. Clearing the denominator and using the independence just proved, comparison of any nonzero denominator coefficient shows that $d \in k(z)$. Since d is constant, $d \in k$. □

Proposition 53.3. *The field F is a Picard–Vessiot field for M over K, with fundamental matrix*

$$\Phi = \begin{pmatrix} 1 & 0 & 0 \\ 0 & E & 0 \\ L & J & 1 \end{pmatrix}. \tag{53.3}$$

Its differential Galois group consists of the transformations

$$E \longmapsto tE, \qquad L \longmapsto L + b, \qquad J \longmapsto tJ + c, \qquad t \in k^{\times}, \quad b, c \in k. \tag{53.4}$$

It is isomorphic to $\mathbb{G}_a \times (\mathbb{G}_a \rtimes \mathbb{G}_m)$, with the action specified below.

Proof. Direct differentiation gives $\Phi' = A\Phi$ and $\det \Phi = E$. Its entries and inverse determinant generate F, which has no new constants by Lemma 53.2. Thus it is a Picard–Vessiot field. Every differential automorphism must have the form (53.4), by successive use of (53.2). Conversely, algebraic independence shows that every transformation in (53.4) is a well-defined differential automorphism.

In the convention $\sigma(\Phi) = \Phi g_\sigma$, its matrix is

$$g(t, b, c) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & t & 0 \\ b & c & 1 \end{pmatrix}.$$

Set $T(t) = g(t, 0, 0)$, $U_b(b) = g(1, b, 0)$, and $U_c(c) = g(1, 0, c)$. Then

$$T(t)U_c(c)T(t)^{-1} = U_c(c/t), \qquad [T, U_b] = [U_b, U_c] = 1. \tag{53.5}$$

These identities specify the asserted semidirect product. □

53.2 The classification

The solution functor in F identifies $\langle M \rangle_{\otimes}$ with the finite-dimensional algebraic representations of this Picard–Vessiot group. In particular, invariant subspaces correspond to differential submodules, and quotients correspond to quotients. This is the standard tensor interpretation of differential Galois theory; see [62, Sections I and IV]. Let $F_W \subset F$ denote the Picard–Vessiot subfield generated by a fundamental matrix for an object W under this identification.

Theorem 53.4. *For an object $W \in \langle M \rangle_{\otimes}$, the following are equivalent:*

1. *W is globally nilpotent.*

2. *The torus T acts trivially on its solution representation.*

3. *Over K, the module W is isomorphic to $Y' = (N/z)Y$ for a constant nilpotent matrix N over k.*

4. *$F_W \subset K(L)$.*

In case (3), a suitable K-basis gives the fundamental matrix $\exp(NL)$*, a finite polynomial in L. In an arbitrary initial basis, a rational gauge factor can also occur.*

Proof. First consider an arbitrary finite-dimensional algebraic representation of the group in Proposition 53.3. The two additive subgroups act as $\exp(bB)$ and $\exp(cC)$ for commuting nilpotent matrices B, C. Their common kernel is nonzero: $\ker B$ is C-stable and the restriction of C to it is nilpotent. By (53.5) this common kernel is T-stable. Since a torus representation is diagonalizable, it contains a T-eigenline. That line is invariant under the whole group. Repeating in the quotient produces a full invariant flag. Each one-dimensional factor has trivial additive action and torus character $t \mapsto t^m$ for some integer m. Under the differential Galois correspondence it is the rank-one module with solution $E^m = e^{-mz}$.

Suppose (1) holds. The flag just constructed, its inclusions and quotients, and the rational changes of basis identifying its one-dimensional factors with the modules E^m, involve only finitely many algebraic coefficients. Choose one number field containing them and a model for W. After removing finitely many places, these matrices, their required nonzero minors, and their connection identities reduce correctly. Consequently the flag survives at the generic point in good characteristic, and p-curvature on a factor is induced from the p-curvature on W. Nilpotence therefore passes to every factor.

For the rank-one connection $\nabla_\partial = \partial + m$, however, its curvature at a place of residue characteristic p is

$$\psi_p = \nabla_\partial^p - \nabla_{\partial^{[p]}} = m^p.$$

Here $\partial^{[p]} = 0$ on the rational function field in characteristic p. If $m \neq 0$, this scalar is nonzero at all places except those over the finitely many rational primes dividing m. Thus every factor has $m = 0$. The torus is diagonalizable, and all of its weights on a representation are its weights on these flag factors. Hence T acts trivially, proving (1)⇒(2).

If (2) holds, (53.5) implies $\rho(U_c(c)) = \rho(U_c(c/t))$ for every $t \neq 0$. The algebraic morphism $c \mapsto \rho(U_c(c))$ is then constant on $k^\times$, hence on k, and its value at zero is the identity. Therefore U_c also acts trivially. The representation factors through the remaining additive group U_b. It has the form $b \mapsto \exp(bN)$ for a nilpotent constant matrix N. The module $Y' = (N/z)Y$ has fundamental matrix

$$\exp(NL) = \sum_{j=0}^{r-1} \frac{N^j L^j}{j!}, \qquad N^r = 0,$$

and precisely that representation. The tensor correspondence now gives (3), and also gives (4). Conversely, (4) implies that the subgroup fixing L, in particular T, acts trivially, so it implies (2).

Finally, reduce a number-field model of N at any good place. For $\nabla_\partial = \partial - N/z$, direct iteration on constant vectors gives

$$\psi_p = z^{-p}(N - N^p).$$

Indeed, in characteristic p the relevant polynomial identity is $\prod_{j=0}^{p-1}(X + j) = X^p - X$; the general formula follows from the linearity of p-curvature over the rational function field. The matrix $N - N^p = N(I - N^{p-1})$ is nilpotent. Thus (3) implies (1), completing the equivalences. □

53.3 Invariants and the remaining comparison constant

Proposition 53.5. *For $\mathcal{R} = K[E, E^{-1}, L, J]$, one has*

$$\mathcal{R}^T = K[L, J/E], \qquad \mathcal{R}^{\langle T, U_c\rangle} = K[L], \qquad F^{\langle T, U_c\rangle} = K(L).$$

Proof. A monomial $E^a J^b L^c$, with $a \in \mathbb{Z}$ and $b, c \geq 0$, has torus weight $a + b$. The weight-zero monomials give the first equality. Invariance under every translation $J \mapsto J + c$ first removes dependence on J; invariance under every dilation of E then removes dependence on E. This proves the second equality. The same translation and dilation arguments for rational functions prove the assertion about F. □

In particular, the single function J/E has torus weight zero, but this does not make its generated representation a trivial torus representation. In fact,

$$(J/E)' = J/E - 1/z, \qquad \sigma(J/E) = J/E + c/(tE).$$

Closure under the group action reintroduces the nonzero-weight function E^{-1}. This distinction prevents a weight-zero normalization of one function from evading Theorem 53.4.

Remark 53.6 (Scope and framing). For every complex constant a, the distinguished analytic solution satisfies

$$\begin{pmatrix} 1 \\ E \\ \mathrm{Ein}(z) - a \end{pmatrix} = \Phi(z) \begin{pmatrix} 1 \\ 1 \\ \gamma - a \end{pmatrix}. \tag{53.6}$$

Thus $\gamma - a$ is a coordinate of an analytic comparison vector, not a coefficient of (53.1). The classification of globally nilpotent objects does not determine this coordinate.

For example, both $K(E, L, J)$ and $K(E, L, J + \gamma)$ are Picard–Vessiot fields with constant field k, by the same independence argument. In their common analytic overfield their compositum contains the constant $(J + \gamma) - J = \gamma$. The assertion that each individual model has no new constants does not assert that their analytically specified compositum has none. Replacing this analytic comparison by an abstract Picard–Vessiot isomorphism would discard the datum under study.

The theorem classifies all the stated finite tensor constructions within this fixed module. It does not cover adjoining independent functions, differentiating an auxiliary parameter, applying a Fourier–Laplace transform, taking an infinite limit, or imposing new arithmetic constraints on a comparison vector. In particular, it proves neither the algebraicity nor the transcendence of γ.

54 Independent samples, exact cancellation, and positivity

Let T be a standard exponential random variable and let $G_a = -\log T + a - \gamma$. Its density is strictly positive on $\mathbb{R}$. The Gamma integral gives $\mathbb{E}\exp(t(-\log T)) = \Gamma(1 - t)$ for $t < 1$; differentiating the logarithmic expansion at $t = 0$ gives

$$\mathbb{E}G_a = a, \qquad \mathrm{Var}(G_a) = v := \zeta(2). \tag{54.1}$$

These identities follow from [59, Sections 5.2 and 5.7]. At $a = \gamma$ the sample is just $-\log T$, with no numerical input of γ. The translated family is an analytical comparison family, not an assertion that all its means equal Euler's constant.

Theorem 54.1 (Unbiased polynomial and its second moment). *Let $P(X) = \sum_{k=0}^{d} p_k X^k \in \mathbb{Q}[X]$ have degree $d \geq 1$. For $N \geq d$, define the symmetric multi-affine polynomial*

$$U_{P,N}(x_1, \ldots, x_N) = \sum_{k=0}^{d} \frac{p_k}{\binom{N}{k}} \sum_{\substack{I \subseteq \{1,\ldots,N\} \\ |I| = k}} \prod_{i \in I} x_i. \tag{54.2}$$

If its arguments are independent copies of G_a, then

$$\mathbb{E}_a U_{P,N} = P(a), \tag{54.3}$$

$$\mathbb{E}_a U_{P,N}^2 = P(a)^2 + \sum_{r=1}^{d} \left(\frac{P^{(r)}(a)}{r!} \right)^2 \frac{v^r}{\binom{N}{r}}. \tag{54.4}$$

The same formula holds for any independent identically distributed sample with mean a and finite variance v.

Proof. Independence makes the expectation of each k-fold product equal to a^k, proving (54.3). Put $Z_i = G_{a,i} - a$ and

$$V_{r,N} = \binom{N}{r}^{-1} \sum_{|I|=r} \prod_{i \in I} Z_i, \qquad V_{0,N} = 1.$$

For a fixed r-element subset, counting the k-element subsets containing it gives $\binom{N-r}{k-r}/\binom{N}{k} = \binom{k}{r}/\binom{N}{r}$. Expanding each factor as $a + Z_i$ therefore yields

$$U_{P,N} = \sum_{r=0}^{d} \frac{P^{(r)}(a)}{r!} V_{r,N}. \tag{54.5}$$

Products belonging to different subsets have a centered variable to its first power, and hence expectation zero. Equal subsets contribute v^r. Thus $\mathbb{E} V_{r,N} V_{s,N} = 0$ for $r \neq s$ and $\mathbb{E} V_{r,N}^2 = v^r / \binom{N}{r}$. Squaring (54.5) proves the result. Only second moments are used, since each variable occurs with exponent at most two after squaring. □

This is a direct U-statistic calculation in the sense of [61]. At a real root a of multiplicity m of P, the first surviving term in (54.4) gives

$$\mathbb{E}_a U_{P,N}^2 \sim \frac{P^{(m)}(a)^2 v^m}{m!} N^{-m}. \tag{54.6}$$

It is strictly positive for every N, even if $P(a) = 0$. For example, $P(X) = qX - p$ and $a = p/q$ give $\mathbb{E}_a U_{P,N}^2 = q^2 v / N$. Positive expectations tending to zero do not contradict rationality of their mean.

Theorem 54.2 (Exact integer jackknife)**.** *For any integer $N_0 \geq d$,*

$$P(a)^2 = \sum_{j=0}^{d} (-1)^{d-j} \binom{d}{j} \binom{N_0 + j}{d} \mathbb{E}_a U_{P,N_0+j}^2. \tag{54.7}$$

The weights are integers and do not depend on a or on the sample variance. For $N_0 = d$, their absolute sum is at most $(3 + 2\sqrt{2})^d$.

Proof. Let $F(N) = \mathbb{E}_a U_{P,N}^2$ and let $(N)_d = N(N-1)\cdots(N-d+1)$. Equation (54.4) implies

$$(N)_d F(N) = P(a)^2 (N)_d + \sum_{r=1}^{d} \left(\frac{P^{(r)}(a)}{r!} \right)^2 v^r \, r! (N-r)_{d-r}.$$

The right side is a polynomial of degree at most d, whose coefficient of N^d is $P(a)^2$. Its dth forward difference is $d! P(a)^2$. Dividing that difference by $d!$ proves (54.7). For $N_0 = d$ the absolute sum satisfies

$$\sum_{j=0}^{d} \binom{d}{j} \binom{d+j}{d} = [t^d](1+t)^d (2+t)^d \leq \left(\frac{(1+\sqrt{2})(2+\sqrt{2})}{\sqrt{2}} \right)^d = (3 + 2\sqrt{2})^d.$$

The coefficient identity follows by summing $\binom{d}{j}(1+t)^{d+j}$ before taking the coefficient. The inequality uses nonnegative coefficients evaluated at $t = \sqrt{2}$. □

For $d = 1, 2, 3, 4$, the weights at sample sizes $d, \dots, 2d$ are respectively

$$(-1, 2), \quad (1, -6, 6), \quad (-1, 12, -30, 20), \quad (1, -20, 90, -140, 70).$$

If $P \in \mathbb{Z}[X]$, multiplication by $((2d)!)^2$ clears all polynomial kernel denominators in (54.7) at $N_0 = d$: each $\binom{N}{k}$ divides $N!$, and $N!$ divides $(2d)!$. This is a nonoptimal denominator bound, not an arithmetic lower bound for the expectation.

Proposition 54.3 (Signs required for universal elimination)**.** *A linear combination of second moments that preserves their constant term and cancels their $1/N$ term identically must use a negative weight.*

Proof. For sample sizes $N_j > 0$, the two coefficient conditions are $\sum_j w_j = 1$ and $\sum_j w_j/N_j = 0$. They cannot hold with every $w_j \geq 0$. These are conditions for universal coefficient cancellation; no independence assumption about special values of $P^{(r)}(a)$ is needed. □

Independent blocks offer a different exact identity:

$$\mathbb{E}_a[U_{P,N}(X)U_{P,N}(Y)] = P(a)^2. \tag{54.8}$$

Its integrand need not be nonnegative. In fact $U_{P,N}$ is nonconstant, since $U_{P,N}(x, \dots, x) = P(x)$. A nonconstant multi-affine polynomial takes both signs: fix all but one variable where the remaining affine slope is nonzero. The positive density of the sample law gives each sign positive probability, so independent blocks make the product negative with positive probability.

Theorem 54.4 (Finite-sample nonnegative-kernel obstruction)**.** *Fix a finite sample size N and a measurable function $K : \mathbb{R}^N \to [0, \infty]$ independent of a. Either $\mathbb{E}_a K = 0$ for every $a \in \mathbb{R}$, or $\mathbb{E}_a K > 0$ for every $a \in \mathbb{R}$. Consequently, if a nonzero real polynomial P has a real root, no such kernel satisfies $\mathbb{E}_a K = P(a)^2$ for every real a.*

Proof. If $\mathbb{E}_r K = 0$, then $K = 0$ almost surely under the law of the sample with mean r. Its density is strictly positive everywhere, so $K = 0$ Lebesgue almost everywhere. Every other sample law is equivalent to Lebesgue measure, proving $\mathbb{E}_a K = 0$ for all a. Apply this at a root of P and then at a point where P is nonzero. □

The proof is an absolute-continuity argument; compare the broader constrained-estimation discussion in [64]. It excludes a single kernel valid across the whole location family. It does not exclude a new arithmetic identity proved only at $a = \gamma$.

Remark 54.5 (The infinite-sample boundary)*.* For infinitely many independent samples the strong law gives $\overline{G}_N \to a$ almost surely. The nonnegative functional $P(\lim_N \overline{G}_N)^2$, defined as zero if the limit does not exist, has expectation $P(a)^2$ for every a. There is no contradiction with Theorem 54.4: different means give mutually singular infinite product laws. If $P(a) = 0$, this functional is zero almost surely and provides no nonvanishing certificate.

Proposition 54.6 (Quadratic algebraic-coefficient kernels)**.** *Let $K(x) = x^{\mathsf{T}} A x + b^{\mathsf{T}} x + c$ have real algebraic coefficients, with A symmetric, and suppose $K \geq 0$ on $\mathbb{R}^N$. If the polynomial $a \mapsto \mathbb{E}_a K$ has algebraic coefficients, then K is a nonnegative constant.*

Proof. With $\mathbf{1} = (1, \dots, 1)^{\mathsf{T}}$,

$$\mathbb{E}_a K = a^2 \mathbf{1}^{\mathsf{T}} A \mathbf{1} + a b^{\mathsf{T}} \mathbf{1} + c + \zeta(2) \operatorname{tr} A.$$

Since $\zeta(2) = \pi^2/6$ is transcendental, the constant coefficient can be algebraic only when $\operatorname{tr} A = 0$. Nonnegativity of K forces A to be positive semidefinite, so $A = 0$. A globally nonnegative affine function has $b = 0$, and then $c \geq 0$. □

55 Exact denominator growth after division of Euler's series

The distinction between an analytic zero and arithmetic divisibility can be made explicit for Euler's factorial series. The construction in this section has two purposes: to compute the denominator loss exactly, and to identify the additional assertion that would be needed to deduce transcendence from a Borel-summed zero. The weighted prime nonagreement result below is also a consequence of an existing global relation theorem; no priority claim is made for that consequence.

55.1 Normalization and exact common denominators

Put

$$f(z)=\sum_{n\ge 0}(-1)^n n!z^n, \qquad S_n=\sum_{j=0}^{n}(-1)^j j!, \qquad g_a(z)=\frac{f(z)-a}{z-1}=\sum_{n\ge 0}(a-S_n)z^n, \tag{55.1}$$

where $a\in\overline{\mathbb{Q}}$ and the quotient is taken in $\overline{\mathbb{Q}}[[z]]$. Its formal Borel transform is

$$G_a(z)=\sum_{n\ge 0}b_n(a)z^n, \qquad b_n(a)=\frac{a-S_n}{n!}. \tag{55.2}$$

For an algebraic number b, let $\operatorname{den}(b)$ be the least positive integer D for which Db is an algebraic integer. Write $D_N(a)$ for the least positive integer that makes all of $b_0(a),\dots,b_N(a)$ algebraic integers.

We use the usual arithmetic Gevrey convention: a formal series $\sum c_n z^n$ is an arithmetic Gevrey series of order 1 when $\sum(c_n/n!)z^n$ is a G-function; see [14] for this class and its summation conventions. In particular, a bound of the form $|c_n|\le C^{n+1}n!$ alone does not establish this arithmetic condition. A G-function must also have algebraic coefficients, exponentially bounded conjugates and common denominators, and satisfy a linear differential equation over $\overline{\mathbb{Q}}(z)$.

Proposition 55.1 (Exact denominator formula)**.** *For $n\ge 1$,*

$$nb_n(a)=b_{n-1}(a)-n(-1)^n, \qquad b_0(a)=a-1. \tag{55.3}$$

Consequently,

$$D_N(a)=\operatorname{den}(b_N(a)), \qquad D_{N-1}(a)\mid D_N(a). \tag{55.4}$$

If $a=A/B\in\mathbb{Q}$, where $A,B\in\mathbb{Z}$, $B>0$, and $\gcd(A,B)=1$, then

$$D_N(A/B)=\frac{BN!}{\gcd(BN!,A-BS_N)}. \tag{55.5}$$

The convention $\gcd(m,0)=m$ makes the formula valid when $A-BS_N=0$.

Proof. The identity $S_n-S_{n-1}=(-1)^n n!$ gives (55.3). If $Db_n(a)$ is an algebraic integer, then

$$Db_{n-1}(a)=nDb_n(a)+Dn(-1)^n$$

is an algebraic integer as well. Applying this implication successively shows that the denominator of $b_N(a)$ clears all earlier coefficients. The converse is part of the definition of a common denominator. The divisibility assertion follows, and reduction of $(A-BS_N)/(BN!)$ gives (55.5). □

Theorem 55.2 (Factorial denominators for nonrational algebraic parameters)**.** *For every* $a \in \overline{\mathbb{Q}} \setminus \mathbb{Q}$,

$$\log D_N(a) = \log N! + O_a(1). \tag{55.6}$$

More explicitly, let $K = \mathbb{Q}(a)$*, choose an integral basis* $1, \omega_1, \dots, \omega_{d-1}$ *of* $\mathcal{O}_K$*, and write*

$$a = \frac{c_0 + c_1\omega_1 + \cdots + c_{d-1}\omega_{d-1}}{B}, \qquad B > 0, \quad c_i \in \mathbb{Z}.$$

Then

$$D_N(a) = \frac{BN!}{\gcd(BN!, c_0 - BS_N, c_1, \dots, c_{d-1})}. \tag{55.7}$$

In particular, for $h = \gcd(c_1, \dots, c_{d-1}) > 0$,

$$\frac{BN!}{h} \le D_N(a) \le BN!. \tag{55.8}$$

Proof. First, an integral basis containing 1 exists. Indeed, $\mathcal{O}_K/\mathbb{Z}$ is torsion-free: if $m\beta \in \mathbb{Z}$ with $\beta \in \mathcal{O}_K$ and $m \in \mathbb{Z} \setminus \{0\}$, then β is a rational algebraic integer and therefore belongs to $\mathbb{Z}$. Thus 1 is a primitive vector in the free abelian group $\mathcal{O}_K$ and extends to a basis.

In that basis the coordinates of $b_N(a)$ are

$$\frac{c_0 - BS_N}{BN!}, \frac{c_1}{BN!}, \dots, \frac{c_{d-1}}{BN!}.$$

Integrality is coordinatewise, so Proposition 55.1 gives (55.7). Since a is not rational, at least one c_i with $i \ge 1$ is nonzero. The gcd in the denominator of (55.7) is therefore bounded above by the fixed positive integer h. This proves (55.8) and (55.6). □

55.2 The Borel differential equation and its monodromy

Proposition 55.3. *The germ* G_a *is holomorphic at* 0 *and satisfies*

$$G_a'(z) = G_a(z) + \frac{1}{(1+z)^2}, \qquad G_a(0) = a - 1. \tag{55.9}$$

Analytic continuation once counterclockwise around -1 *changes it by*

$$\Delta G_a(z) = -2\pi i e^{z+1}. \tag{55.10}$$

For every $a \in \overline{\mathbb{Q}}$*, the function* G_a *is not a G-function, and* g_a *is not an arithmetic Gevrey series of order* 1*.*

Proof. The estimate $\sum_{j=0}^{n} j! \le 3n!$ shows that, for every embedding $\sigma : \mathbb{Q}(a) \hookrightarrow \mathbb{C}$,

$$|\sigma(b_n(a))| \le |\sigma(a)| + 3. \tag{55.11}$$

Thus the germ converges for $|z| < 1$. Summing (55.3) gives (55.9), hence

$$G_a(z) = e^z \left(a - 1 + \int_0^z \frac{e^{-t}}{(1+t)^2}\, dt \right) \tag{55.12}$$

along any path avoiding -1. The integrand has residue $-e$ there; the residue theorem gives (55.10).

Suppose that G_a were a G-function. The minimal differential operator annihilating a G-function is regular singular at every point, including infinity; see [13, Section 1] for this standard consequence of the theory of G-operators. Analytic continuations of G_a, and therefore their

differences, solve the same homogeneous equation. Equation (55.10) would make e^z a solution. This is incompatible with regular singularity at infinity: after finite ramification, such solutions have power–logarithm local expansions, whereas e^z has exponential growth on a ray to infinity. The contradiction proves the assertion about G_a; the statement about g_a follows from the Borel normalization. □

Remark 55.4 (An explicit minimal operator). The homogeneous operator

$$\left(\partial + \frac{2}{1+z}\right)(\partial - 1), \qquad \partial = \frac{d}{dz}, \tag{55.13}$$

annihilates G_a and has minimal order 2, even over $\mathbb{C}(z)$. For if $G_a' = rG_a$ with $r \in \mathbb{C}(z)$, then $(r-1)G_a = (1+z)^{-2}$. The case $r = 1$ is impossible, and otherwise G_a would be rational. A rational solution R of $R' - R = (1+z)^{-2}$ can have no finite pole except -1. At -1 its principal part would have to be $-1/(1+z)$, leaving a nonzero simple pole in $R' - R$, a contradiction. This also gives a direct check of minimality. The coefficient of ∂ in (55.13) is $-1 + 2/(1+z)$, which fails the Fuchs condition at infinity.

Remark 55.5 (Cancellation on one ray does not remove monodromy). For $x > 0$ put $E_1(x) = \int_x^\infty e^{-t}\,dt/t$ and $\delta = eE_1(1)$. Formula (55.12) gives, on the positive real axis,

$$G_a(z) = e^z(a - \delta) - \frac{1}{1+z} + e^{z+1}E_1(z+1). \tag{55.14}$$

The term $e^z(a-\delta)$ vanishes at $a = \delta$. The monodromy (55.10) does not. Thus removal of an exponentially growing term on one branch is insufficient to establish the arithmetic condition governing all branches and normalized denominators.

The coefficients of G_a belong to a fixed number field, (55.11) supplies the conjugate bound, and (55.13) supplies holonomicity. Hence its failure to be a G-function is exactly the failure of an exponential bound on $D_N(a)$. This already implies $\limsup_{N\to\infty} \log D_N(a)/N = +\infty$. Obtaining a limit rather than a limsup for rational a requires the following argument.

55.3 Rational parameters and weighted prime nonagreement

For each prime p, the alternating series f in (55.1) converges in $\mathbb{Q}_p$ at $z = 1$. Set

$$E_p = \sum_{n\ge 0} (-1)^n n! \in \mathbb{Z}_p, \qquad \mathcal{E}(a) = \{p : E_p \neq a \text{ in } \mathbb{Q}_p\} \quad (a \in \mathbb{Q}). \tag{55.15}$$

Theorem 55.6 (Weighted nonagreement and denominator growth). *For every $a \in \mathbb{Q}$,*

$$\sum_{p \in \mathcal{E}(a)} \frac{\log p}{p-1} = +\infty \tag{55.16}$$

and

$$\lim_{N\to\infty} \frac{\log D_N(a)}{N} = +\infty. \tag{55.17}$$

The first assertion is a consequence of the global relation theorem of Ernvall-Hytönen, Matala-aho, and Seppälä [60, Theorem 1]. A separate proof through Proposition 55.3 is included to connect that assertion directly to the exact denominators.

Proof. Write $a = A/B$ in lowest terms with $B > 0$. If $p \notin \mathcal{E}(a)$, then $a = E_p \in \mathbb{Z}_p$, so $p \nmid B$ and

$$v_p(a - S_N) = v_p(E_p - S_N) \ge v_p((N+1)!). \tag{55.18}$$

Indeed, every term of the p-adically convergent tail is divisible by $(N+1)!$ in $\mathbb{Z}_p$. Such a prime divides none of the $D_N(a)$, and (55.5) yields

$$D_N(a) \mid B \prod_{p \in \mathcal{E}(a)} p^{v_p(N!)}. \tag{55.19}$$

The product is finite for each N. If the sum in (55.16) had finite value C, Legendre's bound $v_p(N!) \le N/(p-1)$ would give

$$\log D_N(a) \le \log B + CN.$$

Together with the conjugate bound and the differential equation, this would make G_a a G-function, contradicting Proposition 55.3. This proves (55.16).

Fix now $p \in \mathcal{E}(a)$ and put $t_p = v_p(A - BE_p) < \infty$. Convergence in $\mathbb{Q}_p$ gives $v_p(A - BS_N) = t_p$ for all sufficiently large N. Increasing N further if necessary, (55.5) therefore gives

$$v_p(D_N(a)) = v_p(B) + v_p(N!) - t_p. \tag{55.20}$$

For every finite $T \subset \mathcal{E}(a)$ it follows that

$$\liminf_{N\to\infty} \frac{\log D_N(a)}{N} \ge \sum_{p \in T} \frac{\log p}{p-1}, \tag{55.21}$$

since $v_p(N!)/N \to 1/(p-1)$ at each fixed prime. The right-hand side can be made arbitrarily large by (55.16). This proves (55.17). □

Remark 55.7 (Derivation from the existing global relation theorem). For completeness, the precise implication from [60, Theorem 1] can be checked without any prime-density claim. Specializing that theorem to the argument -1, a set T of primes has infinitely many members with $A - BE_p \neq 0$ whenever, for every finite $S \subset T$,

$$\limsup_{n\to\infty} 4^n n! \prod_{p \in T \setminus S} |n!|_p^2 = 0. \tag{55.22}$$

If (55.16) were false, with finite sum C, take T to be the complement of $\mathcal{E}(a)$. The product formula and Legendre's bound imply

$$\begin{aligned} 4^n n! \prod_{p \in T \setminus S} |n!|_p^2 &= \frac{4^n}{n!} \prod_{p \in \mathcal{E}(a) \cup S} |n!|_p^{-2} \\ &\le \frac{\exp\big((\log 4 + 2C + 2C_S)n\big)}{n!} \longrightarrow 0, \qquad C_S = \sum_{p \in S} \frac{\log p}{p-1}. \end{aligned}$$

The cited theorem would then give a prime of nonagreement inside the set T of agreement primes, a contradiction. Accordingly, (55.16) is recorded here as a known consequence and as an input to the explicit denominator conclusion, not as a new prime-distribution theorem.

Corollary 55.8. *For every $a \in \overline{\mathbb{Q}}$ and every real $C > 1$, there is $N_0(a, C)$ such that $D_N(a) > C^N$ for every $N \ge N_0(a, C)$. Equivalently,*

$$\lim_{N\to\infty} \frac{\log D_N(a)}{N} = +\infty.$$

Proof. For nonrational algebraic a, use Theorem 55.2. For rational a, use Theorem 55.6. □

Remark 55.9. For rational a, these arguments do not establish $\log D_N(a) \sim \log N!$. The all-N superexponential conclusion must be distinguished from that stronger factorial asymptotic. Nor does the prime p in (55.15) remain fixed when a rational candidate a is changed.

55.4 Finite certificates and the exact scope of the infinite family

Proposition 55.10 (Finite certificates of nonagreement)**.** *Let $a = A/B \in \mathbb{Q}$ be in lowest terms, with $B > 0$. A triple (p, N, r), where p is prime, $N \geq 0$, and $r \geq 1$, certifies $E_p \neq a$ if*

$$v_p(B) + v_p((N+1)!) \geq r, \qquad A - BS_N \not\equiv 0 \pmod{p^r}. \tag{55.23}$$

Conversely, every prime of $\mathcal{E}(a)$ admits such a certificate. For each positive rational R, there is a terminating procedure that produces a finite certified set $T_R \subset \mathcal{E}(a)$ with

$$\sum_{p \in T_R} \frac{\log p}{p-1} > R. \tag{55.24}$$

Proof. The tail bound gives $B(E_p - S_N) \equiv 0 \pmod{p^r}$ under the first condition in (55.23). The second condition therefore implies $A - BE_p \neq 0$. Conversely, if $t_p = v_p(A - BE_p) < \infty$, take $r = t_p + 1$ and then take N large enough that the tail has valuation at least r.

Enumerate primes p, integers N, r, and check (55.23) using integer arithmetic. Remove repeated primes. Simultaneously refine rational lower bounds for the finitely many logarithms already encountered. By (55.16), some finite certified set has total weight strictly greater than R; its sufficiently accurate rational lower bound will eventually certify the same strict inequality. Thus the procedure terminates. □

This is an effective infinite family of finite certificates concerning different p-adic values. It is not a family of certificates proving the irrationality of one fixed E_p, of the real constant δ, or of γ.

55.5 What zero lifting would have to add

The positive-direction Borel sum of f at 1 is

$$f_0(1) = \int_0^\infty \frac{e^{-t}}{1+t}\, dt = \delta.$$

In particular, the restriction of the zero-lifting conjecture [14, Conjecture 2] to this single family is the assertion

$$\text{for every } a \in \overline{\mathbb{Q}}, \qquad \delta = a \implies g_a \text{ is arithmetic Gevrey of order 1.} \tag{55.25}$$

Proposition 55.11. *Assertion (55.25) is equivalent to the transcendence of δ.*

Proof. If δ is transcendental, every antecedent in (55.25) is false. If δ is algebraic, take $a = \delta$. The conclusion of (55.25) contradicts Proposition 55.3. □

Thus the exact denominator calculation establishes the obstruction on the conclusion side of (55.25); it does not establish the implication. Ordinary Gevrey growth and formal division do not supply that missing arithmetic assertion.

For comparison with the earlier mixed-function formulation for γ, define

$$\widehat{D}(z) = \sum_{n \geq 0} (-1)^n n! z^{-n-1}, \qquad F_a(z) = e^z(\mathrm{Ein}(z) - a), \qquad \Psi_a(z) = F_a(z) - \widehat{D}(z), \tag{55.26}$$

where

$$\mathrm{Ein}(z) = \sum_{n \geq 1} \frac{(-1)^{n-1} z^n}{n\, n!}.$$

Here a mixed arithmetic Gevrey function means a bilateral formal sum $H(z) + h(1/z)$ with H an E-function and h arithmetic Gevrey of order 1, as in [14, Definition 1]. Multiplication by a polynomial is well-defined coefficientwise. The expression "division by $z-1$ within the mixed class" means the existence of a mixed function Θ satisfying $(z-1)\Theta = \Psi_a$; it does not presuppose a field structure on all bilateral sums.

On the positive real axis, $\mathrm{Ein}(z) = \gamma + \log z + E_1(z)$ and $\widehat{D}_0(z) = e^z E_1(z)$. Consequently,

$$\Psi_{a,0}(z) = e^z(\gamma + \log z - a), \qquad \Psi_{a,0}(1) = e(\gamma - a). \tag{55.27}$$

Proposition 55.12 (The earlier mixed zero-lifting equivalence)**.** *For every $a \in \overline{\mathbb{Q}}$, there is no mixed arithmetic Gevrey function Θ such that $(z-1)\Theta = \Psi_a$. Consequently the assertion*

$$\textit{for every } a \in \overline{\mathbb{Q}}, \qquad \Psi_{a,0}(1) = 0 \implies \Psi_a \textit{ is divisible by } z-1 \textit{ in the mixed class} \tag{55.28}$$

is equivalent to the transcendence of γ.

Proof. Suppose $\Theta = H(z) + h(1/z)$ exists. Absorb the constant term of h into H and write $h(1/z) = \sum_{j\geq 1} h_j z^{-j}$, where $h_j \in \overline{\mathbb{Q}}$. Comparing the nonnegative powers of z in $(z-1)\Theta = F_a - \widehat{D}$ gives the entire-function identity

$$F_a(z) = (z-1)H(z) + h_1.$$

Hence $F_a(1) = h_1$ is algebraic. However, e and $\mathrm{Ein}(1)$ are algebraically independent, as follows from [63, Section 5(ii)]. Therefore $F_a(1) = e(\mathrm{Ein}(1) - a)$ cannot be algebraic: an algebraic value b would give the nonzero relation $X(Y-a) - b = 0$ at $(X, Y) = (e, \mathrm{Ein}(1))$. This proves nondivisibility.

By (55.27), the antecedent of (55.28) is exactly $\gamma = a$. If γ is transcendental all such antecedents are false; if it is algebraic, choosing $a = \gamma$ contradicts the nondivisibility just proved. □

Proposition 55.12 restates the bridge already identified in the preceding manuscript and specializes the mixed zero-lifting framework of [14]. It is not a new transcendence result. The additional content of the present section is the explicit denominator analysis of the Euler component and its connection with prime nonagreement. Neither equivalence proves its own zero-lifting antecedent-to-conclusion assertion.

56 Remaining arithmetic assertions

The preceding sections complete several auxiliary constructions. Their endpoints do not identify which polynomials vanish at γ. The following distinctions fix the logical status of the program.

1. **Relations and completions.** To obtain a homomorphism $\mathbb{Z}[\gamma] \to \mathbb{Z}_p$ sending γ to a, one must prove $P(\gamma) = 0 \Rightarrow P(a) = 0$ for every $P \in \mathbb{Z}[X]$. The explicitly constructed simultaneous limits do not give this implication. The counterexamples in Proposition 51.3 establish this failure without any assumption about γ.

2. **Modules and comparison vectors.** Theorem 53.4 classifies the stated differential modules. It does not determine the analytic comparison coordinate $\gamma - a$ in (53.6). An arithmetic theorem about that coordinate would be an additional result.

3. **Exact cancellation and nonvanishing.** Theorem 54.2 cancels the second-moment correction terms exactly. Its signed combination does not retain positivity. Theorem 54.4 excludes a nonnegative replacement valid across the whole finite-sample location family; an identity special to the actual Gamma integral would require additional information.

4. **Analytic zeros and arithmetic division.** Corollary 55.8 describes the normalized denominator loss of a formal quotient. The implication from a zero of a Borel sum to division in the arithmetic class is unproved. Propositions 55.11 and 55.12 explain why, for these specific families, proving that implication already settles the corresponding transcendence question.

These are limitations of the specified constructions, not impossibility theorems for every method of proving transcendence. The paper makes no claim that the missing implications must follow after further numerical calculation or after enlarging the number of fields considered. Neither arithmetic alternative for γ has been excluded here.

57 Transfer certificates and reproducibility

The accompanying scripts use only the Python standard library. Their finite checks are separated from the proofs of the infinite assertions. They should be run with assertions enabled:

```
python3 verify_adelic_candidates.py
python3 verify_positive_tensor.py
python3 verify_zero_lifting.py
```

Simultaneous approximation. The first script checks 25 rational candidates for γ at five levels and five prescribed 2-adic targets. It checks the certified real bounds, congruences, compatibility, and the denominator cost for target zero. It also checks 40 rational counterexamples, 20 each for $x = 1/3$ and $x = 1/2$. The proof uses $m = 3n$; the implementation chooses the first $m \leq 3n$ giving sufficient interval width.

Statistical identities. The second script checks 90 exact cancellation identities for degrees up to 12 and 14 exhaustive finite-distribution cases for an asymmetric three-point law. The direct algebraic proofs in Section 54 apply to all degrees and sample sizes.

Factorial denominators. The third script checks 2,173 coefficient and common-denominator cases for 53 rational parameters at $N = 0, \ldots, 40$, 41 denominator cases for $\sqrt{2}$, and 18 explicit finite p-adic nonagreement certificates. The limits in Section 55 are proved there; they are not inferred from these finite computations.

The approximation and zero-lifting scripts write JSON certificates. The statistical script prints its check record. A SHA-256 manifest identifies the delivered versions. Exact arithmetic prevents rounding errors in these checks but does not turn them into formal proofs of the general theorems. Independent mathematical review was performed for the main arguments; no proof-assistant certification or external peer review is claimed.

58 Transfer revision record

Revision 2 integrates the five proof notes into a standalone paper. It makes the computability input, reduced versus common denominator distinction, finite tensor-category restriction, and finite-sample location-family restriction explicit. It separates a denominator limsup from the all-index limit, treats nonrational algebraic parameters using an integral basis, and defines mixed division by a polynomial without assuming a field of unrestricted bilateral series. The prime-discrepancy result is attributed to its existing global-relation source, and the Katz reference is identified as the 1982 paper.

The previously reported finite checks are retained, with the same counts. This revision adds a complete exposition and corrects scope and bibliographic details; it adds no claim of irrationality or transcendence for γ or δ.

Part VIII
Unified targets, corrections, and the source record

59 What would complete an arithmetic argument?

The constructions in the preceding parts have several outputs that must not be identified with one another. A nonzero vector in an integer kernel can have zero image in the target quotient. A nonzero target polynomial can vanish at an algebraic argument. A polynomial that does not vanish can still have an insufficiently small value relative to its primitive height. These are separate assertions, and the surviving targets below keep them separate.

Primitive rational forms. For the positive-moment route, a sufficient completed output would be coprime integers p_n, q_n, with $q_n \to \infty$, and one fixed $\varepsilon > 0$ such that

$$0 < |q_n\gamma - p_n| < q_n^{-1-\varepsilon} \quad \text{for infinitely many } n. \tag{59.1}$$

The strict inequality excludes rational γ once the denominators are large, and Roth's theorem excludes every irrational algebraic γ. Thus this is a transcendence criterion. The one certified exponent 103/100 in the positive-moment part is an instance at a finite target; it is not an infinite family satisfying (59.1). The denominator here is the reduced target denominator, after cancellation of every common factor. The size of an auxiliary residue vector is not a substitute for it.

Small polynomial values. Part V states the degree-sensitive polynomial criteria and proves what algebraicity would force for the corresponding total selection rule. Producing nonzero *polynomials* under that rule is not the unresolved step: their *values* may eventually vanish if γ is algebraic. A successful completion must force escape from the relation ideal with the stated degree and height budgets, or independently certify sufficiently many nonzero values. Neither finite modular tests nor selection from a shrinking real interval automatically supplies this.

Local arithmetic. Part VI turns the two alternatives into exact statements about maps from $\mathbb{Z}[\gamma]$ to residue rings. Those statements are useful because they specify an observable which would distinguish the cases. Part VII constructs prescribed real and 2-adic limits, but also proves that these two limits need not preserve polynomial relations. A completion by this route must establish the missing well-defined map from the actual ring $\mathbb{Z}[\gamma]$, with enough depth or unbounded counts to apply the local criterion. Choosing residues freely in a rational approximation does not establish that map.

Differential and mixed-series arithmetic. The differential reductions classify objects in a specified tensor category; the factorial-quotient theorems classify denominators of a specified division operation. They are completed results about those objects. Their relation to (1.1) still requires an arithmetic specialization or zero-lifting assertion. In particular, a common analytic value of an E-series and a summed divergent series cannot simply be divided out in the arithmetic category. Assuming that operation would assume the very bridge the construction is intended to prove.

The two-case strategy. Let A denote "γ is algebraic" and B denote "γ is transcendental". Deductions made under A can exclude A only when every additional premise in the contradiction

has independent justification. An attractive property obtained in B cannot be imported as an established fact into A. Conversely, the fact that many finite observations can be modeled in both cases does not establish that either case is consistent with every theorem. The local camouflage results quantify a particular finite limitation; they do not settle the original alternative.

60 Corrections carried into the cumulative edition

1. **Construction tests have a specified scope.** Independence from unknown digits, a fixed target space, favorable error versus height, and value nonvanishing are useful tests for an explicit integer-form strategy. They are not necessary conditions for every possible proof of transcendence. A construction from certified rational intervals is not circular merely because the intervals locate γ. It still needs its own arithmetic nonvanishing argument.

2. **Dirichlet gives existence, not an upper ceiling.** A Dirichlet exponent cannot be used as an unconditional upper bound on the approximation exponent of an unknown number. In particular, an asserted universal bound such as $\tau_R \leq 2$ does not follow from the pigeonhole principle. Any elimination estimate that uses such a bound must state it as an additional hypothesis, not as a completed obstruction.

3. **Three nonvanishing statements differ.** An integer coefficient vector, its primitive target, and the evaluated linear form can vanish independently at successive stages. Positivity is a value certificate only when it applies to the actual final form. Siegel or Minkowski bounds on a kernel vector do not alone locate that vector in a positive cone or outside the target kernel.

4. **Finite improvements have not been extrapolated.** The finite 103/100 exponent, interval contractions, numerical asymptotic experiments, and exploratory large-N thresholds are retained with their actual status. A fixed-margin infinite family with unbounded primitive denominator is not inferred from a finite run. Explicit horizon claims require effective inequalities and constants, not only an asymptotic comparison.

5. **Target height is used after reduction.** A large common denominator or a large representation coefficient is not by itself a lower bound for reduced rational height or minimal-polynomial height. Upper bounds, lower bounds, and exact asymptotics are distinguished throughout the moment, Todd, and moving-parameter constructions.

6. **A strict Stokes transition is distinguished from a comparison entry.** The elementary lateral jump of the exponential integral is determined by $2\pi i$. The appearance of γ in a regular-to-asymptotic comparison requires the relevant normalizations or framing. The phrase "γ is a Stokes constant" is not used to infer an arithmetic theorem for the unframed jump matrix.

7. **Finite camouflage uses the sharp degree.** The earlier degree p^L construction is replaced in the operative statement by the minimum $\mu(p^L) = \min\{m : p^L \mid m!\}$. The older construction remains a valid, nonoptimal example in its source record. The exact root-count theorem concerns the relation ideal, not a selected polynomial with an unrelated algebraic root.

8. **The four transfer mechanisms retain their corrected hypotheses.** Approximation denominators are positive; the statistical polynomial has exact degree $d \geq 1$; the cleared second-moment polynomial has degree *at most* d; factorial-series sign and argument conventions are fixed explicitly. Mixed divisibility means existence of a series in the stated class satisfying the product identity. No closure under analytic division is silently added.

These corrections do not turn an unfinished transcendence proof into a finished one. They

ensure that the positive results and the limits of each construction can be assessed without inheriting an earlier overstatement.

61 Coverage and reproducibility

The companion Korean guide contains a file-by-file coverage table. Each entry identifies its destination in this manuscript or the preserved source calculation and records whether the content is proved, conditional, finite, experimental, or superseded. The accompanying archive contains:

1. this single-file LaTeX source and its compiled PDF;
2. the Korean reading guide, source-claim ledgers, and a machine-readable manifest with SHA-256 digests;
3. all 59 identified source files unchanged, including the 27 original research ZIP packages, their proof notes, programs, and certificates;
4. the editorial section fragments and assembly script used for this edition, together with a record of compilation and document checks.

The new LaTeX file compiles independently: it requires no external image, `input` file, or bibliography database. To rebuild it with standard TeX Live packages, run `pdflatex` repeatedly until the references and table of contents stabilize. To reproduce an older numerical certificate, first unpack its named source ZIP and follow that package's README; paths in the preserved mathematical parts are relative to those unpacked packages. An old verification record is identified as inherited evidence, not described as a new execution performed for this edition.

The integration checks cover source inventory, section and citation resolution, formula layout, and targeted mathematical comparisons. They do not constitute independent external peer review of every theorem in every unpublished source package. In particular, preserving a source file in the archive does not endorse its superseded wording. The corrected statements and explicit status labels in this manuscript govern the cumulative edition.

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